

DYNAMIC GROUND WATER RESOURCES OF ANDHRA PRADESH STATE 2024



Central Ground Water Board
*Department of Water Resources,
River Development & Ganga Rejuvenation
Ministry of Jal Shakti
Government of India*

Visakhapatnam
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Message of the Concerned Member

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MESSAGE

Groundwater plays a crucial role in sustaining Andhra Pradesh's agricultural, domestic, and industrial needs. Given the state's dependence on monsoonal recharge, periodic assessments are necessary to ensure long-term sustainability. The 2024 groundwater resource assessment, carried out by the Central Ground Water Board, SR & SUO, Visakhapatnam, in collaboration with the State Ground Water and Water Audit Department (GW & WAD), Government of Andhra Pradesh, provides valuable insights into the state's groundwater scenario. Using the GEC 2015 methodology, this study estimates Andhra Pradesh annual groundwater recharge at 27.8 bcm, with an annual extractable groundwater resource of 26.41 bcm. The total annual extraction has been recorded at 7.88 bcm, resulting in a statewide stage of groundwater extraction of 29.83%.

An evaluation of 679 assessment units (Mandals/Blocks) indicates that 9 units (1.32%) fall under the 'Over-exploited' category, while 2 units (0.29%) are classified as 'Critical,' 38 units (5.59%) as 'Semi-critical,' and 39 units (5.74%) as 'Saline.' The remaining 591 units (87.03%) are categorized as 'Safe,' highlighting the variation in groundwater availability across different regions. Furthermore, an analysis of total annual extractable groundwater resources (26.41 bcm) shows that 0.262 bcm (0.9%) is from Over-exploited units, 0.066 bcm (0.25%) from Critical units, 0.897 bcm (3.4%) from Semi-critical units, while 25.06 bcm (94.87%) is within Safe zones, and 2.301 bcm (8.05%) is from Saline units.

I extend my appreciation to Shri G. Krishnamurthy, Regional Director, CGWB, SR Hyderabad, and Shri Lakshmi Narayana Damodara, Scientist-C & OIC, SUO, Visakhapatnam, and their dedicated team for their pivotal role in groundwater resource assessment. I also commend Shri M. John Satya Raju, Director, GW & WAD, Ground Water Department, and Shri N. Srinivasu, Joint Director, GW & WAD, Ground Water Department, Government of Andhra Pradesh, and their team for their tireless efforts in conducting groundwater assessments in the village level. The State Level Committee (SLC), under the chairmanship of the Special Chief Secretary, Water Resources Department, Government of Andhra Pradesh, has meticulously reviewed and approved this assessment. The guidance and leadership of Shri G. Sai Prasad, IAS, Special Chief Secretary, Water Resources Department, Government of Andhra Pradesh, have been instrumental in the successful completion of this assessment.

I hope this compilation will serve as a vital resource for planners, decision-makers, and stakeholders, supporting sustainable groundwater management and conservation efforts.


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Preface

Groundwater is a vital resource for Andhra Pradesh, supporting agriculture, domestic use, and industry. However, uncontrolled exploitation of this resource, particularly in deltaic and hard rock regions. This warrants us to estimate precisely the ground water resources available for better management of the resources to sustainably meet the ever increase in its demand with the growth of population, increase in agriculture and modernization and enlargement of industrial sectors. To effectively quantify groundwater resources, it is imperative to periodically assess groundwater availability. In line with this objective, the Central Ground Water Board, SR & SUO, Visakhapatnam and the State Ground Water Department and Water Audit Department (GW & WAD), Government of Andhra Pradesh, undertook the task of estimating the dynamic groundwater resources of Andhra Pradesh State for the year 2024. This assessment was conducted using the GEC 2015 methodology.

The 2024 assessment of ground water resources in Andhra Pradesh state reveals that the annual ground water recharge is 27.8 bcm, with an annual extractable ground water resource of 26.41 bcm. The annual groundwater extraction reaches 7.88 bcm, with an average stage of ground water extraction of the state is 29.83%. Out of the total 679 assessment units (Blocks/ Mandals/), 09 (1.33%) has been categorized as 'Over-exploited', 2 (<1%) as 'Critical', 38 (5.60%) as 'Semi-critical', 591 (87%) units as 'Safe' and 39 (5.74%) units as Saline. Also out of 26.41 bcm of Total Annual Extractable Resources of the Andhra State, 0.262 bcm (0.9 %) are under 'Over-Exploited', 0.066 bcm (0.25%) are under 'Critical', 0.897 bcm (3.4 %) are under 'Semi-Critical', 25.06 bcm (94.87%) are under 'Safe' and 2.301 bcm (8.05%) are from saline units.

The estimates were reviewed and approved by the State Level Committee, chaired by the Secretary, Sri. G. Sai Prasad, IAS, Water Resources Department (WRD), Government of Andhra State. The dedicated efforts of the team, led by Sri. M. John Satya Raju, Director, and Sri. N Srinivasu, Joint Director, GW & WA Department, Government of Andhra State, are highly commendable. The contributions of CGWB SUO Visakhapatnam officers in conducting this assessment as per the time frame fixed by MOJS are deeply appreciated. Furthermore, the constant guidance and support provided by Shri G. Sai Prasad, IAS, Chief Secretary, Water Resources Department (WRD), Government of Andhra State, are gratefully acknowledged. It is my sincere hope that this report will serve as a valuable resource for administrators, planners, and policymakers, enabling informed decision-making for the planning and regulation of groundwater schemes and the sustainable management of this vital resource.


G. Krishnamurthy
Regional Director &
Member Secretary SLC

Chapter	Contents	Page
	EXECUTIVE SUMMARY	
1	INTRODUCTION	1-4
	1.1 Overview of Groundwater Resource Assessment in India	1
	1.2 Overview of Groundwater Resource Assessment in Andhra Pradesh State	2
	1.2.1 Groundwater Resource Assessment in Andhra Pradesh State-2024	2-4
2	GROUND WATER RESOURCE ESTIMATION METHODOLOGY	5-25
	2.1 Groundwater assessment of unconfined aquifer	5
	2.1.1 Assessment of Annually Replenishable or Dynamic Ground Water Resources	5
	2.1.1.1 Rainfall Recharge	6
	2.1.1.1.1 Ground Water Level Fluctuation Method	6-8
	2.1.1.1.2 Rainfall Infiltration Factor Method	8
	2.1.1.1.3 Percent Deviation	8-9
	2.1.1.2 Recharge from Other Sources	9-10
	2.1.1.3 Evaporation and Transpiration	10
	2.1.1.4. Recharge During Monsoon Season	10
	2.1.1.5. Recharge During Non-Monsoon Season	10
	2.1.1.6. Total Annual Ground Water Recharge	10
	2.1.1.7 Annual Extractable Ground Water Resource (EGR)	10
	2.1.1.8 Estimation of Ground Water Extraction	11-12
	2.1.1.9. Stage of Ground Water Extraction	12
	2.1.1.10 Validation of Stage of Ground Water Extraction	12
	2.1.1.11. Categorization of Assessment Unit	12-13
	2.1.1.12 Allocation of Ground Water Resource for Utilisation	13
	2.1.1.13 Net Annual Ground Water Availability for Future Use	13-14
	2.1.1.14 Apportioning of Ground Water Assessment from Watershed to Development Unit	14-15
	2.2 Norms has been used in the assessment	15
	2.2.1 Specific Yield	15-19
	2.2.2 Rainfall Infiltration Factor	19-23
	2.2.3 Norms for Canal Recharge	23-24
	2.2.4 Norms for Recharge Due to Irrigation	24
	2.2.5 Norms for Recharge due to Tanks & Ponds	24
	2.2.6 Norms for Recharge due to Water Conservation Structures	24
	2.2.7 Unit Draft	25
	2.3 India -groundwater resource estimation system (in-gres)	25
3	RAINFALL	26-31
	3.1 Climate	26
	3.2 State Normal Rainfall and Actual Rainfall	26
	3.3 District wise Normal and Actual Rainfall	27
	3.3.1 District wise Monsoon and Non-monsoon period rainfall during Assessment Year 2023-24	30

4	HYDROGEOLOGICAL SETUP OF ANDHRA PRADESH STATE		32-34
	4.1	Hydrogeological Setup of Andhra Pradesh State	32
	4.1.1	Archaean to Lower Proterozoic	33
	4.1.2	Middle to Upper Proterozoic	33
	4.1.3	Gondwana Super Group	33
	4.1.4	Deccan Traps	33
	4.1.5	Tertiaries	33
	4.1.6	Quaternary	33
	4.2	Hydrogeology	34
5	GROUND WATER LEVEL SCENARIO IN THE ANDHRA PRADESH STATE		35-40
	5.1	Groundwater level data for Pre- Monsoon 2023	35
	5.1.1	Depth to Water Levels (Unconfined & Confined/Semi-Confined)	35
	5.2	Groundwater level data for post-monsoon 2023	35
	5.2.1	Depth to Water Levels (Unconfined & Confined/Semi-Confined)	35
	5.3	Fluctuation of Groundwater Level	36
	5.3.1	Comparison of Pre-monsoon 2023 to Pre-monsoon 2022	36
	5.3.2	Comparison of November 2023 to November 2022	36
	5.3.3	Comparison of Pre-Monsoon 2023 with decadal mean of Pre-Monsoon (2013 to 2022)	36
	5.3.4	Comparison of post-monsoon 2023 with decadal mean of post-monsoon (2013 to 2022)	37
6	GROUND WATER RESOURCES OF THE ANDHRA PRADESH STATE		41-45
	6.1	Annual ground water recharge	41
	6.2	Annual extractable ground water resources	41
	6.3	Annual total ground water extraction	41
	6.4	Stage of ground water extraction	42
	6.5	Categorization of assessment units	43
	6.6	Comparison with previous assessment	43
7	DISTRICT WISE DYNAMIC GROUNDWATER RESOURCES-2024		47-72
8	CONCLUSIONS		73-74
References			
List of Tables			
	1.1	Assessment year wise categorization of number of Mandals of Andhra Pradesh	2
	1.2	Categorization of different assessment units of Andhra Pradesh-GWRA-2024	3
	1.3	Ground water Resources assessment in Andhra Pradesh State in 2017 to 2024	3
	1.4	Ground water Resources assessment in Andhra Pradesh State in 2017 to 2024	4
	2.1	Norms Recommended for Specific Yield	15-19
	2.2	Norms Recommended for Rainfall Infiltration Factor	19-23
	2.3	Norms Recommended for Recharge due to Canals	23
	2.4	Norms Recommended for Recharge from Irrigation	24

	3.1	District-wise monthly and annual normal and actual rainfall data	28-29
	3.2	District-wise Monsoon and Non-monsoon actual and normal rainfall during Assessment Year 2023	30
	4.1	Geological Succession in Andhra Pradesh	32
List of Figures			
	1.1	Ground water Resources assessment in Andhra Pradesh State in 2017 to 2024	4
	3.1	Isohyetal map of Andhra Pradesh State	27
	3.2	District-wise annual actual and normal rainfall for Andhra Pradesh State for the year 2023	27
	3.3	District-wise Monsoon and Non-monsoon rainfall during Assessment Year 2023-2024	31
	4.1	Geology and Principal Aquifers of Andhra Pradesh	33
	5.1	Depth to Water Level Map of the State/UT (Pre-Monsoon 2023)	37
	5.2	Depth to Water Level Map of the State/UT (Post Monsoon 2023)	38
	5.3	Groundwater Level Fluctuation: Pre-monsoon 22 compared to Pre-monsoon 23	38
	5.4	Groundwater Level Fluctuation: November 22 compared to November 23	39
	5.5	Decadal water level fluctuation with mean Pre-Monsoon (2013 to 22) and Pre-Monsoon 23	39
	5.6	Decadal water level fluctuation with mean post-monsoon (2013 to 22) and post-monsoon 23	40
	6.1	District Wise Ground Water Recharge & Extraction (GWRA 2024)	42
	6.2	District Wise Stage of Ground Water Extraction (GWRA 2024)	42
	6.3	No of Mandals/Blocks in different categories GWRA-2024	43
	6.4	Dynamic Ground Water Resource Assessment 2023 to 2024	43
	6.5	Annual Ground Water Recharge 2024 of Andhra Pradesh (Unit Map in m)	44
	6.6	Ground Water Recharge Distribution (MCM), GWRA-2024	44
	6.7	Annual Ground Water Extraction 2024 of Andhra Pradesh (Unit Map in m)	45
	6.8	Ground Water Extraction Distribution, GWRA-2024	45
	6.9	Block wise Categorization of Assessment Unit in Andhra Pradesh state	46
	7.1-7.26	District Wise Dynamic Groundwater Resources-2024	47-72
List of Annexures			
Annexure-I	Ground water resources availability, utilization and stage of extraction (as in 2024)		75
Annexure-II	District-wise ground water resources availability, utilization and stage of extraction (as in 2024)		76
Annexure-III (A)	Categorization of blocks/ mandals/ taluks in Andhra Pradesh State (as in 2024)		77
Annexure III (B)	District Wise Categorization of blocks/ mandals/ taluks for the State/UT (as in 2024)		78
Annexure III (C)	Annual Extractable Ground Water Resource of Assessment Units under Different Category for the State/UT (as in 2024)		79
Annexure- III (D)	District Wise Annual Extractable Ground Water Resource of Assessment Units		80

	under Different Category for the State	
Annexure- III (E)	Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)	81
Annexure III (F)	District Wise Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)	82
Annexure IV (A)	Categorization of Over Exploited, Critical and Semi Critical blocks/ mandals/ taluks (as in 2024)	83-84
Annexure IV (C)	Saline Assessment units (as in 2024)	85
Annexure V (A)	Summary of Assessment units improved or deteriorated from 2023 to 2024 assessment	86
Annexure V (B)	Comparison of categorization of assessment units (2023 to 2024)	87-88
Annexure VI	Assessment Unit Wise Report (Attribute Table)	89-115
Annexure VII	Minutes of the Meeting of the SLC Committee	116-117

DYNAMIC GROUND WATER RESOURCES OF ANDHRA PRADESH STATE 2024

AT A GLANCE

1.	Total Annual Ground Water Recharge (in bcm)	27.8
2.	Annual Extractable Ground Water Resources (in bcm)	26.41
3.	Annual Ground Water Extraction (in bcm)	7.88
4.	Stage of Ground Water Extraction (in bcm)	29.83

CATEGORIZATION OF ASSESSMENT UNITS

(Blocks/ Mandals/ Taluks)

Sl.No	Category	Number of Assessment Units		Recharge worthy Area		Annual Extractable Ground Water Resource	
		Number	%	in lakh sq. km	%	(in bcm)	%
1	Safe	591	87.03	123819.54	88.2	25.05	94.87
2	Semi Critical	38	5.59	7947.83	5.66	0.89	3.4
3	Critical	02	0.29	236.68	0.17	0.06	0.25
4	Over-Exploited	09	1.32	2278.23	1.62	0.26	0.99
5	Saline	39	5.74	6096.33	4.34	2.30	8.05
	TOTAL	679	100	140378.61	100	28.58	

EXECUTIVE SUMMARY

Ground Water Resource Assessment in Andhra Pradesh is conducted periodically by the State Ground Water Department in collaboration with the Central Ground Water Board under the Ministry of Jal Shakti. Since 2022, this assessment has been carried out annually, following the Ground Water Estimation Committee (GEC) 2015 Methodology using INGRES software. The latest assessment for the base year 2023-24 provides critical insights into groundwater availability and utilization across the state.

Andhra Pradesh, the 7th largest state in India, has a 974 km coastline and a geographical area of 1.63 lakh km², comprising 26 districts, 679 mandals, and 17,467 villages. With a population of ~4.96 crores, the state is primarily drained by the Godavari, Krishna, and Pennar rivers. It receives an average annual rainfall of 966 mm, ranging from 552 mm in Anantapur to 1217 mm in East Godavari. Geographically, AP has three zones: coastal plains, eastern ghats, and western pediplains. Of the total 162.76 lakh ha, 54% (87.90 lakh ha) is cultivable, and 22% (36.63 lakh ha) is forested. The cropped area is 74.51 lakh ha, with 22.37 lakh ha irrigated by surface water and 15.75 lakh ha (42%) by groundwater, leaving an ayacut gap of 36.38 lakh ha.

The state has 14 principal aquifer systems, with 85% underlain by hard rock formations like granites, gneiss, and metamorphics (Archaean), quartzites and limestones (Proterozoic), and Deccan Trap basalts (Mesozoic). The remaining area consists of Gondwana sedimentaries (Tertiary) and alluvium (Recent).

Andhra Pradesh is divided into 40 drainage basins and 74 sub-basins, encompassing both major and minor rivers. These 74 sub-basins are further segmented into 748 micro-basins, each ranging from 100 to 300 square kilometers, based on drainage patterns, geomorphology, and hydrogeology for groundwater resource estimation. The assessment covers all 748 basins, 679 mandals, and 17,467 villages, with villages serving as the fundamental unit of evaluation. Apart from Telangana, Andhra Pradesh is the only state conducting this assessment at the village level, whereas other states limit their evaluations to the Block, Firka, or Mandal level.

The 2024 assessment of dynamic groundwater resources in Andhra Pradesh estimates that the state's total annual groundwater recharge is 27.8 (bcm). Of this, 1.39 bcm is allocated for natural discharge, leaving 26.41 bcm as the annual extractable groundwater resource. Recharge from other sources, including canal seepage, return flow from irrigation, recharge from tanks, ponds, and water conservation structures, contributes 17.97 bcm (65%) of the total recharge. The remaining 9.82 bcm (35%) comes from rainfall. Seasonal variations indicate that monsoon recharge accounts for 39% and non-monsoon recharge for 26% from other sources, while rainfall recharge is 32% in the monsoon season and 3% in the non-monsoon period.

The total annual groundwater extraction in 2024 is estimated at 7.88 bcm. Irrigation accounts for 86% (6.75 bcm) of this extraction, domestic use accounts for 13% (1.0 bcm), and industrial use represents 1% (0.13 bcm). In districts such as Anaparthi, Chittoor, Eluru, Kurnool, Palnadu, Prakasam, Sri Potti Srimulolu Nellore, Sri Sathya Sai, Srikakulam, and Y.S.R Kadapa, groundwater extraction for domestic use exceeds 65%.

The overall stage of groundwater extraction in Andhra Pradesh is 29.83%, ranging from 1.5% in Alluri Sitharama Raju to 60.7% in Sri Sathya Sai. Across all 26 districts, groundwater extraction remains below 70%.

Among the 679 assessment units (mandals), 9 units (1.32%) are classified as 'Over-exploited', 2 units (0.29%) as 'Critical', 38 units (5.59%) as 'Semi-Critical', 591 units (87.03%) as 'Safe', and 39 units (5.74%) as 'Saline'.

Regarding spatial distribution, out of 140,378.61 sq km of recharge-worthy area, 1.62% (2,278.23 sq km) falls under the 'Over-Exploited' category, 0.17% (236.68 sq km) under 'Critical', 5.66% (7,947.83 sq km) under 'Semi-Critical', 88.2% (123,819.54 sq km) under 'Safe', and 4.34% (6,096.33 sq km) under 'Saline'.

Out of total 26.41bcm annual extractable ground water resources of the State, 0.26 bcm (0.99%) are under 'Over-exploited', 0.066 bcm (0.25 %) under 'Critical', 0.89 bcm (3.4 %) under 'Semi-critical' and 25.05 bcm (94.87 %) are under 'Safe' categories of assessment units.

A comparison with 2023 shows a negligible reduction in total annual groundwater recharge, decreasing from 27.83 bcm to 27.80 bcm (0.03 bcm decrease). This decline is attributed to reduced recharge from surface water sources and increased groundwater extraction. This assessment provides critical insights for sustainable groundwater management and policy formulation in Andhra Pradesh.

CHAPTER 1

Introduction

Water is an essential resource for life, and groundwater has become a critical component in meeting the freshwater needs of various sectors in India. Its sustainable development and efficient management present significant challenges. Groundwater serves as the backbone of India's agriculture and drinking water systems, contributing approximately 62% to irrigation, 85% to rural water supply, and 50% to urban water supply. While groundwater is replenished annually, primarily through rainfall, its availability varies significantly across regions and over time. The groundwater in the zone of water level fluctuation is recharged annually, underscoring the need for a realistic and scientifically sound quantitative assessment of this resource for sustainable utilization.

The **National Water Policy, 2012**, emphasizes the periodic scientific assessment of groundwater resources. It also highlights the need to consider trends in water availability influenced by various factors, including climate change, as part of water resource planning. To address the growing demand for water, the policy advocates for the direct use of rainfall, desalination, and minimizing inadvertent evapotranspiration to enhance utilizable water resources.

Moreover, the policy underscores that safe water for drinking and sanitation should be prioritized above all other needs. Following this, high priority should be given to other domestic requirements (including those for animals), ensuring food security, supporting subsistence agriculture, and meeting minimum ecosystem demands. Any remaining water should then be allocated in ways that encourage conservation and efficient usage.

1.1 Overview of Groundwater Resource Assessment in India

The assessment of India's water resources began in 1901 with the First Irrigation Commission estimating surface water resources at 144-million-hectare meters (M.ham). By 1949, Dr. A. N. Khosla expanded the scope to include both surface and groundwater resources, estimating the total average annual runoff at 167 M.ham. Over time, various committees and task forces were constituted to refine these assessments in response to growing developmental needs, reflecting the increasing importance of groundwater as a critical resource for agriculture, drinking water, and industrial use.

Significant progress was made in 1976 when the National Commission on Agriculture assessed total groundwater resources at 67 M.ham, with 35 M.ham identified as utilizable, including 26 M.ham for irrigation. A milestone was achieved in 1979 with the Ground Water Over-Exploitation Committee, which developed the first systematic methodology for groundwater assessment. This effort estimated Gross Ground Water Recharge at 47 M.ham and Net Recharge at 32 M.ham. The Ground Water Estimation Committee (GEC), constituted in 1982, introduced the dynamic GEC 1984 methodology, which became the basis for a comprehensive national groundwater assessment published in 1995. The report estimated India's Total Replenishable Ground Water at 432 billion cubic meters (bcm), with 361 bcm available for irrigation and a groundwater development level of 32%.

Subsequent revisions further refined these methodologies. The GEC 1997 methodology addressed challenges such as groundwater estimation in hard rock terrains. Recognizing the need for modern approaches, the GEC 2015 methodology introduced more robust, scientific techniques for dynamic groundwater assessments. Since 2017, this revised methodology has been the standard for evaluating groundwater resources, reflecting

advancements in data acquisition and analysis while addressing emerging challenges like climate change and sustainable water management.

1.2 Overview of Groundwater Resource Assessment in Andhra Pradesh State

Andhra Pradesh, formed on June 2, 2014, after the bifurcation of Andhra Pradesh, began its journey as an independent state with distinct challenges in managing natural resources, particularly groundwater. Groundwater plays a critical role in meeting the state's agricultural, domestic, and industrial water needs. Recognizing its importance, Andhra Pradesh initiated systematic assessments of its groundwater resources to ensure sustainable utilization and management.

1.2.1 Groundwater Resource Assessment in Andhra Pradesh State-2024

Groundwater plays a crucial role in agriculture in Andhra Pradesh, accounting for 41% of the state's gross irrigated area, supported by approximately 19 lakh agricultural bore wells. The Ground Water and Water Audit Department of the Government of Andhra Pradesh, in collaboration with the Central Ground Water Board under the Ministry of Jal Shakti, Government of India, conducts periodic technical assessments of groundwater resources. In line with the National Water Policy, these assessments were carried out in 2004, 2008, 2011, 2013, 2017, 2019, 2022, and 2023 across all states, including Andhra Pradesh, following the guidelines and methodologies issued by the Ministry of Jal Shakti. The Ministry has now proposed a re-estimation for the base year 2023-24. Assessment year wise categorization of number of Mandals of Andhra Pradesh given in table no 1.1.

Table No 1.1 Assessment year wise categorization of number of Mandals of Andhra Pradesh.

S. No	Category	Assessment year wise number of Mandals										Change (2023-24 vs 2022-23)
		2004-05	2006-07	2008-09	2010-11	2012-13	2016-17	2019-20	2021-22	2022-23	2023-24	
1	Over Exploited	211	111	84	41	61	45	23	6	10	9	-1
2	Critical	71	60	26	7	17	24	15	5	3	2	-1
3	Semi Critical	167	160	93	42	54	60	40	19	18	38	20
4	Safe	221	339	467	580	538	541	592	640	639	630	-9
Total		670	670	670	670	670	670	670	670	670	679	

The state is divided in to 40 drainage basins and 74 sub basins of major and minor rivers. These 74 sub basins are further sub divided into 748 micro basins of 100 to 300 square kilometers size based on drainage, geomorphology and hydrogeology for the preparation of Groundwater Resources Estimation. Assessment is being done for all the 748 basins, 679 mandals and 17467 villages taking village as the basic assessment unit. Andhra Pradesh is the only State other than Telangana doing this exercise taking village as basic unit of assessment, remaining states are restricted the assessment to Block/Firka/Mandal level only. Categorization of different assessment units of Andhra Pradesh-GWRA-2024 is given below table no 1.2.

The assessment involves computation of Dynamic Groundwater Resources or Annual Extractable Ground Water Resource, Annual Ground Water Extraction (utilization) and the percentage of utilization (stage of Ground Water Extraction) with respect to annual extractable resources.

The total annual groundwater recharge is estimated at 27.8 billion cubic meters (bcm), with extractable resources at 26.41 bcm. The total groundwater extraction stands at 7.88 bcm, with irrigation accounting for 86% (6.75 bcm), domestic use 13% (1.01 bcm), and industrial use 1% (0.13 bcm). The stage of groundwater development is 29.8%, with 37.40% in non-command areas and 22.58% in command areas. Out of 679 assessed mandals, 9 (1.32%) are categorized as over-exploited, 2 (0.29%) as critical, 38 (5.59%) as semi-critical, 591 (87.03%) as safe, and 39 (5.74%) as saline. In terms of recharge-worthy area, 2,278.23 sq km (1.62%) falls under over-exploited, 236.68 sq km (0.17%) under critical, 7,947.83 sq km (5.66%) under semi-critical, 123,819.54 sq km (88.2%) under safe, and 6,096.33 sq km (4.34%) under saline categories.

Table No 1.2 Categorization of different assessment units of Andhra Pradesh-GWRA-2024

Categorization of different assessment units of Andhra Pradesh-GWRA-2024			
Category	Villages	Mandals	Basins
Over Exploited	830	9	20
Critical	450	2	7
Semi Critical	1271	38	33
Safe	14020	591	656
Saline	896	39	32
Total	17467	679	748
GW Stress units	2551	49	60

The assessment units are categorized based on Stage of Ground Water Extraction, which are then validated with long-term water level trends. The resources are assessed following the Groundwater Estimation Methodology,2015 (GEC-2015) using INGRES software. The recent assessment for the base year 2023-24 has been completed and abstract of the results are given below in table no 1.3.

Table No 1.3 Ground water Resources assessment in Andhra Pradesh State 2024

Total annual GW recharge from all sources	:	2779945 Ham / 27,799 MCM / 982 TMC / 27.8 Bcm
Total Extractable GW resources (Available resource)	:	2641144 Ham / 26,411 MCM / 932 TMC/ 26.41 Bcm
Total GW extraction for all uses	:	787981 Ham / 7,879 MCM / 278 TMC / 7.88 Bcm
Stage of GW development	:	29.8 %
(37.40 % in non-command and 22.58 % in Command)		

The table 1.4 & Fig 1.1 presents groundwater assessment data for the years 2017, 2020, 2022, 2023, and 2024. Total Annual Groundwater Recharge has shown a steady increase from 21.22 BCM in 2017 to 27.8 BCM in 2024, indicating improved replenishment. Similarly, Annual Extractable Groundwater Resources have risen from 20.15 BCM in 2017 to 26.41 BCM in 2024. However, Annual Groundwater Extraction has fluctuated, decreasing from 8.90 BCM in 2017 to 7.45 BCM in 2022, followed by a slight rise to 7.88 BCM in 2024. The steady improvement in groundwater availability and the reduction in extraction levels highlight Andhra Pradesh's proactive approach to sustainable resource management. These independent assessments reinforce the state's dedication to scientific and adaptive groundwater management, ensuring long-term sustainability in meeting its water needs.

Table No 1.4: Ground water Resources assessment in Andhra Pradesh State in 2017 to 2024

Year of Assessment	2017	2020	2022	2023	2024
Component (BCM)					
Total Annual GW Recharge	21.22	24.15	27.22	27.81	27.8
Annual Extractable GW Resources	20.15	22.94	25.86	26.42	26.41
Annual GW Extraction	8.90	7.6	7.45	7.48	7.88
Stage of GW Extraction (%)	44.15	33.13	28.81	28.31	29.84

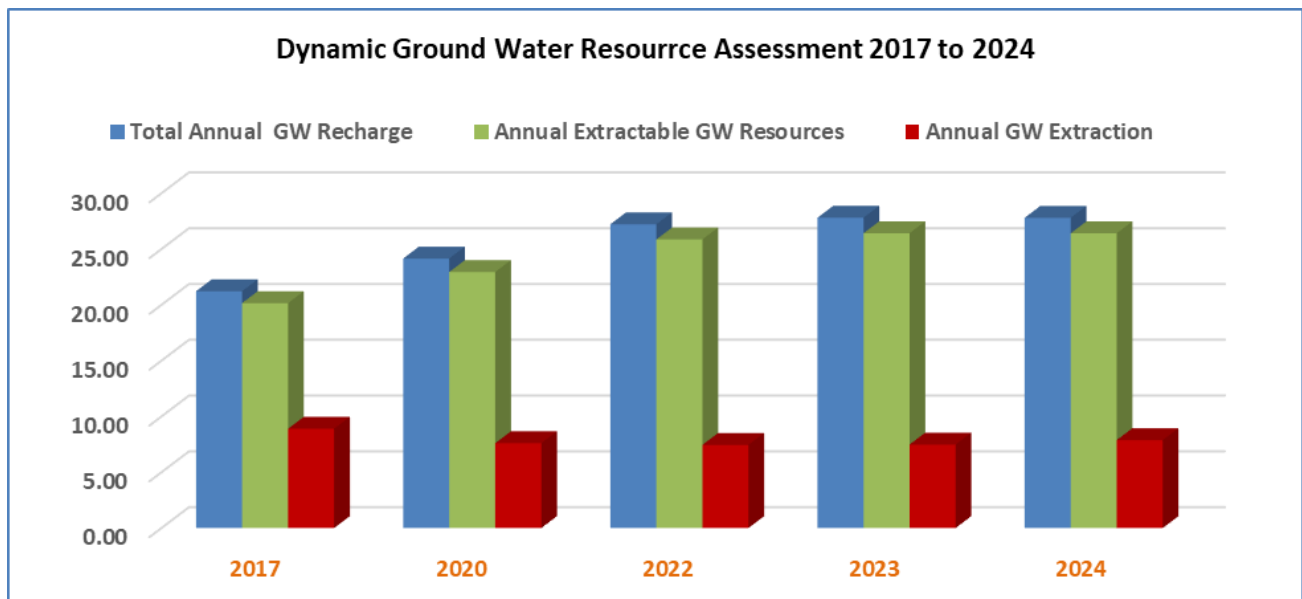


Fig No 1.1 Ground water Resources assessment in Andhra Pradesh State in 2017 to 2024

CHAPTER 2

GROUND WATER RESOURCE ESTIMATION METHODOLOGY

Ground water resource as in 2024 have been estimated following the guidelines mentioned in the GEC 2015 methodology using appropriate assumptions depending on data availability. The principal attributes of GEC 2015 methodology are given below:

It is also important to add that as it is advisable to restrict the groundwater development as far as possible to annual replenishable resources, the categorization also considers the relation between the annual replenishment and groundwater development. An area devoid of ground water potential may not be considered for development and may remain safe whereas an area with good groundwater potential may be developed and may become over exploited over a period. Thus, water augmentation efforts can be successful in such areas, where the groundwater potential is high and there is scope for augmentation.

2.1. GROUND WATER ASSESSMENT OF UNCONFINED AQUIFER

Though the assessment of ground water resources includes assessment of dynamic and in-storage resources, the development planning should mainly focus on dynamic resource as it gets replenished on an annual basis. Changes in static or in-storage resources normally reflect long-term impacts of ground water mining. Such resources may not be replenishable annually and may be allowed to be extracted only during exigencies with proper planning for augmentation in the succeeding excess rainfall years.

2.1.1. Assessment of Annually Replenishable or Dynamic Ground Water Resources

The methodology for ground water resources estimation is based on the principle of water balance as given below

$$\text{Inflow} - \text{Outflow} = \text{Change in Storage (of an aquifer)} \dots \dots \dots (1)$$

Equation (1) can be further elaborated as –

$$\Delta S = R_{RF} + R_{STR} + R_C + R_{SWI} + R_{GWI} + R_{TP} + R_{WCS} \pm VF \pm LF - GE - T - E - B \dots \dots \dots (2)$$

Where,

- ΔS - Change in storage
- R_{RF} - Rainfall recharge
- R_{STR} - Recharge from stream channels
- R_C - Recharge from canals
- R_{SWI} - Recharge from surface water irrigation
- R_{GWI} - Recharge from ground water irrigation
- R_{TP} - Recharge from Tanks & Ponds
- R_{WCS} - Recharge from water conservation structures
- VF - Vertical flow across the aquifer system
- LF - Lateral flow along the aquifer system (through flow)
- GE - Ground Water Extraction
- T - Transpiration
- E - Evaporation
- B - Base flow

Due to lack of data for all the components in most of the assessment units, at present the water budget has been assessed based on major components only, taking into consideration certain reasonable assumptions. The estimation has been carried out using lumped parameter estimation approach keeping in mind that data from many more sources if available may be used for refining the assessment.

2.1.1.1. Rainfall Recharge

Ground water recharge has been estimated on ground water level fluctuation and specific yield approach since this method considers the response of ground water levels to ground water input and output components. In units or subareas where adequate data on ground water level fluctuations are not available, ground water recharge is estimated using rainfall infiltration factor method only. The rainfall recharge during non-monsoon season has been estimated using rainfall infiltration factor method only.

2.1.1.1.1. Ground Water Level Fluctuation Method

The ground water level fluctuation method is used for assessment of rainfall recharge in the monsoon season. The ground water balance equation in non-command areas is given by

$$\Delta S = R_{RF} + R_{STR} + R_{SWI} + R_{GWI} + R_{TP} + R_{WCS} \pm VF \pm LF - GE - T - E - B \dots \dots \dots (3)$$

Where,

- ΔS - Change in storage
- R_{RF} - Rainfall recharge
- R_{STR} - Recharge from stream channels
- R_{SWI} - Recharge from surface water irrigation
- R_{GWI} - Recharge from ground water irrigation
- R_{TP} - Recharge from Tanks & Ponds
- R_{WCS} - Recharge from water conservation structures
- VF - Vertical flow across the aquifer system
- LF - Lateral flow along the aquifer system (through flow)
- GE - Ground water extraction
- T - Transpiration
- E - Evaporation
- B - Base flow

Whereas the water balance equation in command area have another term i.e., Recharge due to canals (R_C) and the equation is as follows:

$$\Delta S = R_{RF} + R_{STR} + R_C + R_{SWI} + R_{GWI} + R_{TP} + R_{WCS} \pm VF \pm LF - GE - T - E - B \dots \dots \dots (4)$$

The change in storage has been estimated using the following equation:

$$\Delta S = \Delta h \times A \times S_y \dots \dots \dots (5)$$

Where,

- ΔS - Change in storage
- Δh - rise in water level in the monsoon season
- A - Area for computation of recharge
- S_y - Specific Yield

Substituting the expression in equation (5) for storage increase ΔS in terms of water level fluctuation and specific yield, the equations (3) & (4) becomes (6) & (7) for non-command and command sub-units,

$$R_{RF} = \Delta h \times A \times S_Y - R_{STR} - R_{SWI} - R_{GWI} - R_{TP} - R_{WCS} \pm VF \pm LF + GE + T + E + B \dots \dots \dots (6)$$

$$R_{RF} = \Delta h \times A \times S_Y - R_{STR} - R_C - R_{SWI} - R_{GWI} - R_{TP} - R_{WCS} \pm VF \pm LF + GE + T + E + B \dots \dots \dots (7)$$

Where base flow/ recharge to/from streams have not been estimated, the same is assumed to be zero. The rainfall recharge obtained by using equation (6) and (7) provides the recharge in any particular monsoon season for the associated monsoon season rainfall. This estimate has been normalized for the normal monsoon season rainfall as per the procedure indicated below.

Normalization of Rainfall Recharge

Let R_i be the rainfall recharge and r_i be the associated rainfall. The subscript "i" takes values 1 to N where N is the number of years for which data is available. This should be at least 5. The rainfall recharge, R_i is obtained as per equation (6) & equation (7) depending on the sub-unit for which the normalization is being done.

After the pairs of data on R_i and r_i have been obtained as described above, a normalisation procedure is carried out for obtaining the rainfall recharge corresponding to the normal monsoon season rainfall. Let $r(\text{normal})$ be the normal monsoon season rainfall obtained as the average of recent 30 to 50 years of monsoon season rainfall. Two methods are possible for the normalisation procedure. The first method is based on a linear relationship between recharge and rainfall of the form

$$R = ar \dots \dots \dots (8)$$

Where,

R = Rainfall recharge during monsoon season

r = Monsoon season rainfall

a = a constant

The computational procedure is followed in the first method is as given below:

$$R_{RF}(\text{normal}) = \frac{\sum_{i=1}^N \left[R_i \frac{r(\text{normal})}{r_i} \right]}{N} \dots \dots \dots (9)$$

Where,

$R_{RF}(\text{normal})$ - Normalized Rainfall Recharge in the monsoon season

R_i - Rainfall Recharge in the monsoon season for the i^{th} year

$r(\text{normal})$ - Normal monsoon season rainfall

r_i - Rainfall in the monsoon season for the i^{th} year

N - No. of years for which data is available

The second method is also based on a linear relation between recharge and rainfall. However, this linear relationship is of the form,

$$R_{RF}(\text{normal}) = a \times r(\text{normal}) + b \dots \dots \dots (10)$$

Where,

$R_{RF}(\text{normal})$ - Normalized Rainfall Recharge in the monsoon season

$r(\text{normal})$ - Normal monsoon season rainfall

a and b - Constants.

The two constants 'a' and 'b' in the above equation are obtained through a linear regression analysis. The computational procedure has been followed in the second method is as given below:

$$a = \frac{NS_4 - S_1S_2}{NS_3 - S_1^2} \dots \dots \dots (11)$$

$$b = \frac{S_2 - aS_1}{N} \dots \dots \dots (12)$$

Where,

$$S_1 = \sum_{i=1}^N r_i, \quad S_2 = \sum_{i=1}^N R_i, \quad S_3 = \sum_{i=1}^N r_i^2, \quad S_4 = \sum_{i=1}^N R_i r_i$$

2.1.1.1.2. Rainfall Infiltration Factor Method

The rainfall recharge estimation based on Water level fluctuation method reflects actual field conditions since it takes into account the response of ground water level. However, the ground water extraction estimation included in the computation of rainfall recharge using water level fluctuation approach is often subject to uncertainties. Therefore, the rainfall recharge obtained from water level fluctuation approach has been compared with that estimated using rainfall infiltration factor method. Recharge from rainfall is estimated by using the following relationship –

$$R_{RF} = RFIF \times A \times \frac{(R - a)}{1000} \dots \dots \dots (13)$$

Where,

R_{RF} - Rainfall recharge in ham

A - Area in hectares

RFIF - Rainfall Infiltration Factor

R- Rainfall in mm

a - Minimum threshold value above which rainfall induces ground water recharge in mm

The threshold limit of minimum and maximum rainfall event which can induce recharge to the aquifer is considered while estimating ground water recharge using rainfall infiltration factor method. The minimum threshold limit is in accordance with the relation shown in equation (13) and the maximum threshold limit is based on the premise that after a certain limit, the rate of storm rain is too high to contribute to infiltration and they will only contribute to surface runoff. Thus, 10% of Normal annual rainfall has been taken as minimum rainfall threshold and 3000 mm as maximum rainfall limit. While computing the rainfall recharge, 10% of the normal annual rainfall has been deducted from the monsoon rainfall and balance rainfall is considered for computation of rainfall recharge. The same recharge factor is used for both monsoon and non-monsoon rainfall, with the condition that the recharge due to non-monsoon rainfall is taken as zero, if the normal rainfall during the non-monsoon season is less than 10% of normal annual rainfall. In using the method based on the specified norms, recharge due to both monsoon and non-monsoon rainfall has been estimated for normal rainfall, based on recent 30 to 50 years of data.

2.1.1.1.3. Percent Deviation

After computing the rainfall recharge for normal monsoon season rainfall using the ground water level fluctuation method and rainfall infiltration factor method these two estimates is compared with each other. A term, Percent Deviation (PD) which is the difference between the two expressed as a percentage of the later is computed as

$$PD = \frac{R_{RF}(\text{normal}, wtfm) - R_{RF}(\text{normal}, rfm)}{R_{RF}(\text{normal}, rfm)} \times 100 \dots \dots \dots (14)$$

Where,

$R_{RF}(\text{normal}, wtfm)$ = Rainfall recharge for normal monsoon season rainfall estimated by the ground water level fluctuation method

$R_{RF}(\text{normal}, rfm)$ = Rainfall recharge for normal monsoon season rainfall estimated by the rainfall infiltration factor method

The rainfall recharge for normal monsoon season rainfall is finally adopted as per the criteria given below:

- If PD is greater than or equal to -20%, and less than or equal to +20%, $R_{RF}(\text{normal})$ is taken as the value estimated by the ground water level fluctuation method.
- If PD is less than -20%, $R_{RF}(\text{normal})$ is taken as equal to 0.8 times the value estimated by the rainfall infiltration factor method.
- If PD is greater than +20%, $R_{RF}(\text{normal})$ is taken as equal to 1.2 times the value estimated by the rainfall infiltration factor method.

2.1.1.2. Recharge from Other Sources

Recharge from other sources constitutes recharges from canals, surface water irrigation, ground water irrigation, tanks & ponds and water conservation structures in command areas where as in non-command areas it constitutes the recharge due to surface water irrigation, ground water irrigation, tanks & ponds and water conservation structures. The methods of estimation of recharge from different sources are used in the assessment as follows.

Sl. No.	Source	Estimation Formula	Parameters
1	Recharge from Canals	$R_C = WA \times SF \times Days$	R_C = Recharge from Canals WA = Wetted Area SF = Seepage Factor Days = Number of Canal Running Days
2	Recharge from Surface Water Irrigation	$R_{SWI} = AD \times Days \times RFF$	R_{SWI} = Recharge due to applied surface water irrigation AD = Average Discharge Days = Number of days water is discharged to the Fields RFF = Return Flow Factor
3	Recharge from Ground Water Irrigation	$R_{GWI} = GE_{IRR} \times RFF$	R_{GWI} = Recharge due to applied ground water irrigation GE_{IRR} = Ground Water Extraction for Irrigation RFF = Return Flow Factor

Sl. No.	Source	Estimation Formula	Parameters
4	Recharge due to Tanks & Ponds	$R_{TP} = AWSA \times N \times RF$	R_{TP} = Recharge due to Tanks & Ponds AWSA = Average Water Spread Area N = Number of days Water is available in the Tank/Pond RF = Recharge Factor
5	Recharge due to Water Conservation Structures	$R_{WCS} = GS \times RF$	RWCS = Recharge due to Water Conservation Structures GS = Gross Storage = Storage Capacity multiplied by number of fillings. RF = Recharge Factor

2.1.1.3. Evaporation and Transpiration

Evaporation is estimated for the aquifer in the assessment unit if water levels in the aquifer are within the capillary zone. For areas with water levels within 1.0mbgl, evaporation is estimated using the evaporation rates available for other adjoining areas. If depth to water level is more than 1.0mbgl, the evaporation losses from the aquifer is taken as zero.

Transpiration through vegetation has been estimated if water levels in the aquifer are within the maximum root zone of the local vegetation. If water levels are within 3.5mbgl, transpiration is estimated using the transpiration rates available for other areas. If it is greater than 3.5m bgl, the transpiration has been taken as zero.

2.1.1.4. Recharge During Monsoon Season

The sum of normalized monsoon rainfall recharge and the recharge from other sources and lateral and vertical flows into & out of the sub unit and stream inflows & outflows during monsoon season is the total recharge/ accumulation during monsoon season for the sub unit. Similarly, this is to be computed for all the sub units available in the assessment unit.

2.1.1.5. Recharge During Non-Monsoon Season

The rainfall recharge during non-monsoon season is estimated using rainfall infiltration factor Method only when the non-monsoon season rainfall is more than 10% of normal annual rainfall. The sum of non-monsoon rainfall recharge and the recharge from other sources and lateral and vertical flows into & out of the sub unit and stream inflows & outflows during non-monsoon season is the total recharge/ accumulation during non-monsoon season for the sub unit. Similarly, this is to be computed for all the sub units available in the assessment unit.

2.1.1.6. Total Annual Ground Water Recharge

The sum of the recharge/ accumulations during monsoon and non-monsoon seasons is the total annual ground water recharge/ accumulations for the sub unit. Similarly, this is computed for all the sub units available in the assessment unit.

2.1.1.7. Annual Extractable Ground Water Resource (EGR)

The Annual Extractable Ground Water Resource (EGR) is computed by deducting the Total Annual Natural Discharge from Total Annual Ground Water Recharge.

In the water level fluctuation method, a significant portion of base flow is already accounted for by taking the post monsoon water level one month after the end of rainfall. The base flow in the remaining non-monsoon period is likely to be small, especially in hard rock areas. In the assessment units, where river stage data are not available and neither the detailed data for quantitative assessment of the natural discharge are available, allocation of unaccountable natural discharges to 5% or 10% of annual recharge is considered. If the rainfall recharge is assessed using water level fluctuation method this has been taken 5% of the annual recharge and if it is assessed using rainfall infiltration factor method, 10% of the annual recharge is considered. The balance is account for Annual Extractable Ground Water Resources (EGR).

2.1.1.8. Estimation of Ground Water Extraction

Ground water draft or extraction is assessed as follows.

$$GE_{ALL} = GE_{IRR} + GE_{DOM} + GE_{IND} \dots \dots \dots (15)$$

Where,

- GE_{ALL} = Ground water extraction for all uses
- GE_{IRR} = Ground water extraction for irrigation
- GE_{DOM} = Ground water extraction for domestic uses
- GE_{IND} = Ground water extraction for industrial uses

2.1.1.8.1. Ground Water Extraction for Irrigation (GE_{IRR})

The methods for estimation of ground water extraction are as follows.

Unit Draft Method: – In this method, season-wise unit draft of each type of well in an assessment unit is estimated. The unit draft of different types (eg. Dug well, Dug cum bore well, shallow tube well, deep tube well, bore well etc.) is multiplied with the number of wells of that particular type to obtain season-wise ground water extraction by that particular structure.

Crop Water Requirement Method: – For each crop, the season-wise net irrigation water requirement is determined. This is then multiplied with the area irrigated by ground water abstraction structures. The database on crop area is obtained from Revenue records in Tehsil office, Agriculture Census and also by using Remote Sensing techniques.

Power Consumption Method: –Ground water extraction for unit power consumption (electric) is determined. Extraction per unit power consumption is then multiplied with number of units of power consumed for agricultural pump sets to obtain total ground water extraction for irrigation.

2.1.1.8.2. Ground Water Extraction for Domestic Use (GE_{DOM})

There are several methods for estimation of extraction for domestic use(GEDOM). Some of the commonly adopted methods are described here.

Unit Draft Method: – In this method, unit draft of each type of well is multiplied by the number of wells used for domestic purpose to obtain the domestic ground water extraction.

Consumptive Use Method: – In this method, population is multiplied with per capita consumption usually expressed in litre per capita per day (lpcd). It can be expressed using following equation.

$$GE_{DOM} = Population \times Consumptive Requirement \times L_g \dots \dots \dots (16)$$

Where,

L_g = Fractional Load on Ground Water for Domestic Water Supply.

The Load on Ground water can be obtained from the Information based on Civic water supply agencies in urban areas.

2.1.1.8.3. Ground Water Extraction for Industrial Use (GE_{IND})

The commonly adopted methods for estimating the extraction for industrial use are as below:

Unit Draft Method: - In this method, unit draft of each type of well is multiplied by the number of wells used for industrial purpose to obtain the industrial ground water extraction.

Consumptive Use Pattern Method: – In this method, water consumption of different industrial units is determined. Numbers of Industrial units which are dependent on ground water are multiplied with unit water consumption to obtain ground water extraction for industrial use.

$$GE_{IND} = \text{Number of Industrial Units} \times \text{Unit Water Consumption} \times L_g \dots \dots \dots (17)$$

Where,

L_g = Fractional load on ground water for industrial water supply.

The load on ground water for industrial water supply can be obtained from water supply agencies in the Industrial belt.

Ground water extraction obtained from different methods need to be compared and based on field checks, the seemingly best value may be adopted. At times, ground water extraction obtained by different methods may vary widely. In such cases, the value matching the field situation should be considered. The storage depletion during a season, where other recharges are negligible can be taken as ground water extraction during that particular period.

2.1.1.9. Stage of Ground Water Extraction

The stage of ground water extraction is defined by,

$$\text{Stage of GW Extraction} = \frac{\text{Existing Gross GW Extraction for all Uses}}{\text{Annual Extractable GW Resources}} \times 100 \dots \dots \dots (18)$$

The existing gross ground water extraction for all uses refers to the total of existing gross ground water extraction for irrigation and all other purposes. The stage of ground water extraction should be obtained separately for command areas, non-command areas and poor ground water quality areas.

2.1.1.10. Validation of Stage of Ground Water Extraction

The assessment based on the stage of ground water extraction has inherent uncertainties. In view of this, it is desirable to validate the 'Stage of Ground Water Extraction' with long term trend of ground water levels.

Long term Water Level trends are prepared for a minimum period of 10 years for both pre-monsoon and post-monsoon period. If the ground water resource assessment and the trend of long term water levels contradict each other, this anomalous situation requires a review of the ground water resource computation, as well as the reliability of water level data. The mismatch conditions are enumerated below.

SOGWE	Ground Water Level Trend	Remarks
$\leq 70\%$	Significant decline in trend in both pre-monsoon and post-monsoon	Not acceptable and needs reassessment
$> 100\%$	No significant decline in both pre-monsoon and post-monsoon long term trend	Not acceptable and needs reassessment

2.1.1.11. Categorisation of Assessment Unit

2.1.1.11.1. Categorisation of Assessment Unit Based on Quantity

The categorisation based on status of ground water quantity is defined by Stage of Ground Water Extraction as given below:

Stage of Ground Water Extraction	Category
≤ 70%	Safe
> 70% and ≤90%	Semi-critical
> 90% and ≤100%	Critical
> 100%	Over Exploited

2.1.1.11.2. Categorisation of Assessment Unit Based on Quality

As it is not possible to categorize the assessment units in terms of the extent of quality hazard, based on the available water quality monitoring mechanism and database on ground water quality, the Committee recommends that each assessment unit, in addition to the Quantity based categorization (safe, semi-critical, critical and over-exploited) should bear a quality hazard identifier. If any of the three quality hazards in terms of Arsenic, Fluoride and Salinity are encountered in the assessment sub unit in mappable units, the assessment sub unit has been tagged with the particular Quality hazard.

2.1.1.12. Allocation of Ground Water Resource for Utilisation

The Annual Extractable Ground Water Resources are to be apportioned between domestic, industrial and irrigation uses. Among these, as per the National Water Policy, requirement for domestic water supply is to be accorded priority. This requirement based on population has been projected to the year 2025, per capita requirement of water for domestic use, and relative load on ground water for urban and rural water supply. In situations where adequate data is not available to make this estimate, the following empirical relation has been utilized.

$$Alloc = 22 \times N \times L_g \text{ mm per year} \dots \dots \dots (19)$$

Where,

Alloc = Allocation for domestic water requirement

N = population density in the unit in thousands per sq. km.

L_g = fractional load on ground water for domestic water supply (≤ 1.0)

2.1.1.13. Net Annual Ground Water Availability for Future Use

The water available for future use is obtained by deducting the allocation for domestic use and current extraction for Irrigation and Industrial uses from the Annual Extractable Ground Water Recharge. The resulting ground water potential is termed as the net annual ground water availability for future use. The Net annual ground water availability for future use is calculated separately for non-command areas and command areas. As per the recommendations of the R&D Advisory committee, the ground water available for future use can never be negative. If it becomes negative, the future allocation of Domestic needs can be reduced to current extraction for domestic use. Even then if it is still negative, then the ground water available for future uses has been projected as zero.

2.1.1.13.1 Apportioning of Groundwater Assessment from Watershed to Mandal:

Where the assessment unit is a watershed, there is a need to convert the groundwater assessment in terms of an administrative unit such as block/taluka/mandal. A block may comprise of one or more watersheds, in part or full. The apportionment was done as per standard procedure as given below. First, the groundwater assessment in the sub-areas, command and non-command areas of the watershed was converted into depth unit (mm), by dividing the annual recharge by the respective area. The contribution of this sub-area of the watershed to the block, is then calculated by multiplying this depth with the area in the block occupied by this sub-area. This procedure was followed to calculate the contribution from the sub-areas of all watersheds occurring in the block, to work out the total groundwater resource of the block. The total groundwater resource of the block was presented separately for each type of sub-area, namely for command areas and non-command areas

2.1.1.13.2 Potential Resource in Waterlogged and Shallow Water Table Areas

In the area where the ground water level is less than 5m below ground level or in waterlogged areas, the resources up to 5m below ground level are potential and would be available for development in addition to the annual recharge in the area. The computation of potential resource to ground water reservoir in shallow water table areas has been done by adopting the following equation:

$$\text{Potential groundwater resource in shallow water table areas} \\ = (5 - D) \times A \times S_Y \dots \dots \dots (21)$$

Where,

D = Depth to water table below ground surface in pre-monsoon period in shallow aquifers.

A = Area of shallow water table zone.

S_Y = Specific Yield

2.1.1.13.3 Potential Resource in Flood Prone Areas

Ground water recharge from a flood plain is mainly the function of the following parameters-

- Areal extent of flood plain
- Retention period of flood
- Type of sub-soil strata and silt charge in the river water which gets deposited and controls seepage

Since collection of data on all these factors is time taking and difficult, in the meantime, the potential resource from flood plain may be estimated on the same norms as for ponds, tanks and lakes. This has been calculated over the water spread area and only for the retention period using the following formula.

$$\text{Potential groundwater resource in Flood Prone Areas} \\ = 1.4 \times N \times \frac{A}{1000} \dots \dots \dots (22)$$

Where,

N = No. of Days Water is Retained in the Area

A = Flood Prone Area

2.1.1.14 Apportioning of Ground Water Assessment from Watershed to Development Unit

Where the assessment unit is a watershed, there is a need to convert the ground water assessment in terms of an administrative unit such as block/ taluka/ mandal. This has been done as follows.

A block may comprise of one or more watersheds, in part or full. First, the ground water assessment in the subareas, command, non-command and poor ground water quality areas of the watershed has been converted into depth unit (mm), by dividing the annual recharge by the respective area. The contribution of this subarea of the watershed to the block, is now calculated by multiplying this depth with the area in the block occupied by this sub-area.

The total ground water resource of the block has been presented separately for each type of sub-area, namely for command areas, non-command areas and poor ground water quality areas, as in the case of the individual watersheds.

2.2 NORMS HAS BEEN USED IN THE ASSESSMENT

2.2.1 Specific Yield

Recently under Aquifer Mapping Project, Central Ground Water Board has classified all the aquifers into 14 Principal Aquifers which in turn were divided into 42 Major Aquifers. Hence, it is required to assign Specific Yield values to all these aquifer units. The values recommended in the **Table-2.1** has been followed in the present assessments, unless sufficient data based on field studies are available to justify the minimum, maximum or other intermediate values.

Table-2.1: Norms Recommended for Specific Yield

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
1	Alluvium	AL01	Younger Alluvium (Clay/Silt/Sand/ Calcareous concretions)	Quaternary	10	8	12
2	Alluvium	AL02	Pebble / Gravel/ Bazada/ Kandi	Quaternary	16	12	20
3	Alluvium	AL03	Older Alluvium (Silt/Sand/Gravel/Lithomargic clay)	Quaternary	6	4	8
4	Alluvium	AL04	Aeolian Alluvium (Silt/ Sand)	Quaternary	16	12	20
5	Alluvium	AL05	Coastal Alluvium (Sand/Silt/Clay)	Quaternary	10	8	12
6	Alluvium	AL06	Valley Fills	Quaternary	16	12	20
7	Alluvium	AL07	Glacial Deposits	Quaternary	16	12	20
8	Laterite	LT01	Laterite / Ferruginous concretions	Quaternary	2.5	2	3
9	Basalt	BS01	Basic Rocks (Basalt) - Weathered, Vesicular or Jointed	Mesozoic to Cenozoic	2	1	3
10	Basalt	BS01	Basic Rocks (Basalt) - Massive Poorly Jointed	Mesozoic to Cenozoic	0.35	0.2	0.5

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
11	Basalt	BS02	Ultra Basic - Weathered, Vesicular or Jointed	Mesozoic to Cenozoic	2	1	3
12	Basalt	BS02	Ultra Basic - Massive Poorly Jointed	Mesozoic to Cenozoic	0.35	0.2	0.5
13	Sandstone	ST01	Sandstone/Conglomerate	Upper Palaeozoic to Cenozoic	3	1	5
14	Sandstone	ST02	Sandstone with Shale	Upper Palaeozoic to Cenozoic	3	1	5
15	Sandstone	ST03	Sandstone with shale/ coal beds	Upper Palaeozoic to Cenozoic	3	1	5
16	Sandstone	ST04	Sandstone with Clay	Upper Palaeozoic to Cenozoic	3	1	5
17	Sandstone	ST05	Sandstone/Conglomerate	Proterozoic to Cenozoic	3	1	5
18	Sandstone	ST06	Sandstone with Shale	Proterozoic to Cenozoic	3	1	5
19	Shale	SH01	Shale with limestone	Upper Palaeozoic to Cenozoic	1.5	1	2
20	Shale	SH02	Shale with Sandstone	Upper Palaeozoic to Cenozoic	1.5	1	2
21	Shale	SH03	Shale, limestone and sandstone	Upper Palaeozoic to Cenozoic	1.5	1	2
22	Shale	SH04	Shale	Upper Palaeozoic	1.5	1	2

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
				to Cenozoic			
23	Shale	SH05	Shale/Shale with Sandstone	Proterozoic to Cenozoic	1.5	1	2
24	Shale	SH06	Shale with Limestone	Proterozoic to Cenozoic	1.5	1	2
25	Limestone	LS01	Miliolitic Limestone	Quaternary	2	1	3
26	Limestone	LS01	Karstified Miliolitic Limestone	Quaternary	10	5	15
27	Limestone	LS02	Limestone / Dolomite	Upper Palaeozoic to Cenozoic	2	1	3
28	Limestone	LS02	Karstified Limestone / Dolomite	Upper Palaeozoic to Cenozoic	10	5	15
29	Limestone	LS03	Limestone/Dolomite	Proterozoic	2	1	3
30	Limestone	LS03	Karstified Limestone/Dolomite	Proterozoic	10	5	15
31	Limestone	LS04	Limestone with Shale	Proterozoic	2	1	3
32	Limestone	LS04	Karstified Limestone with Shale	Proterozoic	10	5	15
33	Limestone	LS05	Marble	Azoic to Proterozoic	2	1	3
34	Limestone	LS05	Karstified Marble	Azoic to Proterozoic	10	5	15
35	Granite	GR01	Acidic Rocks (Granite, Syenite, Rhyolite etc.) - Weathered, Jointed	Mesozoic to Cenozoic	1.5	1	2
36	Granite	GR01	Acidic Rocks (Granite, Syenite, Rhyolite etc.) - Massive or Poorly Fractured	Mesozoic to Cenozoic	0.35	0.2	0.5
37	Granite	GR02	Acidic Rocks (Pegmatite, Granite, Syenite, Rhyolite etc.) - Weathered, Jointed	Proterozoic to Cenozoic	3	2	4
38	Granite	GR02	Acidic Rocks (Pegmatite, Granite, Syenite, Rhyolite	Proterozoic to Cenozoic	0.35	0.2	0.5

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
			etc.) - Massive, Poorly Fractured				
39	Schist	SC01	Schist - Weathered, Jointed	Azoic to Proterozoic	1.5	1	2
40	Schist	SC01	Schist - Massive, Poorly Fractured	Azoic to Proterozoic	0.35	0.2	0.5
41	Schist	SC02	Phyllite	Azoic to Proterozoic	1.5	1	2
42	Schist	SC03	Slate	Azoic to Proterozoic	1.5	1	2
43	Quartzite	QZ01	Quartzite - Weathered, Jointed	Proterozoic to Cenozoic	1.5	1	2
44	Quartzite	QZ01	Quartzite - Massive, Poorly Fractured	Proterozoic to Cenozoic	0.3	0.2	0.4
45	Quartzite	QZ02	Quartzite - Weathered, Jointed	Azoic to Proterozoic	1.5	1	2
46	Quartzite	QZ02	Quartzite- Massive, Poorly Fractured	Azoic to Proterozoic	0.3	0.2	0.4
47	Charnockite	CK01	Charnockite - Weathered, Jointed	Azoic	3	2	4
48	Charnockite	CK01	Charnockite - Massive, Poorly Fractured	Azoic	0.3	0.2	0.4
49	Khondalite	KH01	Khondalites, Granulites - Weathered, Jointed	Azoic	1.5	1	2
50	Khondalite	KH01	Khondalites, Granulites - Massive, Poorly Fractured	Azoic	0.3	0.2	0.4
51	Banded Gneissic Complex	BG01	Banded Gneissic Complex - Weathered, Jointed	Azoic	1.5	1	2
52	Banded Gneissic Complex	BG01	Banded Gneissic Complex - Massive, Poorly Fractured	Azoic	0.3	0.2	0.4
53	Gneiss	GN01	Undifferentiated metasedimentaries/ Undifferentiated metamorphic - Weathered, Jointed	Azoic to Proterozoic	1.5	1	2
54	Gneiss	GN01	Undifferentiated metasedimentaries/ Undifferentiated	Azoic to Proterozoic	0.3	0.2	0.4

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
			metamorphic - Massive, Poorly Fractured				
55	Gneiss	GN02	Gneiss -Weathered, Jointed	Azoic to Proterozoic	3	2	4
56	Gneiss	GN02	Gneiss-Massive, Poorly Fractured	Azoic to Proterozoic	0.3	0.2	0.4
57	Gneiss	GN03	Migmatitic Gneiss - Weathered, Jointed	Azoic	1.5	1	2
58	Gneiss	GN03	Migmatitic Gneiss - Massive, Poorly Fractured	Azoic	0.3	0.2	0.4
59	Intrusive	IN01	Basic Rocks (Dolerite, Anorthosite etc.) - Weathered, Jointed	Proterozoic to Cenozoic	2	1	3
60	Intrusive	IN01	Basic Rocks (Dolerite, Anorthosite etc.) - Massive, Poorly Fractured	Proterozoic to Cenozoic	0.35	0.2	0.5
61	Intrusive	IN02	Ultrabasics (Epidiorite, Granophyre etc.) - Weathered, Jointed	Proterozoic to Cenozoic	2	1	3
62	Intrusive	IN02	Ultrabasics (Epidiorite, Granophyre etc.) - Massive, Poorly Fractured	Proterozoic to Cenozoic	0.35	0.2	0.5

2.2.2 Rainfall Infiltration Factor

The values mentioned in **Table-2.2** has been used in the present assessment. The recommended Rainfall Infiltration Factor values has been used for assessment, unless sufficient data based on field studies are available to justify the minimum, maximum or other intermediate values.

Table-2.2: Norms Recommended for Rainfall Infiltration Factor

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
1	Alluvium	AL01	Younger Alluvium (Clay/Silt/Sand/ Calcareous concretions)	Quaternary	22	20	24
2	Alluvium	AL02	Pebble / Gravel/ Bazada/ Kandi	Quaternary	22	20	24

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
3	Alluvium	AL03	Older Alluvium (Silt/Sand/Gravel/Lithomargic clay)	Quaternary	22	20	24
4	Alluvium	AL04	Aeolian Alluvium (Silt/Sand)	Quaternary	22	20	24
5	Alluvium	AL05	Coastal Alluvium (Sand/Silt/Clay) -East Coast	Quaternary	16	14	18
5	Alluvium	AL05	Coastal Alluvium (Sand/Silt/Clay) - West Coast	Quaternary	10	8	12
6	Alluvium	AL06	Valley Fills	Quaternary	22	20	24
7	Alluvium	AL07	Glacial Deposits	Quaternary	22	20	24
8	Laterite	LT01	Laterite / Ferruginous concretions	Quaternary	7	6	8
9	Basalt	BS01	Basic Rocks (Basalt) - Vesicular or Jointed	Mesozoic to Cenozoic	13	12	14
9	Basalt	BS01	Basic Rocks (Basalt) - Weathered	Mesozoic to Cenozoic	7	6	8
10	Basalt	BS01	Basic Rocks (Basalt) - Massive Poorly Jointed	Mesozoic to Cenozoic	2	1	3
11	Basalt	BS02	Ultra Basic - Vesicular or Jointed	Mesozoic to Cenozoic	13	12	14
11	Basalt	BS02	Ultra Basic - Weathered	Mesozoic to Cenozoic	7	6	8
12	Basalt	BS02	Ultra Basic - Massive Poorly Jointed	Mesozoic to Cenozoic	2	1	3
13	Sandstone	ST01	Sandstone/Conglomerate	Upper Palaeozoic to Cenozoic	12	10	14
14	Sandstone	ST02	Sandstone with Shale	Upper Palaeozoic to Cenozoic	12	10	14
15	Sandstone	ST03	Sandstone with shale/ coal beds	Upper Palaeozoic	12	10	14

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
				to Cenozoic			
16	Sandstone	ST04	Sandstone with Clay	Upper Palaeozoic to Cenozoic	12	10	14
17	Sandstone	ST05	Sandstone/Conglomerate	Proterozoic to Cenozoic	6	5	7
18	Sandstone	ST06	Sandstone with Shale	Proterozoic to Cenozoic	6	5	7
19	Shale	SH01	Shale with limestone	Upper Palaeozoic to Cenozoic	4	3	5
20	Shale	SH02	Shale with Sandstone	Upper Palaeozoic to Cenozoic	4	3	5
21	Shale	SH03	Shale, limestone and sandstone	Upper Palaeozoic to Cenozoic	4	3	5
22	Shale	SH04	Shale	Upper Palaeozoic to Cenozoic	4	3	5
23	Shale	SH05	Shale/Shale with Sandstone	Proterozoic to Cenozoic	4	3	5
24	Shale	SH06	Shale with Limestone	Proterozoic to Cenozoic	4	3	5
25	Limestone	LS01	Miliolitic Limestone	Quaternary	6	5	7
27	Limestone	LS02	Limestone / Dolomite	Upper Palaeozoic to Cenozoic	6	5	7
29	Limestone	LS03	Limestone/Dolomite	Proterozoic	6	5	7
31	Limestone	LS04	Limestone with Shale	Proterozoic	6	5	7

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
33	Limestone	LS05	Marble	Azoic to Proterozoic	6	5	7
35	Granite	GR01	Acidic Rocks (Granite, Syenite, Rhyolite etc.) - Weathered, Jointed	Mesozoic to Cenozoic	7	5	9
36	Granite	GR01	Acidic Rocks (Granite, Syenite, Rhyolite etc.) - Massive or Poorly Fractured	Mesozoic to Cenozoic	2	1	3
37	Granite	GR02	Acidic Rocks (Pegmatite, Granite, Syenite, Rhyolite etc.) - Weathered, Jointed	Proterozoic to Cenozoic	11	10	12
38	Granite	GR02	Acidic Rocks (Pegmatite, Granite, Syenite, Rhyolite etc.) - Massive, Poorly Fractured	Proterozoic to Cenozoic	2	1	3
39	Schist	SC01	Schist - Weathered, Jointed	Azoic to Proterozoic	7	5	9
40	Schist	SC01	Schist - Massive, Poorly Fractured	Azoic to Proterozoic	2	1	3
41	Schist	SC02	Phyllite	Azoic to Proterozoic	4	3	5
42	Schist	SC03	Slate	Azoic to Proterozoic	4	3	5
43	Quartzite	QZ01	Quartzite - Weathered, Jointed	Proterozoic to Cenozoic	6	5	7
44	Quartzite	QZ01	Quartzite - Massive, Poorly Fractured	Proterozoic to Cenozoic	2	1	3
45	Quartzite	QZ02	Quartzite - Weathered, Jointed	Azoic to Proterozoic	6	5	7
46	Quartzite	QZ02	Quartzite - Massive, Poorly Fractured	Azoic to Proterozoic	2	1	3
47	Charnockite	CK01	Charnockite - Weathered, Jointed	Azoic	5	4	6
48	Charnockite	CK01	Charnockite - Massive, Poorly Fractured	Azoic	2	1	3
49	Khondalite	KH01	Khondalites, Granulites - Weathered, Jointed	Azoic	7	5	9
50	Khondalite	KH01	Khondalites, Granulites - Massive, Poorly Fractured	Azoic	2	1	3

Sl. No.	Principal Aquifer	Major Aquifers		Age	Recommended (%)	Minimum (%)	Maximum (%)
		Code	Name				
51	Banded Gneissic Complex	BG01	Banded Gneissic Complex - Weathered, Jointed	Azoic	7	5	9
52	Banded Gneissic Complex	BG01	Banded Gneissic Complex - Massive, Poorly Fractured	Azoic	2	1	3
53	Gneiss	GN01	Undifferentiated metasedimentaries/ Undifferentiated metamorphic - Weathered, Jointed	Azoic to Proterozoic	7	5	9
54	Gneiss	GN01	Undifferentiated metasedimentaries/ Undifferentiated metamorphic - Massive, Poorly Fractured	Azoic to Proterozoic	2	1	3
55	Gneiss	GN02	Gneiss -Weathered, Jointed	Azoic to Proterozoic	11	10	12
56	Gneiss	GN02	Gneiss-Massive, Poorly Fractured	Azoic to Proterozoic	2	1	3
57	Gneiss	GN03	Migmatitic Gneiss - Weathered, Jointed	Azoic	7	5	9
58	Gneiss	GN03	Migmatitic Gneiss - Massive, Poorly Fractured	Azoic	2	1	3
59	Intrusive	IN01	Basic Rocks (Dolerite, Anorthosite etc.) - Weathered, Jointed	Proterozoic to Cenozoic	7	6	8
60	Intrusive	IN01	Basic Rocks (Dolerite, Anorthosite etc.) - Massive, Poorly Fractured	Proterozoic to Cenozoic	2	1	3
61	Intrusive	IN02	Ulra Basics (Epidiorite, Granophyre etc.) - Weathered, Jointed	Proterozoic to Cenozoic	7	6	8
62	Intrusive	IN02	Ulra Basics (Epidiorite, Granophyre etc.) - Massive, Poorly Fractured	Proterozoic to Cenozoic	2	1	3

2.2.3 Norms for Canal Recharge

The Norms suggested in **Table-2.3** has been used for estimating the recharge from Canals, where sufficient data based on field studies are not available.

Table-2.3: Norms Recommended for Recharge due to Canals

Formation	Canal Seepage factor ham/day/million square meters of wetted area		
	Recommended	Minimum	Maximum
Unlined canals in normal soils with some clay content along with sand	17.5	15	20
Unlined canals in sandy soil with some silt content	27.5	25	30
Lined canals in normal soils with some clay content along with sand	3.5	3	4
Lined canals in sandy soil with some silt content	5.5	5	6
All canals in hard rock area	3.5	3	4

2.2.4 Norms for Recharge Due to Irrigation

The Recommended Norms are presented in **Table-2.4**.

Table-2.4: Norms Recommended for Recharge from Irrigation

DTW m bgl	Ground Water		Surface Water	
	Paddy	Non-paddy	Paddy	Non-paddy
≤ 10	45.0	25.0	50.0	30.0
11	43.3	23.7	48.3	28.7
12	40.4	22.1	45.1	26.8
13	37.7	20.6	42.1	25.0
14	35.2	19.2	39.3	23.3
15	32.9	17.9	36.7	21.7
16	30.7	16.7	34.3	20.3
17	28.7	15.6	32.0	18.9
18	26.8	14.6	29.9	17.6
19	25.0	13.6	27.9	16.4
20	23.3	12.7	26.0	15.3
21	21.7	11.9	24.3	14.3
22	20.3	11.1	22.7	13.3
23	18.9	10.4	21.2	12.4
24	17.6	9.7	19.8	11.6
≥ 25	20.0	5.0	25.0	10.0

2.2.5 Norms for Recharge due to Tanks & Ponds

As the data on the field studies for computing recharge from Tanks & Ponds are very limited, for Seepage from Tanks & Ponds has been used as 1.4 mm / day in the present assessment.

2.2.6 Norms for Recharge due to Water Conservation Structures

The data on the field studies for computing recharge from Water Conservation Structures are very limited, hence, the norm recommended by GEC-2015 for the seepage from Water Conservation Structures is 40% of gross storage during a year which means 20% during monsoon season and 20% during non-monsoon Season is adopted.

2.2.7 Unit Draft

The methodology recommends to use well census method for computing the ground water draft. The norm used for computing ground water draft is the unit draft. The unit draft can be computed by field studies. This method involves selecting representative abstraction structure and calculating the discharge from that particular type of structure and collecting the information on how many hours of pumping is being done in various seasons and number of such days during each season. The Unit Draft during a particular season is computed using the following equation:

$$\text{Unit Draft} = \frac{\text{Discharge in } m^3/hr \times \text{No. of pumping hours in a day}}{\text{No. of days}} \dots \dots \dots (29)$$

But the procedure that is being followed for computing unit draft does not have any normalization procedure. Normally, if the year in which one collects the draft data in the field is an excess rainfall year, the abstraction from ground water will be less. Similarly, if the year of the computation of unit draft is a drought year the unit draft will be high. Hence, there is a requirement to devise a methodology that can be used for the normalization of unit draft figures. The following are the two simple techniques, which are followed for normalization of Unit Draft. Areas where, unit draft values for one rainfall cycle are available for at least 10 years second method shown in equation 31 is followed or else the first method shown in equation 30 has been used.

$$\text{Normalised Unit Draft} = \frac{\text{Unit Draft} \times \text{Rainfall for the year}}{\text{Normal Rainfall}} \dots \dots \dots (30)$$

$$\text{Normalised Unit Draft} = \frac{\sum_{i=1}^n \text{Unit Draft}_i}{\text{Number of Years}} \dots \dots \dots (31)$$

2.3 INDIA -GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)

“INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES) is a Software/Web-based Application developed by CGWB in collaboration with IIT-Hyderabad. It provides common and standardized platform for Ground Water Resource Estimation for the entire country and its pan-India operationalization (Central and State Governments). The system takes ‘Data Input’ through Excel as well as Forms, compute various ground water components (recharge, extraction etc.) and classify assessment units into appropriate categories (safe, semi-critical, critical and over-exploited). The Software uses GEC 2015 Methodology for estimation and calculation of Groundwater resources. It allows for unique and homogeneous representation of groundwater fluxes as well as categories for all the assessment units (AU) of the country.

URL of IN-GRES → <http://ingres.iith.ac.in>

CHAPTER 3

3.1 Climate

The climate of the state is tropical in nature and is influenced by the topographical variations and maritime influence. The Eastern Ghats in Visakhapatnam and its neighborhood play a significant role, which acts as a barrier to easterly winds in association with depression from Bay of Bengal during the southwestern monsoon. The Agro-climatic classification (Agricultural Department) of the state is given below.

Region	Classification
Rayalaseema	Scarce rainfall zone
	Southern Zone
Coastal Andhra Pradesh	Krishna-Godavari Zone
	North Coastal zone
	South Coastal Zone
	High Altitude
	Tribal Zone
	Scarce rainfall zone

3.2. State Normal Rainfall and Actual Rainfall

District-wise monthly and annual rainfall of both normal and actual of the year 2023 is compiled from daily and weekly weather reports of Andhra Pradesh Water Resources Information & Management System (APWRIMS) and is presented in **Table-3.1**. Isohyetal map for annual normal rainfall for 2023 is presented in **Fig 3.1**. The salient features are given below:

- The normal annual rainfall of the state is 975.9 mm. Season-wise normal rainfall is 575 mm, 285.4 mm, 15.90 mm and 99.6 mm in southwest monsoon (June-Sept), northeast monsoon (Oct-Dec), winter (Jan-Feb) and summer (March-May) respectively, contributing 59% of annual in southwest monsoon, 29% of annual rainfall in northeast monsoon and 12% in non-monsoon season.
- Annual normal rainfall ranges from 512.5 mm in Ananthpuramu district to 1295 mm in Konaseema district
- Annual rainfall in 2023 ranges from 569.52 mm (-21 % below normal) in Nandyal district to 1215.22 mm (36% above normal) in Guntur.
- District wise annual actual and normal rainfall 2023 is presented in **Fig. 3.2**

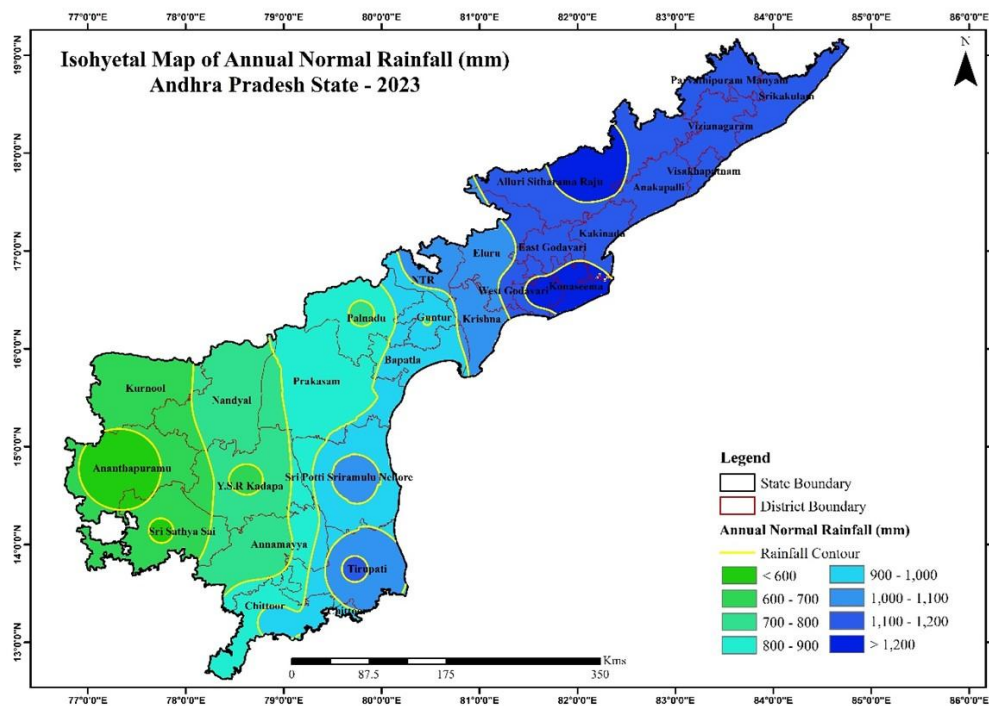


Fig.3.1: Isohyetal map of Andhra Pradesh State (normal annual rainfall in mm).

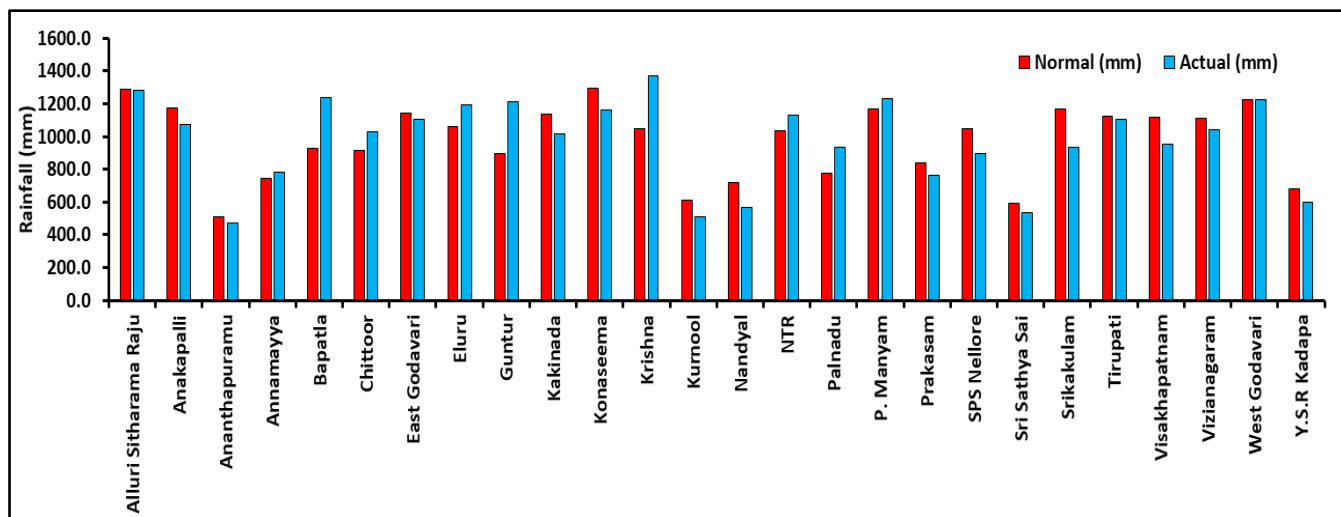


Fig.3.2: District-wise annual actual and normal rainfall for Andhra Pradesh State for the year 2023

3.3. District wise Normal and Actual Rainfall

The district-wise monthly and annual normal and actual rainfall data are presented in Table 3.1. The annual normal rainfall for the state is recorded at 976.2 mm, while the actual rainfall for the year 2023 amounted to 975.9 mm, indicating a 0% surplus compared to the normal. The highest normal rainfall was observed in July, followed by September 2023, whereas the lowest normal rainfall was recorded in February and January 2023.

DISTRICT	Jan		Feb		Mar		Apr		May		Jun		Jul	
	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)
A.S. Raju	7.6	0.5	10.2	0.0	19.1	79.3	42.9	41.95	107.64	105.37	162.4	101.2	295.12	352.42
Anakapalli	9.6	0.7	13.8	0.0	20.6	107.8	27.3	13.19	116.28	87.65	123.2	56	175.92	239.38
Ananthapuramu	1.8	0.0	0.8	0.0	2.1	20.6	12.1	25.51	36.68	88.93	61.1	52.33	63.87	79.6
Annamayya	5.6	0.4	3.2	0.0	6.8	56.4	14.1	11.36	45.97	76.58	75.1	42.85	90.45	65.13
Bapatla	11.3	1.7	7.0	0.1	6.0	64.6	10.2	12.16	63.68	128.99	71.0	95.57	118.54	239.28
Chittoor	4.9	0.1	2.8	0.0	9.4	52.5	19.5	24.99	67.16	115.74	81.0	93.44	103.46	75.72
East Godavari	9.6	5.7	7.5	0.0	15.2	56.7	18.4	19.66	85.57	116.14	117.0	64.88	247.06	298
Eluru	9.8	0.4	6.8	0.0	14.6	53.7	19.9	24.37	63.52	97.64	112.3	96.91	242.52	353.95
Guntur	8.0	0.9	9.5	0.5	9.9	53.9	12.4	32.54	62.18	116.8	97.2	114.05	164.36	311.65
Kakinada	9.5	8.3	11.9	0.0	12.5	89.5	17.6	18.36	118.02	115.72	112.8	74.14	198.01	221.96
Konaseema	9.7	9.9	10.5	0.0	12.0	93.9	13.0	51.2	74.83	156.65	111.4	89.86	241	181.18
Krishna	7.1	2.6	10.1	0.0	11.4	76.1	13.1	33.17	57.04	135.46	91.6	106.84	196.12	351.35
Kurnool	3.2	0.0	1.5	0.0	5.9	10.4	15.5	48.95	40.22	75.35	77.6	60.99	89.46	142.47
Nandyal	2.3	0.0	2.3	0.0	5.6	27.0	18.2	28.89	36.93	70.11	76.8	49.73	141.17	122.41
NTR	10.1	0.1	4.2	0.0	9.0	77.1	14.5	34.16	60.26	102.36	109.4	87.62	233.81	355.99
Palnadu	8.4	0.8	9.0	0.0	8.9	55.7	12.1	11.78	52.69	101.23	80.9	80.05	130.95	158.33
P. Manyam	9.1	1.2	17.0	0.0	30.6	98.9	36.1	77.04	117.09	106.5	140.7	93.64	208.46	321.79
Prakasam	11.4	0.8	4.3	0.0	10.7	33.4	12.6	28.1	51.27	96.7	57.9	49.36	87.84	97.31
SPS Nellore	12.5	0.8	5.5	0.0	5.0	55.7	9.3	30.62	54.46	42.77	51.9	25.53	81.58	73.07
S.S. Sai	1.5	0.0	1.9	0.0	4.4	33.8	13.5	15.86	42.45	87.35	66.6	65.45	70.87	51.48
Srikakulam	6.1	2.5	19.5	0.1	19.7	85.5	29.8	28.46	100.4	75.32	134.9	66.34	184.41	303.89
Tirupati	16.0	0.4	6.3	0.0	5.0	66.9	12.7	16.02	55	56.75	72.8	57.28	103.24	64.25
Visakhapatnam	12.2	3.4	13.0	0.2	16.9	64.7	23.6	12.12	88.4	74.84	110.9	32.94	138.99	309.71
Vizianagaram	9.2	1.6	16.4	0.5	17.7	64.5	28.4	25.66	111.62	95.6	123.5	52.03	170.84	324.39
West Godavari	9.8	2.5	8.7	0.0	14.3	69.3	16.5	30.13	72.69	113.32	110.6	114.27	246.25	197.57
Y.S.R Kadapa	1.7	0.1	0.9	0.0	2.5	47.0	11.7	8.95	36.48	68.99	68.2	35.75	98.46	78.7
State Mean	8.0	1.7	7.9	0.1	11.4	61.3	18.3	27.1	69.9	96.5	96.1	71.5	158.6	206.6

DISTRICT	Aug		Sept		Oct		Nov		Dec		Annual		
	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)	Departure (%)
A.S. Raju	252.51	136.79	185.0	251.4	157.78	19.03	47.47	26.56	2.92	167.12	1290.6	1281.59	-1%
Anakapalli	179.47	77.61	203.3	223.03	214.05	26.11	87.33	19.99	4.57	220.18	1175.44	1071.69	-9%
Ananthapuramu	83.85	25.04	110.9	145.48	100.89	8.32	28.64	23.38	9.81	6.12	512.46	475.28	-7%
Annamayya	99.78	53.63	114.7	178.01	139.18	20.62	109.21	106.95	39.71	171.02	743.67	782.94	5%
Bapatla	157.23	133.7	157.8	223.55	171.92	34.41	122.63	90.37	28.02	212.52	925.33	1236.97	34%
Chittoor	121.19	116.99	153.1	252.29	156.63	51.42	141.21	117.09	54.44	129.08	914.8	1029.3	13%
East Godavari	238.57	118.22	173.2	154.91	169.08	10.56	56.82	29.67	6.73	228.61	1144.78	1103.01	-4%
Eluru	239.34	121.91	171.1	164.8	134.36	19.57	42.04	35.34	7.64	226.88	1063.85	1195.46	12%
Guntur	165.36	138.29	146.1	201.64	128.8	25.91	77.03	50.8	15.89	168.23	896.65	1215.22	36%
Kakinada	179.85	64.94	179.5	139.86	201.3	14.24	92.14	39.56	7.1	229.23	1140.16	1015.77	-11%
Konaseema	229.74	49.55	196.8	156.29	254.41	21.33	122.7	62.07	18.97	287.43	1295.02	1159.36	-10%
Krishna	194.36	131.35	163.4	221.13	182.82	46.34	97.34	58.58	23.44	207.05	1047.68	1369.97	31%
Kurnool	114.88	31.61	116.0	103.93	110.4	1.35	28.38	31.45	8.04	3.89	611.09	510.38	-16%
Nandyal	152.29	49.32	134.0	172.82	117.98	5.84	26.95	24.45	6.99	18.89	721.42	569.52	-21%
NTR	240	65.5	162.4	181.43	142.92	19.29	41.74	45.54	6.92	160.26	1035.14	1129.31	9%
Palnadu	139.75	67.31	134.5	210.32	118.21	22.34	67.91	64.21	12.05	163.27	775.29	935.35	21%
P. Manyam	209	197.7	206.1	248.02	137.39	24.13	54.75	11.68	4.21	49.56	1170.56	1230.17	5%
Prakasam	92.01	42.19	128.5	156.58	205.07	11.37	136.17	122.68	43.45	127.22	841.13	765.75	-9%
SPS Nellore	83.2	41.4	103.9	90.45	249.3	15.49	295.46	235.89	98.45	287.65	1050.55	899.37	-14%
S.S. Sai	93.4	19.2	125.6	183.19	120.11	6.62	40.6	47.29	10.04	26.03	590.9	536.27	-9%
Srikakulam	181.94	125.93	195.2	185.65	192.31	7.76	99.24	16.02	2.78	40.29	1166.26	937.74	-20%
Tirupati	114.03	109.01	124.2	210.63	203.48	34.99	288.94	197.26	122.71	289.58	1124.44	1103.07	-2%
Visakhapatnam	180.6	118.56	162.3	141.22	234.88	14.26	129.1	18.28	7	164.48	1117.75	954.61	-15%
Vizianagaram	187.94	160.37	184.7	190.36	178.7	9.88	79.21	11.07	4.13	109.04	1112.36	1044.95	-6%
West Godavari	250.23	81.33	186.9	209.62	203.04	28.15	93.65	65.09	16.25	314.18	1228.9	1225.38	0%
Y.S.R Kadapa	119.96	46.27	118.4	161.86	130.43	5.91	76.92	58.8	18.62	83.82	684.25	596.15	-13%
State Mean	165.4	89.4	155.3	183.0	167.5	19.4	95.5	61.9	22.3	157.4	976.2	975.9	0%

3.3.1 District wise Monsoon and Non-monsoon period rainfall during Assessment Year 2023-24

The analysis of monsoon and non-monsoon rainfall for the assessment year 2023–2024 is presented in Table 3.2. During the monsoon period, the normal rainfall was recorded as 575.36 mm, while the actual rainfall amounted to 550.47 mm, reflecting a -4% below the normal rainfall. Conversely, for the non-monsoon period, the normal and actual rainfall were recorded as 400.81 mm and 324.17 mm, respectively, indicating a below normal rainfall with a departure of -19%.

Table 3.2 District-wise Monsoon and Non-monsoon actual and normal rainfall during Assessment Year 2023-2024

DISTRICT	Monsoon (June to September 2023)		Non-monsoon (October 2023 to May 2024)	
	Normal (mm)	Actual (mm)	Normal (mm)	Actual (mm)
Alluri Sitharama Raju	894.96	841.81	395.62	328.4
Anakapalli	681.89	596.02	493.55	364.05
Ananthapuramu	319.69	302.45	192.77	143.68
Annamayya	380.00	339.62	363.67	370.05
Bapatla	504.61	692.1	420.72	439.5
Chittoor	458.74	538.44	456.06	394.53
East Godavari	775.88	636.01	368.9	346.1
Eluru	765.17	737.57	298.68	341.2
Guntur	573	765.63	323.65	353.43
Kakinada	670.13	500.9	470.03	382.54
Konaseema	778.95	476.88	516.07	446.98
Krishna	645.45	810.67	402.23	390.22
Kurnool	398.02	339.49	213.09	137.29
Nandyal	504.24	394.28	217.18	133.65
NTR	745.55	690.54	289.59	310.96
Palnadu	486.08	516.01	289.21	333.09
Parvathipuram Manyam	764.3	861.15	406.26	206.41
Prakasam	366.24	345.44	474.89	332.1
Sri Potti Sriramulu Nellore	320.55	230.45	730	583.46
Sri Sathya Sai	356.46	319.32	234.42	160.91
Srikakulam	696.44	681.81	469.82	170.75
Tirupati	414.26	441.17	710.18	577.68
Visakhapatnam	592.73	602.43	525.01	253.05
Vizianagaram	666.98	727.15	445.38	210.78
West Godavari	793.98	602.79	434.92	486.1
Y.S.R Kadapa	405.12	322.58	279.25	231.57
State Mean	575.36	550.47	400.81	324.17

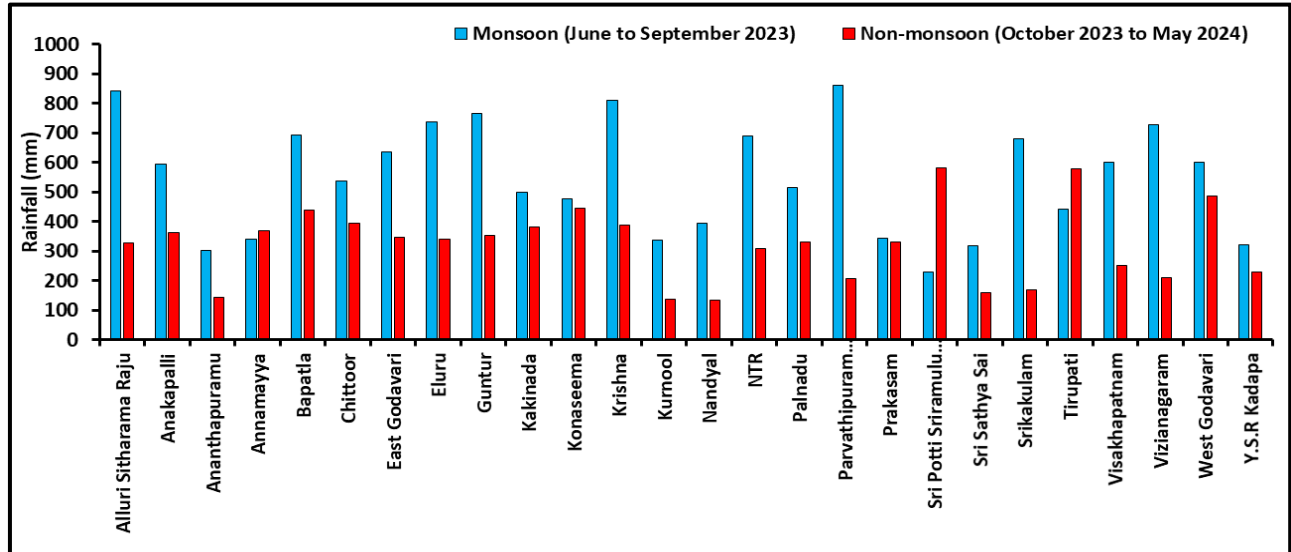


Fig.3.3: District-wise Monsoon and Non-monsoon rainfall during Assessment Year 2023-2024

CHAPTER 4

4.1 Geology/Hydrogeological Setup of Andhra Pradesh State

The State is characterized by various geological formations which forms the disposition of all 14 Principal aquifer (**Fig.4.1**) systems ranging in age from Archaean to Recent. Nearly 85% of the State is underlain by hard rock formations consisting of banded gneissic complex, granites, metamorphic and intrusives (Archaean), Precambrian quartzites, shales and limestones of Cuddapahs & Kurnools (Proterozoic), Deccan Trap basalts (Mesozoic). The remaining area is underlain by Gondwana Sedimentaries (Tertiary) and Alluvium (Sub Recent-Recent). The generalized litho- stratigraphic succession of the state is presented in **Table-4.1**.

Table-4.1 Geological Succession in Andhra Pradesh

<u>Age</u>	<u>System</u>	<u>Formation</u>	<u>Lithology</u>
Recent Recent to Sub-Recent		Alluvium	Gravel, sand, silt and clay. Laterite
----- Unconformity -----			
Mio-pliocene		Rajahmundry	Conglomerates, ferruginous gritty variegated sandstone and clays
----- Unconformity -----			
L.Eocene to U Cretaceous		Deccan Trap	Basalts with inter and infratrapeans, marls and shales
----- Unconformity -----			
L.Cretaceous to L. Tertiary	Upper Gondwana	Tirupati sandstones Raghavapuram shales Gollapally Sandstones Chintalpudi Sandstones	Gritty and ferruginous sandstone and clays Sandstones, shale and clays Sandstone, shale and clays
	Lower Gondwana		
----- Unconformity -----			
Late Proterozoic	Kurnool Super group		Nargi Limestones and shales, Bunganapalli Quartzites, Auk shales
Middle Proterozoic	Cuddapah Super Group		Shales, limestones, quartzites
Dharwar Super Group	Dharwars		Pegmatites, schists, phyllites, Dolerite dykes, Khondalites, charnockite
Archaean	Peninsular Gneissic Complex		Granites and gneiss

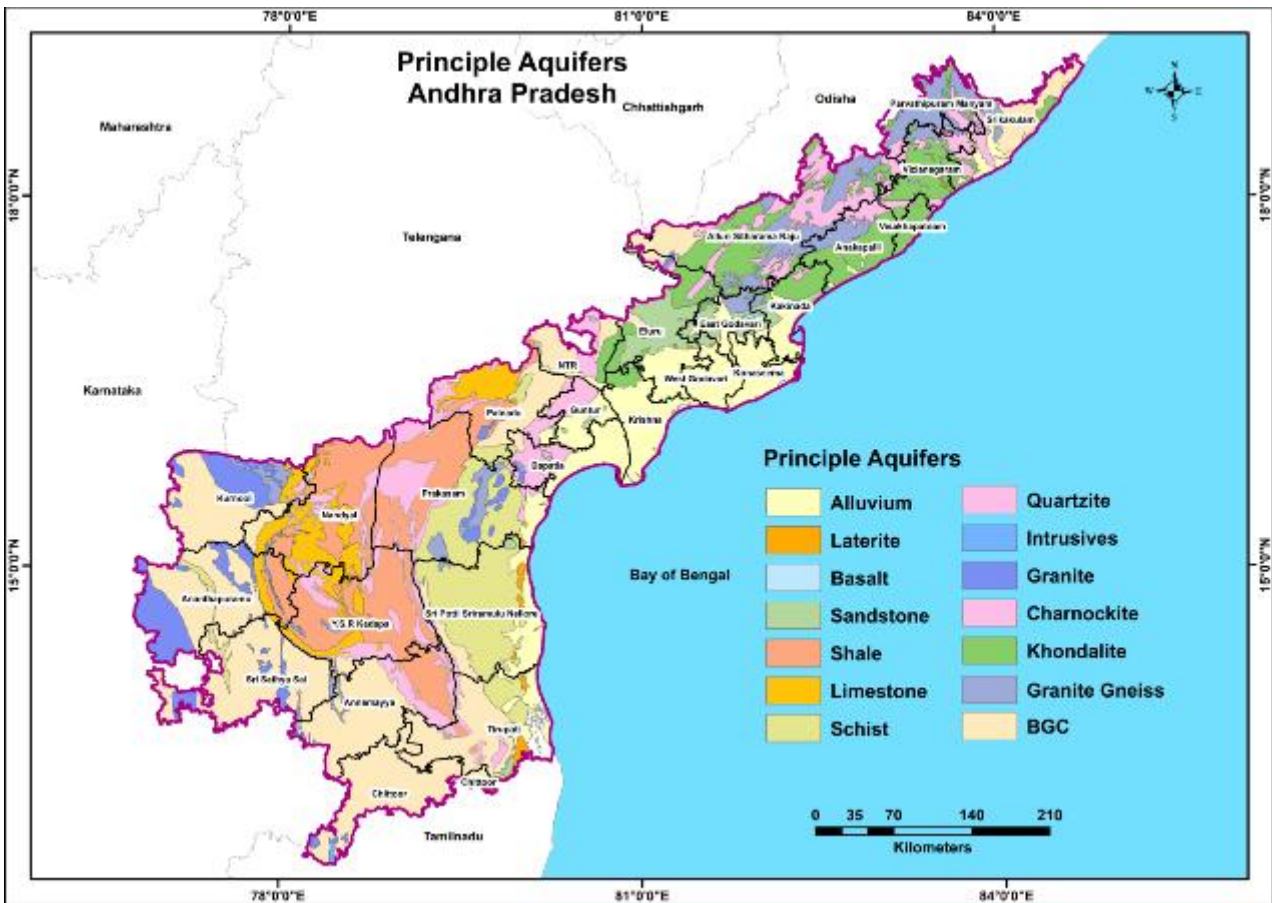


Fig. 4.1 Geology and Principal Aquifers of Andhra Pradesh

4.1.1 Archaean to Lower Proterozoic: Peninsular Gneissic complex comprising granites, gneisses and associated rocks occupy about 30 percent of the state. Volcano sediments assemblage occurs as linear belts in the gneissic terrain. These are called “Supra Crustals”, “Supra Crustal Association”, “Schist Belts”, “Green Stone Belts”, “Older Metamorphics”, etc. The Eastern Ghat mobile belt extends in NE–SW direction traversing through the eastern coastal districts. The mobile belt includes Khondalites and Charnockites group of rocks.

4.1.2 Middle to Upper Proterozoic: The crescent shaped Cuddapah basin lies in southcentral part of the state in the districts of Chittoor, Anantapur, Kurnool, Guntur and Krishna consisting mostly shales and quartzites. Kurnool Group of rocks consisting of quartzites, sandstones, shales and lime stones unconformably overlies the Cuddapah sediments in Kurnool, Cuddapah, Anantapur, Krishna and Guntur districts.

4.1.3 Gondwana Super Group: Gondwana Super Group spanning from Upper Carboniferous to Lower Cretaceous occupy along the NW–SE extend in parts of West Godavari district and their equivalents occur in coastal districts of Prakasam, Guntur, Krishna, West Godavari and Visakhapatnam. The Lower Gondwanas are represented by Chintalapudi sand stones and the Upper Gondwanas are represented by Gollapalli sandstones, Raghavapuram shales and Tirupathi sandstones. These formations comprise sandstones, shales and conglomerates.

4.1.4 Deccan Traps: In East Godavari and West Godavari districts beneath tertiaries, the Deccan Traps are exposed as a narrow belt.

4.1.5 Tertiaries: Tertiary rocks represented by Rajahmundry sandstones and their equivalents of Mio-Pliocene age occur in West Godavari, East Godavari, Chittoor, Nellore and Prakasam districts.

4.1.6 Quaternary: The Sub-Recent deposits are represented by laterite, which occur in the plain and the foot hills in Visakhapatnam, Vizianagaram and Srikakulam districts capping Khondalites. The Recent deposits in the form of unconsolidated sediments occur as coastal alluvium, deltaic and river alluvium occupying about 28,000 sq. km. The alluvial deposits occupy all along the coast line, in deltas of Godavari, Krishna and Pennar Rivers and also along the courses of major rivers. Alluvial deposits attain a thickness of more than 600 m in the Godavari and Krishna deltas. In Srikakulam and Visakhapatnam districts, the thickness is upto a maximum of 30 m, in Nellore district it varies from 50 m to more than 120 m.

4.2 HYDROGEOLOGY

Andhra Pradesh is characterized by various geological formations ranging from Archaeans to Recent age. In ground water point of view, the geological formations occurring in the state can broadly be grouped in to three categories namely the consolidated formations, semi-consolidated formations and unconsolidated formations. The consolidated formations comprise rocks of Peninsular Gneissic Complex, Dharwar and Eastern Ghats Super Group of Archaean Age, Cuddapah Super Group and Kurnool Group and their equivalents and the Deccan Trap Basalts. Gondwanas and Tertiaries are grouped under semi-consolidated formations while coastal deltaic and river alluvium constitute the unconsolidated formations.

Consolidated Formations rocks lack primary porosity. Weathering, fracturing, development of solution cavities and channels and vesicles make these rocks water bearing and water yielding. In consolidated formations, in general depth of weathering varies from 5 to 10 m bgl occasionally up to 30 m bgl. The general yield of the bore wells varies from 1 and 3 lps, though higher discharges upto 20 lps are noticed along lineaments, contact zones and solution cavities. Semi-consolidated formations represented by Gondwanas and Tertiaries form important multiple aquifers. Deccan Traps are hard and compact and possess meager primary porosity by virtue of vesicles. Ground water in the traps occur in vesicular, weathered and fractured zones and inter-trappean and infra-trappean beds. Fracturing is significant in the traps often lead to columnar joints. Depth of weathering varies from 5 to 20 m bgl. Ground water in coastal alluvial deposits is shallow, and quality of water is usually either brackish or saline. The yield of these wells range from <1 to as high as 44 lps. The drilling in deltaic alluvium revealed that there is a lot of heterogeneity in hydrogeological conditions both spacially and vertically. Fresh water is generally limited to shallow to moderate depths only. Deeper aquifers are generally saline. Salinity in these aquifers could be related to entrapped sea water at the time of deposition of sediments. In Paleo-channels ground water is noticed fresh.

CHAPTER-5

GROUND WATER LEVEL SCENARIO IN THE ANDHRA PRADESH STATE

Groundwater Level Scenario (2023)

Groundwater level is one of the basic data elements, which reflects the groundwater regime in an area. Central Ground Water Board (CGWB), SR Hyderabad monitors groundwater levels four times a year during January, May, August and November through a network of fixed observation wells spreading throughout Andhra Pradesh State. The primary objective of monitoring the groundwater level is to record the response of groundwater regime to the natural and anthropogenic stresses on recharge and discharge components which are governed by geology, climate, physiography, land use pattern and hydrologic characteristics. Natural conditions affecting the regime include climatic parameters like rainfall, evapotranspiration etc. Anthropogenic influences include pumpage from the aquifer, recharge due to irrigation systems and other practices like waste disposal etc. Water level data of CGWB and Andhra Pradesh State Ground Water department have been used for assessment of groundwater resources. An outline of groundwater scenario during the period of assessment is given below.

5.1 Groundwater level data for Pre- Monsoon 2023

5.1.1 Depth to Water Levels (Unconfined & Confined/Semi-Confined)

Analysis of depth to water level data shows that water levels range between 0.04 m bgl (Sri Sathya sai district) to 65.93 m bgl (Prakasam district). Depth to water level map of May, 2023 (**Fig.5.1**) shows that, shallow water level of less than 2 m bgl is noticed in 1.3 % area, mainly observed as isolated patches in Ananthapuramu, Eluru, Konaseema, Krishna, Palnadu, SPS Nellore, Sri Sathya Sai and West Godavari districts. Water level between 2 to 5 m bgl is covered in 39 % of area predominantly in Sri Sathya Sai, Guntur, SPS Nellore, Palnadu, Srikakulam, Anakapalli, Annamayya, Alluri Sitharama Raju, Tirupati, Kurnool, Bapatla, and Konaseema districts and water level of 5 to 10 m bgl in 45% of area predominantly in Sri Sathya Sai, Prakasam, SPS Nellore, Alluri Sitharama Raju, Chittoor, Vizianagaram and YSR Kadapa districts. Water level of 10 to 20 m bgl is covered in 12 % of the area of the state, mainly in Annamayya, Prakasam, Eluru, Chittoor, SPS Nellore and Srikakulam districts. Deeper water levels of more than 20 m are observed in 4% area of the state.

5.2 Groundwater level data for post-monsoon 2023

5.2.1 Depth to Water Levels (Unconfined & Confined/Semi-Confined)

Analysis of depth to water level data of unconfined and confined/semi-confined aquifers) shows that water levels range between 0.3 m bgl (Tirupati district) to 68 m bgl (Sri Sathya Sai district). Depth to water level map of November, 2023 (**Fig.5.2**) shows that, shallow water level of less than 2 m bgl is noticed in 2% of State area, mainly observed in parts of Bapatla, Guntur Srikakulam, Nandyal, West Godavari, Palnadu, Alluri Sitharam Raju, Vizianagaram, Anakapalli, Konaseema, East Godavari, Sri Sathya Sai and Eluru districts. Water level between 2 to 5 m bgl is noticed in 34 % of State area, observed mainly in coastal districts of Andhra Pradesh and also YSR Kadapa, Tirupati, Kurnool, Annamaya, Ananthapuramu, Alluri Sitharam Raju, Parvathipuram Manayam, Palnadu, NTR, Sri Satya Sai districts. Water level between 5 to 10 m bgl observed in 45% of State area, 10 to 20 m bgl in 16% of State area. Deeper water levels of more than 20 m bgl are observed in 3% area covering mainly Prakasam, Annamaya, Chittoor, West Godavari, Sri Sathya Sai, Eluru and Nandyal districts.

5.3. Fluctuation of Groundwater Level:

5.3.1 Comparison of Pre-monsoon 2023 to Pre-monsoon 2022

In the state, about 51% of the area experienced rise in water levels compared with pre-monsoon period (May, 2022) (**Fig 5.3**). Rise in water level of less than 2 m is mainly observed parts of Palnadu, Alluri Sitharama Raju, Krishna, Prakasam, Srikakulam, Konaseema, Ananthaparamu, Eluru and NTR districts. Water level rise of 2 to 4 m is predominantly observed in parts of Alluri Sitharama Raju, Prakasam, Ananthaparamu, Nandyal and Sri Sathya Sai districts. Rise in water level of more than 4 m is observed in parts in Prakasam, Sri Sathya Sai, Eluru, Alluri Sitharama Raju and Ananthaparamu districts. In the state, 49% of the area experienced fall in water levels compared to the pre-monsoon period (May, 2022). Less than 2m fall observed as small patches in Prakasam, Vizianagaram, Annamayya, YSR Kadapa, Bapla, Palnadu, Alluri Sitharama Raju and Anakapalli districts. Water Level in the range of 2 to 4 m fall mainly observed in Rayalseema region and in parts of SPS Nellore, Eluru, Palnadu and Vizianagaram districts. Water level > 4m fall observed in small patches in Eluru, Chittoor, Prakasam, SPS Nellore and Annamaya districts.

5.3.2 Comparison of November 2023 to November 2022

In the state about 4% of the area experienced rise in water levels compared November, 2022 (**Fig 5.4**). Rise in water level of 0 to 2 m is observed throughout the district, mainly in Prakasam, Bapla, Chittoor, Annamaya, Krishna and Vizianagaram districts. Water level rise of 2 to 4 m is observed parts of Krishna, Bapla, Alluri Sitharam Raju, Chittoor and Prakasam districts. Rise in water level of more than 4 m is observed mainly in Krishna, Eluru and Annamaya districts. In the state about 96% of the area experienced fall in water levels compared to the November 2022. Less than 2m fall observed throughout the district, mainly in Alluri Sitharam Raju, YSR Kadapa, East Godavari, Srikakulam, Palnadu, Krishna, Konaseema, Prakasam, Chittoor, West Godavari, Eluru, Bapla, Guntur and Sri Sathya Sai districts. 2 to 4 m water level fall observed significantly in Chittoor, Sri Sathya Sai, Prakasam, SPS Nellore, Vizianagaram, Ananthaparamu, Palnadu, Tirupati, Anakapalli, Eluru, YSR Kadapa and Kurnool districts. Water level > 4m fall observed significantly in Rayalseema region

The monsoon in 2023 witnessed significant fluctuations in rainfall pattern across the state. The evident decline in annual ground water level decline during November 2023 in Andhra Pradesh can be attributed to a substantial -36% deficit rainfall when comparing 2023 to 2022. This drastic reduction in rainfall has far-reaching implications, leading to a further depletion of groundwater resources. Rainfall distribution varied with some part of coastal Andhra Pradesh received good rainfall during July 2023 while Rayalseema region experienced comparatively weaker rainfall.

5.3.3 Comparison of Pre-Monsoon 2023 with decadal mean of Pre-Monsoon (2013 to 2022)

In the state about 90% of the area experienced rise in water levels compared with decadal mean (**Fig 5.5**). Rise in water level of less than 2 m is observed in most of the districts, significantly in Konaseema, Alluri Sita Rama Raju Palnadu, Prakasam, Bapla, NTR, Krishna, YSR Kadapa, East Godavari and West Godavari districts. Water level rise of 2 to 4 m is recorded mainly in Rayalseema region such as Ananthapuramu, Palnadu, Prakasam, Tirupati, Sri Sathya Sai, Chittoor, YSR Kadapa, Kurnool, SPS Nellore and parts of Prakasam and Nandyal districts. Rise of water level more than 4 m is mainly observed in Prakasam, Kurnool, Sri Sathya Sai, Annamayya and Tirupati districts. The notable rise in the water level in 90% of the area with respect to decadal mean can be

attributed to the above normal rainfall received during June 2020 to May 2023. In the state about 10% of the area experienced fall in water levels compared with decadal mean (2013-2022). Fall in water level of less than 2 m is observed mainly in parts of Parvathipuram Manyam, Prakasam, Srikakulam, Vizianagaram and Visakhapatnam districts. Fall of 2 to 4 m is recorded as isolated patches in parts of Ananthapuramu, Annamayya, Prakasam, Visakhapatnam and Parvathipuram Manyam districts. Fall of water level more than 4 m is observed as isolated patches in Parvathipuram Manyam, Srikakulam, Nandyal, Kakinada, SPS Nellore and NTR districts.

5.3.4 Comparison of Post-Monsoon 2023 with decadal mean of Post-Monsoon (2013 to 2022)

In the state about 33% of the area experienced rise in water levels when compared with decadal mean (**Fig 5.6**). Out of 153 wells, water level rise of less than 2 m is recorded in 60% wells, in the range of 2 to 4 m in 18% wells and rise of more than 4 m is recorded in 22% wells. Rise in water level of less than 2 m is seen in all the districts, significantly in Rayalseema region. Water level rise of 2 to 4 m is observed mainly in YSR Kadapa, Ananthapuramu, Sri Sathya Sai, Prakasam and Chittoor districts. Rise in water level of more than 4 m is significantly observed in YSR Kadapa, Sri Sathya Sai and Annamaya districts. In the state about 67% of the area experienced fall in water levels when compared with decadal mean (2013-2022). Fall in water level of less than 2 m is observed in all districts mainly in parts of Palnadu, Konaseema, Alluri Sitharam Raju, Krishna, SPS Nellore, Srikakulam, East Godavari, Krishna and Guntur districts. Fall of 2 to 4 m is recorded in Vizianagaram, Srikakulam, Eluru, SPS Nellore, Parvathipuram Maniyam, Palnadu, Visakhapatnam, Tirupati, Kakinada, and Kurnool districts. Fall in water level of more than 4 m is recorded mainly in Ananthapuramu, Nandyal, SPS Nellore, Palnadu, Srikakulam, Anakapalli, Prakasam, West Godavari and Kakinada districts.

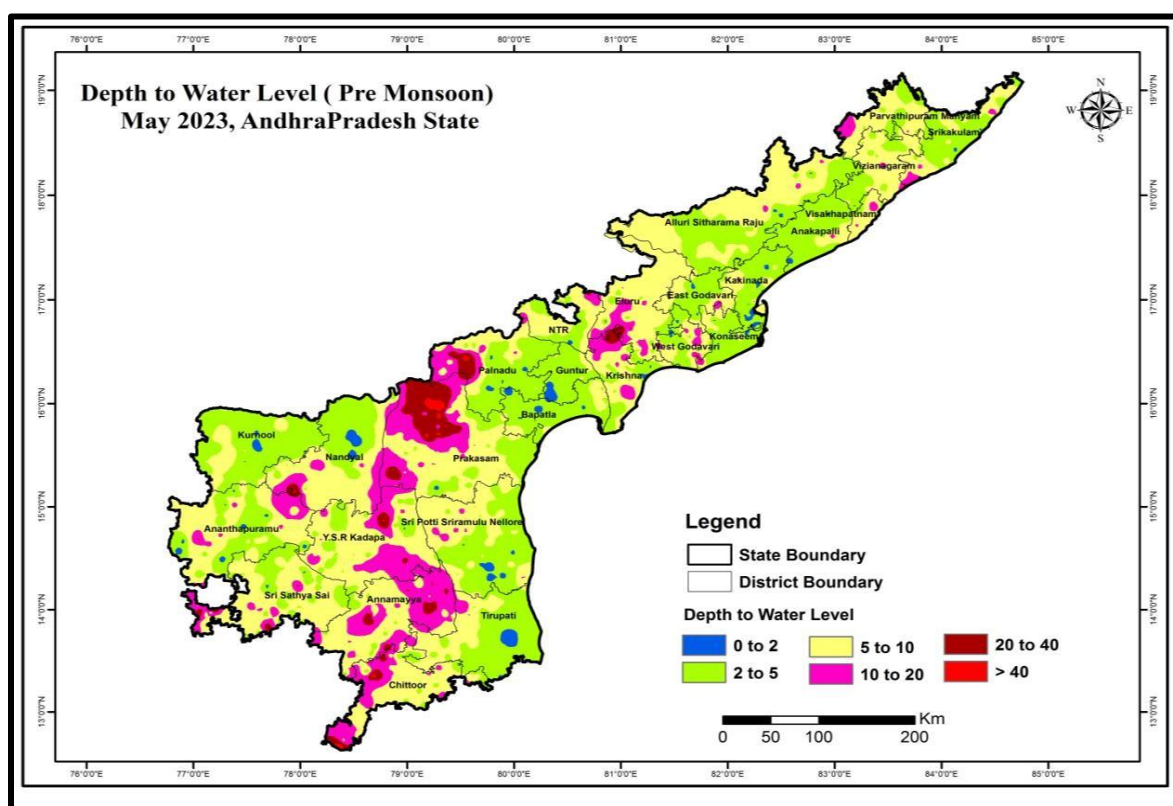


Fig 5.1 : Depth to Water Level Map of the State/UT (Pre Monsoon 2023)

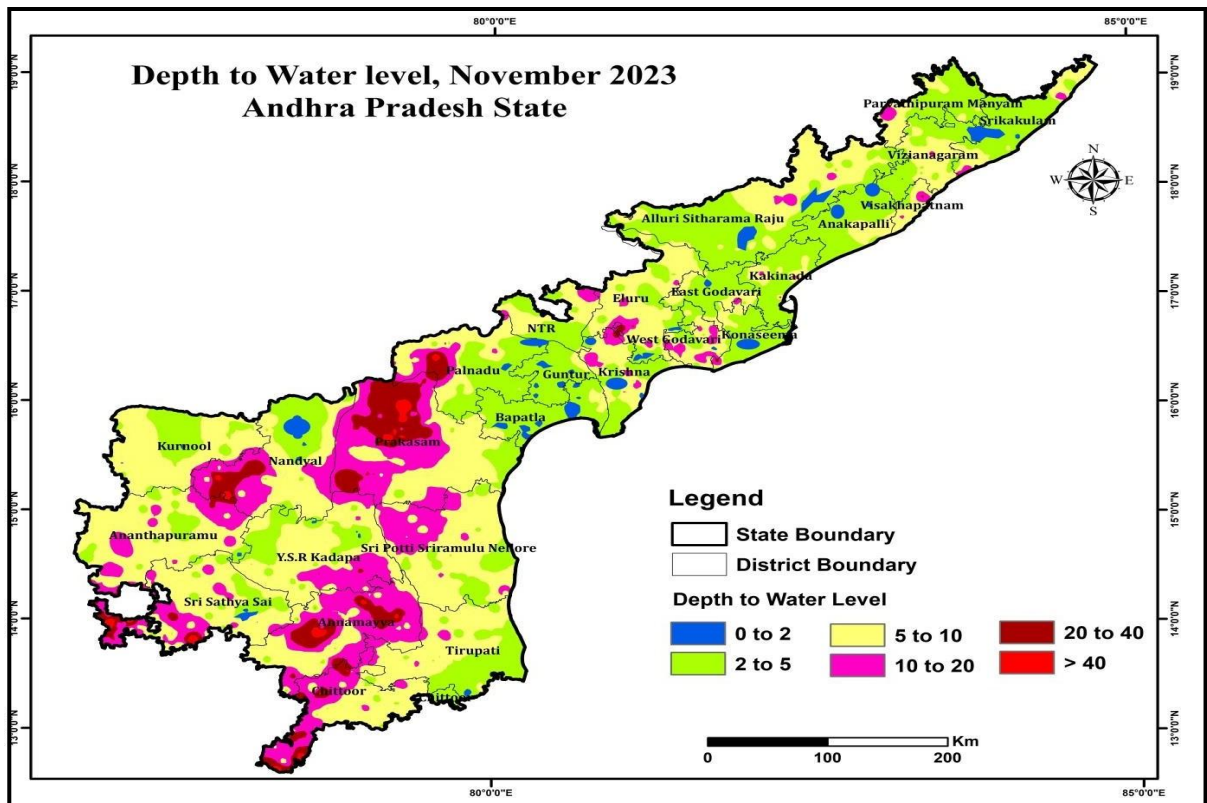


Fig 5.2 : Depth to Water Level Map of the State/UT (Post Monsoon 2023)

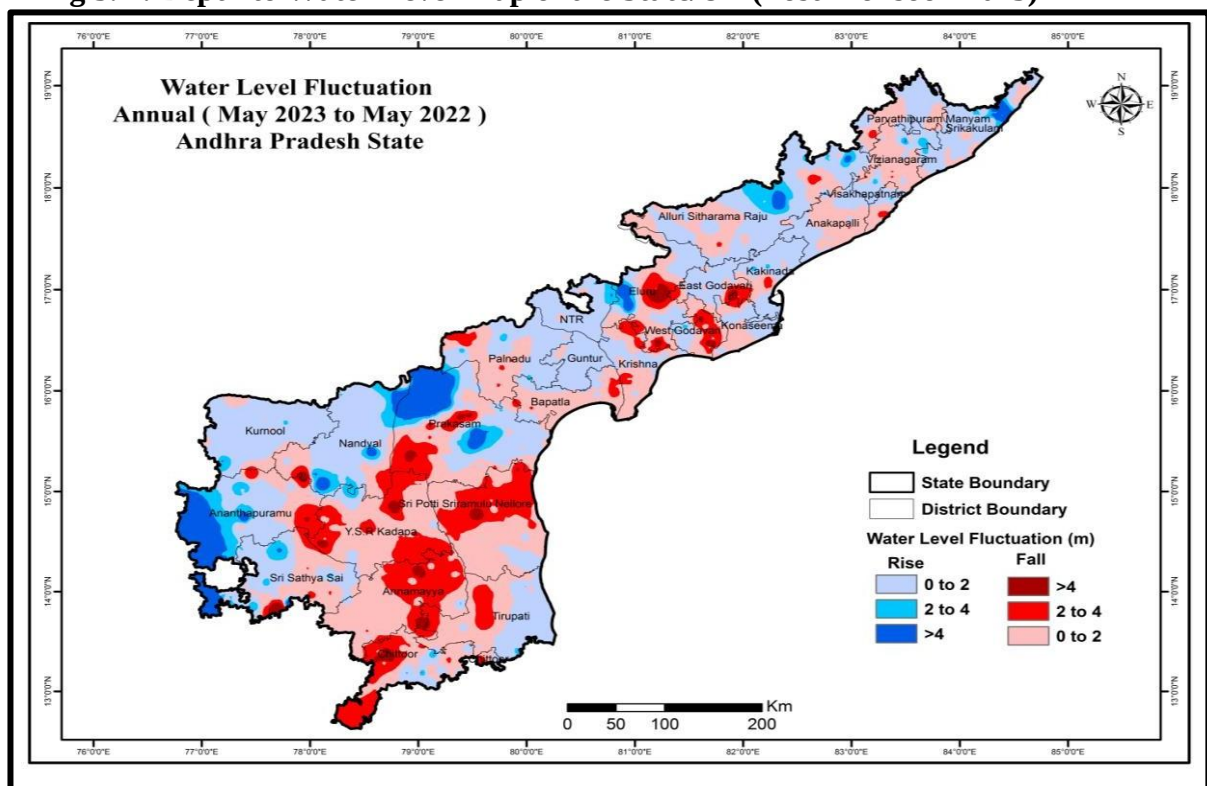


Fig 5.3 : Groundwater Level Fluctuation: Pre-monsoon 22 compared to Pre-monsoon 2

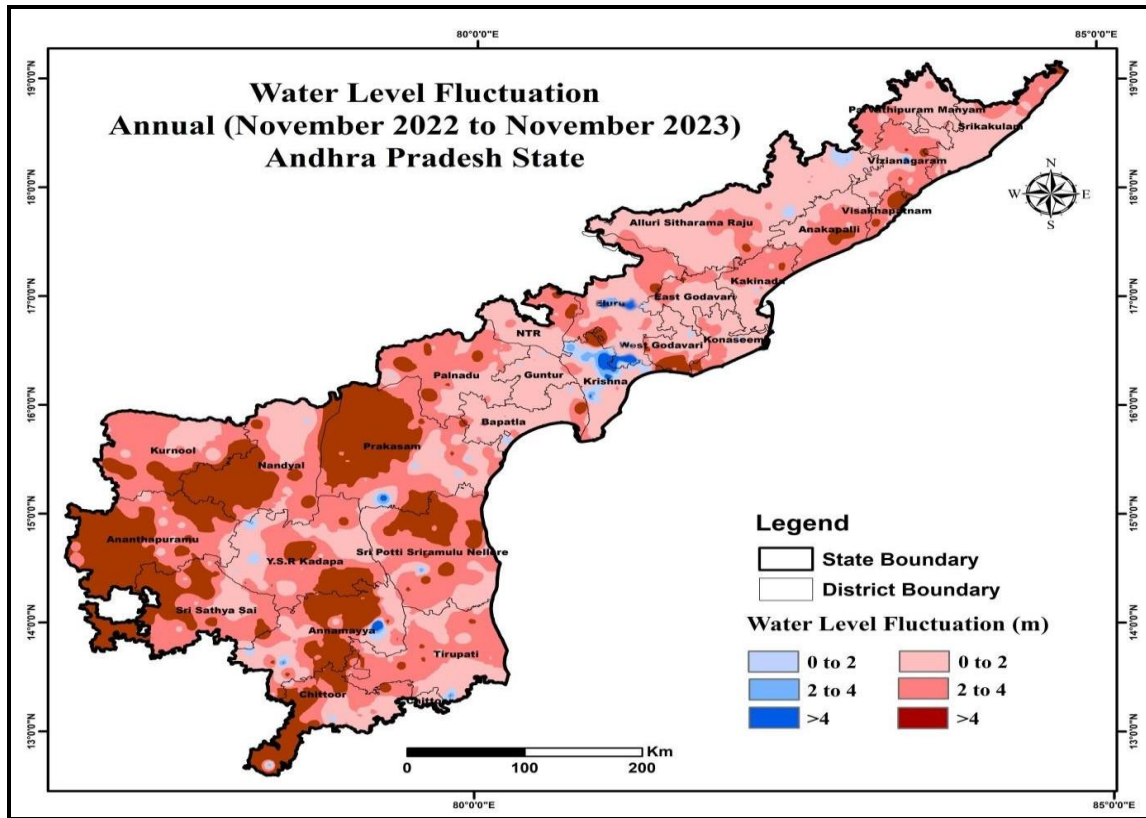


Fig 5.4: Groundwater Level Fluctuation: November 22 compared to November 23

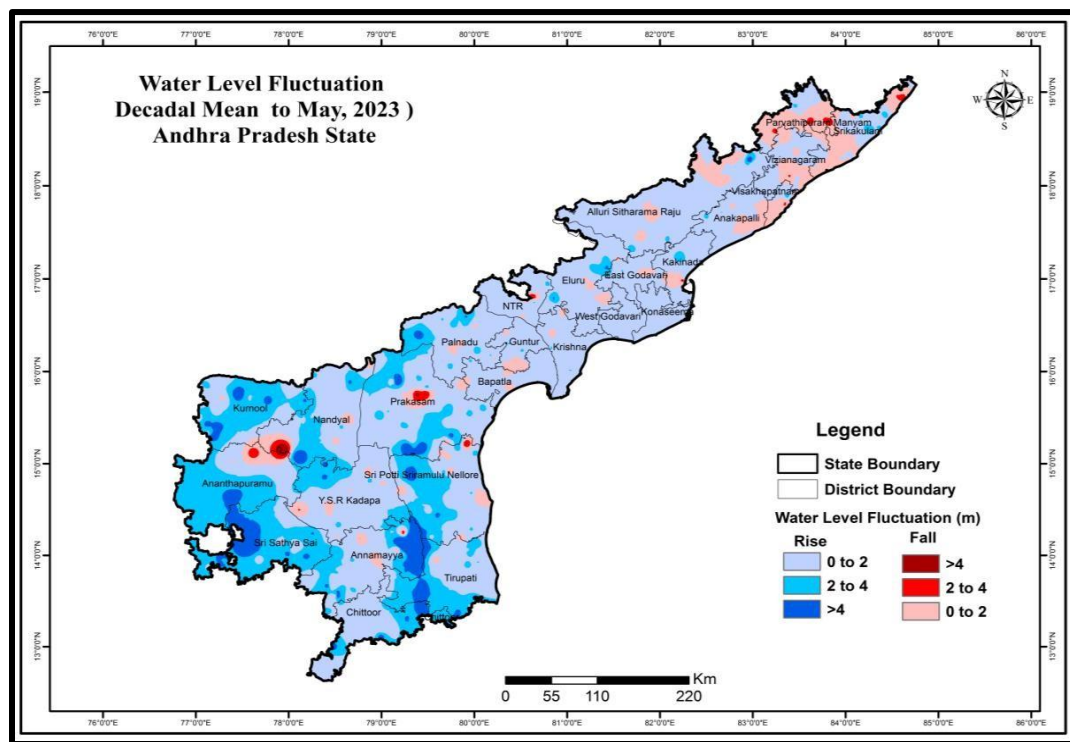


Fig 5.5: Decadal water level fluctuation with mean Pre-Monsoon (2013 to 22) and Pre-Monsoon 23

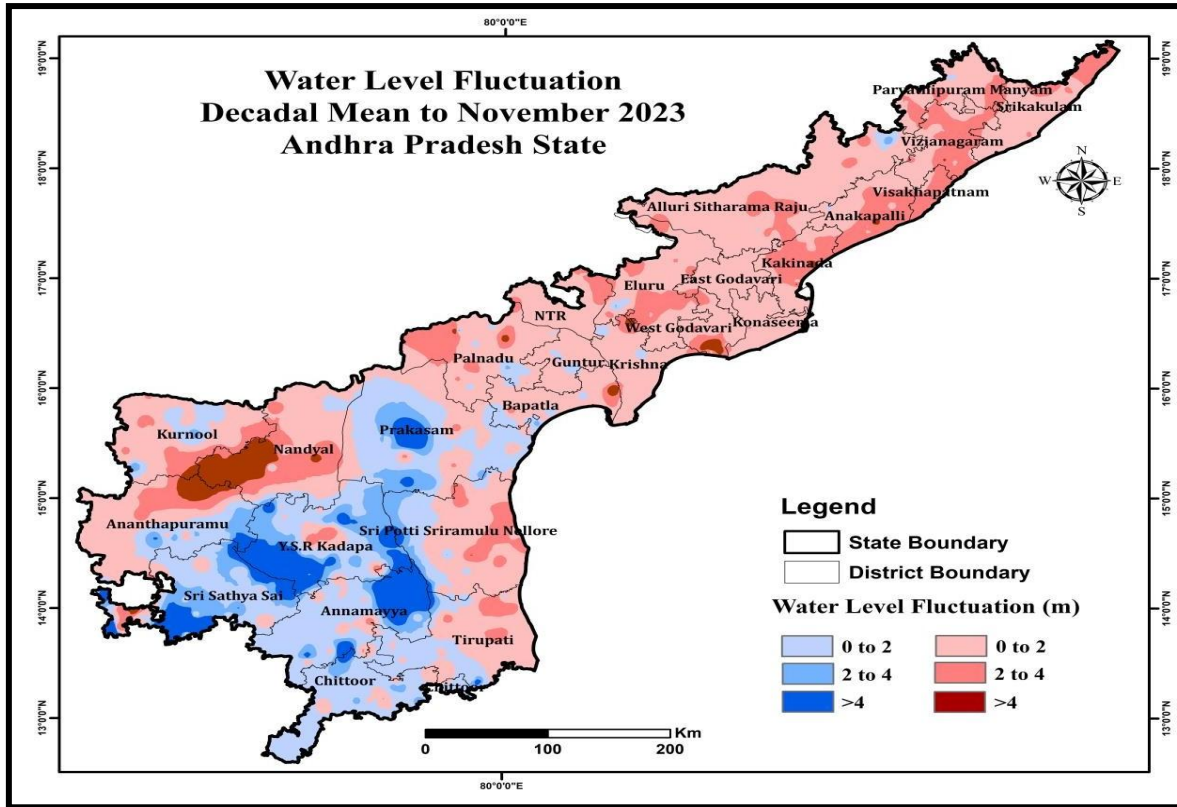


Fig 5.6: Decadal water level fluctuation with mean post-monsoon (2013 to 22) and post-monsoon 23

CHAPTER 6

GROUND WATER RESOURCES OF THE ANDHRA PRADESH STATE

The Dynamic ground water resources (as in 2024) of Andhra Pradesh state have been assessed jointly by CGWB and State Ground Water Department of Andhra Pradesh under the supervision of the State level Committee. The dynamic ground water resources are also known as Annual Ground Water Recharge, since it gets recharged every year from rainfall and other sources (secondary sources) such as applied irrigation water, surface water bodies, water conservation structures, etc. Methodology adopted for the assessment has been outlined in Chapter-2 of this report. This section provides a summary of the Ground water Resources Assessment 2024 (GWRA-2024) of the Andhra Pradesh State.

6.1. ANNUAL GROUND WATER RECHARGE

As per the 2024 assessment of Dynamic Ground Water Resources, the Total Annual Ground Water Recharge for the entire state has been assessed as 27.8 (bcm) and Total natural discharges works out to be 1.39 bcm. Hence, Annual Extractable Ground Water Resources for the entire state is 26.41 bcm. Recharge from Other sources 'viz. canal seepage, return flow from irrigation, recharge from tanks, ponds and water conservation structures taken together during monsoon and non-monsoon period is the major contributor of total annual groundwater recharge in the Andhra Pradesh state, which is 17.97 bcm or 65% of the total recharge (Monsoon season: 39%, non-monsoon season: 26%) and the remaining 35% (Monsoon season: 32%, Non monsoon season: 3%) or 9.82 bcm is recharge from rainfall. Ground Water Recharge Distribution (MCM), GWRA-2024 (Fig-6.6). In order to compare the unit recharge of different assessment units, the volumetric estimates of annual ground water recharge had been converted to depth units (m) by dividing the annual ground water recharge (ha.m) by the area of the respective assessment units (hectare). Spatial variation in annual ground water recharge (m) is shown in Fig-6.5.

6.2. ANNUAL EXTRACTABLE GROUND WATER RESOURCES

The total annual extractable groundwater resource of Andhra Pradesh State for the year 2024 has been estimated at 26.41 (bcm). The total annual groundwater recharge of the state has been assessed at 27.8 bcm, with a provision of 1.39 bcm allocated for natural discharge. In districts such as Alluri Sitharama Raju, Ananthapuramu, Eluru, Kakinada, Konaseema, Krishna, Sri Potti Sriramulu Nellore, Sri Sathya Sai, Srikakulam, Tirupati and Vizianagaram, the extractable groundwater resources constitute approximately 58%.

6.3. AANUAL TOTAL GROUND WATER EXTRACTION

The assessment of ground water extraction is carried out considering the Minor Irrigation Census data by the Andhra Pradesh State Ground Water Department. The Total Annual Ground Water Extraction of the Andhra Pradesh for the year 2024 has been estimated as 7.88 bcm. Irrigation accounts for 86% of the total annual groundwater extraction, which amounts to 6.75 bcm. The domestic use accounts for 13% (01 bcm), while industrial use represents 1% (0.13 bcm) of total annual groundwater extraction of the State. In Anakapalli, Chittoor , Eluru , Kurnool , Palnadu Prakasam , Sri Potti Sriramulu Nellore , Sri Sathya Sai , Srikakulam and Y.S.R Kadapa districts where the ground water extraction for domestic uses is more than 65 % (Fig-6.7 & 6.8).

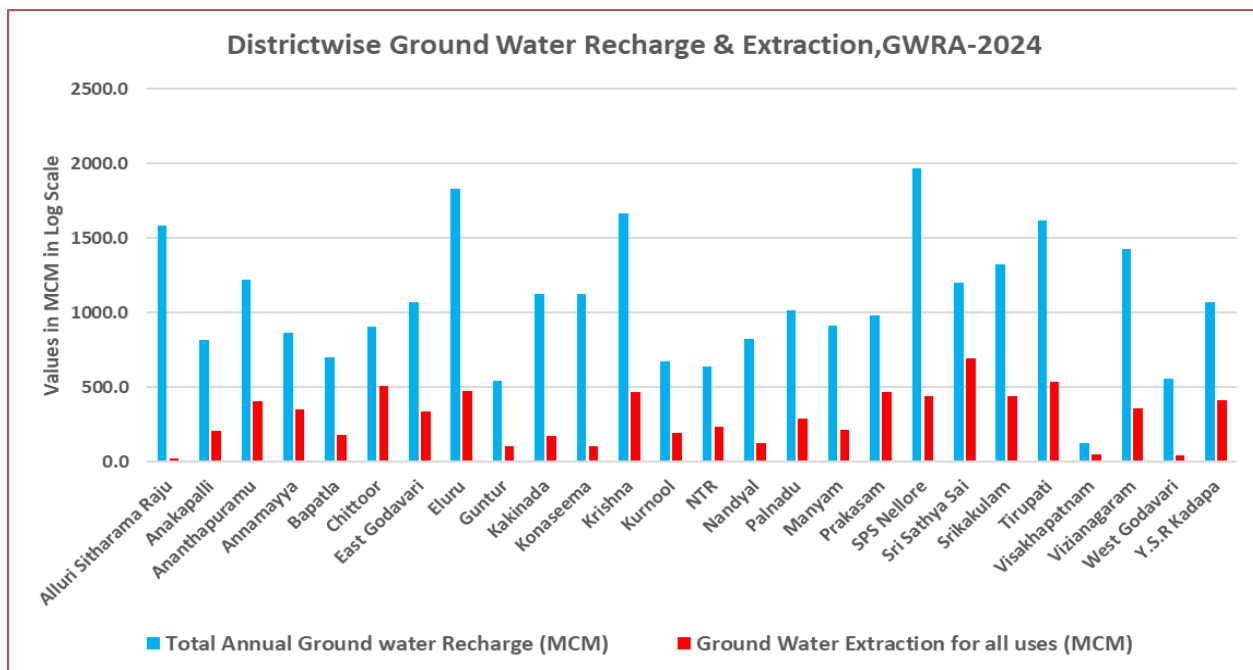


Fig 6.1: District Wise Ground Water Recharge & Extraction (GWRA 2024)

6.4. STAGE OF GROUND WATER EXTRACTION

The overall stage of groundwater extraction in Andhra Pradesh is 29.83%. It ranges from 1.5% in Alluri Sitharama Raju to 60.7% in Sri Sathya Sai. Across all 26 districts, groundwater extraction remains below 70%. (Fig-6.2)

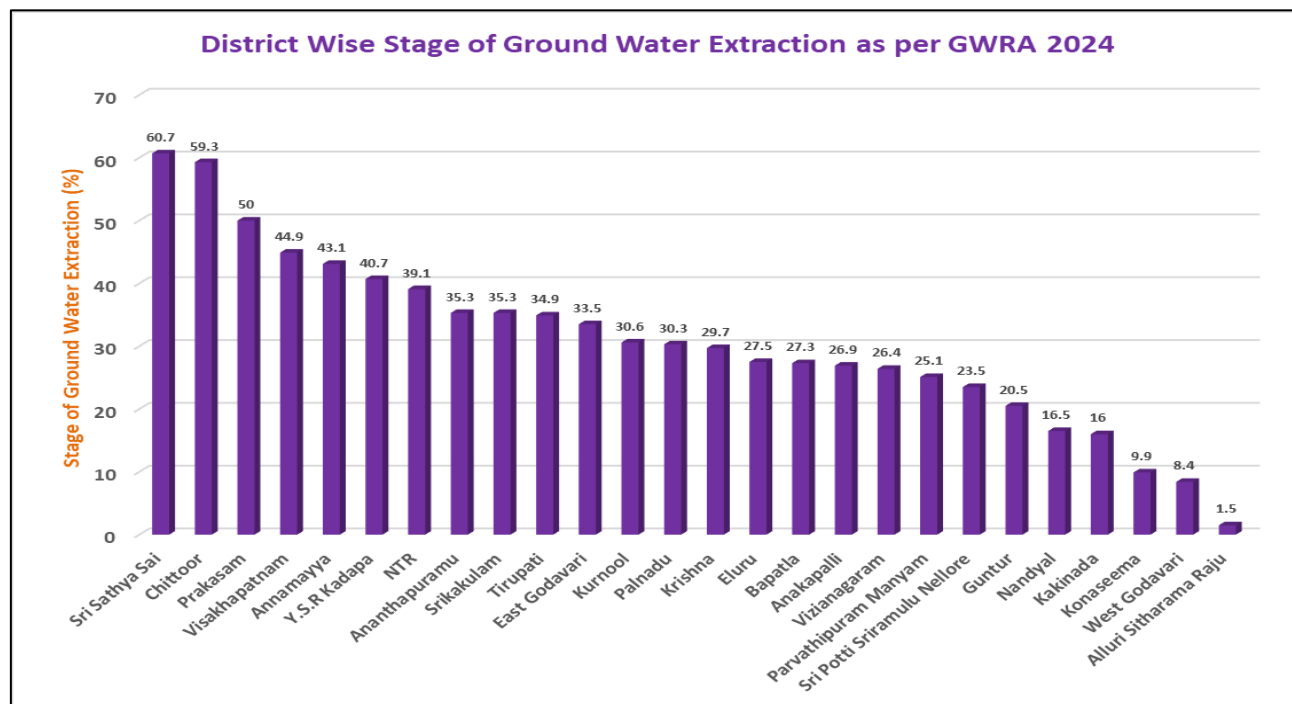


Fig 6.2: District Wise Stage of Ground Water Extraction (GWRA 2024)

6.5. CATEGORIZATION OF ASSESSMENT UNITS

Out of 679 assessment units (mandals), 09 (1.32%) units have been categorized, as 'Over-exploited', 02 units (0.29%) as 'Critical', 38 units (5.59%) as 'Semi-Critical', 591 units (87.03 %) as 'Safe' and 39 units categorized as 'Saline' (5.74%). Similarly, out of 140378.61 sq km recharge worthy area of the State, 2278.23 sq km (1.62 %) area are under 'Over-Exploited', 236.68sq km (0.17 %) under 'Critical', 7947.83 sq km (5.66 %) under 'Semi-critical', 123819.54 sq km (88.2 %) under 'Safe' category of assessment units. 6096.33 sq km (4.34%) area is under 'Saline' category of assessment units. Out of total 26411.44 mcm annual extractable ground water resources of the State, 261.89 mcm (0.99%) are under 'Over-exploited', 66.12 mcm (0.25 %) under 'Critical', 897.37 mcm (3.4 %) under 'Semi-critical' and 25056.01 mcm (94.87 %) are under 'Safe' categories of assessment units. (Fig-6.3 & 6.9)

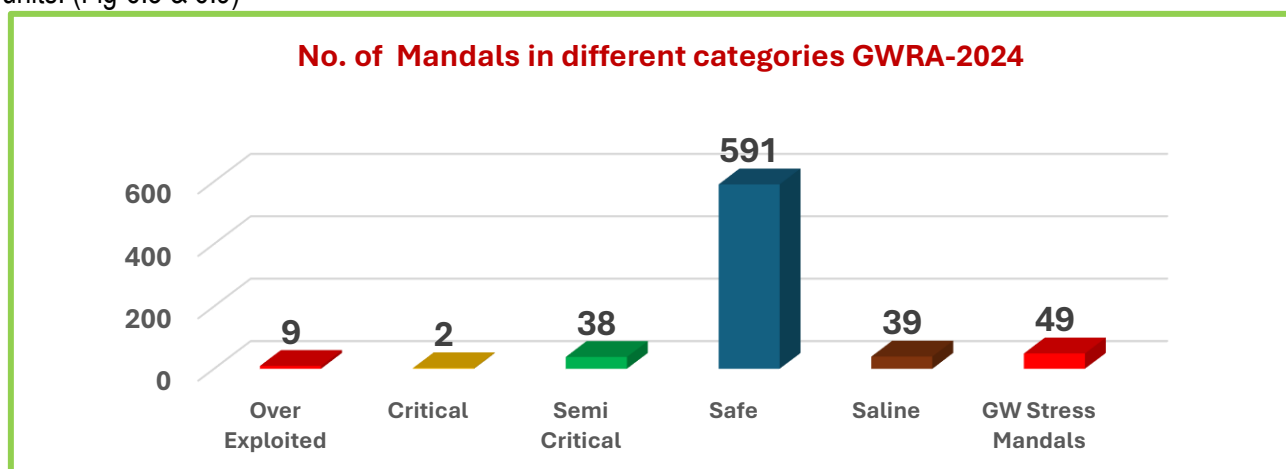


Fig No 6.3 No of Mandals/Blocks in different categories GWRA-2024

6.6. COMPARIOSN WITH PREVIOUS ASSESSMENT

As compared to the 2023 assessment, the total Annual Ground Water Recharge for the state shows a negligible difference, decreasing slightly from 27.83 bcm to 27.80 bcm and a difference of just 0.03 bcm. This minimal change is because reduce recharge from other sources like Canals, Surface water irrigation, Ground water irrigation, Tanks, ponds and water conservation structures. Increase in groundwater extraction. (Fig-6.4)

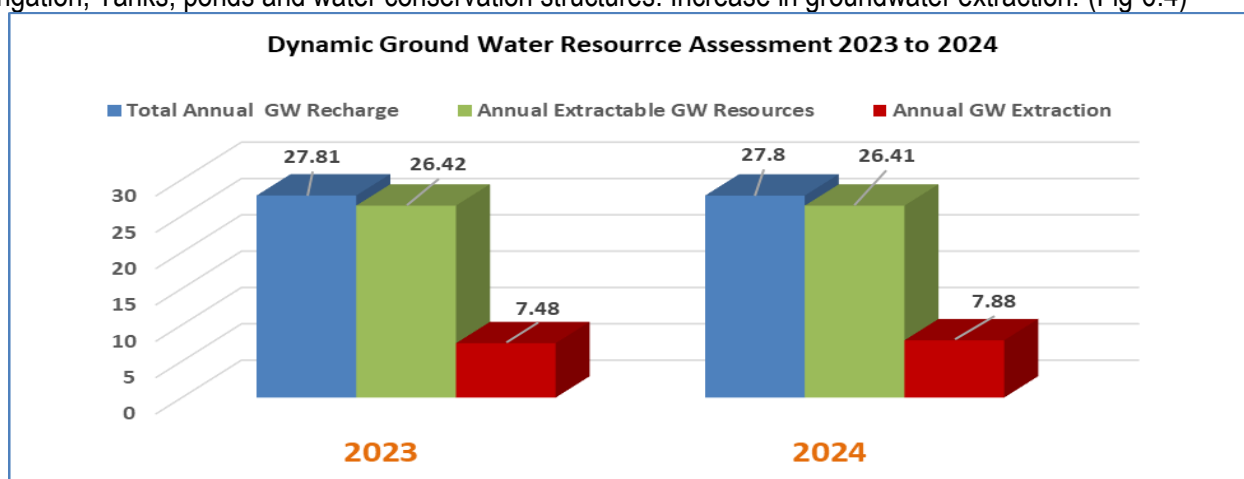


Fig No 6.4 Dynamic Ground Water Resource Assessment 2023 to 2024

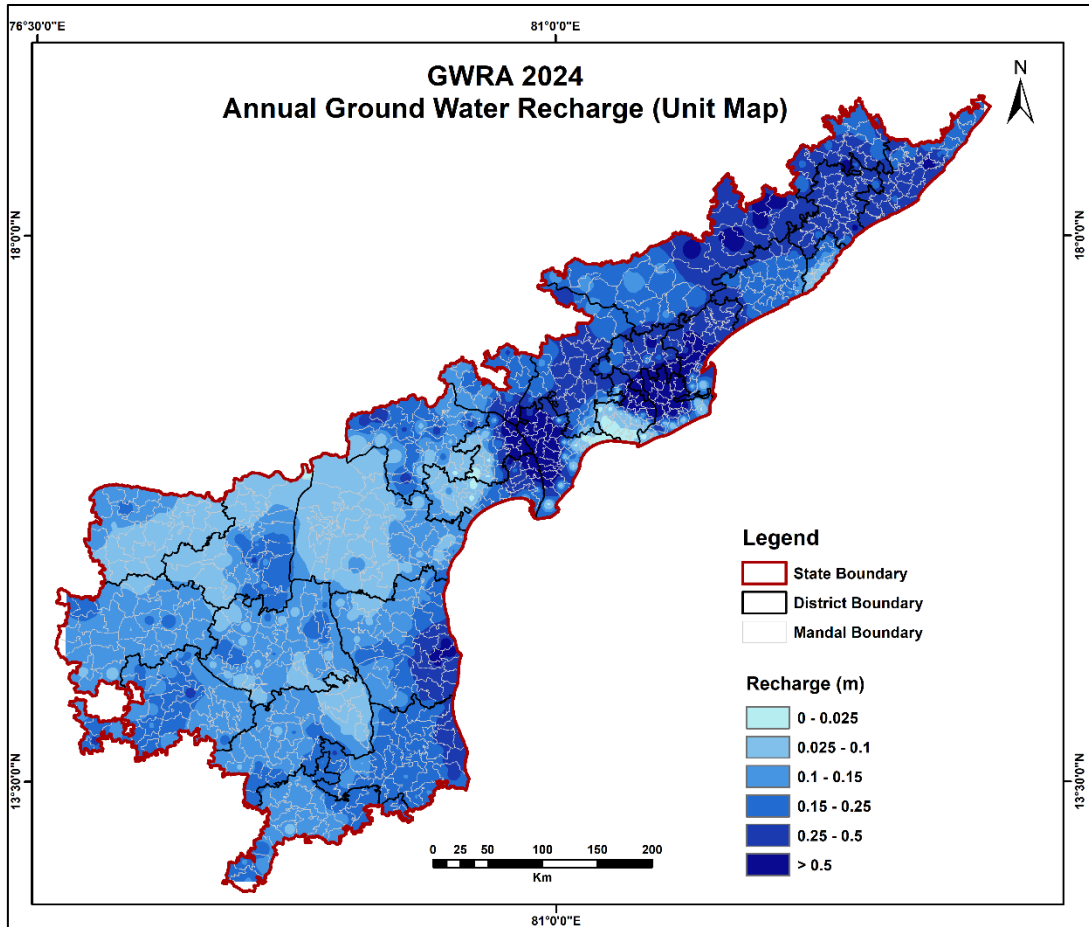


Fig 6.5: Annual Ground Water Recharge 2024 of Andhra Pradesh (Unit Map in m)

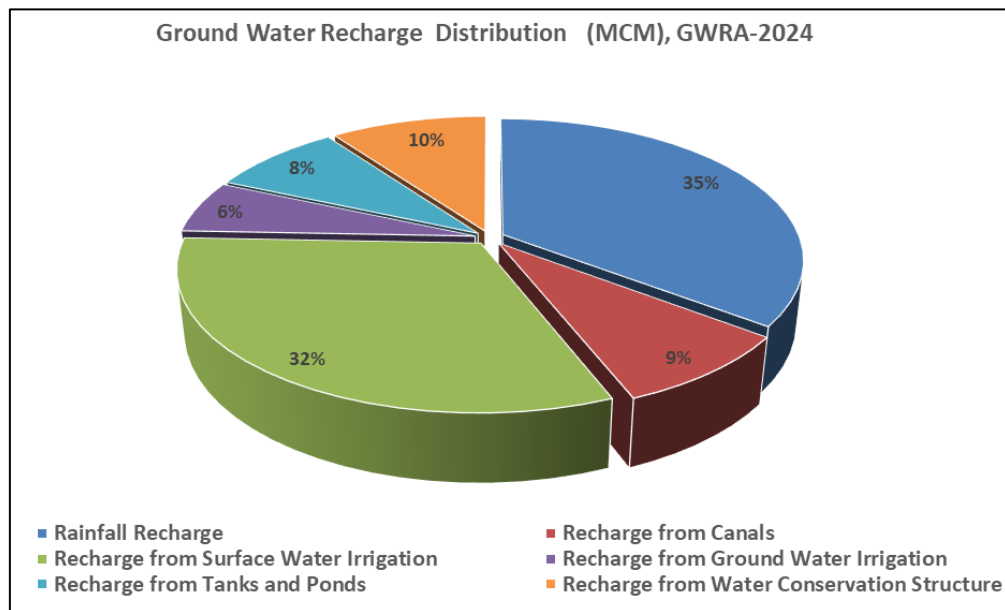


Fig 6.6 Ground Water Recharge Distribution (MCM), GWRA-2024

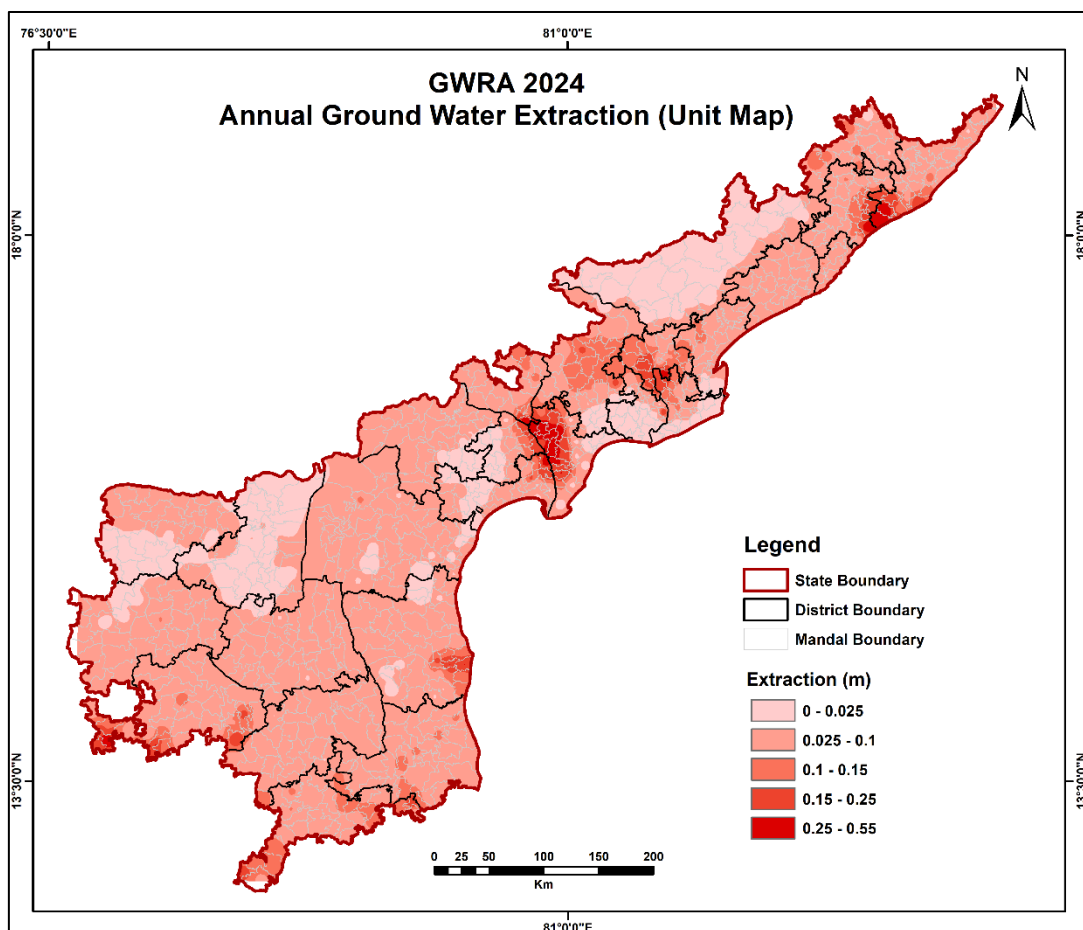


Fig 6.7: Annual Ground Water Extraction 2024 of Andhra Pradesh (Unit Map in m)

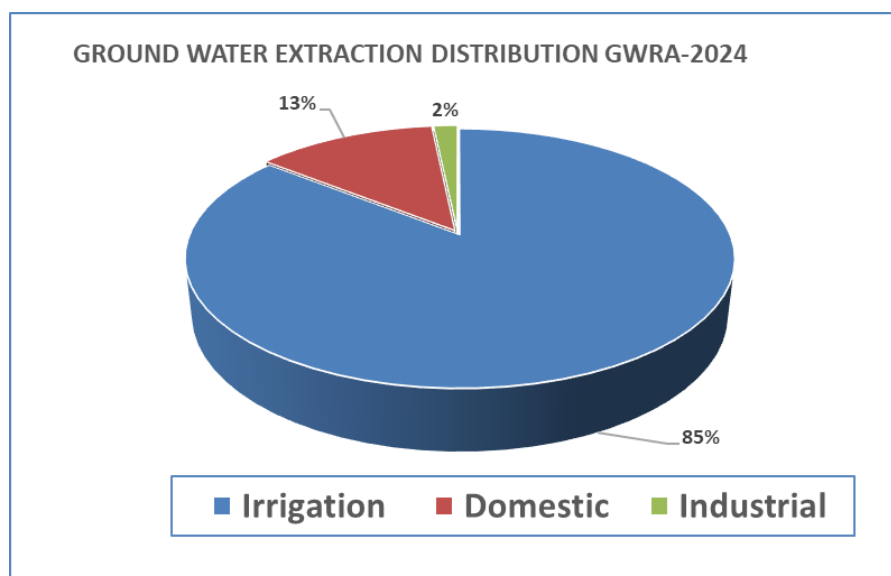


Fig 6.8 Ground Water Extraction Distribution, GWRA-2024

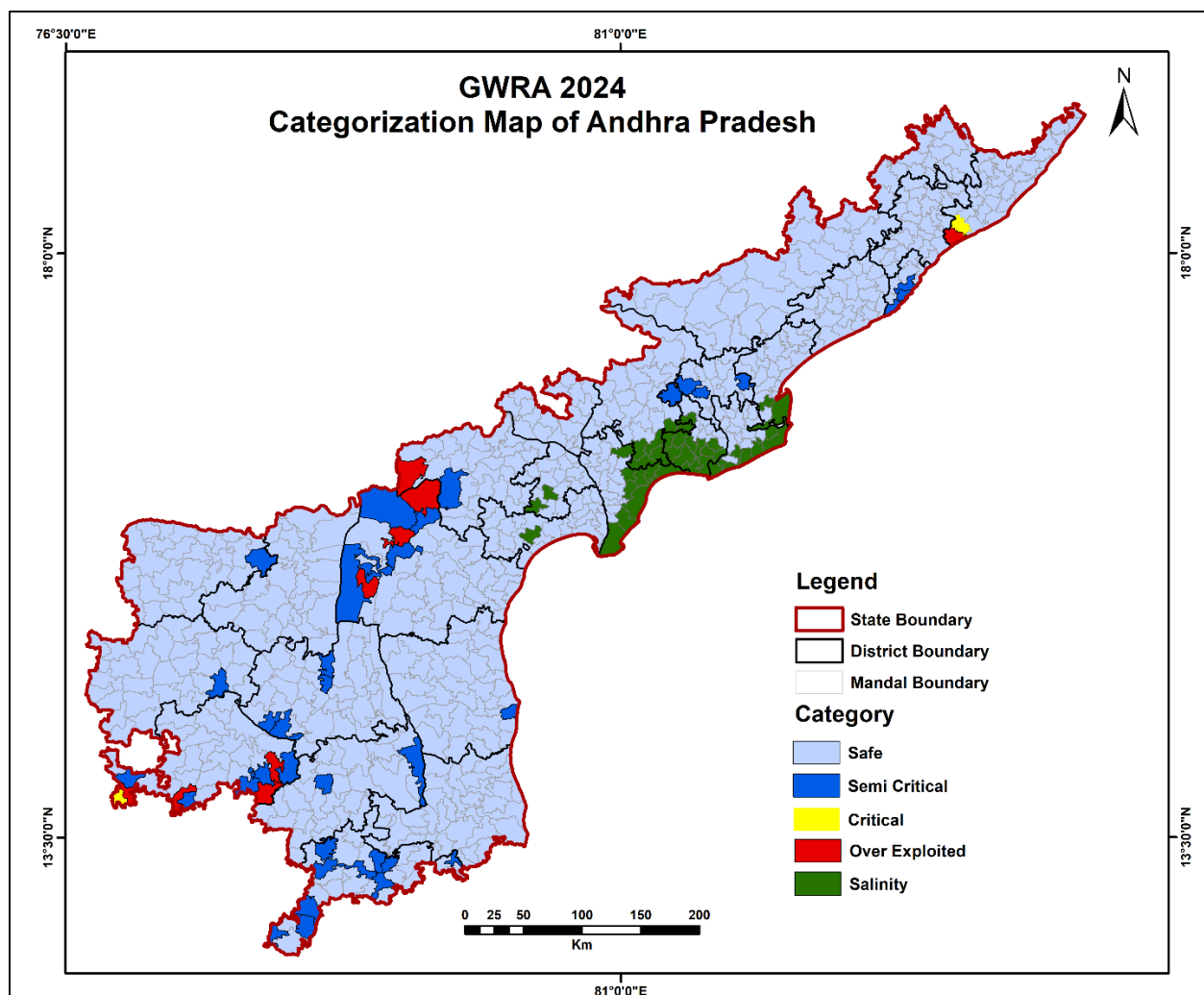


Fig 6.9: Block wise Categorization of Assessment Unit in Andhra Pradesh state

CHAPTER 7

7.0 District Wise Dynamic Groundwater Resources-2024

7.1 Alluri Sitharama Raju District

The Alluri Sitharama Raju district, Andhra Pradesh having geographical area of 12253.42 km², and lies between 17° 17' and 18° 21' of the Northern Latitude and 80° 53' to 82° 50' of the Eastern Longitude. Administratively, the district is governed by 02 Revenue Divisions viz., Paderu and Rampachodavaram with 22 Revenue mandals and 3326 villages. The total population of the district as per 2011 population census is ~9.54 lakhs. The normal rainfall of the district is 1286 mm. The main river that drains the Alluri Sitharama Raju district is Godavari River.

The geology of Alluri Sita Ramaraju district is primarily composed of rocks from the Eastern Ghat Mobile Belt, including the Khondalite Group, Charnockite Group, and Migmatite complex, with sediments of Cenozoic laterite and Quaternary deposits overlying these formations. The district's lithology features Khondalite, Charnockite, Gneiss, and Banded Gneissic Complex as the major units, along with smaller occurrences of Granite and Alluvium.

Total Annual Groundwater recharge of the Alluri Sitharama Raju district has been assessed as 1.58 bcm and Annual extractable Ground Water resource as 1.50 bcm. The Annual Ground Water Extraction is 0.02 bcm and Stage of Ground Water Extraction is 1.50 %. Out of 22 assessment units (Mandals), all mandals in the district are classified as safe category and there is no 'Saline' category of assessment unit in the district. Similarly, out of 6392 sq km recharge worthy area of the district, 6392 sq km (100 %) under 'Safe' categories of assessment units. Out of total 1503 mcm annual extractable ground water resources of the district, 1503 mcm (100 %) are under 'Safe' categories of assessment units.

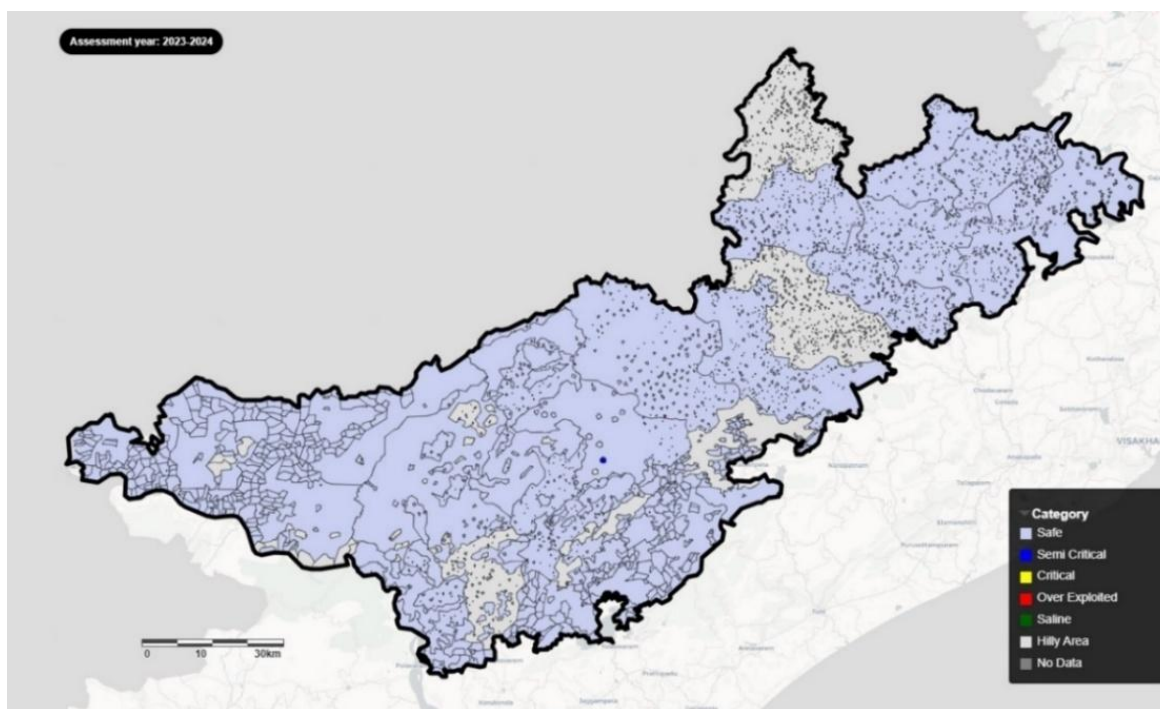


Fig 7.1: Categorization Map of Alluri Sitharama Raju District (Village level).

7.2 Anakapalli District

The Anakapalli District, Andhra Pradesh State having geographical area of 4292 km² and lies between 17° 21' and 17° 59' N latitude and 82°49' and 83°20' East longitude. Administratively, the district is governed by 24 Revenue mandals and 753 villages. The total population of the district as per 2011 population census is ~17.5 lakhs. The annual rainfall of the district is 883.2 mm. The main river that drains the Annakapalli district is tributaries of Nagavalli river.

The geology of Anakapalli district is primarily composed of Archaean group of rocks includes Khondalites and Charnockites of Eastern Ghat super group and Granitic gneisses of Migmatite group. The recent alluvium is prevalent along the rivers and coastal areas.

Total Annual Groundwater recharge of the Anakapalli district has been assessed as 0.82 bcm and Annual extractable Ground Water resource as 0.78 bcm. The Annual Ground Water Extraction is 0.21 bcm and Stage of Ground Water Extraction is 28.35 %. Out of 24 assessment units (Mandals), all mandals in the district are classified as safe category and there is no 'Saline' category of assessment unit in the district. Similarly, out of 3459 sq km recharge worthy area of the district, 3459 sq km (100 %) under 'Safe' categories of assessment units. Out of total 778 mcm annual extractable ground water resources of the district, 778 mcm (100 %) are under 'Safe' categories of assessment units.

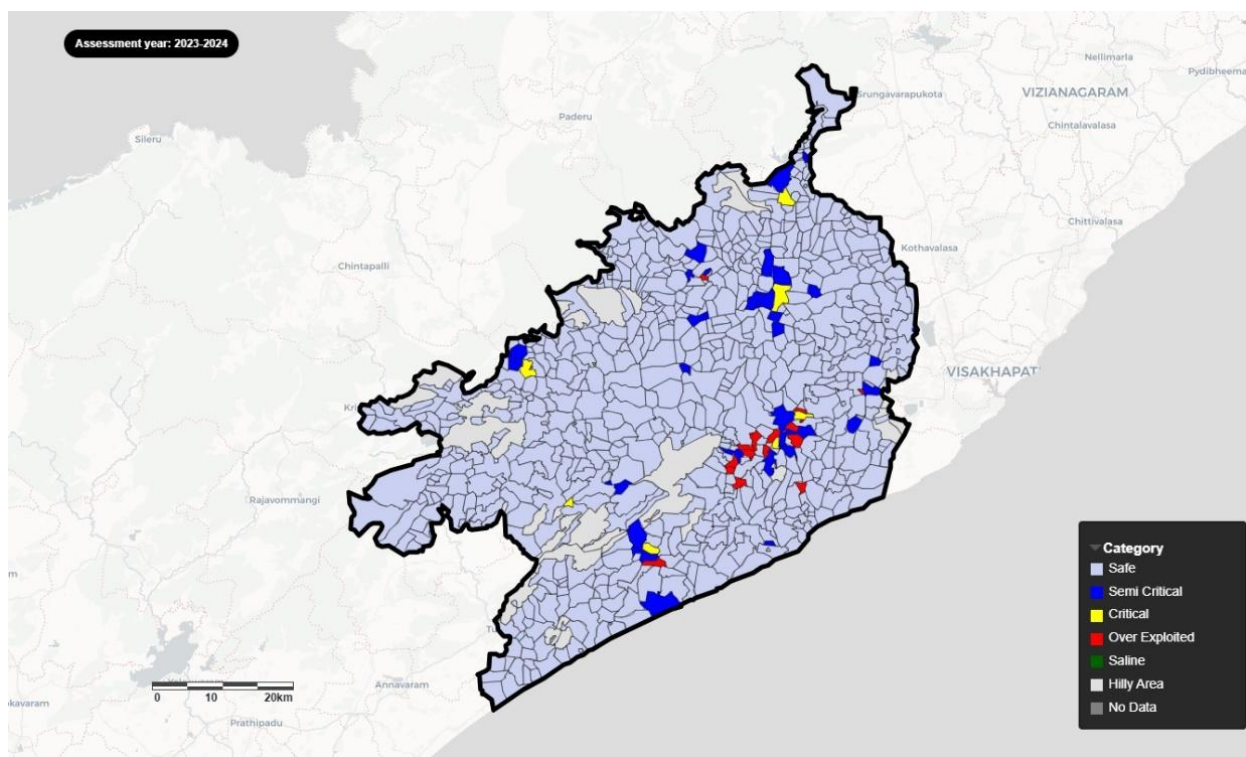


Fig 7.2: Categorization Map of Anakapalli District (Village level)

7.3 Ananthapuramu District

The Ananthapuramu district, Andhra Pradesh covering 10,443 sq.km and lies between north latitude of 14°12'42" - 15°15'16" and east longitude of 76°45'43" - 78°08'47". The district is bounded on the north by Kurnool, on the east by Nandyal and Y.S.R. Kadapa, on the south by Sri Sathya Sai and Pavagada taluk of Tumkur district, Karnataka state and on the west by Chitradurga district of Karnataka state. Administratively, the area is governed by 3 Revenue Divisions (Anantapuramu, Guntakal and Kalyandurg), 32 Revenue Mandals and 506 revenue villages with a population of 22,26,878 (2011 census) (Rural: 67%, Urban: 33%) with average density of 218 persons/sq.km. Administratively, the area is governed by 3 Revenue Divisions, 31 Revenue Mandals and 506 Revenue villages with a population of 22.26 lakhs (2011 census) with average density of 218 persons/sq.km. The annual normal average rainfall of the district is 508 mm.

The district falls under Krishna basin and Krishna Middle Sub-Basin. The major river flowing in the district is Pennar, Pedda Hagari, Chinna Hagari, Tadakaleru and Pandameru river. The major ground water flow is towards S-N and E-W. Geologically, about 74 % of the area is underlain by crystalline rocks namely Banded gneissic complex and granites of Archaean to Proterozoic age and remaining 26 % of the area of the district is underlain by schist, metasediments of Cuddapah supergroup of Proterozoic age, namely shales, limestone and intrusive as well. Pediplain is the major landform covering about 63% of the district. The other landforms observed are pediment, structural hills, denudational hills, residual hills etc.

Total Annual Groundwater recharge of the Ananthapuramu district has been assessed as 1.22 bcm and Annual extractable Ground Water resource as 1.16 bcm. The Annual Ground Water Extraction is 0.41 bcm and Stage of Ground Water Extraction is 37.68 %. Out of 32 assessment units (Mandals), 1 unit have been categorized as 'Semi-critical' and 31 units as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 9465 sq km recharge worthy area of the district, 248.73 sq km area are under 'Semi-Critical' and 9216 sq.km under 'Safe' categories of assessment units. Out of total 1158 mcm annual extractable ground water resources of the district, 29.17 mcm are under 'Semi-Critical' and 1129 mcm are under 'Safe' categories of assessment units.

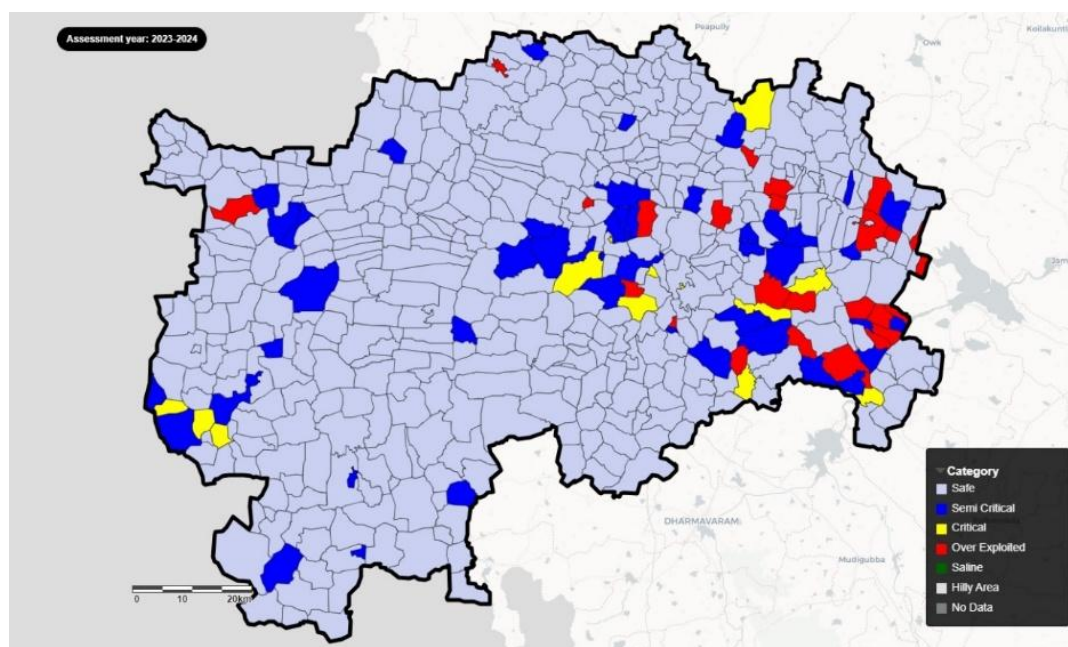


Fig 7.3: Categorization Map of Ananthapuramu District (Village level)

7.4 Annamayya District

The Annamayya district, Andhra Pradesh having geographical area of 7,951 km² and situated within the geographical co-ordinate of 13°19'50" and 14°42'32" of northern latitude and 78°18'55" and 79°20'26" Eastern longitude. The latitude varies from 269 to 3787 meters above sea level. The district is bounded on North by YSR Annamayya District, South by Chittoor District, West by Anantapur District and East by Nellore & Chittoor Districts. It is part of the River Pennar basin wherein Cheyyeru River joins Penneru at Gundlamada in Vontimitta mandal. Administratively the district is governed by 3 revenue Division, 30 mandals with 462 Gram Panchayats and 463 revenue villages and 5,673 habitations with a population of ~16.97 lakhs (2011 census). The density of population in the district is 213 per Sq.km. The average annual rainfall of the district is 743.13 mm, of which 373.41 mm falls as 50% South-West (June-September) monsoon and 285.2 mm as 38% North-East (October-December) monsoon.

The Annamayya District is underlain by various rock types belong to Late Archaean or Early Proterozoic era which are succeeded by rocks of Dharwarian Age and both are traversed by dolerite dykes. The older rocks are overlain by rocks of Cuddapah Super group and Kurnool Group belonging to Middle and Upper Proterozoic Age. The major rock types are quartzites, shales, limestones, phyllites, granites, granodiorites and granite gneiss. The Archaean comprises the Peninsular Gneissic Complex, represented by granite, granodiorite, granite-gneiss and migmatite. These rock types occur in the southwestern part of the district. Both the Archaean and Dharwar are traversed by dolerite dykes and quartz reefs. Alluvium consisting of gravel, sand, silt and clay occur along the river courses

Total Annual Groundwater recharge of the Annamayya district has been assessed as 0.87 bcm and Annual extractable Ground Water resource as 0.82 bcm. The Annual Ground Water Extraction is 0.35 bcm and Stage of Ground Water Extraction is 46.02 %. Out of 30 assessment units (Mandals), 2 units have been categorized as 'Semi-critical' and 28 units as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 6661 sq km recharge worthy area of the district, 412.58 sq km area are under 'Semi-Critical', and 6248 sq.km under 'Safe' categories of assessment units. Out of total 822 mcm annual extractable ground water resources of the district, 28.57 mcm are under 'Semi-Critical', 793 mcm are under 'Safe' categories of assessment units.

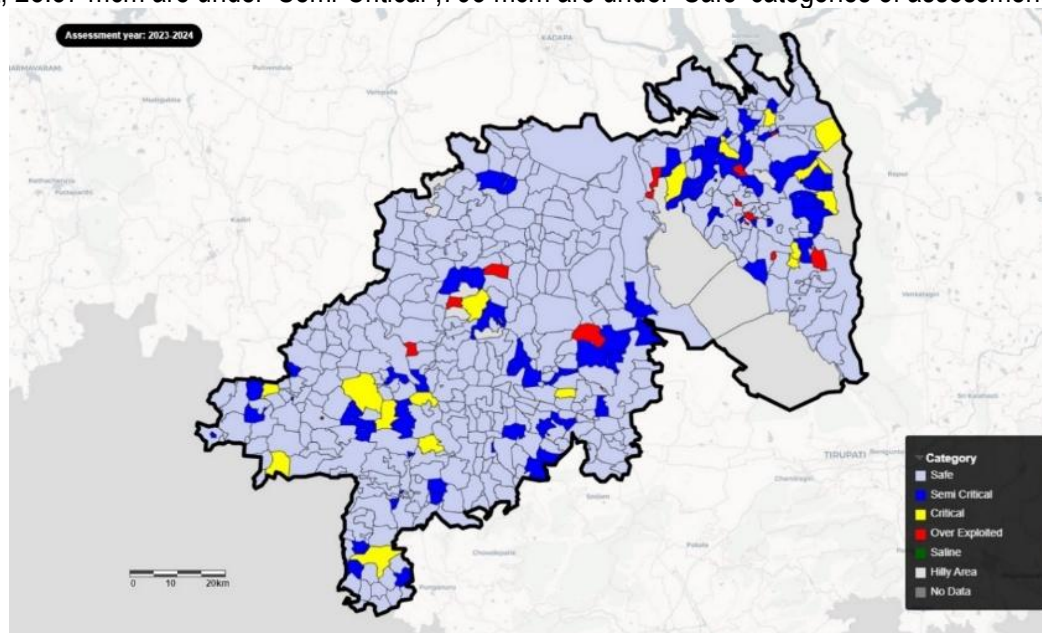


Fig 7.4: Categorization Map of Annamayya District (Village Level)

7.5 Bapatla District

The Bapatla district, Andhra Pradesh having geographical area of 3,828.84 km², lies between north latitudes from 16°15'0"N to 17°47'42"N and east longitudes from 79°57'22"E to 80°55'0"E located in the middle eastern part of Andhra Pradesh. The district is bounded on the north by Guntur District, on the west by Palnadu and Prakasam Districts, on the south by Bay of Bengal, on the east by Krishna western delta. Administratively the district is governed by 25 revenue mandals and 380 villages with a population of ~15.86 lakhs (2011 census). The population density is 414 per Sq.km. The annual normal rainfall of Bapatla district is 923 mm, which ranges from 738.22 mm at Santhamanguluru mandal to 1059.37 mm at Bhattiprolu Mandal. The area receives more than 85 % of the annual rainfall by southwest monsoon between June and September and the rest during the northeast monsoon from October to November. Major rivers in the district include the Krishna River, Romperu, and Vogeru Vagu.

The geology of the area is primarily characterized by recent to sub-recent alluvium, which includes sand, gravel, clay, and silt. Upper Gondwana formations, specifically the Gollapalli sandstones, appear as isolated linear outcrops trending in a northeast-southwest direction in the northwestern region near Chebrolu. In the eastern deltaic area, the alluvium is primarily underlain by Rajahmundry sandstones, while in the western deltaic region, it is predominantly underlain by Gollapalle sandstones and, in some locations, directly by crystalline rocks. The area comprises rock units of the Khondalite Group and Charnockite Group of the Eastern Ghats Super Group and Unclassified Metamorphics and Dharwar Super Group of Archaean age, Younger Igneous Intrusives, Cuddapah Super Group and Kurnool Group of Proterozoic age, Upper Gondwana sequence of Upper Cretaceous age, Cainozoic laterite and Recent Alluvium.

Total Annual Groundwater recharge of the Bapatla district has been assessed as 0.70 bcm and Annual extractable Ground Water resource as 0.67 bcm. The Annual Ground Water Extraction is 0.18 bcm and Stage of Ground Water Extraction is 25.19 %. Out of 25 assessment units (Mandals), 1 unit have been categorized as 'Saline' and 24 units as 'Safe' units. Similarly, out of 3801 sq km recharge worthy area of the district, 162.25 sq km area are under 'Saline' and 3639 sq.km under 'Safe' categories of assessment units. Out of total 667.64 mcm annual extractable ground water resources of the district, 667.64 mcm are under 'Safe' categories of assessment units.

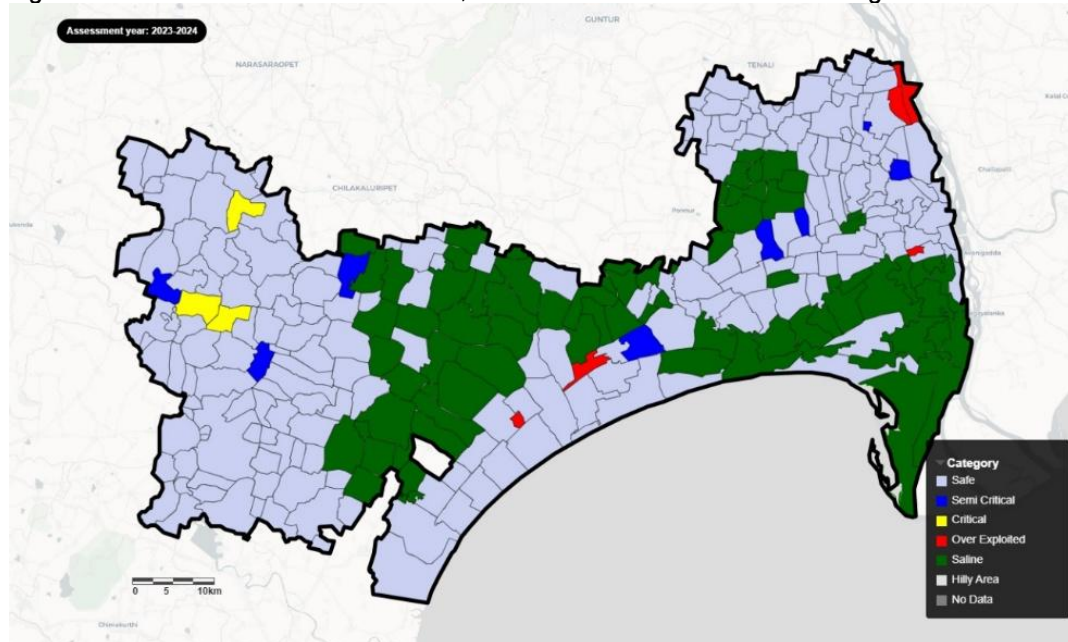


Fig 7.5: Categorization Map of Bapatla District (Village Level)

7.6 Chittoor District

The Chittoor district, Andhra Pradesh having geographical area of 6,860 sq.km, lies between Northern Latitudes of 12°44'42" and 13°39'21" and between the Eastern Longitudes 78°2'2" and 79°41'52". It is bounded on the east by Tirupati District of Andhra Pradesh and Tamilnadu State, on the west by Annamayya District and Karnataka State, on the North by Annamayya and Tirupati Districts of Andhra Pradesh and on the south by Tamilnadu State. Administratively the district is governed by 31 revenue mandals and 822 villages with a population of ~18.73 lakhs (2011 census). The normal annual rainfall for Chittoor district is 918.1 mm. The river Ponnai (non-perennial) flows towards south-east and joins river Palar in Tamil nadu. Swarnamukhi is another important river originates in Eastern Ghats and flows in NE direction before flowing into Nellore district. Other important rivers are river Bahuda, river Pincha, river Kushastali, river Kalyani, river Bheema, river Araniyar and river Pedderu etc.

The district's geology ranges from Archaean to recent age. The oldest rocks, part of the Migmatite Complex, include migmatized quartzo-felspathic gneiss in the northeast. Older metamorphics like amphibolites, schists, quartzites, calc-silicate rocks, and marble form enclaves within the Peninsular/Banded Gneissic Complex (PGC/BGC), covering ~90% of the area. Dharwar Supergroup rocks, including schists and banded ferruginous quartzite, occur as N-S trending belts overlying PGC. Proterozoic intrusives include granites and quartz veins, while dolerite dykes trend E-W, N-S, and NW-SE. The Cuddapah Supergroup (shale, quartzite, limestone) appears in the northeast, while Gondwana rocks (Satyavedu Formation) and laterite occur in the southeast. Recent alluvium is found along major streams.

Total Annual Groundwater recharge of the Chittoor district has been assessed as 0.90 bcm and Annual extractable Ground Water resource as 0.86 bcm. The Annual Ground Water Extraction is 0.51 bcm and Stage of Ground Water Extraction is 59.95 %. Out of 31 assessment units (Mandals), 10 unit have been categorized as 'Semi-critical' and 21 units as 'Safe' units. There is no 'Saline' category of assessment unit in the district. Similarly, out of 6005 sq km recharge worthy area of the district, 1781 sq km area are under 'Semi-Critical', and 4224 sq.km under 'Safe' categories of assessment units. Out of total 859.03 mcm annual extractable ground water resources of the district, 275.16 mcm are under 'Semi-Critical' and 583.9 mcm are under 'Safe' categories of assessment units.

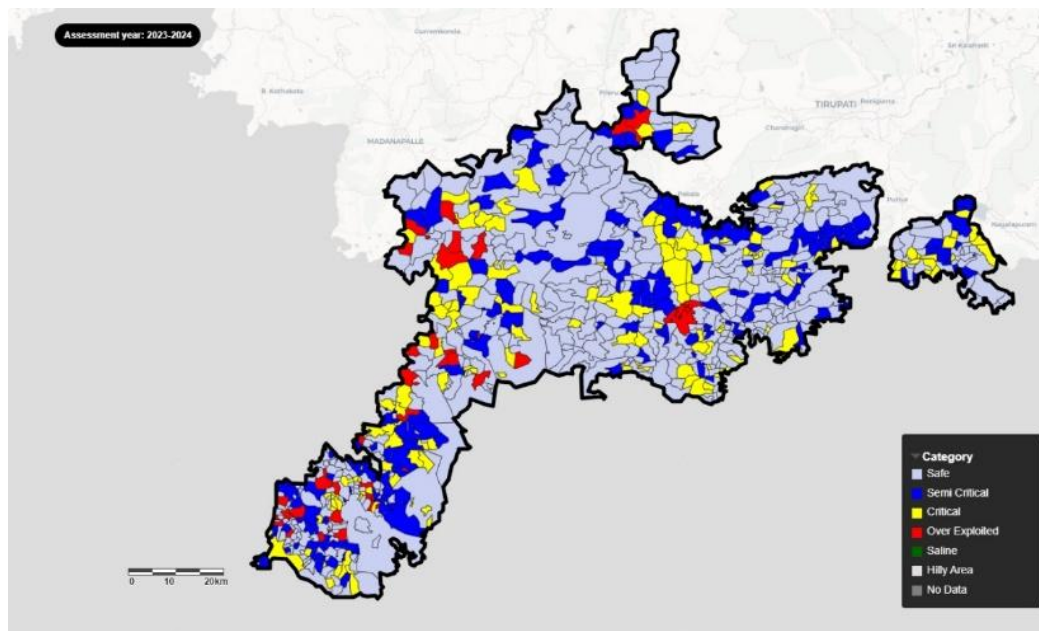


Fig 7.6: Categorization Map of Chittoor District (Village Level)

7.7 East Godavari District

The East Godavari district, Andhra Pradesh having geographical area of 2560 km², lies between north latitude 16°40' to 17°20' and east longitude 81°18' to 82°6'. The total forest area in the district accounts for 90 km² forming 4 % of the total Geographical area. The district is bounded by West Godavari district in the south, Kakinada in the east, Eluru district in the west and Alluri Sitharama Raju district in the North. Administratively the district is governed by 19 revenue mandals. There are 303 villages with a population of ~18.32 lakhs (2011 census). The density of population is 477 persons/ km² whereas it is 304 per Sq.km. for the State. The annual normal rainfall of district is 1142 mm, which ranges from 1007 mm (Devarapalle) and 1230 mm (Undrajavaram) Mandal. The SW monsoon contributes 66%, NE monsoon contributes 23% while the rest by winter and summer rainfall. Major part of the district falls under Godavari basin.

Godavari river flows N to S direction in the area. The district is divided into 3 major drainage basins namely Godavari, Yerrakalva and Yeleru.

Geologically the area is mainly underlain by sandstone of Gondwana, Rajahmundry sandstone, Alluvium and Pre-cambrian crystalline formations. The Archaean crystallines form the basement for the younger formations i.e., sandstone and alluvium formation in the area.

Total Annual Groundwater recharge of the East Godavari district has been assessed as 0.89 bcm and Annual extractable Ground Water resource as 0.84 bcm. The Annual Ground Water Extraction is 0.29 bcm and Stage of Ground Water Extraction is 42.28 %. Out of 19 assessment units (Mandals), 4 units have been categorized as 'Semi-critical Safe' and 15 units as 'Safe' units. There is no 'Saline' category of assessment unit in the district. Similarly, out of 2545 sq km recharge worthy area of the district, 714 sq km area are under 'Semi-Critical', and 1831 sq.km under 'Safe' categories of assessment units. Out of total 841.42 mcm annual extractable ground water resources of the district, 146.96 mcm are under 'Semi-Critical' and 694.5 mcm are under 'Safe' categories of assessment units.

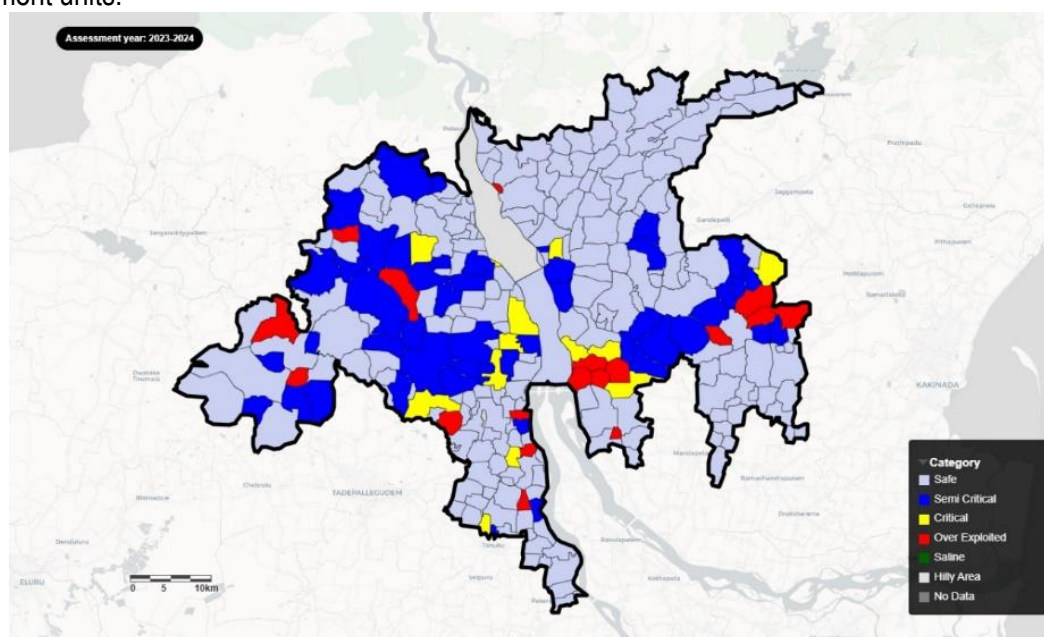


Fig 7.7: Categorization Map of East Godavari District (Village Level)

7.8 Eluru District

The Eluru district, Andhra Pradesh having geographical area of 6412 km², lies between 16° 22' and 17° 40' N of the Northern Latitude and 80° 40' and 81° 45' of the Eastern Longitude. The district is bounded by East Godavari in the east, NTR district and Telangana state in the west, Alluri Sitharamraju district in the north and Krishna & West Godavri Districts in the South. Administratively, the district is governed by 03 Revenue Divisions viz., Eluru, Nuzvidu, and Jangareddigudem, 27 revenue mandals and there are as many as 655 Revenue villages exist in the district. As per 2011 census, the total population of the district is 20.02 lakh and density of population according 2011 census is 312 per Sq.km. The normal rainfall of the district is 1064 mm. The district is mainly drained by Godavari and Tammileru rivers. The river Godavari enters the district near the northeastern corner. The other prominent rivers/streams in the district are Peddavagu, Kalperu Vagu, Pamuleru, Pedrala Kaluva, Kovvada Kaluva, Raja Madugu, and Kinnerasani joins Godavari River while Tammileru join Kolleru lake.

Eluru district is underlain by variety of geological formations comprising from the oldest Archaean to Recent Alluvium. The oldest rock types are of Archaean age, belonging to the Eastern Ghat Super group represented by khondalite, charnockite group and granite gneiss (migmatite). Talchir, Barakar and Kamthi formations of Lower Gondwana, and Kota and Gangapur formations of Upper Gondwana represent Gondwana sediments of U. Carboniferous to L. Cretaceous age. Gollapalli, Raghavapuram and Tirupati formations represent East coast Gondwanas. The Deccan Traps overly these Gondwana rocks. Deposition of Rajahmundry Sandstone indicates marine transgression during Mio-Pliocene. Laterite cappings on khondalite and charnockite are seen on hilltops. Quaternary sediments occur along the rivers.

Total Annual Groundwater recharge of the district has been assessed as 1.83 bcm and Annual extractable Ground Water resource as 1.74 bcm. The Annual Ground Water Extraction is 0.48 bcm and Stage of Ground Water Extraction is 24.84 %. Out of 27 assessment units (Mandals), 5 units have been categorized as 'Saline' and 22 units (92.31 %) as 'Safe' category. Similarly, out of 6492 sq km recharge worthy area of the district, 5627sq km (86.67 %) under 'Safe' categories of assessment units. Out of total 1737 mcm annual extractable ground water resources of the district, 130.4 mcm (7.6%) are under 'Saline' and 1606.6 mcm (92.4 %) are under 'Safe' categories of assessment units.

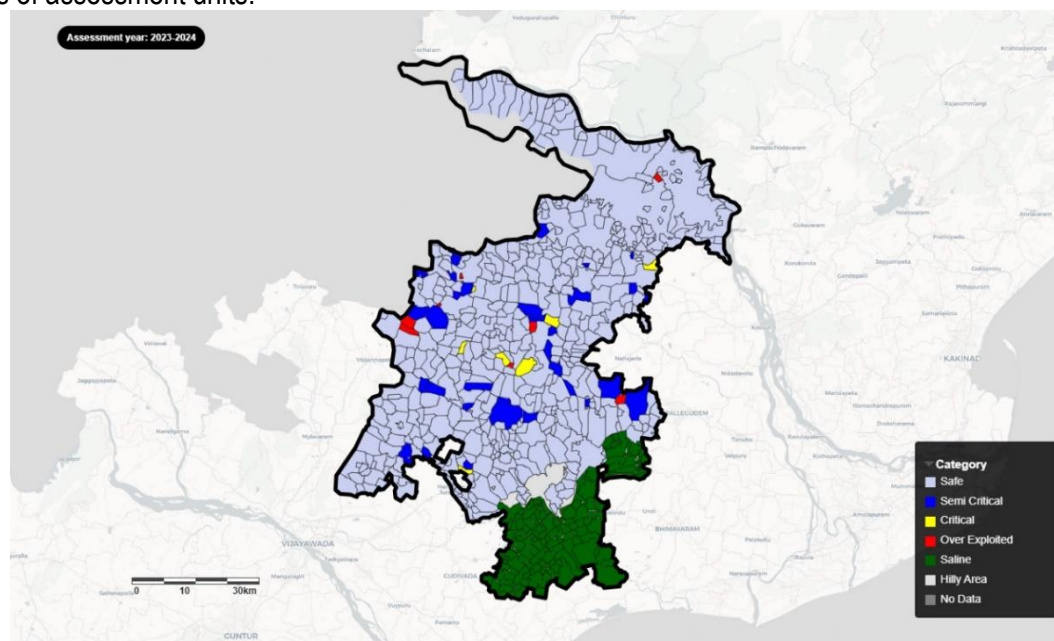


Fig 7.8: Categorization Map of Eluru District (Village Level)

7.9 Guntur District

The district having geographical area of 2447 sq.km, lies between north latitude 16°17' 57.18" and east longitude 80°25' 55" with a mappable area of 2447 sq. km. The district is falling in krishna river basins and their tributaries.

The Guntur district area is underlain by various geological formations of different age groups ranging from Archean to Recent. The Archean basement complex comprising the Charnockites granite-gneisses, Schists, and basic dykes of dolerites form the predominant rock types in northern part. The southern area is mainly underlain by the recent to sub-recent alluvium comprising Sand, Gravel, Clay and Silt.

Total Annual Groundwater recharge of the district has been assessed as 0.54 bcm and Annual extractable Ground Water resource as 0.52 bcm. The Annual Ground Water Extraction is 0.11 bcm and Stage of Ground Water Extraction is 20.5%. Out of 18 assessment units (Mandals), no units have been categorized as 'Over Exploited' and 16 units (88.89%) as 'Safe'. There are two 'Saline' category of assessment units observed in the district. Similarly, out of 24.23.33 sq km recharge worthy area of the district, 2165.85 sq km (89.38%) area are under 'Safe', and 257.48 sq km (10.62%) under 'Saline' categories of assessment units.

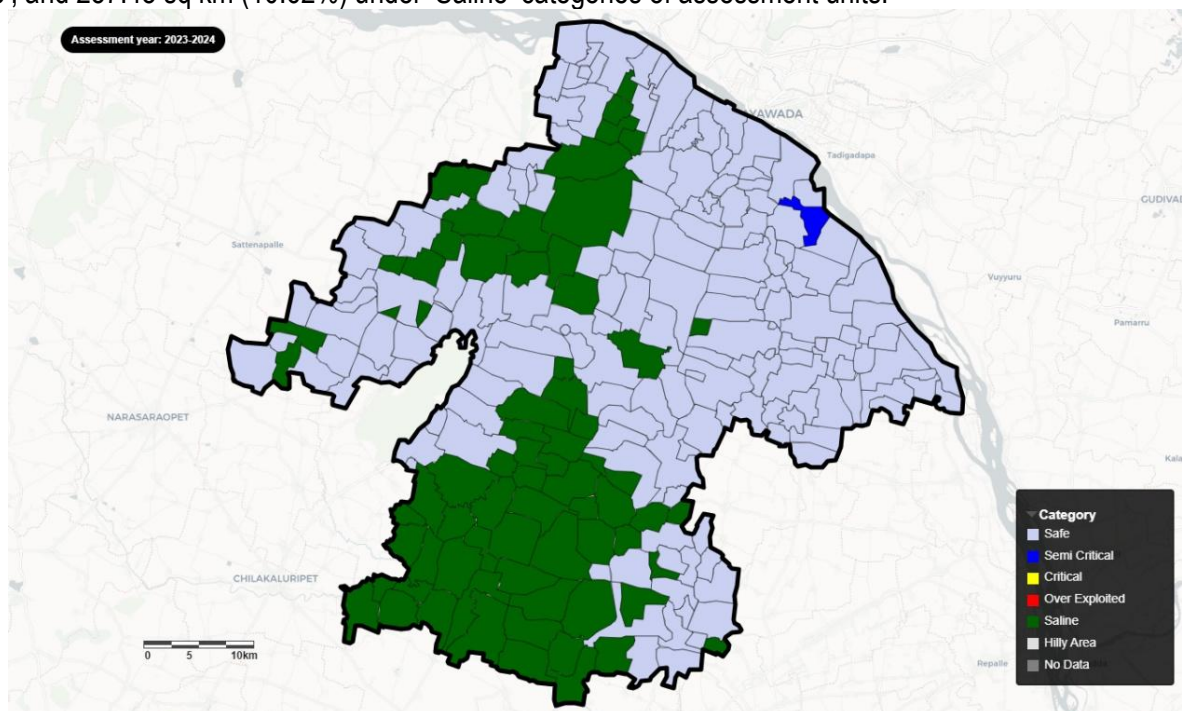


Fig 7.9: Categorization Map of Guntur District (Village level).

7.10 Kakinada District

The district having geographical area of 3020 sq.km, lies between north latitude 17° 15' and 19° 15' and east longitude 83° 0' to 83° 45' with a mappable area of 3020 sq. km. The district is falling in Godavari and Pushkara basins and their tributaries.

The district area is underlain by various geological formations of different age groups ranging from Archean to Recent. The Archean basement complex comprising the Khondalite (37% area), Charnockite (4% area) granite-gneisses (3% area), Sandstone (14% area), Basalt (1% area), and Alluvium (41% area).

Total Annual Groundwater recharge of the district has been assessed as 1.12 bcm and Annual extractable Ground Water resource as 1.07 bcm. The Annual Ground Water Extraction is 0.17 bcm and Stage of Ground Water Extraction is 16%. Out of 21 assessment units (Mandals), no units have been categorized as 'Over Exploited' and 19 units (90.48%) as 'Safe'. There are two 'Saline' category of assessment units observed in the district. Similarly, out of 2881.24 sq km recharge worthy area of the district, 2362.46 sq km (81.99%) area are under 'Safe', and 518.78 sq km (18.01%) under 'Saline' categories of assessment units.

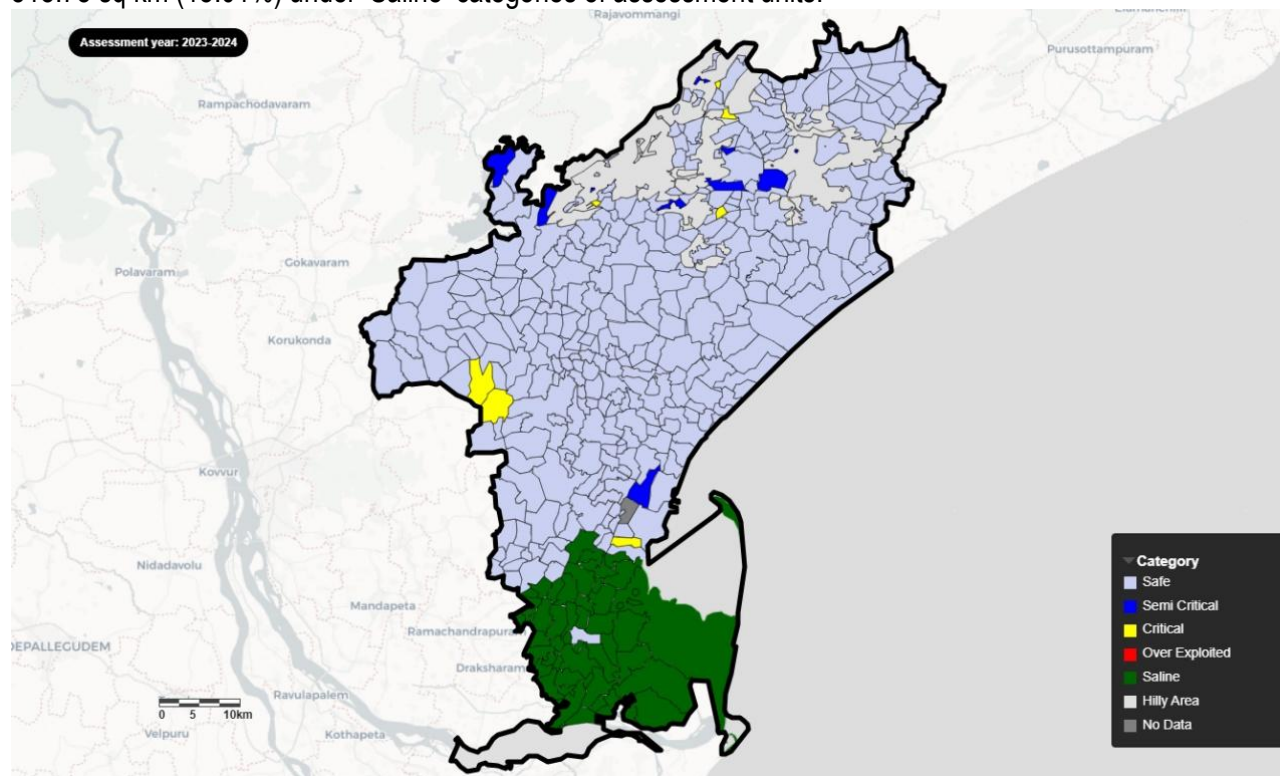


Fig 7.10: Categorization Map of Kakinada District (Village level)

7.11 Konaseema District

The district having geographical area of 2081 sq.km, lies between north latitude 16°15' to 17°00' and east longitude 81°35' to 82°25' with a mappable area of 2081 sq. km. The district is falling in Godavari basin and their tributaries.

The district area is underlain by various geological formations of different age groups. Geologically the area is underlain by the recent to sub-recent alluvium comprising Sand, Gravel, Clay and Silt. The alluvium is underlined by sandstones. The deltaic plains Alluvium covering 99% of area and rest is Rajahmundry Sandstone (1% area).

Total Annual Groundwater recharge of the district has been assessed as 1.12 bcm and Annual extractable Ground Water resource as 1.07 bcm. The Annual Ground Water Extraction is 0.10 bcm and Stage of Ground Water Extraction is 9.9%. Out of 22 assessment units (Mandals), no units have been categorized as 'Over Exploited' and 15 units (68.18%) as 'Safe'. There are seven 'Saline' category of assessment units observed in the district. Similarly, out of 2078.95 sq km recharge worthy area of the district, 1320.07 sq km (63.47%) area are under 'Safe', and 758.88 sq km (36.53%) under 'Saline' categories of assessment units.

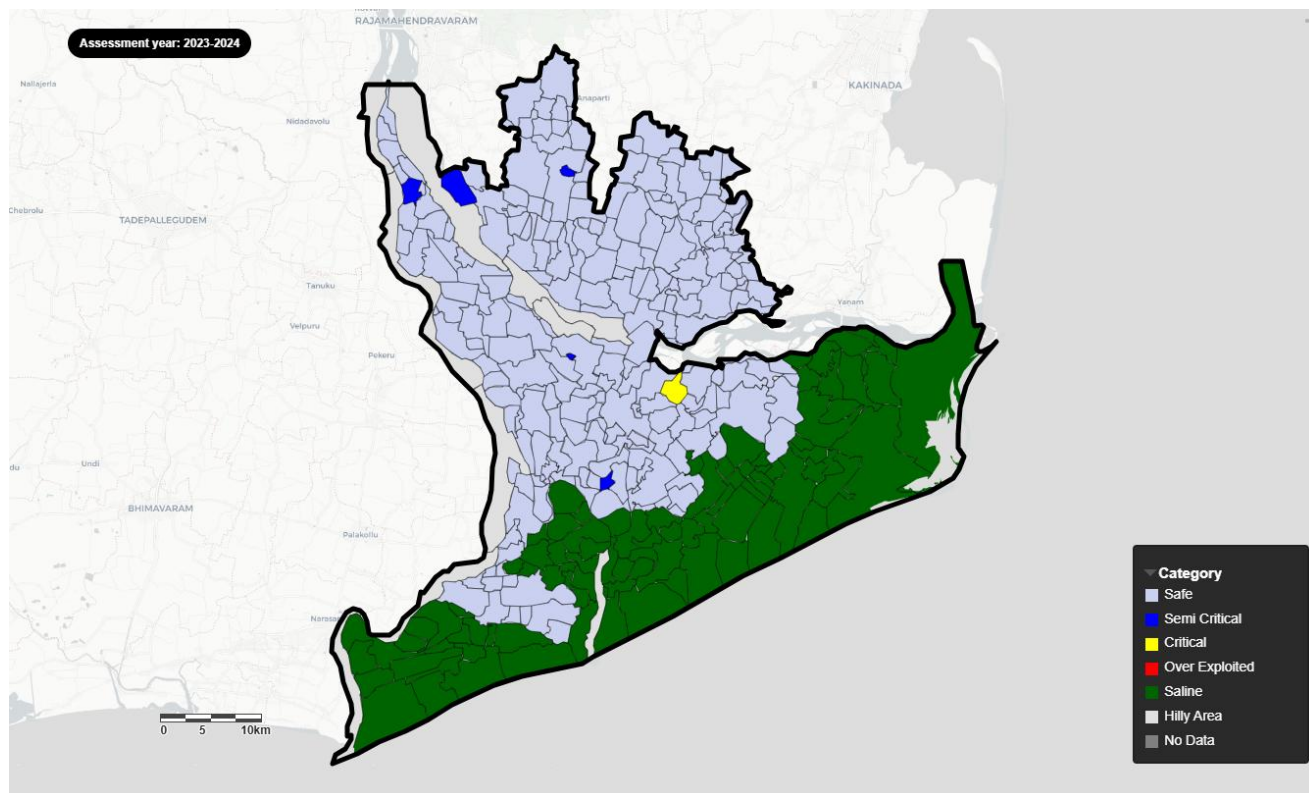


Fig 7.11: Categorization Map of Konaseema District (Village level)

7.12 Krishna District

The district having geographical area of 3880 sq.km, lies between north latitude 15° 45' N to 16° 45' N and east longitude 80° 45' E to 81° 45' E with a mappable area of 3880 sq. km. The district is falling in Krishna basin and their tributaries.

The district is underlain by deltaic and coastal alluvium consisting of fine to medium sand, silt and gravel with intercalations of clay of recent age followed by sandstones of Mio Pliocene age followed by crystalline and at places alluvium directly underlain by crystalline/ basement from the oldest Archaeans to Recent. The consolidated formations include crystallines (Khondalites, Charnockites and granitic gneisses). The semi consolidated formations are represented by Tertiary formations (Rajahmundry & Gollapalli sandstones) and unconsolidated formations comprise deltaic alluvial deposits of Quaternary period comprising Sand, Gravel, Clay and Silt.

Total Annual Groundwater recharge of the district has been assessed as 1.66 bcm and Annual extractable Ground Water resource as 1.58 bcm. The Annual Ground Water Extraction is 0.47 bcm and Stage of Ground Water Extraction is 29.7%. Out of 25 assessment units (Mandals), no units have been categorized as 'Over Exploited' and 16 units (64%) as 'Safe'. There are nine 'Saline' category of assessment units observed in the district. Similarly, out of 3880 sq km recharge worthy area of the district, 1935.06 sq km (49.87%) area are under 'Safe', and 1945.5 sq km (50.13%) under 'Saline' categories of assessment units.

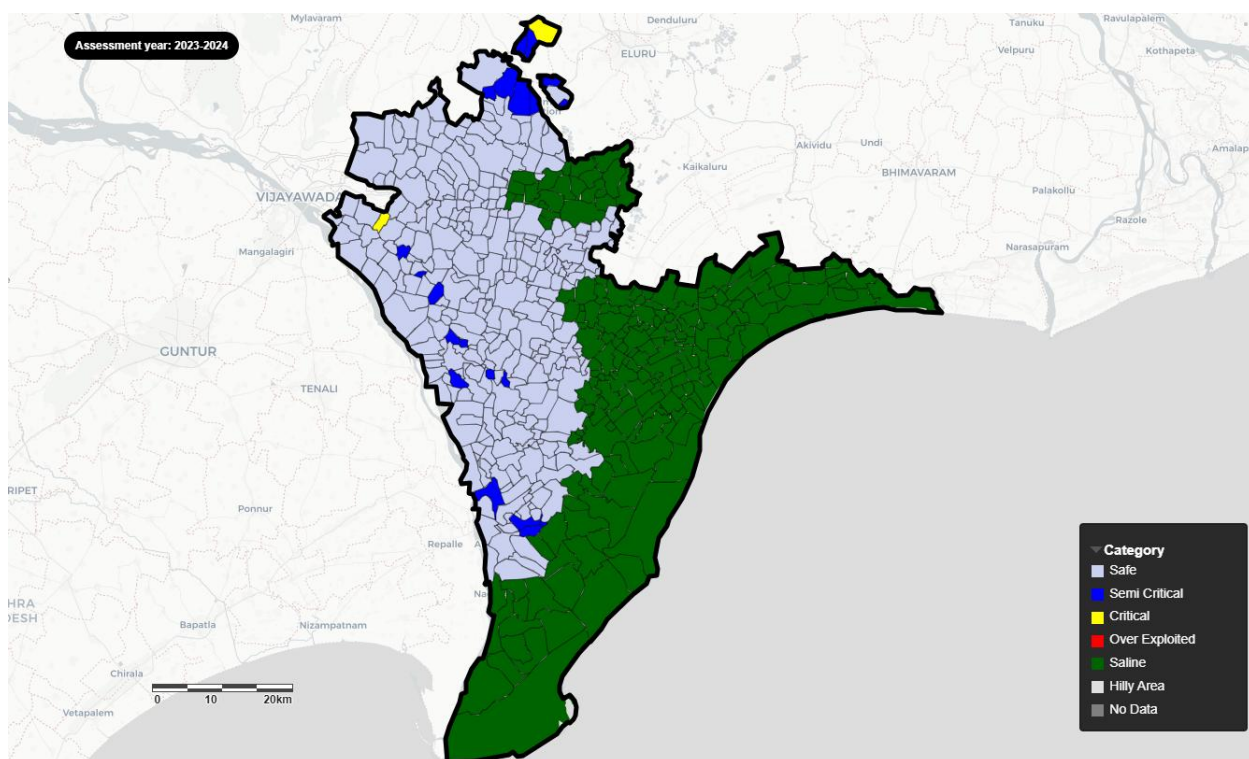


Fig 7.12: Categorization Map of Krishna District (Village Level)

7.13 Kurnool District

The district having geographical area of 3880 sq.km, lies between north latitude 15°07'58"-15°57'58" and east longitude 76°58'12"-78°16'06" E with a mappable area of 7977sq. km. The district is falling in Tungabhadra and Krishna basin and their tributaries.

Geologically the area is underlain by the hard-crystalline formation of Archaean to Lower Proterozoic age and sedimentary formation of Middle to Upper Proterozoic Age. The Banded gneiss is covered about 56% and granite is occupied 26% of the total geographic area of the district. The remaining area of 18% of the area is covered by sedimentary formation like Shale (1.12%), Limestone (6.11%), Quartzites (1.73%) in the district.

Total Annual Groundwater recharge of the district has been assessed as 0.67 bcm and Annual extractable Ground Water resource as 0.64 bcm. The Annual Ground Water Extraction is 0.19 bcm and Stage of Ground Water Extraction is 30.6%. Out of 26 assessment units (Mandals), no units have been categorized as 'Over Exploited', 01 assessment unit is falling in Semi-critical category and 25 units (64%) as 'Safe'. Similarly, out of 7522.47 sq km recharge worthy area of the district, 7191.65 sq km (95.6%) area are under 'Safe', and 330.82sq km (4.4%) under 'Semi-critical' categories of assessment units.

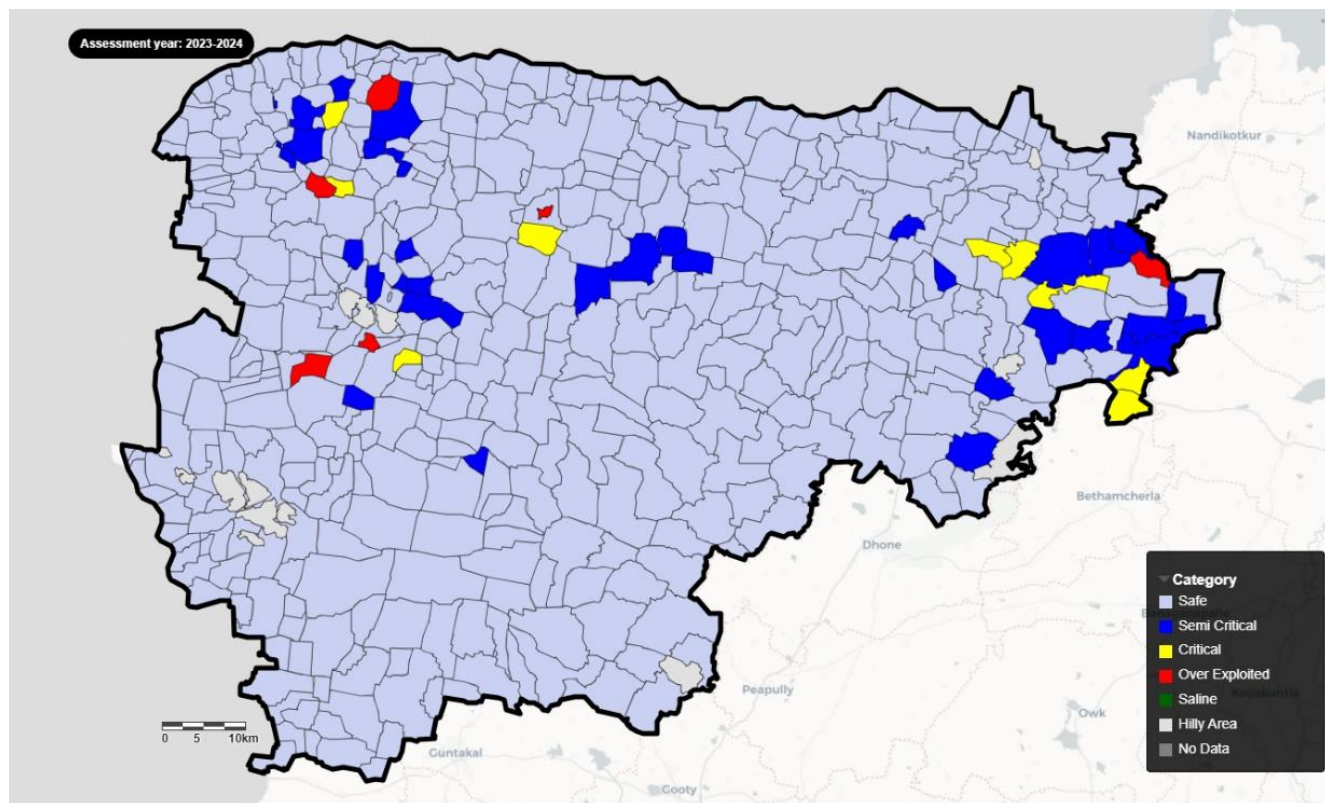


Fig 7.13: Categorization Map of Kurnool District (Village Level)

7.14 NTR District

The district having geographical area of 3315 sq.km, lies between north latitude 16° 30' and 17° 50' N and east longitude 80° 0' and 80° 45' E with a mappable area of 3315 sq. km. The district is falling in Krishna basin and their tributaries.

The study area is underlain by various geological formation from Archaean to Tertiary age, with some isolated pockets of Recent to Sub-recent alluvium. Archean to proterozoic Banded Gneiss, occupies 41% of the area. The Archean Charnockite (19% of the area) and Khondalite (30% of the area) overlies the Archean Granitic Gneiss (41% of the area). The Precambrian metasedimentary formation covers 4% of the area and remaining 6% is alluvium sediments along the river channels.

Total Annual Groundwater recharge of the district has been assessed as 0.64 bcm and Annual extractable Ground Water resource as 0.61 bcm. The Annual Ground Water Extraction is 0.24 bcm and Stage of Ground Water Extraction is 39.10%. Out of 20 assessment units (Mandals), no units have been categorized as 'Over Exploited'/ Critical/ Semi-critical, and all assessment units (100%) as 'Safe'. Similarly, out of 3063.19 sq km recharge worthy area of the district, 3063.19 sq km (100%) area are under 'Safe'.

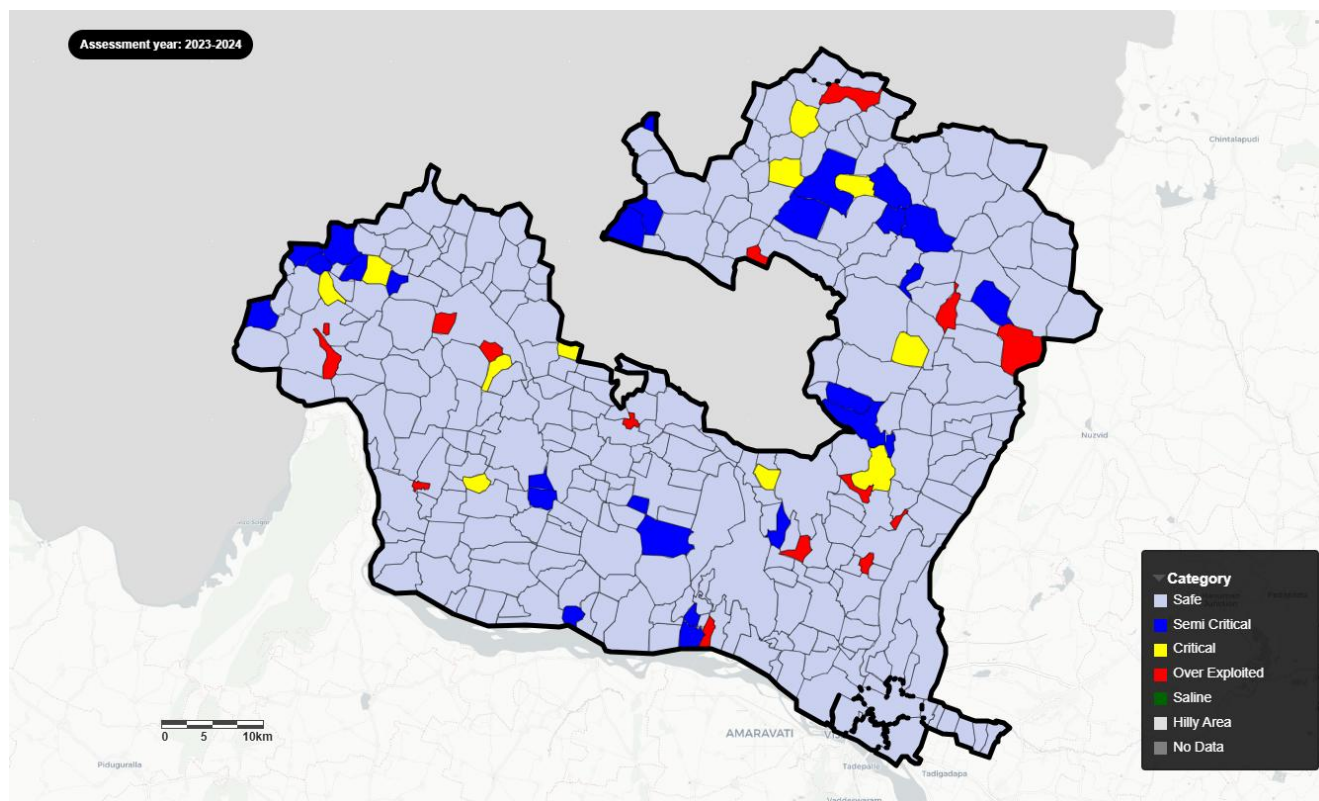


Fig 7.14: Categorization Map of NTR District (Village Level)

7.15 Nandyal District

The district having geographical area of 9681 sq.km, lies between north latitude 14° 54' 9"N to 16° 9' 4"N and east longitude 77° 38' 54"E to 78° 55' 47"E with a mappable area of 7061 sq. km. The district is falling in Krishna basin and their tributaries.

The area is underlain by various rock types that belong to Late Archaean or Early Proterozoic era which are succeeded by rocks of Dharwarian Age and both are traversed by dolerite dykes. The older rocks are overlain by rocks of Cuddapah Super group and Kurnool Group belonging to Middle and Upper Proterozoic Age. The Archaeans comprise the Peninsular Gneissic Complex, represented by granite, granodiorite, granite-gneiss and migmatite. The metasediment rocks of Cuddapah and Kurnool Group include mostly shales, quartzites, limestones and dolomites. The recent alluvium is confined to the major stream and river courses like Krishna and Kunderu.

Total Annual Groundwater recharge of the district has been assessed as 0.82 bcm and Annual extractable Ground Water resource as 0.78 bcm. The Annual Ground Water Extraction is 0.13 bcm and Stage of Ground Water Extraction is 16.5%. Out of 29 assessment units (Mandals), no units have been categorized as 'Over Exploited'/ Critical/ Semi-critical, and all assessment units (100%) as 'Safe'. Similarly, out of 7061.82 sq km recharge worthy area of the district, 7061.82 sq km (100%) area are under 'Safe'.

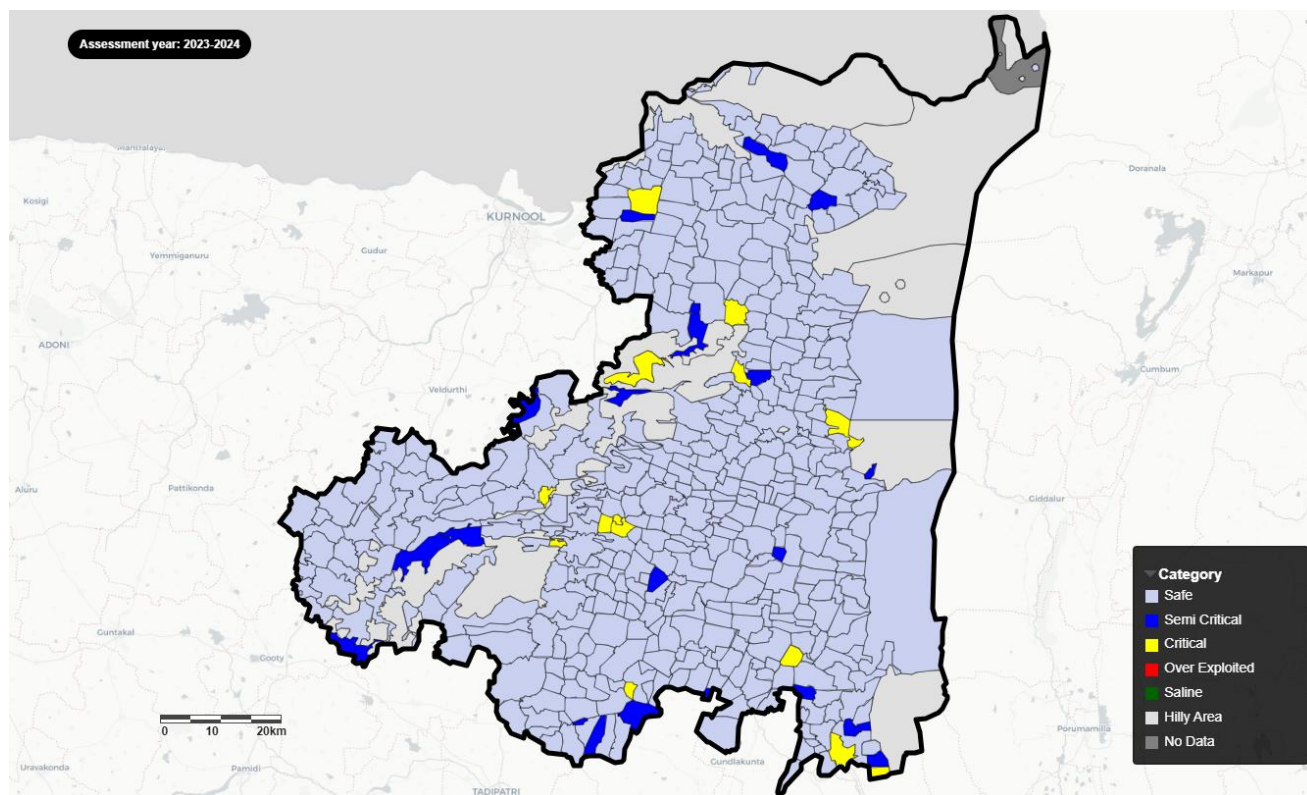


Fig 7.15: Categorization Map of Nandyal District (Village Level)

7.16 Palnadu District

The district having geographical area of 7281 sq.km, lies between north latitude 16°03'-16°37' and east longitude 79°22'-80°21'E with a mappable area of 7281 sq. km. The district is falling in Krishna, Gundlakamma basin and their tributaries.

The district is underlain by various geological formation from Archean to Recent. The Archean basement complex comprising the granite gneisses, migmatites, schists and is overlain by Cuddapah Super Group of rocks and Kurnool Group of rocks. The Archean Granitic Gneiss and granites covers 30% of the area and Charnockites covers 8%. The Precambrian metasedimentary formation of Kurnool and Cuddapah system covers 58% of the area.

Total Annual Groundwater recharge of the district has been assessed as 1.01 bcm and Annual extractable Ground Water resource as 0.96 bcm. The Annual Ground Water Extraction is 0.29 bcm and Stage of Ground Water Extraction is 30.3%. Out of 28 assessment units (Mandals), one unit each have been categorized as 'Over Exploited' and Semi-critical, and 26 units (92.86%) as 'Safe'. Similarly, out of 6516.56 sq km recharge worthy area of the district, 5641.54 sq km (86.57%) area are under 'Safe', 539.1 sq km (8.27%) under 'Semi-critical' and 335.92 sq km (5.15%) under 'Over Exploited' categories of assessment units.

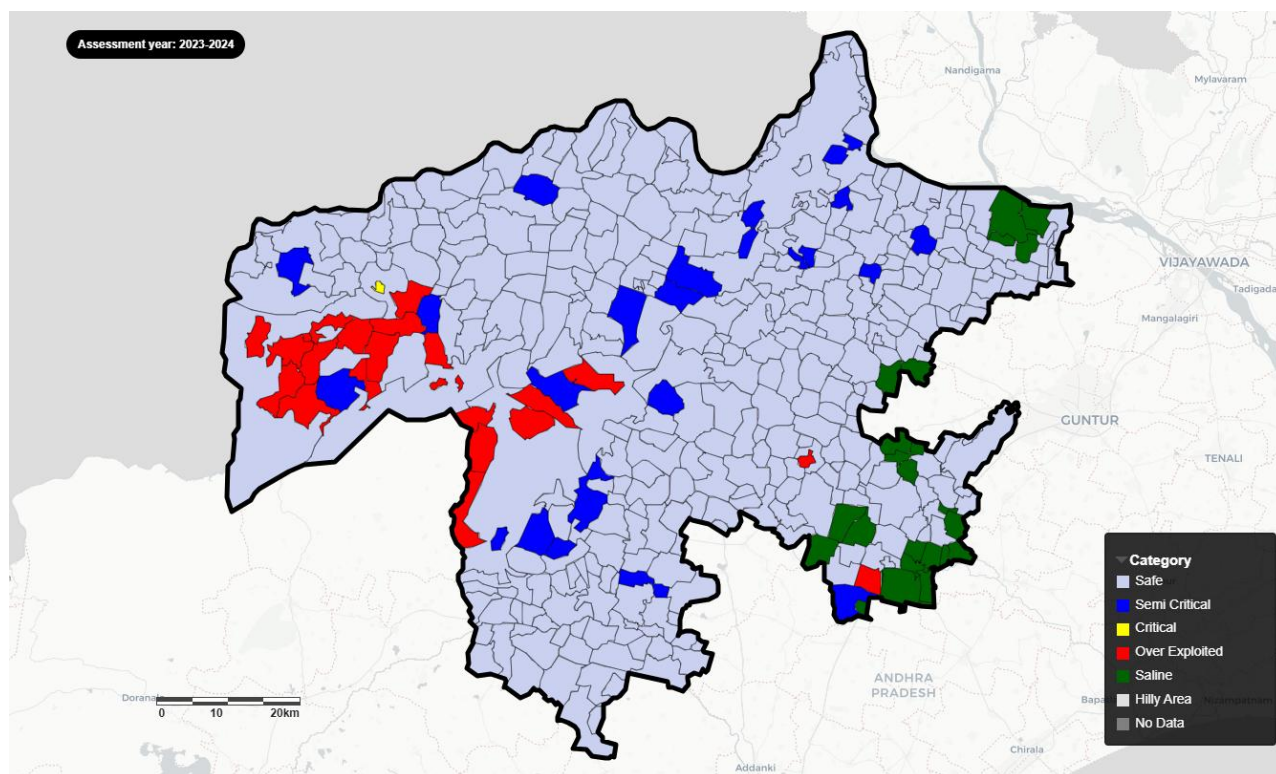


Fig 7.16: Categorization Map of Palnadu District (Village Level)

7.17 Parvathipuram Manyam District

The district having geographical area of 3559 sq.km, lies between north latitude $17^{\circ} 15'$ and $19^{\circ} 15'$ N and east longitude $83^{\circ} 0'$ to $83^{\circ} 45'$ E with a mappable area of 2404 sq. km. The district is falling in Nagavali, Suvarnamukhi, Vegavathi, Gomukhi and its tributaries.

The district is underlain by Eastern Ghat Mobile Belt of Archaean age. It includes rocks of Khondalite Group, Charnockite Group and Migmatite complex. Sediments of Cenozoic laterite and Quaternary deposits overlie the Archaeans. The lithological units of Khondalite Group (24.55%) include quartzite, talc- granulite and talc-silicate rock. The Charnockite Group (30.23%) include pyroxene granulite (basic Charnockite) and charnockite (acid/intermediate). The Migmatite complex includes hypersthene-biotite gneiss, hypersthene-quartz-feldspar augen gneiss, granitoid gneiss (38.20%). Gneissosity in the granite gneiss is more pronounced in the vicinity of Khondalite. Quaternary alluvial sediments (1.31%), including brown, residual soil of fluvial origin occur along the river courses.

Total Annual Groundwater recharge of the district has been assessed as 0.91 bcm and Annual extractable Ground Water resource as 0.87 bcm. The Annual Ground Water Extraction is 0.22 bcm and Stage of Ground Water Extraction is 25.10%. Out of 15 assessment units (Mandals), no units have been categorized as 'Over Exploited'/ Critical/ Semi-critical, and all assessment units (100%) as 'Safe'. Similarly, out of 2948.81 sq km recharge worthy area of the district, 2948.81 sq km (100%) area are under 'Safe'.

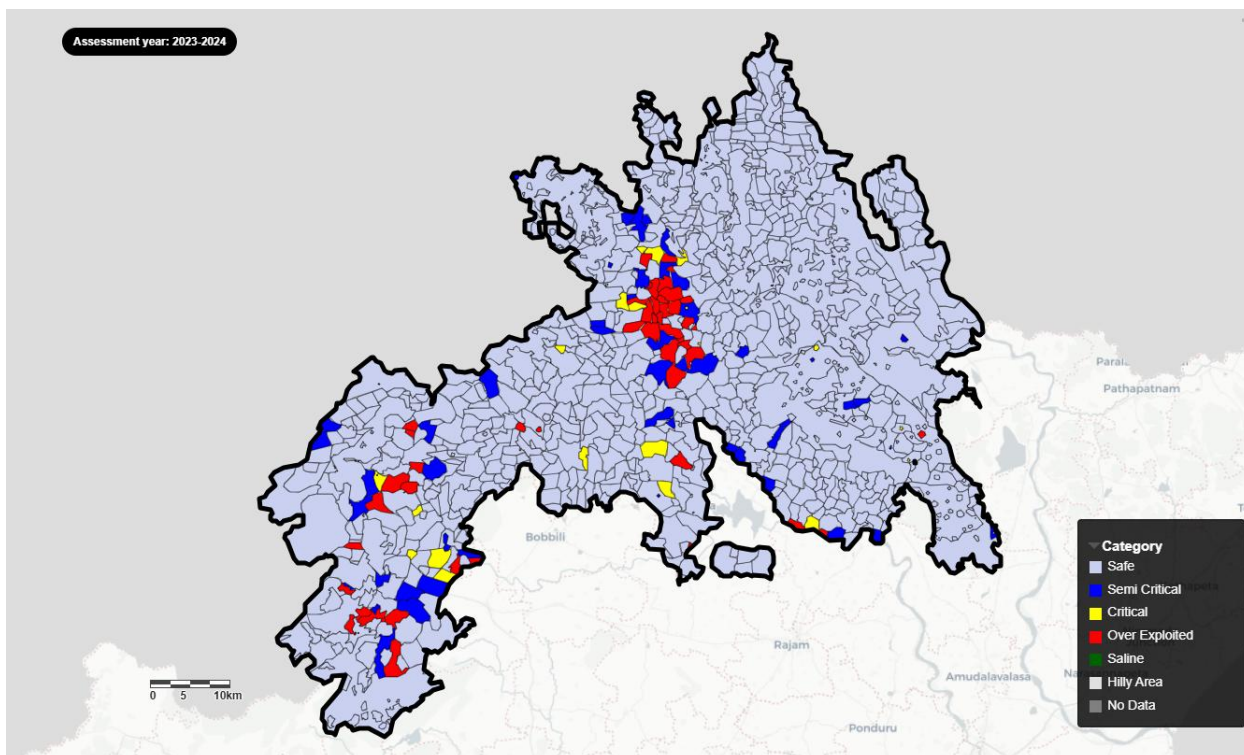


Fig 7.17: Categorization Map of Parvathipuram Manyam District (Village Level)

7.18 Prakasam District

The Prakasam district having geographical area of 14,323 km², lies between north latitude of 14° 57' and 16° 17' and east longitudes of 78° 43' and 80° 25' with a mappable area of 9903 km². The Gundlakamma, Musi and Paleru River are the major rivers flowing in the district and they further split into smaller arms before it debouches into the Bay of Bengal. Besides small rivers like Thammileru, Sagileru and Gudisileru and streams like Ogeru vagu, Nallavagu and Vedimangala vagu flowing in the district.

The district comprises rock units of the Khondalite and Charnockite Group of Eastern Ghats Super Group and Unclassified Metamorphics and Dharwar Super Group of Archaean age, Younger Igneous Intrusives, Cuddapah Super Group and Kurnool Group of Proterozoic age, Upper Gondwana sequence of Upper Cretaceous age, Cenozoic laterite and Recent Alluvium.

Total Annual Groundwater recharge of the Prakasam district has been assessed as 0.98 bcm and Annual extractable Ground Water resource as 0.93 bcm. The Annual Ground Water Extraction is 0.46 bcm and Stage of Ground Water Extraction is 54 %. Out of 38 assessment units (Mandals), 3 units (7.89 %) have been categorized as 'Over Exploited', 5 units (13.1 %) as 'semi critical' and 30 units (78.9 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 11381 sq km recharge worthy area of the district, 932.64 sq km (8.19 %) area are under 'Over-Exploited', 1898.51 sq.km (16.6%) area is under 'semi critical' and 8549.8 sq km (75.12%) under 'Safe' categories of assessment units. Out of total 934.2 mcm annual extractable ground water resources of the district, 47.6 mcm (5.09 %) are under 'Over-exploited', 115.1 mcm (12.3%) are under 'semi-critical' and 771.5 mcm (82.5 %) are under 'Safe' categories of assessment units.

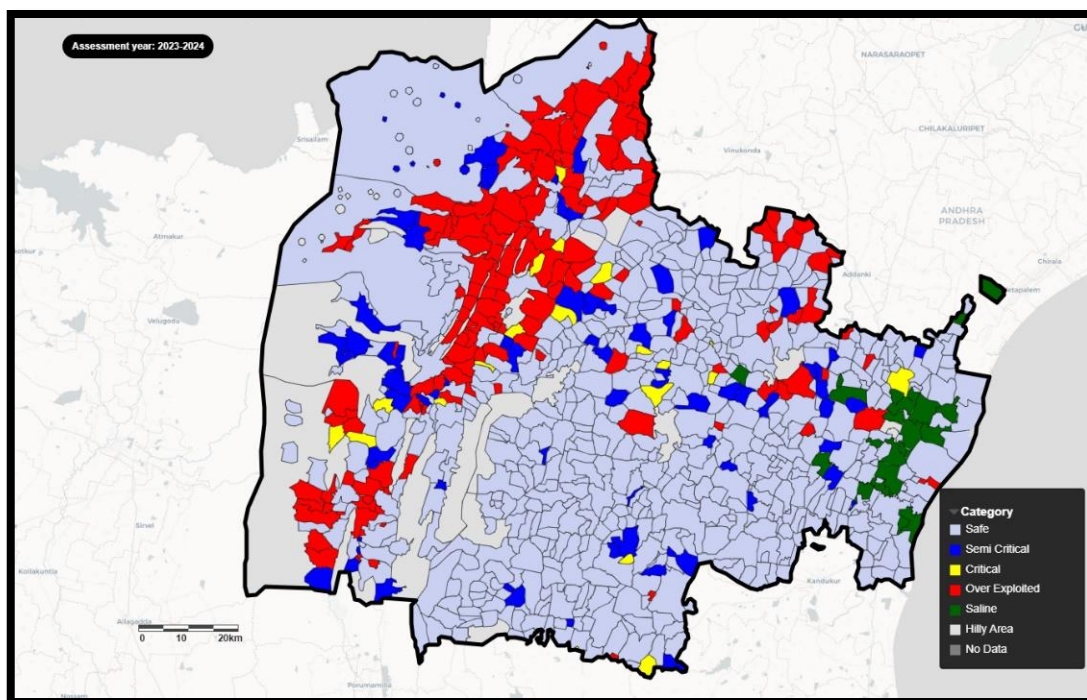


Fig 7.18: Categorization Map of Prakasam District (Village Level)

7.19 SPSR Nellore District

The SPSR Nellore district having geographical area of 10,440 km², lies between north latitude of 14° 6' and 15° 51' N and 79° 0' and 80° 42' of the Eastern Longitude. The main River that drains the Nellore district is Pennar River, which is ephemeral in nature. The other streams are Kandaleru and Boggeru which are occasionally torrential in character during rainy season. The rivers and rivulets remain dry for major part of the year and carry floods only during rainy season. Pennar River is the most important one and flows for about 112 Kms in the district which serve as a boon to the eastern half of the taluks of Nellore and Kovur. Two anicuts are situated one at Sangam and the other at Nellore.

The district is underlain by oldest Archaean Super Group to Recent alluvium. Major part of the district is underlain by gneisses, schists of Peninsular Gneissic Complex and Dharwar Super Group of rocks. All along, the western boundary of the district is underlain by quartzites, shales of Cuddapah Super Group of rocks. A small patch of Gondwana sandstone and shales occur in northern part of the district. Sandstones of Tertiary age overlain by laterites occur as isolated patches in and around Nellore town. Along Pennar river and eastern part of the district is underlain by the deltaic, river and coastal alluvium.

Total Annual Groundwater recharge of the SPSR Nellore district has been assessed as 1.97 bcm and Annual extractable Ground Water resource as 1.87 bcm. The Annual Ground Water Extraction is 0.44 bcm and Stage of Ground Water Extraction is 24.9 %. Out of 37 assessment units (Mandals), 1 unit (2.70 %) have been categorized as 'Semi-Critical' and 36 units (97.3 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 9953 sq km recharge worthy area of the district, 144.20 sq km (1.4 %) area are under 'Semi-Critical' and 9808 sq.km (98.6 %) area is under 'Safe' categories of assessment units. Out of total 1871 mcm annual extractable ground water resources of the district, 52 mcm (2.77 %) are under 'Semi-Critical' and 1819 mcm (97.23 %) are under 'Safe' categories of assessment units.

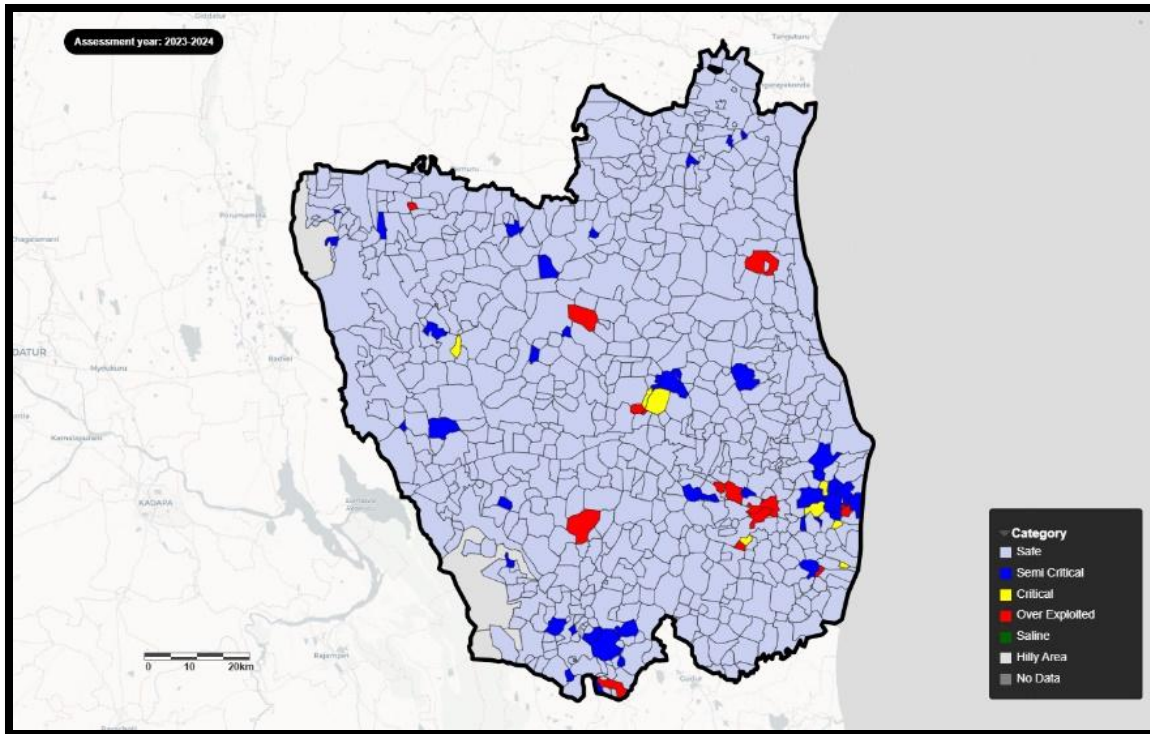


Fig 7.19: Categorization Map of SPSR Nellore District (Village Level)

7.20 Sri Sathya Sai District

The district having geographical area of 8925 km², lies between 13°40' and 14° 6' Northern Latitude and 76°88' and 78°30' Eastern Longitude. The district is mainly drained by the rivers Pennar and its tributaries Jayamangala, Chitravathi, Vedavathi Maddileru, Papagni, Vandamanneru, Tadakaleru and Pandameru.

About 90 % of the area is underlain by crystalline rocks, namely Banded gneissic complex, granites of Archaean to Proterozoic age and remaining 10 % of the area in the northeastern and northwestern parts of the district is underlain by metasediments of Cuddapah supergroup of Proterozoic age, namely shales and limestone.

Total Annual Groundwater recharge of the Sri Sathya Sai district has been assessed as 1.19 bcm and Annual extractable Ground Water resource as 1.13 bcm. The Annual Ground Water Extraction is 0.69 bcm and Stage of Ground Water Extraction is 64.3 %. Out of 32 assessment units (Mandals), 4 units (13 %) have been categorized as 'Over-Exploited', 5 units (16 %) as semi-critical, 1 unit (3 %) as critical and 22 units (69 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 7636 sq km recharge worthy area of the district, 795 sq km (10 %) area are under 'Over-exploited', 931 sq.km (12 %) area are under semi-critical, 123 sq.km (2 %) are under critical and 5787 sq.km (76 %) area are under 'Safe' categories of assessment units. Out of total 1139 mcm annual extractable ground water resources of the district, 142 mcm (12 %) are under 'over-exploited', 127 mcm (11 %) are under semi-critical, 22 mcm (2 %) are under critical and 848 mcm (75 %) are under 'Safe' categories of assessment units.

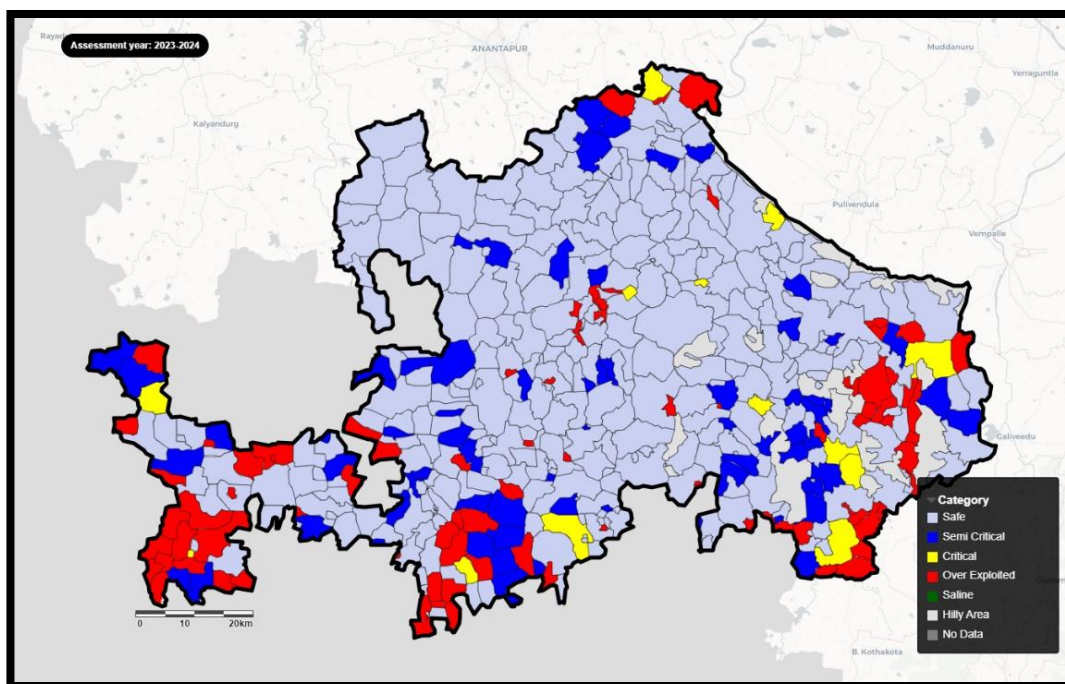


Fig 7.20: Categorization Map of Sri Sathya Sai District (Village Level)

7.21 Srikakulam District

The district having geographical area of 4576 km², lies between north latitude 18°04'46" to 19°09'55" and east longitude 83°35'22" to 84°45'52". The main rivers in the district are Vamsadhara and Nagavali which are perennial in nature. The other important rivers flowing are Mahendra Tanya and Bahuda.

Srikakulam district is underlain by geological formation ranging from older Archaean to Recent alluvium. The Pre-Cambrian rock include Banded Gneiss (43 %), Gneiss (9%), Khondalite (13%) and Charnockites (15%). The Gondwanas are represented by sandstones. The Sub-Recent to Recent formations comprise laterites (1%) to Alluvium (19%).

Total Annual Groundwater recharge of the district has been assessed as 1.32 bcm and Annual extractable Ground Water resource as 1.25 bcm. The Annual Ground Water Extraction is 0.44 bcm and Stage of Ground Water Extraction is 35 %. Out of 30 assessment units (Mandals), 1 unit (3.3 %) have been categorized as 'Over-Exploited', 1 unit (3.3 %) as critical and 28 units (93.3 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 4483 sq km recharge worthy area of the district, 214.8 sq km (4.8 %) area are under 'Over-exploited', 113.7 sq.km (2.5 %) are under critical and 4155 sq.km (92.7 %) area is under 'Safe' categories of assessment units. Out of total 1254 mcm annual extractable ground water resources of the district, 54.45 mcm (4.3 %) are under 'over-exploited', 44.31 mcm (3.5 %) are under critical and 1157 mcm (92.1 %) are under 'Safe' categories of assessment units.

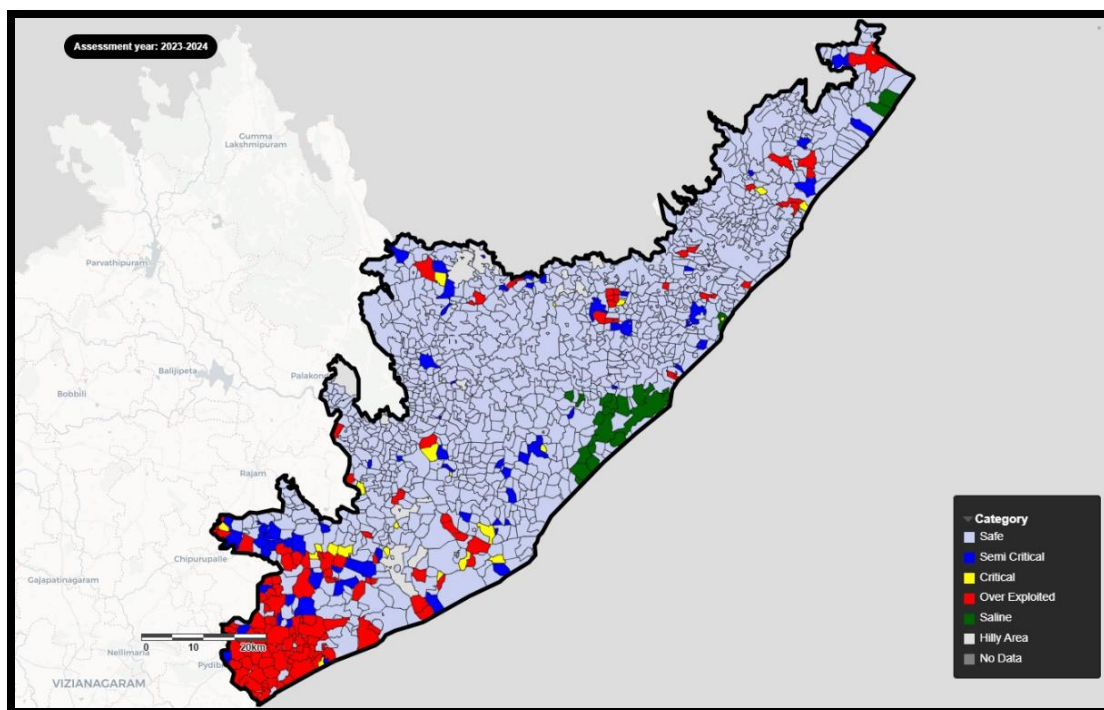


Fig 7.21: Categorization Map of Srikakulam District (Village Level)

7.22 Tirupati District

The Tirupati district having geographical area of 8229 km², lies between Northern Latitudes of 13°21'54" and 14°30'40" and between the Eastern Longitudes 79°05'42" and 80°04'10". Swarnamukhi is important river originates in Eastern Ghats and flows in NE direction and this is non-perennial in nature other important rivers are Kushastali and Kalyani River.

The district is underlain by formations of Archaean, Proterozoic, Jurassic, Cretaceous, Tertiary and Quaternary ages. The oldest rocks in the area belong to Migmatite Complex, represented by migmatized quartzo-felspathic gneiss and are exposed in the northwestern part of the district. Acid intrusive of Proterozoic Age comprises granite and quartz veins. The granite plutons are exposed as patches and linear bodies in southwestern and northern parts of the district respectively. The basic dykes include dolerite. The southern tip of the well-known Cuddapah Basin falls in the northeastern part of the district. Laterite capping occurs over Gondwana formations. Large tracts of Alluvium occur eastern part along coastal tract and the major rivers/streams, which belong to Recent Age.

Total Annual Groundwater recharge of the Tirupati district has been assessed as 1.61 bcm and Annual extractable Ground Water resource as 1.53 bcm. The Annual Ground Water Extraction is 0.54 bcm and Stage of Ground Water Extraction is 39.78 %. Out of 34 assessment units (Mandals), all units (100 %) have been categorized as 'Safe'. There is no 'Saline' category of assessment unit in the district.

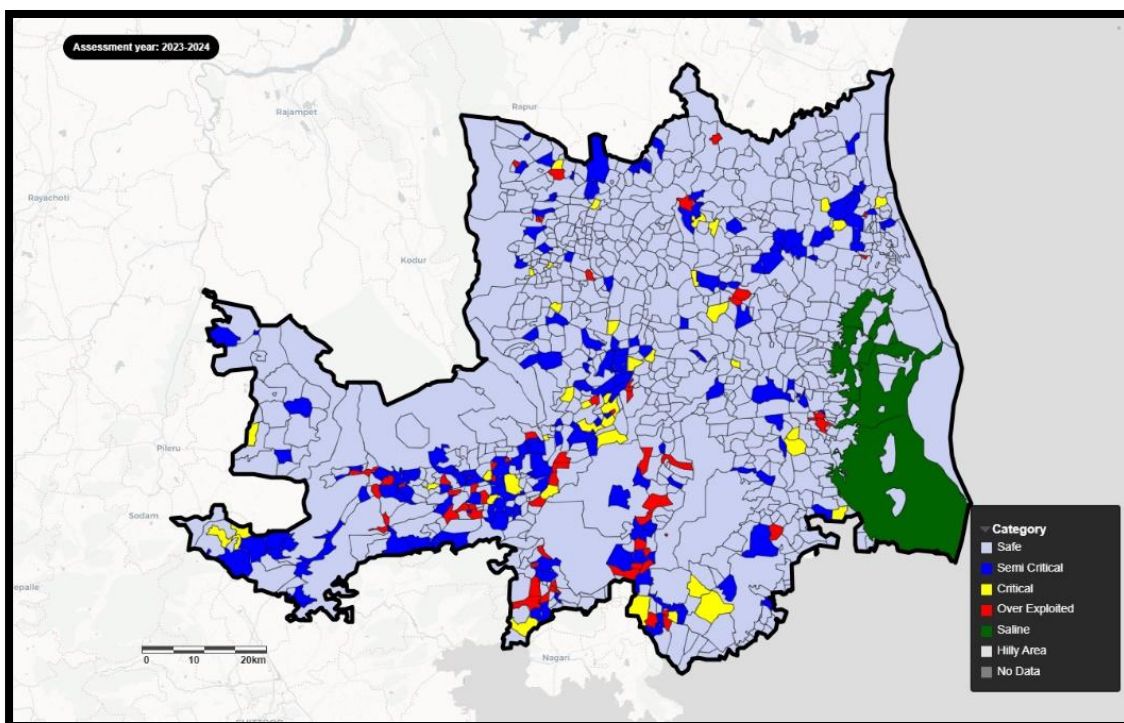


Fig 7.22: Categorization Map of Tirupati District (Village Level)

7.23 Visakhapatnam District

The Visakhapatnam district having geographical area of 1049 km², lies north latitude 17°50'52"-18°23'36" and east longitude 81°12'19"-83°37'7". Major part of the district falls under Godavari and Vamsadhara major basins and sub-basins are lower Godavari and Nagavali. The district is divided into four local sub drainage basins namely Anakapalli, Narvagadda, Madhuvada & Gosthani. No major river/perennial rivers are flowing across the district.

The district is underlain by geological formations ranging from oldest Archaeans to Recent Alluvium. The Archaean group of rocks includes Khondalites and Charnockites of Eastern Ghat super group and Granitic gneisses of Migmatite group. The recent alluvium is prevalent along the rivers and coastal areas.

Total Annual Groundwater recharge of the Visakhapatnam district has been assessed as 0.126 bcm and Annual extractable Ground Water resource as 0.119 bcm. The Annual Ground Water Extraction is 0.05 bcm and Stage of Ground Water Extraction is 56.6 %. Out of 11 assessment units (Mandals), 04 units (36 %) have been categorized as 'Semi-Critical' and 07 units (64 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 1002 sq km recharge worthy area of the district, 189.6 sq km (19 %) area are under 'Semi-Critical' and 812 sq.km (81 %) area is under 'Safe' categories of assessment units. Out of total 119.91 mcm annual extractable ground water resources of the district, 14.44 mcm (12 %) are under 'Semi-Critical' and 105.47 mcm (88 %) are under 'Safe' categories of assessment units.

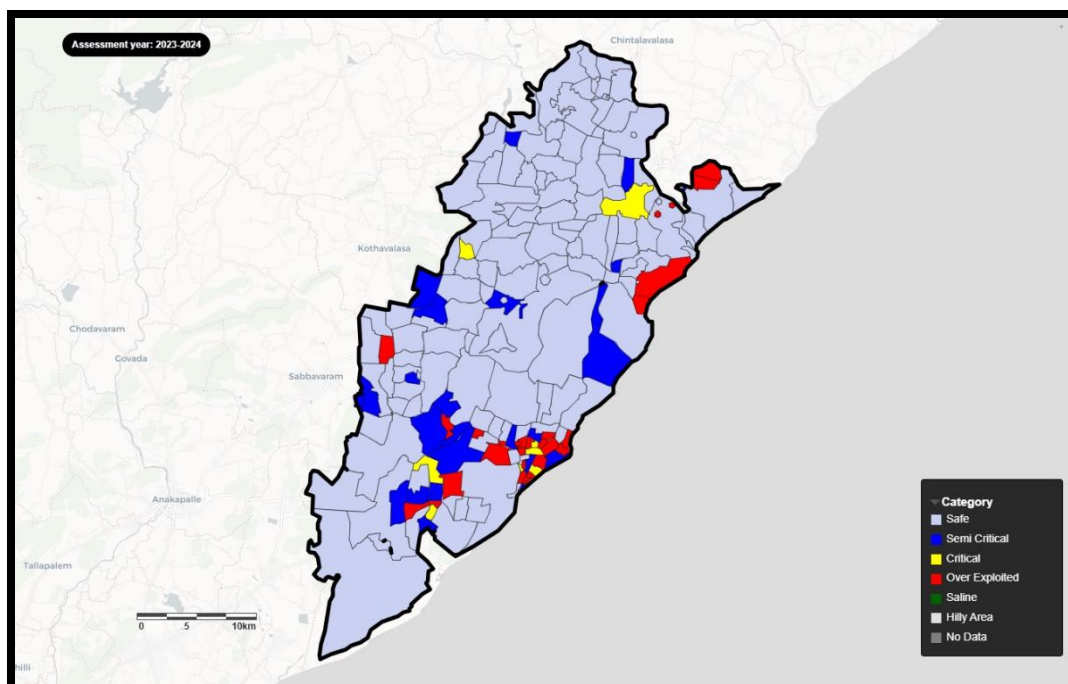


Fig 7.23: Categorization Map of Visakhapatnam District (Village Level)

7.24 Vizianagaram District

The Vizianagaram district having geographical area of 4123 km², lies between north latitude of 17° 83' and 18° 70' and east longitudes of 83° 00' and 83° 85'. There are six rivers draining the district, viz. Nagavali, Gosthani, Suvarnamukhi, Champavathi, Vegavathi, and Gomukhi. The Nagavali, Champavathi and Gosthani basin covering major portion of the district; Peddagedda and Kandivalasagedda basin occur in SE part and Sarada basin in SW part of the district.

The Vizianagaram district is underlain by various geological formations, namely Khondalite (60%), Granite gneiss (13%), Charnockites (24%), Alluvium (2%), Quartzite (1%) and Laterite (1%). The Upper Gondwanas and Tertiaries are represented by sandstones and shales, but these formations are not exposed in the study area as laterite occur as capping on the Khondalite. The alluvium occurs along the river courses consisting of sandy clay, sands and gravel.

Total Annual Groundwater recharge of the Vizianagaram district has been assessed as 1.42 bcm and Annual extractable Ground Water resource as 1.35 bcm. The Annual Ground Water Extraction is 0.35 bcm and Stage of Ground Water Extraction is 28.10 %. Out of 27 assessment units (Mandals), all units (100 %) have been categorized as 'Safe'. There is no 'Saline' category of assessment unit in the district.

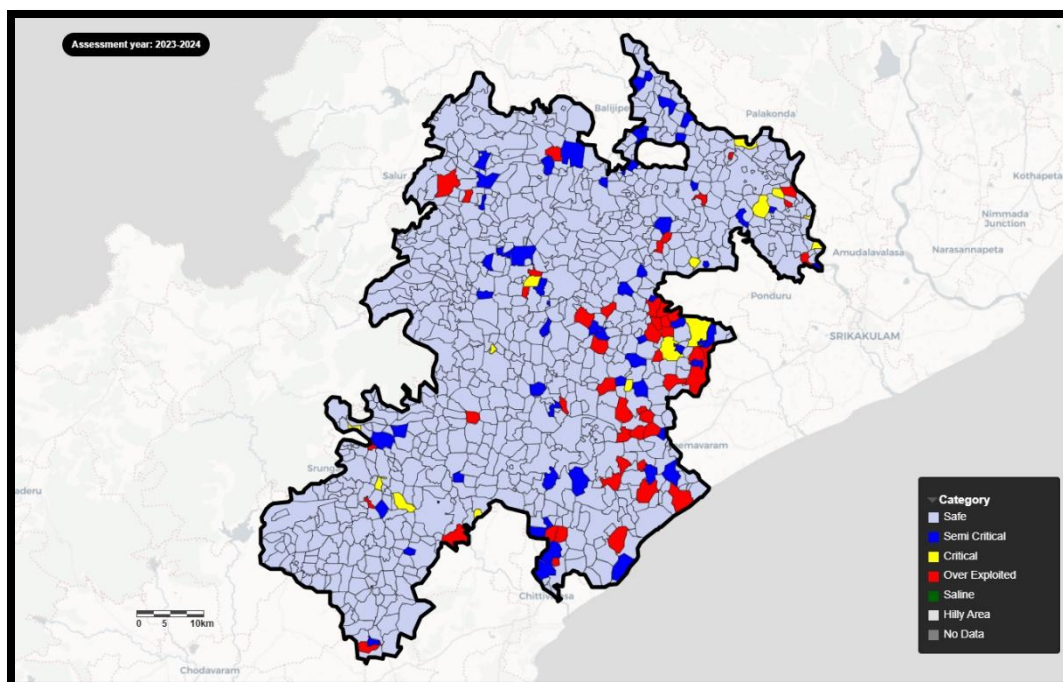


Fig 7.24: Categorization Map of Vizianagaram District (Village Level)

7.25 West Godavari District

The West Godavari district having geographical area of 2346 km², lies between north latitude 16°19' to 16°58' and east longitude 81°19' to 81°48'. Major part of the district falls under Godavari basin. Godavari river flows N to S direction in the area. The district is divided into 2 major drainage basins namely Godavari and Yerrakalva. The drainage density in the area is sparse due to high permeability of alluvial soils.

Geologically the area is underlain by the recent to sub-recent alluvium comprising Sand, Gravel, Clay and Silt. The alluvium is underlined by sandstones.

Total Annual Groundwater recharge of the West Godavari district has been assessed as 0.55 bcm and Annual extractable Ground Water resource as 0.53 bcm. The Annual Ground Water Extraction is 0.044 bcm and Stage of Ground Water Extraction is 8.68 %. Out of 20 assessment units (Mandals), 7 units (35 %) have been categorized as 'Safe' and 13 units have been categorized as 'Saline' category in the district.

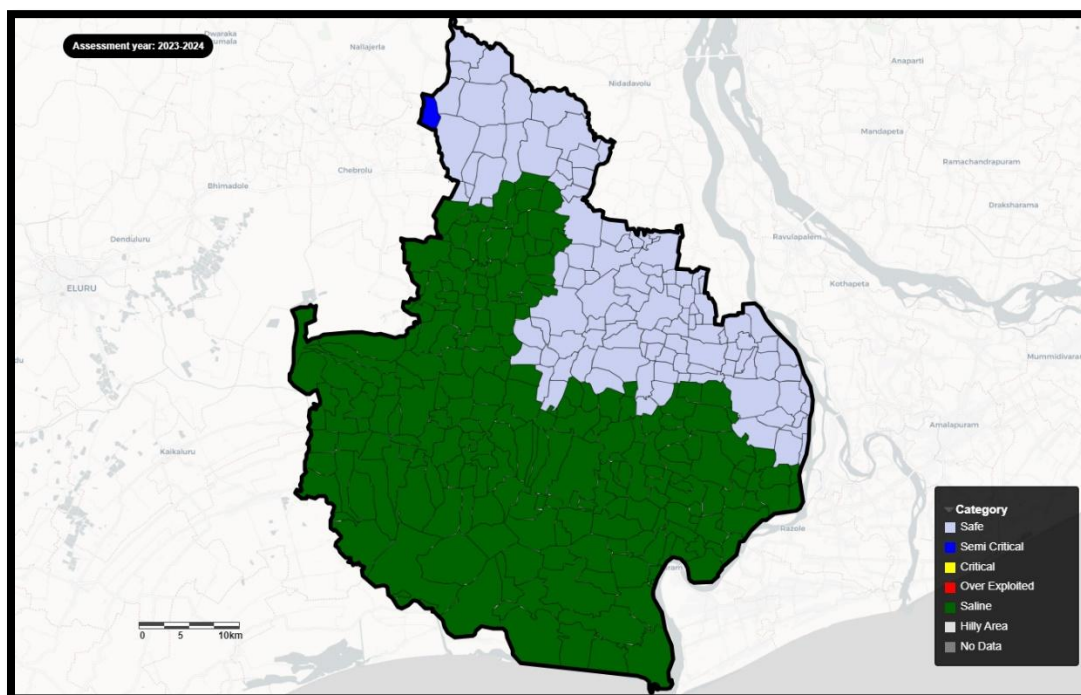


Fig 7.25: Categorization Map of West Godavari District (Village Level)

7.26 YSR Kadapa District

The YSR Kadapa district having geographical area of 11,228 km², lies between 14°9'40" and 16°0'57" of Northern Latitude and 78° 07'02" and 79° 59'58" Eastern Longitude. Y.S.R Kadapa District is drained by the Pennar Basin and its tributaries. The chief Northern tributaries to Pennar are the Kunderu (Kundu), Sagileru and the southern tributaries – the Cheyyeru, Papagni and the Chitravathi.

The area is underlain by various rock types that belong to Late Archaean or Early Proterozoic era which are succeeded by rocks of Dharwarian Age and both are traversed by dolerite dykes. The older rocks are overlain by rocks of Cuddapah Super group and Kurnool Group belonging to Middle and Upper Proterozoic Age. The Archaean comprises the Peninsular Gneissic Complex, represented by granite, granodiorite, granite-gneiss and migmatite. The metasediment rocks of Cuddapah and Kurnool Group include mostly shales, quartzites and limestones.

Total Annual Groundwater recharge of the YSR Kadapa district has been assessed as 1.07 bcm and Annual extractable Ground Water resource as 1.01 bcm. The Annual Ground Water Extraction is 0.41 bcm and Stage of Ground Water Extraction is 45.29 %. Out of 36 assessment units (Mandals), 4 units (11.1 %) have been categorized as 'Semi-Critical' and 32 units (88.9 %) as 'Safe'. There is no 'Saline' category of assessment unit in the district. Similarly, out of 8808 sq km recharge worthy area of the district, 758 sq km (8.6 %) area are under 'Semi-Critical' and 8050 sq.km (91.4 %) area is under 'Safe' categories of assessment units. Out of total 1018 mcm annual extractable ground water resources of the district, 63.3 mcm (6.2 %) are under 'Semi-Critical' and 955.5 mcm (93.8 %) are under 'Safe' categories of assessment units.

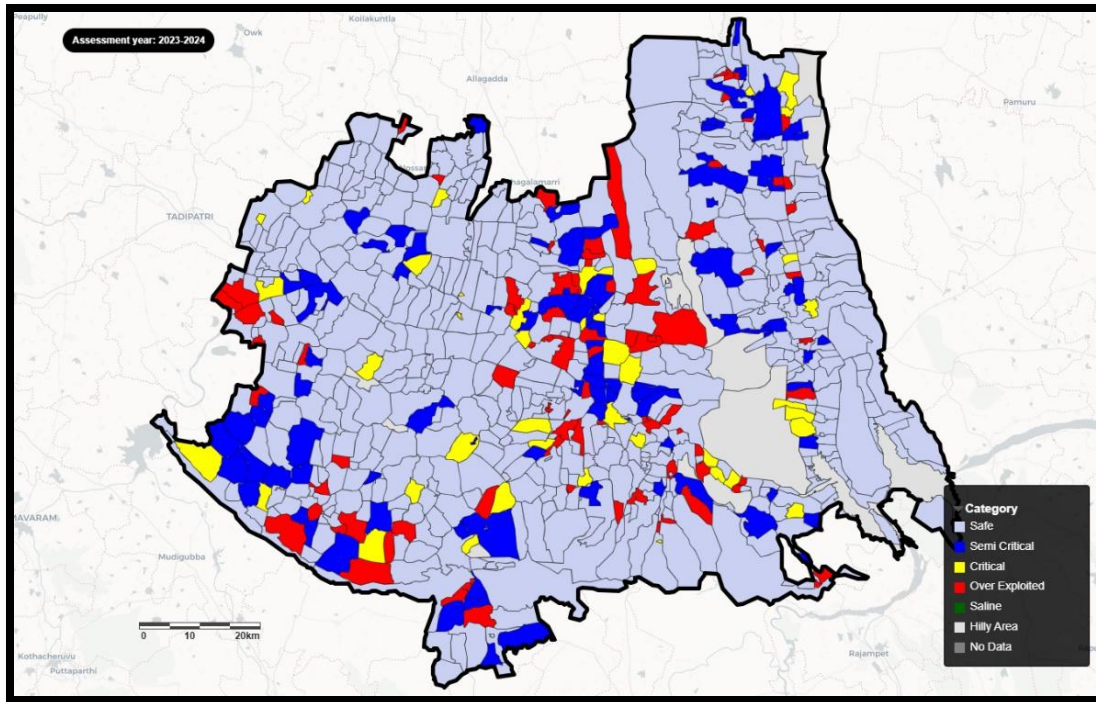


Fig 7.26: Categorization Map of YSR Kadapa District (Village Level)

CHAPTER 8

CONCLUSIONS

Andhra Pradesh, the 7th largest state in India, spans a geographical area of 1.63 lakh km², with a 974 km coastline. It comprises 26 districts, 679 mandals, and 17,467 villages, with a population of approximately 4.96 crores. The state is primarily drained by the Godavari, Krishna, and Pennar rivers and receives an average annual rainfall of 966 mm, varying from 552 mm in Anantapur to 1217 mm in East Godavari. The state is divided into three geographical zones: Coastal Plains, Eastern Ghats, and Western Pediplains. Of the total 162.76 lakh hectares, 54% (87.90 lakh ha) is cultivable land, while 22% (36.63 lakh ha) is forested. The cropped area spans 74.51 lakh ha, with 22.37 lakh ha irrigated by surface water and 15.75 lakh ha (42%) by groundwater, leaving an ayacut gap of 36.38 lakh ha.

The state has 14 principal aquifer systems, with 85% of the area underlain by hard rock formations, including granites, gneiss, and metamorphics (Archaean), quartzites and limestones (Proterozoic), and Deccan Trap basalts (Mesozoic). The remaining area consists of Gondwana sedimentaries (Tertiary) and alluvium (Recent). Andhra Pradesh is divided into 40 drainage basins and 74 sub-basins, which are further segmented into 748 micro-basins for groundwater estimation. The state is among the few to conduct assessments at the village level, whereas most other states assess groundwater at the Block, Firka, or Mandal levels.

The total annual groundwater recharge in 2024 is estimated at 27.8 bcm, with 1.39 bcm allocated for natural discharge, leaving 26.41 bcm as the annual extractable groundwater resource. Recharge from canal seepage, irrigation return flow, recharge from tanks, ponds, and conservation structures contributes 17.97 bcm (65%) of the total recharge, while 9.82 bcm (35%) comes from rainfall. Seasonal variations show that monsoon recharge accounts for 39% from other sources and 32% from rainfall, while non-monsoon recharge includes 26% from other sources and 3% from rainfall.

Total annual groundwater extraction in 2024 is estimated at 7.88 bcm, with irrigation accounting for 86% (6.75 bcm), domestic use 13% (1.0 bcm), and industrial use 1% (0.13 bcm). In districts such as Anakapalli, Chittoor, Eluru, Kurnool, Palnadu, Prakasam, Sri Potti Sriramulu Nellore, Sri Sathya Sai, Srikakulam, and Y.S.R Kadapa, groundwater extraction for domestic use exceeds 65%.

The overall stage of groundwater extraction in Andhra Pradesh is 29.83%, ranging from 1.5% in Alluri Sitharama Raju to 60.7% in Sri Sathya Sai. Across all 26 districts, groundwater extraction remains below 70%.

Among the 679 assessment units (mandals), 9 units (1.32%) are classified as 'Over-exploited,' 2 units (0.29%) as 'Critical,' 38 units (5.59%) as 'Semi-Critical,' 591 units (87.03%) as 'Safe,' and 39 units (5.74%) as 'Saline.' In terms of spatial distribution, out of 140,378.61 sq km of recharge-worthy area, 1.62% (2,278.23 sq km) falls under the 'Over-Exploited' category, 0.17% (236.68 sq km) under 'Critical,' 5.66% (7,947.83 sq km) under 'Semi-Critical,' 88.2% (123,819.54 sq km) under 'Safe,' and 4.34% (6,096.33 sq km) under 'Saline.'

Out of the total 26.41 bcm annual extractable groundwater resources, 0.26 bcm (0.99%) are under 'Over-exploited,' 0.066 bcm (0.25%) under 'Critical,' 0.89 bcm (3.4%) under 'Semi-critical,' and 25.05 bcm (94.87%) under 'Safe' categories of assessment units.

A comparison with 2023 shows a negligible reduction in total annual groundwater recharge, decreasing from 27.83 bcm to 27.80 bcm (0.03 bcm decrease). This decline is attributed to reduced recharge from surface water sources and increased groundwater extraction. The 2024 groundwater assessment provides critical insights for sustainable groundwater management in Andhra Pradesh. Although the overall groundwater situation remains safe, targeted interventions are necessary in over-exploited and semi-critical zones to ensure long-term sustainability. The state's detailed village-level assessment methodology continues to serve as a model for other regions, ensuring efficient planning and policy formulation for groundwater conservation and sustainable utilization.

ANNEXURES	
Annexure-I	Ground water resources availability, utilization and stage of extraction (as in 2024)
Annexure-II	District-wise ground water resources availability, utilization and stage of extraction (as in 2024)
Annexure-III(A)	Categorization of blocks/ mandals/ taluks in Andhra Pradesh State (as in 2024)
Annexure III (B)	District Wise Categorization of blocks/ mandals/ taluks for the State/UT (as in 2024)
Annexure III (C)	Annual Extractable Ground Water Resource of Assessment Units under Different Category for the State/UT (as in 2024)
Annexure- III (D)	District Wise Annual Extractable Ground Water Resource of Assessment Units under Different Category for the State
Annexure- III (E)	Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)
Annexure III (F)	District Wise Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)
Annexure IV (A)	Categorization of Over Exploited, Critical and Semi Critical blocks/ mandals/ taluks (as in 2024)
Annexure IV (C)	Saline Assessment units (as in 2024)
Annexure V (A)	Summary of Assessment units improved or deteriorated from 2023 to 2024 assessment
Annexure V (B)	Comparison of categorization of assessment units (2023 to 2024)
Annexure VI	Assessment Unit Wise Report (Attribute Table)
Annexure VII	Minutes of the Meeting of the SLC Committee

Annexure-I: Ground water resources availability, utilization and stage of extraction (as in 2024)

S.N O	States / Union Territori es	Ground Water Recharge					Total Natural Discharg es	Annual Extracta ble Ground Water Resourc e	Current Annual Ground Water Extraction				Annual GW Allocati on for Domesti c use as on 2025	Net Ground Water Availabili ty for future use	Stage of Ground Water Extracti on (%)
		Monsoon Season		Non-Monsoon Season		Total Annual Ground Water Rechar ge									
		Rechar ge from rainfall	Rechar ge from other Source s	Rechar ge from Rainfall	Rechar ge from other Source s										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	ANDHR A PRADES H	9	10.79	0.82	7.18	27.8	1.39	26.41	6.75	0.13	1	7.88	1.24	18.83	29.83
	Total(bc m)	9	10.79	0.82	7.18	27.8	1.39	26.41	6.75	0.13	1	7.88	1.24	18.83	29.83

Annexure-II: District-wise ground water resources availability, utilization and stage of extraction (as in 2024)

S.NO	Name of District	Ground Water Recharge					Total Natural Discharges	Annual Extractable Ground Water Resource	Current Annual Ground Water Extraction				Annual GW Allocation for Domestic use as on 2025	Net Ground Water Availability for future use	Stage of Ground Water Extraction (%)
		Monsoon Season		Non-Monsoon Season		Total Annual Ground Water Recharge			Irrigation	Industrial	Domestic	Total			
		Recharge from rainfall	Recharge from other Sources	Recharge from Rainfall	Recharge from other Sources										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Alluri Sitharama Raju	43956.31	102516.48	2860.84	8890.6	158224.23	7911.84	150312.39	577.32	0	1661.26	2239.02	1810.56	148119.42	1.5
2	Anakapalli	22443.69	38195.76	14423.44	6889.74	81952.63	4097.83	77854.81	15719.33	478.55	4769.1	20967.06	5470.01	56646.98	26.9
3	Ananthapuramu	40986.63	44150.14	1323.64	35512.65	121973.06	6098.76	115874.29	36312.74	727.29	3886.21	40926.43	6064.03	74925.11	35.3
4	Annamayya	44599.19	19166.03	2836.15	19968.94	86570.31	4328.54	82241.76	31820.22	137.75	3505.52	35463.41	5137.07	45491.54	43.1
5	Bapatla	23318.2	33572.17	1322.8	12065.45	70278.62	3513.96	66764.66	15491.49	55.24	2709.38	18256.09	2805.45	48843.3	27.3
6	Chittoor	55198.12	15130.71	1246.99	18849.65	90425.47	4521.59	85903.88	45253.99	1366.52	4283.36	50903.85	5638.47	34698.53	59.3
7	East Godavari	25369.55	41713.88	2604.84	37601.73	107290	5364.62	101925.38	30888.78	1120.8	2157.51	34167.05	2246.87	68798.43	33.5
8	Eluru	50680.82	88891.13	1620.74	41619.91	182812.6	9140.99	173671.73	37003.69	501.84	10198.63	47703.69	11572.37	126197.81	27.5
9	Guntur	12414.87	26259.86	2064.53	13762.42	54501.68	2725.11	51776.57	7827.2	218.34	2561.57	10607.13	2788.24	40942.74	20.5
10	Kakinada	16664.61	59653.79	3486.88	32788.46	112593.74	5629.83	107164.96	13340.42	1878.77	1876.95	17096.02	1941.94	90026.1	16.0
11	Konaseema	15932.71	48720.13	2077.68	45519	112249.52	5612.65	106636.87	9257.82	36.32	1292.5	10586.63	1338.38	96004.32	9.9
12	Krishna	18162.93	90343.85	10846.76	47391.01	166744.55	8337.3	158407.25	43910.82	1011.94	2037.46	46960.08	2692.79	110791.85	29.7
13	Kurnool	38342.94	17638.7	566.17	10907.07	67454.88	3372.87	64082.01	13993.36	71.27	5532.47	19597.28	5671.97	44662.31	30.6
14	NTR	20780.78	28090.65	1227.81	14159.92	64259.16	3213.03	61046.13	20329.46	1133.88	2432.47	23895.84	2471.08	37302.85	39.1
15	Nandyal	22651.71	40388.65	66.02	19114.07	82220.45	4111.14	78109.31	12220.88	128.4	505.41	12854.15	505.86	65255.16	16.5
16	Palnadu	25660.06	55379.59	1215.19	19221.55	101476.39	5073.93	96402.46	24865.33	65.92	4304.85	29236.42	4393.82	68904.95	30.3
17	Parvathipuram Manyam	29763.31	35032.83	3586.72	23061.67	91444.53	4572.57	86871.96	20015.69	12.96	1766.91	21795.33	2330.44	66218.62	25.1
18	Prakasam	63974.15	19744.85	1072.98	13544.2	98336.18	4917.08	93419.1	41469.89	509.48	4734.19	46714.02	4958.01	55899.08	50.0
19	Sri Potti Sriramulu Nellore	68345.39	36991.28	20510.69	71127.23	196974.59	9849.09	187125.49	37110.91	987.25	5790.41	43888.77	7173.37	144094.45	23.5
20	Sri Sathya Sai	43077.78	31974.68	595.26	44246.68	119894.4	5994.87	113899.53	60054.71	132.17	8904.83	69091.82	11143.26	53422.23	60.7
21	Srikakulam	34587.38	66648.48	2623.08	28333.77	132192.71	6609.82	125582.86	36702.64	369.91	7287.27	44359.44	11559.9	86698.19	35.3
22	Tirupati	81366.63	27763.1	87.94	52318.38	161536.05	8077.11	153458.94	48299.63	1496.79	3806.18	53602.53	9084.71	100140.9	34.9
23	Visakhapatnam	7995.13	3262.26	285.34	1078.97	12621.7	631.09	11990.61	2880.58	50.57	2451.06	5382.2	3184.42	6078.37	44.9
24	Vizianagaram	24997.08	66545.39	2538.33	48662.74	142743.54	7137.43	135606.07	33889.28	110.26	1808.8	35808.31	1808.8	101992.74	26.4
25	West Godavari	12011.91	21549.41	235.36	22132.19	55928.87	2796.42	53132.45	3861.02	101.52	473.89	4436.22	1130.41	48039.7	8.4
26	Y.S.R Kadapa	57104.54	20117.22	549.87	29473.8	107245.43	5362.48	101882.95	31716.54	430.43	9295.14	41442.26	9295.16	62494.21	40.7
	Total (Ham)	900386.42	1079441.02	81876.05	718241.8	2779945.29	139001.95	2641144.42	674813.73	13134.15	100033.31	787981.05	124217.39	1882689.89	29.8
	Total (Bcm)	9	10.79	0.82	7.18	27.8	1.39	26.41	6.75	0.13	1	7.88	1.24	18.83	29.8

Annexure-III(A): Categorization of blocks/ mandals/ taluks in India (as in 2024) for the State/UT

CATEGORIZATION OF BLOCKS/ MANDALS/ TALUKAS IN INDIA (2024)												
S.No	States / Union Territories	Total No. of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
			Nos.	%	Nos.	%	Nos.	%	Nos.	%	Nos.	%
1	ANDHRA PRADESH	17464	14031	80.3	1271	7.3	450	2.6	830	4.8	882	5.1
	Total	17464	14031	80.3	1271	7.3	450	2.6	830	4.8	882	5.1
	Grand Total	17464	14031	80.3	1271	7.3	450	2.6	830	4.8	882	5.1

Annexure III (B): District Wise Categorization of blocks/ mandals/ taluks for the State/UT (as in 2024)

S.No	Name of District	Total No. of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
			No	%	No.	%	No.	%	No.	%	No.	%
1	Sri Potti Sriramulu Nellore	37	36	97.3	1	2.7	-	-	-	-	-	-
2	Nandyal	29	29	100.0	-	-	-	-	-	-	-	-
3	Srikakulam	30	28	93.33	-	-	1	3.33	1	3.33	-	-
4	Vizianagaram	27	27	100.0	-	-	-	-	-	-	-	-
5	Sri Sathya Sai	32	22	68.75	5	15.63	1	3.13	4	12.5	-	-
6	Alluri Sitharama Raju	22	22	100.0	-	-	-	-	-	-	-	-
7	Konaseema	22	15	68.18	-	-	-	-	-	-	7	31.82
8	Eluru	27	22	81.48	-	-	-	-	-	-	5	18.52
9	West Godavari	20	7	35.0	-	-	-	-	-	-	13	65.0
10	Bapatla	25	24	96.0	-	-	-	-	-	-	1	4.0
11	Palnadu	28	26	92.86	1	3.57	-	-	1	3.57	-	-
12	Kakinada	21	19	90.48	-	-	-	-	-	-	2	9.52
13	Y.S.R Kadapa	36	32	88.89	4	11.11	-	-	-	-	-	-
14	Ananthapuramu	32	31	96.88	1	3.13	-	-	-	-	-	-
15	Chittoor	31	21	67.74	10	32.26	-	-	-	-	-	-
16	East Godavari	19	15	78.95	4	21.05	-	-	-	-	-	-
17	Guntur	18	16	88.89	-	-	-	-	-	-	2	11.11
18	Krishna	25	16	64.0	-	-	-	-	-	-	9	36.0
19	Kurnool	26	25	96.15	1	3.85	-	-	-	-	-	-
20	Prakasam	38	30	78.95	5	13.16	-	-	3	7.89	-	-
21	Visakhapatnam	11	7	63.64	4	36.36	-	-	-	-	-	-
22	Anakapalli	24	24	100.0	-	-	-	-	-	-	-	-
23	Annamayya	30	28	93.33	2	6.67	-	-	-	-	-	-
24	NTR	20	20	100.0	-	-	-	-	-	-	-	-
25	Parvathipuram Manyam	15	15	100.0	-	-	-	-	-	-	-	-
26	Tirupati	34	34	100.0	-	-	-	-	-	-	-	-
	Total	679	591	87.04	38	5.6	2	0.29	9	1.33	39	5.74

Annexure III (C): Annual Extractable Ground Water Resource of Assessment Units under Different Category for the State/UT (as in 2024)

ANNUAL EXTRACTABLE RESOURCE OF ASSESSMENT UNITS UNDER DIFFERENT CATEGORIES, 2024												
S.No	State/Union Territories	Total Annual Extractable Resource of Assessed Units (in mcm)	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
			Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%
1	ANDHRA PRADESH	26411.44	25056.01	94.87	897.37	3.4	66.12	0.25	261.89	0.99	2300.84	8.05
	Total	26411.44	25056.01	94.87	897.37	3.4	66.12	0.25	261.89	0.99	2300.84	8.05
	Grand Total	26411.44	25056.01	94.87	897.37	3.4	66.12	0.25	261.89	0.99	2300.84	8.05

Annexure- III (D) : District Wise Annual Extractable Ground Water Resource of Assessment Units under Different Category for the State (as in 2024)

S.No	Name of District	Total Annual Extractable Resource of Assessed Units (in mcm)	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
			Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%	Total Annual Extractable Resource (in mcm)	%
1	Sri Potti Sriramulu Nellore	1871.25	1819.33	97.22	51.93	2.78	-	-	-	-	-	-
2	Nandyal	781.09	781.09	100	-	-	-	-	-	-	-	-
3	Srikakulam	1255.83	1157.07	92.14	-	-	44.31	3.53	54.45	4.34	-	-
4	Vizianagaram	1356.06	1356.06	100	-	-	-	-	-	-	-	-
5	Sri Sathya Sai	1139	848.51	74.5	127.08	11.16	21.82	1.92	141.59	12.43	-	-
6	Alluri Sitharama Raju	1503.12	1503.12	100	-	-	-	-	-	-	-	-
7	Konaseema	1066.37	1066.37	100	-	-	-	-	-	-	299	28.04
8	Eluru	1736.72	1606.67	92.51	-	-	-	-	-	-	296.73	17.09
9	West Godavari	531.32	531.32	100	-	-	-	-	-	-	871.32	163.99
10	Bapatla	667.65	667.65	100	-	-	-	-	-	-	39.77	5.96
11	Palnadu	964.02	914.48	94.86	31.28	3.24	-	-	18.27	1.89	-	-
12	Kakinada	1071.65	1071.65	100	-	-	-	-	-	-	158.36	14.78
13	Y.S.R Kadapa	1018.83	955.49	93.78	63.34	6.22	-	-	-	-	-	-
14	Ananthapuramu	1158.74	1129.57	97.48	29.18	2.52	-	-	-	-	-	-
15	Chittoor	859.04	583.88	67.97	275.16	32.03	-	-	-	-	-	-
16	East Godavari	1019.25	872.29	85.58	146.96	14.42	-	-	-	-	-	-
17	Guntur	517.77	517.77	100	-	-	-	-	-	-	48.46	9.36
18	Krishna	1584.07	1584.07	100	-	-	-	-	-	-	587.2	37.07
19	Kurnool	640.82	626.39	97.75	14.43	2.25	-	-	-	-	-	-
20	Prakasam	934.19	771.61	82.6	115	12.31	-	-	47.58	5.09	-	-
21	Visakhapatnam	119.91	105.47	87.96	14.44	12.04	-	-	-	-	-	-
22	Anakapalli	778.55	778.55	100	-	-	-	-	-	-	-	-
23	Annamayya	822.42	793.85	96.53	28.57	3.47	-	-	-	-	-	-
24	NTR	610.46	610.46	100	-	-	-	-	-	-	-	-
25	Parvathipuram Manyam	868.72	868.72	100	-	-	-	-	-	-	-	-
26	Tirupati	1534.59	1534.59	100	-	-	-	-	-	-	-	-
	Total	26411.44	25056.01	94.87	897.37	3.4	66.12	0.25	261.89	0.99	2300.84	8.05
	Grand Total	26411.44	25056.01	94.87	897.37	3.4	66.12	0.25	261.89	0.99	2300.84	8.05

Annexure- III (E) : Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)

AREA OF ASSESSMENT UNITS UNDER DIFFERENT CATEGORIES IN INDIA (2024)													
S.No	States / Union Territories	Total Geographical Area of Assessed Units (in sq km)	Recharge Worthy Area (in sq km)	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
				Recharge Worthy Area in sq km	%	Recharge Worthy Area in sq km	%	Recharge Worthy Area in sq km	%	Recharge Worthy Area in sq km	%	Recharge Worthy Area in sq km	%
1	ANDHRA PRADESH	163448.46	140378.61	123819.54	88.2	7947.83	5.66	236.68	0.17	2278.23	1.62	6096.33	4.34
	Total	163448.46	140378.61	123819.54	88.2	7947.83	5.66	236.68	0.17	2278.23	1.62	6096.33	4.34
	Grand Total	163448.46	140378.61	123819.54	88.2	7947.83	5.66	236.68	0.17	2278.23	1.62	6096.33	4.34

Annexure III (F) : District Wise Recharge Worthy Area of Assessment unit under Different Category for the State/UT (as in 2024)

S.No	Name of District	Total Recharge Worthy Area of Assessed Units (in sq.km)	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
			Recharge Worthy Area of Assessed Units (in sq.km)	%	Recharge Worthy Area of Assessed Units (in sq.km)	%	Recharge Worthy Area of Assessed Units (in sq.km)	%	Recharge Worthy Area of Assessed Units (in sq.km)	%	Recharge Worthy Area of Assessed Units (in sq.km)	%
1	Sri Potti Sriramulu Nellore	9953.26	9809.06	98.55	144.2	1.45	-	-	-	-	-	-
2	Nandyal	7061.82	7061.82	100.0	-	-	-	-	-	-	-	-
3	Srikakulam	4483.89	4155.38	92.67	-	-	113.71	2.54	214.81	4.79	-	-
4	Vizianagaram	4011.79	4011.79	100.0	-	-	-	-	-	-	-	-
5	Sri Sathya Sai	7636.24	5787.13	75.79	931.28	12.2	122.97	1.61	794.86	10.41	-	-
6	Alluri Sitharama Raju	6392.35	6392.35	100.0	-	-	-	-	-	-	-	-
7	Konaseema	2079.95	1320.07	63.47	-	-	-	-	-	-	759.88	36.53
8	Eluru	6491.9	5627.11	86.68	-	-	-	-	-	-	864.79	13.32
9	West Godavari	2278.37	690.71	30.32	-	-	-	-	-	-	1587.66	69.68
10	Bapatla	3801.18	3638.93	95.73	-	-	-	-	-	-	162.25	4.27
11	Palnadu	6516.56	5641.54	86.57	539.1	8.27	-	-	335.92	5.15	-	-
12	Kakinada	2881.24	2362.46	81.99	-	-	-	-	-	-	518.78	18.01
13	Y.S.R Kadapa	8808.08	8050.04	91.39	758.04	8.61	-	-	-	-	-	-
14	Ananthapuramu	9464.79	9216.06	97.37	248.73	2.63	-	-	-	-	-	-
15	Chittoor	6004.51	4223.14	70.33	1781.37	29.67	-	-	-	-	-	-
16	East Godavari	2545.53	1831.91	71.97	713.61	28.03	-	-	-	-	-	-
17	Guntur	2423.33	2165.85	89.38	-	-	-	-	-	-	257.48	10.62
18	Krishna	3880.56	1935.06	49.87	-	-	-	-	-	-	1945.5	50.13
19	Kurnool	7522.47	7191.65	95.6	330.82	4.4	-	-	-	-	-	-
20	Prakasam	11381.9	8550.75	75.13	1898.51	16.68	-	-	932.64	8.19	-	-
21	Visakhapatnam	1001.67	812.09	81.07	189.58	18.93	-	-	-	-	-	-
22	Anakapalli	3459.23	3459.23	100.0	-	-	-	-	-	-	-	-
23	Annamayya	6660.66	6248.08	93.81	412.58	6.19	-	-	-	-	-	-
24	NTR	3063.19	3063.19	100.0	-	-	-	-	-	-	-	-
25	Parvathipuram Manyam	2948.81	2948.81	100.0	-	-	-	-	-	-	-	-
26	Tirupati	7625.31	7625.31	100.0	-	-	-	-	-	-	-	-
	Total	140378.61	123819.54	88.2	7947.83	5.66	236.68	0.17	2278.23	1.62	6096.33	4.34

Annexure IV (A): Categorization of Over Exploited, Critical and Semi Critical blocks/ mandals/ taluks (as in 2024)

S.NO	Name of District	S.NO	Name of Semi-Critical Assessment Units	S.NO	Name of Critical Assessment Units	S.NO	Name of Over-Exploited Assessment Units
1	Ananthapuramu	1	NARPALA				
2	Annamayya	1	CHITVEL				
		2	CHINNAMANDEM				
3	Chittoor	1	PEDDA PANJANI				
		2	CHITTOOR MANDAL				
		3	RAMA KUPPAM				
		4	VENKATAGIRI KOTA				
		5	THAVANAMPALLE				
		6	CHOWDEPALLE				
		7	GUDI PALLE				
		8	NINDRA				
		9	PENUMURU				
		10	PUTHALAPATTU				
4	East Godavari	1	NALLAJERLA				
		2	CHAGALLU				
		3	DEVARAPALLE				
		4	RANGAMPETA				
5	Kurnool	1	ORVAKAL				
6	Palnadu	1	BOLLAPALLE			1	VELDURTHI
7	Prakasam	1	MARKAPUR			1	RACHERLA
		2	TRIPURANTHAKAM			2	PULLALACHERUVU
		3	YERRAGONDAPALEM			3	PEDAARAVEEDU
		4	CUMBUM				

		5	GIDDALURU				
S.NO	Name of District	S.NO	Name of Semi-Critical Assessment Units	S.NO	Name of Critical Assessment Units	S.NO	Name of Over-Exploited Assessment Units
8	Sri Potti Sriramulu Nellore	1	INDUKURPET				
9	Sri Sathya Sai	1	NAMBULIPULIKUNTA	1	AGALI	1	TANAKAL
		2	GUDIBANDA			2	HINDUPUR
		3	LEPAKSHI			3	ROLLA
		4	NALLACHERUVU			4	GANDLAPENTA
		5	AMADAGUR				
10	Srikakulam			1	LAVERU	1	RANASTALAM
11	Visakhapatnam	1	SEETHAMMADHARA				
		2	VISAKHAPATNAM RURAL				
		3	MAHARANIPETA				
		4	PEDDAGANTYADA				
12	Y.S.R Kadapa	1	VEMULA				
		2	DUVVUR				
		3	PULIVENDLA				
		4	CHAPAD				
ABSTRACT							
Total No. of Assessed Units		Number of Semicritical Assessment Units		Number of Critical Assessment Units		Number of Over Exploited Assessment Units	
640		38		2		9	

Annexure IV (C): List of Saline Assessment units

District		Number of Assessment Units affected by Salinity	
1	West Godavari	1	AKIVEEDU
2	West Godavari	2	UNDI
3	West Godavari	3	GANAPAVARAM
4	West Godavari	4	PALACOLE
5	West Godavari	5	VEERAVASARAM
6	West Godavari	6	BHEEMAVARAM
7	West Godavari	7	KALLA
8	West Godavari	8	MOGALTHUR
9	West Godavari	9	NARASAPURAM
10	West Godavari	10	PALAKODERU
11	West Godavari	11	PENTAPADU
12	West Godavari	12	PODURU
13	West Godavari	13	YELAMANCHILI
14	Krishna	14	NAGAYALANKA
15	Krishna	15	MACHILIPATNAM
16	Krishna	16	GUDUR (KRISHNA)
17	Krishna	17	BANTUMILLI
18	Krishna	18	KRUTHIVENNU
19	Krishna	19	NANDIVADA
20	Krishna	20	GUDLAVALLERU
21	Krishna	21	PEDANA
22	Krishna	22	KODURU
23	Bapatla	23	KARAMCHEDU
24	Guntur	24	VATTICHERUKURU
25	Guntur	25	PEDANANDIPADU
26	Konaseema	26	I POLAVARAM
27	Konaseema	27	ALLAVARAM
28	Konaseema	28	KATRENKONA
29	Konaseema	29	MALIKIPURAM
30	Konaseema	30	MAMIDIKUDURU
31	Konaseema	31	SAKHINETIPALLE
32	Konaseema	32	UPPALAGUPTAM
33	Kakinada	33	KARAPA
34	Kakinada	34	THALLAREVU
35	Eluru	35	KALIDINDI
36	Eluru	36	KAIKALUR
37	Eluru	37	NIDAMARRU
38	Eluru	38	MANDAVALLI
39	Eluru	39	MUDINAPALLE

Annexure V (A): Summary of Assessment units improved or deteriorated from 2023 to 2024 assessment

S.No	Name of States / Union Territories	Number of Assessment Units Improved	Number of Assessment Units Deteriorated	Number of Assessment Units with No Change	
1	ANDHRA PRADESH	20	16	643	

Annexure V (B) : Comparison of categorization of assessment units (2023 to 2024)

S.No	Name of District	Name of Assessment Unit	Stage of Ground Water Extraction (%)2023	Categorization in2023	Name of District	Name of Assessment Unit	Stage of Ground Water Extraction (%)2024	Categorization in2024	Remark
1	Sri Potti Sriramulu Nellore	INDUKURPET	54.21	safe	Sri Potti Sriramulu Nellore	INDUKURPET	70.82	semi_critical	Deteriorated
2	Srikakulam	GARA	76.85	semi_critical	Srikakulam	GARA	55.61	safe	Improved
3	Sri Sathya Sai	LEPAKSHI	69.51	safe	Sri Sathya Sai	LEPAKSHI	76.80	semi_critical	Deteriorated
4	Sri Sathya Sai	NAMBULIPULIKUNTA	67.50	safe	Sri Sathya Sai	NAMBULIPULIKUNTA	77.56	semi_critical	Deteriorated
5	Palnadu	BOLLAPALLE	91.77	critical	Palnadu	BOLLAPALLE	87.73	semi_critical	Improved
6	Y.S.R Kadapa	CHAPAD	20.91	safe	Y.S.R Kadapa	CHAPAD	79.75	semi_critical	Deteriorated
7	Y.S.R Kadapa	DUVVUR	37.71	safe	Y.S.R Kadapa	DUVVUR	74.07	semi_critical	Deteriorated
8	Y.S.R Kadapa	PULIVENDLA	102.23	over_exploited	Y.S.R Kadapa	PULIVENDLA	81.81	semi_critical	Improved
9	Ananthapuramu	ANANTAPUR URBAN	-	-	Ananthapuramu	ANANTAPUR URBAN	26.81	safe	Improved
10	Chittoor	PENUMURU	69.86	safe	Chittoor	PENUMURU	72.04	semi_critical	Deteriorated
11	Chittoor	CHITTOOR MANDAL	55.07	safe	Chittoor	CHITTOOR MANDAL	72.52	semi_critical	Deteriorated
12	Chittoor	CHOWDEPALLE	62.25	safe	Chittoor	CHOWDEPALLE	80.38	semi_critical	Deteriorated
13	East Godavari	CHAGALLU	66.76	safe	East Godavari	CHAGALLU	71.87	semi_critical	Deteriorated
14	East Godavari	RANGAMPETA	59.18	safe	East Godavari	RANGAMPETA	74.15	semi_critical	Deteriorated
15	East Godavari	DEVARAPALLE	63.00	safe	East Godavari	DEVARAPALLE	70.79	semi_critical	Deteriorated
16	Guntur	GUNTUR EAST	-	-	Guntur	GUNTUR EAST	11.00	safe	Improved
17	Guntur	GUNTUR WEST	-	-	Guntur	GUNTUR WEST	36.16	safe	Improved
18	Kurnool	ORVAKAL	65.87	safe	Kurnool	ORVAKAL	70.15	semi_critical	Deteriorated
19	Kurnool	KURNOOL RURAL	-	-	Kurnool	KURNOOL RURAL	34.84	safe	Improved
20	Kurnool	KURNOOL URBAN	-	-	Kurnool	KURNOOL URBAN	21.69	safe	Improved
21	Prakasam	PULLALACHERUVU	89.36	semi_critical	Prakasam	PULLALACHERUVU	192.03	over_exploited	Deteriorated
22	Prakasam	MARKAPUR	66.08	safe	Prakasam	MARKAPUR	89.64	semi_critical	Deteriorated
23	Prakasam	TRIPURANTHAKAM	28.44	safe	Prakasam	TRIPURANTHAKAM	81.90	semi_critical	Deteriorated
24	Prakasam	YERRAGONDAPALEM	29.65	safe	Prakasam	YERRAGONDAPALEM	87.00	semi_critical	Deteriorated
25	Prakasam	CUMBUM	106.37	over_exploited	Prakasam	CUMBUM	83.68	semi_critical	Improved
26	Visakhapatnam	GOPALAPATNAM	-	-	Visakhapatnam	GOPALAPATNAM	42.53	safe	Improved
27	Visakhapatnam	MULAGADA	-	-	Visakhapatnam	MULAGADA	14.89	safe	Improved
28	Visakhapatnam	PEDDAGANTYADA	-	-	Visakhapatnam	PEDDAGANTYADA	86.51	semi_critical	Improved
29	Visakhapatnam	VISAKHAPATNAM RURAL	-	-	Visakhapatnam	VISAKHAPATNAM RURAL	81.95	semi_critical	Improved
30	Visakhapatnam	GAJUWAKA	-	-	Visakhapatnam	GAJUWAKA	64.54	safe	Improved
31	Visakhapatnam	MAHARANIPETA	-	-	Visakhapatnam	MAHARANIPETA	71.67	semi_critical	Improved

S.No	Name of District	Name of Assessment Unit	Stage of Ground Water Extraction (%)2023	Categorization in2023	Name of District	Name of Assessment Unit	Stage of Ground Water Extraction (%)2024	Categorization in2024	Remark
32	Visakhapatnam	SEETHAMMADHARA	-	-	Visakhapatnam	SEETHAMMADHARA	86.36	semi_critical	Improved
33	NTR	VIJAYAWADA NORTH	-	-	NTR	VIJAYAWADA NORTH	18.11	safe	Improved
34	NTR	VIJAYAWADA WEST	-	-	NTR	VIJAYAWADA WEST	65.98	safe	Improved
35	NTR	VIJAYAWADA EAST	-	-	NTR	VIJAYAWADA EAST	39.19	safe	Improved
36	NTR	VIJAYAWADA CENTRAL	-	-	NTR	VIJAYAWADA CENTRAL	22.13	safe	Improved

Annexure VI: Assessment Unit Wise Report (Attribute Table)

Sl.No	District	Assessment Unit Name	Recharge from Rainfall-I-MON	Recharge from Other Source s-MON	Recharge from Rainfall-I-NM	Recharge from Other Source s-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
1	Parvathipuram Manyam	MAKKUVA	2418.69	2550.63	198.2	1703.33	6870.85	343.58	6527.27	1733.02	1.44	50.64	1785.11	105.23	4687.6	27.35	safe
2	Parvathipuram Manyam	VEERAGHATTAM	1276.23	2107.21	124.67	238.27	3746.38	187.37	3559.01	829.23	0	231.30	1060.53	231.44	2514.74	29.80	safe
3	Parvathipuram Manyam	BHAMINI	1897.73	1251.4	364.01	482.13	3995.27	199.76	3795.51	581.23	0	222.42	803.64	222.48	2991.8	21.17	safe
4	Parvathipuram Manyam	PARVATHIPURAM	2575.85	2960.74	379.47	1501.53	7417.59	370.92	7046.67	1529.01	8.64	81.11	1618.73	138.01	5382.96	22.97	safe
5	Parvathipuram Manyam	GARUGUBILLI	1285.57	2055.86	154.48	735.44	4231.35	211.57	4019.78	1489.77	0	40.10	1529.85	93.96	2457.17	38.06	safe
6	Parvathipuram Manyam	JIYYAMMA VALASA	1601.89	1995.43	178.41	440.37	4216.1	210.86	4005.24	1443.90	0	45.75	1489.63	103.66	2800.1	37.19	safe
7	Parvathipuram Manyam	SALUR	2908.04	1535.55	252.24	1325.29	6021.12	301.07	5720.05	2776.60	0	81.94	2858.47	125.39	3018.94	49.97	safe
8	Parvathipuram Manyam	BALJIPETA	1606.28	6062.04	252.58	5816.24	13737.14	686.91	13050.23	1906.22	0	33.78	1940.01	113.47	11091.17	14.87	safe
9	Parvathipuram Manyam	KURUPAM	3334.99	1472.88	265.99	831.32	5905.18	295.2	5609.98	610.77	0	80.47	691.24	117.36	4904.15	12.32	safe
10	Parvathipuram Manyam	KOMARADA	2841.92	1318.73	375.3	1456.03	5991.98	299.65	5692.33	3061.49	1.44	96.39	3159.3	127.19	3014.47	55.50	safe
11	Parvathipuram Manyam	GUMMALAKSHMIPURAM	1393.92	940.26	198.65	551.6	3084.43	154.22	2930.21	74.65	0	95.12	169.74	128.22	2727.38	5.79	safe
12	Parvathipuram Manyam	PACHIPENTA	1666.5	930.22	227.42	685.55	3509.69	175.52	3334.17	1725.06	0	72.85	1797.86	118.18	1907.95	53.92	safe
13	Parvathipuram Manyam	SEETHANAGARAM (MANYAM)	1821.19	6460.42	239.7	6118.13	14639.44	732.03	13907.41	1600.99	1.44	43.12	1645.52	111.58	12230.07	11.83	safe
14	Parvathipuram Manyam	PALAKONDA	994.29	2423.89	81.83	516.4	4016.41	200.78	3815.63	533.26	0	208.32	741.59	209.04	3100.78	19.44	safe
15	Parvathipuram Manyam	SEETHAMPETA	2140.22	967.57	293.77	660.04	4061.6	203.13	3858.47	120.48	0	383.60	504.11	385.23	3389.34	13.07	safe
16	Alluri Sitharama Raju	MAREDUMILLI	1558.2	1935.34	92.82	134.97	3721.33	186	3535.33	9.58	0	25.15	34.72	27.05	3498.72	0.98	safe
17	Alluri Sitharama Raju	CHINTUR	3804.28	7000.65	0	597.08	11402.01	570.12	10831.89	49.18	0	63.96	113.19	68.83	10713.83	1.04	safe
18	Alluri Sitharama Raju	PEDABAYALU	1534.31	7453.9	119.46	604.37	9712.04	485.63	9226.41	112.45	0	60.74	173.2	65.35	9048.59	1.88	safe
19	Alluri Sitharama Raju	RAMPACHODAVARAM	2759.61	3709.29	102.81	88.87	6660.58	332.98	6327.6	34.56	0	44.99	79.57	48.39	6244.59	1.26	safe
20	Alluri Sitharama Raju	Y RAMAVARAM	3489.2	2787.61	123.41	81.44	6481.66	324.09	6157.57	2.52	0	38.50	41.02	41.38	6113.66	0.67	safe
21	Alluri Sitharama Raju	ADDATEEGALA	2197.95	3183.06	146.1	207.37	5734.48	286.76	5447.72	26.56	0	47.97	74.5	51.64	5369.56	1.37	safe
22	Alluri Sitharama Raju	DEVIPATNAM	2049.21	3100.09	90.27	932.72	6172.29	308.6	5863.69	43.68	0	27.38	71.04	29.45	5790.57	1.21	safe
23	Alluri Sitharama Raju	GANGAVARAM (EAST GODAVARI)	2997.82	2932.8	223.9	153.88	6308.4	315.41	5992.99	11.09	0	36.93	48.03	39.73	5942.16	0.80	safe
24	Alluri Sitharama Raju	KUNAVARAM	1028.29	859.73	0	345.07	2233.09	111.71	2121.38	8.57	0	29.68	38.19	31.96	2080.87	1.80	safe
25	Alluri Sitharama Raju	NELLIPAKA	1931.27	2289.85	0	138.94	4360.06	217.98	4142.08	71.53	0	42.27	113.98	45.47	4024.93	2.75	safe

Sl.No	District	Assessment Unit Name	Recharge from Rainfall-I-MON	Recharge from Other Sources-MON	Recharge from Rainfall-I-NM	Recharge from Other Sources-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
26	Alluri Sitharama Raju	RAJAVOMMANGI	2252.92	4073.87	132.53	106.9	6566.22	328.28	6237.94	16.63	0	46.99	63.56	50.52	6170.85	1.02	safe
27	Alluri Sitharama Raju	VARARAMACHANDRAPUR	2091.38	2415.54	0	97.9	4604.82	230.28	4374.54	12.60	0	47.71	60.35	51.32	4310.59	1.38	safe
28	Alluri Sitharama Raju	ANANTHAGIRI	2559.82	4393.28	301.38	361.54	7616.02	380.89	7235.13	13.61	0	77.73	91.38	83.62	7137.9	1.26	safe
29	Alluri Sitharama Raju	ARAKU VALLEY	1138.49	6846.63	172.21	526.11	8683.44	434.37	8249.07	13.10	0	67.72	80.88	72.89	8163.01	0.98	safe
30	Alluri Sitharama Raju	CHINTAPALLE	1253.92	5524.58	208.07	815.92	7802.49	390.29	7412.2	16.60	0	100.28	116.9	107.84	7287.79	1.58	safe
31	Alluri Sitharama Raju	DUMBRIGUDA	1001.59	6676.58	167.85	551.47	8397.49	419.89	7977.6	52.42	0	77.59	130.11	83.46	7841.63	1.63	safe
32	Alluri Sitharama Raju	GANGARAJU MADUGULA	691.77	6223.64	44.96	680.97	7641.34	382.15	7259.19	15.12	0	58.39	73.64	62.87	7181.05	1.01	safe
33	Alluri Sitharama Raju	GUDEM KOTHAVEEDHI	932.09	5449.71	109.74	912.62	7404.16	370.21	7033.95	3.02	0	132.40	135.39	145.87	6900.1	1.92	safe
34	Alluri Sitharama Raju	HUKUMPETA	1334.02	10104.23	295.09	395.84	12129.18	606.44	11522.74	7.06	0	62.08	69.14	66.78	11448.92	0.60	safe
35	Alluri Sitharama Raju	KOYYURU	3193.32	4084.87	253.79	177.19	7709.17	385.46	7323.71	28.22	0	98.91	127.1	106.47	7189.01	1.74	safe
36	Alluri Sitharama Raju	MUNCHINGIPUTTU	2701.48	4967	0	450.05	8118.53	405.99	7712.54	0.00	0	90.14	90.14	96.96	7615.66	1.17	safe
37	Alluri Sitharama Raju	PADERU	1455.37	6504.23	276.45	529.38	8765.43	438.31	8327.12	29.23	0	383.76	412.99	432.71	8045.43	4.96	safe
38	Ananthapuramu	YADIKI	741.95	1376.11	0	1273.1	3391.16	169.54	3221.62	1173.59	5.328	134.69	1313.62	165.07	1877.64	40.78	safe
39	Ananthapuramu	ANANTAPUR RURAL	1502.73	992.64	23.79	572.4	3091.56	154.57	2936.98	408.89	0	159.44	568.34	237.99	2361.68	19.35	safe
40	Ananthapuramu	RAYADURG	1280.68	1150.73	83.43	1160.98	3675.82	183.82	3492	1223.35	1.8	106.78	1331.92	245.85	2021	38.14	safe
41	Ananthapuramu	VAJRAKARUR	1441.95	1973.53	0	1167.29	4582.77	229.15	4353.62	603.33	11.52	120.53	735.37	156.18	3582.61	16.89	safe
42	Ananthapuramu	BRAHMASAMUDRAM	1820.75	1138.56	81.98	1106.52	4147.81	207.42	3940.39	1075.29	0	82.22	1157.51	114.48	2750.63	29.38	safe
43	Ananthapuramu	SINGANAMALA	1493.45	2220.31	0	1175.69	4889.45	244.49	4644.96	1283.06	21.6	178.89	1483.57	189.03	3157.24	31.94	safe
44	Ananthapuramu	D HIRCHAL	742.22	1267.45	709.77	1392.17	4111.61	205.57	3906.04	1293.77	102.6	80.01	1476.37	124.32	2410.2	37.80	safe
45	Ananthapuramu	GUNTAKAL	1462.83	1151.15	0	851.18	3465.16	173.31	3291.85	844.40	0	482.54	1326.92	518.65	2191.05	40.31	safe
46	Ananthapuramu	GOOTY	1468.12	1039.4	26.29	1090.37	3624.18	181.19	3442.99	620.43	1.8	164.60	786.9	168.32	2652.36	22.86	safe
47	Ananthapuramu	PEDDAVADUGUR	734.92	1433.07	2.22	735.75	2905.96	145.31	2760.65	626.85	0.36	126.16	753.39	134.03	2026.08	27.29	safe
48	Ananthapuramu	KUDAIR	1083.6	1031.5	13.08	1149.45	3277.63	163.9	3113.73	1566.47	0.144	50.79	1617.41	103.17	1443.93	51.94	safe
49	Ananthapuramu	YELLANUR	919.82	1242.88	0	1174.43	3337.13	166.87	3170.26	1950.82	18	103.81	2072.65	121.53	1310.88	65.38	safe
50	Ananthapuramu	SETTUR	1171.52	837.29	0	781.98	2790.79	139.55	2651.24	770.68	0	53.61	824.3	109.33	1771.22	31.09	safe
51	Ananthapuramu	TADPATRI	3017.43	1085.67	0	735.59	4838.69	241.95	4596.74	1849.69	510.14	214.44	2574.29	492.17	2147.36	56.00	safe

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52	Ananthapuramu	GARLADINNE	966.2	1457.72	0	1689.48	4113.4	205.64	3907.76	2473.57	0	105.72	2579.26	156.31	1310.75	66.00	safe
53	Ananthapuramu	ANANTAPUR URBAN	273.05	124.01	4.25	153.08	554.39	27.71	526.68	108.75	21.6	10.86	141.21	664.03	357.57	26.81	safe
54	Ananthapuramu	URAVAKONDA	1524.46	2316.58	31.29	1198.13	5070.46	253.52	4816.94	602.81	0	173.82	776.63	252.59	3961.53	16.12	safe
55	Ananthapuramu	PEDDAPPAPPUR	881.7	1244.73	0	1127.57	3254	162.71	3091.29	1517.31	0	134.41	1651.75	151.83	1477.18	53.43	safe
56	Ananthapuramu	ATMAKUR	962.78	981.45	11.22	885.05	2840.5	142.05	2698.45	1008.28	0	43.28	1051.56	94.47	1595.7	38.97	safe
57	Ananthapuramu	BELUGUPPA	1388.21	1202.86	9.34	1243.8	3844.21	192.18	3652.03	980.57	0	83.88	1064.45	113.79	2557.68	29.15	safe
58	Ananthapuramu	BOMMANAHAL	1091.6	3624.63	9.09	2416.14	7141.46	357.09	6784.37	972.40	12.6	95.64	1080.66	140.89	5658.45	15.93	safe
59	Ananthapuramu	BUKKARAYASAMUDRAM	1315.81	885.92	0	932.49	3134.22	156.69	2977.53	1402.20	19.8	156.29	1578.3	198.32	1360.47	53.01	safe
60	Ananthapuramu	GUMMAGATTA	871.71	837.5	76.46	845.23	2630.9	131.55	2499.35	1335.21	0	64.46	1399.7	121.23	1056.87	56.00	safe
61	Ananthapuramu	KALYANDURG	1945.22	1630.6	12.94	1603.72	5192.48	259.61	4932.87	1304.44	0	146.26	1450.71	161.12	3467.3	29.41	safe
62	Ananthapuramu	KAMBADUR	1625.15	926.4	52.26	935.15	3538.96	176.95	3362.01	981.60	0	128.91	1110.49	137.91	2242.52	33.03	safe
63	Ananthapuramu	KANEKAL	2120.09	2815.93	137.01	2629.57	7702.6	385.13	7317.47	746.17	0	130.67	876.83	157.16	6414.15	11.98	safe
64	Ananthapuramu	KUNDURPI	1170.02	753.44	19.69	827.73	2770.88	138.53	2632.35	1057.54	0	100.25	1157.79	132.66	1442.14	43.98	safe
65	Ananthapuramu	NARPALA	978.83	1048.76	0	1043.76	3071.35	153.6	2917.75	2047.96	0	90.18	2138.14	159.09	899.34	73.28	semi_critical
66	Ananthapuramu	PAMIDI	925.22	885.43	1.39	989.14	2801.18	140.07	2661.11	784.35	0	103.12	887.46	174.09	1776.92	33.35	safe
67	Ananthapuramu	PUTLUR	1342.95	985.36	0	1090.39	3418.7	170.92	3247.78	2058.78	0	67.32	2126.09	97.1	1228.71	65.46	safe
68	Ananthapuramu	RAPTADU	1222.51	959.69	3.72	1057.96	3243.88	162.19	3081.69	1063.71	0	118.43	1182.14	133.74	1884.24	38.36	safe
69	Ananthapuramu	VIDAPANAKAL	1499.15	3528.84	14.42	477.36	5519.77	275.98	5243.79	576.49	0	74.20	650.7	137.58	4529.71	12.41	safe
70	Visakhapatnam	GOPALAPATNAM	439.38	4.32	12.22	4.32	460.24	23.01	437.23	0.00	0.716	185.25	185.96	201.44	235.08	42.53	safe
71	Visakhapatnam	BHEEMUNIPATNAM	1537.54	736.97	73.61	293.76	2641.88	132.07	2509.81	670.03	1.4512	757.11	1428.62	844.32	1016.03	56.92	safe
72	Visakhapatnam	MULAGADA	697.08	9.98	25.3	6.33	738.69	36.93	701.76	0.00	0	104.48	104.48	113.62	588.14	14.89	safe
73	Visakhapatnam	PEDDAGANTYADA	81.42	8.64	2.96	8.64	101.66	5.09	96.57	0.00	0	83.54	83.54	90.84	7.65	86.51	semi_critical
74	Visakhapatnam	VISAKHAPATNAM RURAL	203.66	29.97	4.99	29.4	268.02	13.4	254.62	34.93	12.2968	161.44	208.65	175.54	41.46	81.95	semi_critical
75	Visakhapatnam	GAJUWAKA	500.67	34.56	16.11	34.56	585.9	29.3	556.6	0.00	5.4	353.85	359.22	384.8	178.04	64.54	safe
76	Visakhapatnam	ANANDAPURAM	2092.58	629.75	93.39	229.87	3045.59	152.28	2893.31	881.61	5.0129	34.66	921.3	194.84	1814.72	31.84	safe
77	Visakhapatnam	MAHARANIPETA	694.31	159.84	15.49	159.84	1029.48	51.48	978	24.80	16.6544	659.49	700.94	717.15	306.57	71.67	semi_critical

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78	Visakhapatnam	PENDURTHI	177.69	298.9	5.86	49.4	531.85	26.59	505.26	243.93	2.198	5.95	252.08	148.93	170.29	49.89	safe
79	Visakhapatnam	PADMANABHAM	1520.24	1314.77	34.28	228.29	3097.58	154.89	2942.69	1025.27	5.2536	7.78	1038.3	206.91	1705.29	35.28	safe
80	Visakhapatnam	SEETHAMMADHARA	50.56	34.56	1.13	34.56	120.81	6.05	114.76	0.00	1.5825	97.51	99.11	106.03	15.1	86.36	semi_critical
81	Prakasam	PEDACHERLOPALLE	1758.36	416.52	80.57	317.4	2572.85	128.63	2444.22	962.60	0	103.67	1066.28	108.51	1382.02	43.62	safe
82	Prakasam	BESTAVARIPETA	1038.98	1105.36	20.22	563.13	2727.69	136.37	2591.32	746.90	0	94.78	841.69	99.24	1804.88	32.48	safe
83	Prakasam	KONAKANAMITLA	2761.68	474.11	104.92	375	3715.71	185.75	3529.96	1086.70	0	138.07	1224.8	144.53	2320.23	34.70	safe
84	Prakasam	PULLALACHERUVU	934.1	644.37	48.92	377.32	2004.71	100.25	1904.46	3514.30	0	142.71	3657.06	149.39	554.92	192.03	over_exploited
85	Prakasam	CHADRASEKARAPURAM	2457.77	512.44	6.95	347.66	3324.82	166.27	3158.55	675.18	0	114.37	789.6	119.73	2363.58	25.00	safe
86	Prakasam	MARKAPUR	1020.1	892.32	17.27	387.82	2317.51	115.89	2201.62	1828.05	0	145.45	1973.53	152.28	737.9	89.64	semi_critical
87	Prakasam	KURICHEDU	667.13	500.27	26.89	344.65	1538.94	76.95	1461.99	489.40	0	98.06	587.43	102.65	915.83	40.18	safe
88	Prakasam	NAGULUPPALAPADU	2526.63	167.23	26.03	145.03	2864.92	143.27	2721.65	151.76	0	159.43	311.19	166.88	2403.03	11.43	safe
89	Prakasam	VELIGANDLA	2175.94	1021.18	34.38	648.12	3879.62	193.97	3685.65	408.83	0	75.57	484.53	79.11	3197.59	13.15	safe
90	Prakasam	KANIGIRI	3715.31	655.2	121.57	524.74	5016.82	250.85	4765.97	1261.95	0	259.18	1521.16	271.29	3232.69	31.92	safe
91	Prakasam	SINGARAYAKONDA	1944.14	72.4	0	35.4	2051.94	102.61	1949.33	66.39	32.85	170.59	269.82	178.56	1671.53	13.84	safe
92	Prakasam	ZARUGUMILLI	1824.42	127.47	0	123.17	2075.06	103.79	1971.27	348.16	0	121.99	470.17	127.71	1495.38	23.85	safe
93	Prakasam	SANTHANUTHLAPADU	1347.3	490.57	20.05	421.61	2279.53	114	2165.53	986.05	112.2375	110.14	1208.43	117.61	1046.72	55.80	safe
94	Prakasam	GIDDALURU	1561.96	545.12	0	440.26	2547.34	127.35	2419.99	1937.14	0	131.87	2069.03	138.02	894.18	85.50	semi_critical
95	Prakasam	MUNDLAMURU	1708.36	379.33	39.35	330.06	2457.1	122.85	2334.25	1315.94	2.7375	119.37	1438.04	124.96	1263.24	61.61	safe
96	Prakasam	TRIPURANTHAKAM	915.54	573.73	2.99	390.73	1882.99	94.16	1788.83	1328.51	0	136.60	1465.13	143	477.18	81.90	semi_critical
97	Prakasam	PONNALURU	2125.52	475.22	0	280.18	2880.92	144.04	2736.88	318.84	0	120.57	439.47	126.21	2291.77	16.06	safe
98	Prakasam	DORNALA	1769.28	503.07	6.46	301.37	2580.18	129	2451.18	1513.10	0	108.30	1621.4	113.38	1462.65	66.15	safe
99	Prakasam	PODILI	2051.91	797.39	61.7	485.34	3396.34	169.83	3226.51	1493.67	0	169.06	1662.65	176.97	1572.89	51.53	safe
100	Prakasam	HANUMANTHUNIPADU	1477.94	486.19	42.08	345.42	2351.63	117.6	2234.03	568.85	0	86.79	655.64	90.86	1574.32	29.35	safe
101	Prakasam	PEDAARAVEEDU	750.39	384.33	15.58	362.25	1512.55	75.65	1436.9	2583.77	0	118.46	2702.24	124.01	205.3	188.06	over_exploited
102	Prakasam	CHIMAKURTHI	1811.73	761.75	24.3	502.14	3099.92	155.03	2944.89	1505.03	232.6875	163.49	1901.2	171.14	1326.54	64.56	safe
103	Prakasam	KOTHAPATNAM	2220.6	176.01	29.4	274.74	2700.75	135.05	2565.7	770.81	8.2125	100.40	879.46	105.1	1745.01	34.28	safe

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104	Prakasam	YERRAGONDAPALEM	2748.7	691.03	75.44	573.1	4088.27	204.42	3883.85	3201.19	0	177.75	3378.92	186.05	2194.14	87.00	semi_critical
105	Prakasam	THALLUR	1449.1	707.12	22.76	456.32	2635.3	131.77	2503.53	1327.00	0	101.74	1428.75	106.51	1105.96	57.07	safe
106	Prakasam	CUMBUM	687.01	300.97	16.91	264.47	1269.36	63.51	1205.85	872.52	0	136.50	1009.01	142.89	483.27	83.68	semi_critical
107	Prakasam	TANGUTUR	1494.08	122.47	0	72.79	1689.34	84.47	1604.87	217.49	32.85	127.27	377.62	133.22	1221.3	23.53	safe
108	Prakasam	DONAKONDA	1540.8	634.73	22.66	325.21	2523.4	126.18	2397.22	1287.10	0	122.13	1409.27	127.83	1182.52	58.79	safe
109	Prakasam	KONDAPI	1525.02	360.34	2.96	281.67	2169.99	108.53	2061.46	343.65	0	120.54	464.22	126.2	1591.58	22.52	safe
110	Prakasam	RACHERLA	643.96	531.96	0.12	315.32	1491.36	74.57	1416.79	1365.39	0	73.75	1439.16	77.2	297.5	101.58	over_exploited
111	Prakasam	ARDHAVEEDU	1462.38	557.58	0	305.88	2325.84	116.31	2209.53	1219.14	0	101.14	1320.27	105.86	888.68	59.75	safe
112	Prakasam	TARLAPADU	1033.6	336.1	16.33	275.32	1661.35	83.06	1578.29	855.83	0	98.29	954.11	102.9	684.8	60.45	safe
113	Prakasam	MARRIPUDI	2871.4	1133.49	109.02	582.43	4696.34	234.8	4461.54	1368.27	21.9	97.99	1488.18	102.57	2969.43	33.36	safe
114	Prakasam	DARSI	1924.57	856.68	73.48	652.69	3507.42	175.37	3332.05	1078.18	0	189.27	1267.46	198.13	2151.04	38.04	safe
115	Prakasam	KOMAROLU	823.17	508.1	0	303.42	1634.69	81.73	1552.96	721.88	0	113.57	835.44	118.86	759.67	53.80	safe
116	Prakasam	MADDIPADU	1468.51	190.16	0	398.21	2056.88	102.86	1954.02	559.27	66.00635	105.19	730.51	110.11	1242.63	37.38	safe
117	Prakasam	ONGOLE	1094.65	122.12	3.67	80.29	1300.73	65.05	1235.68	133.91	0	3.48	137.39	3.64	1098.13	11.12	safe
118	Prakasam	PAMUR	2642.11	530.42	0	333.54	3506.07	175.29	3330.78	1057.12	0	176.64	1233.76	184.9	2089.05	37.04	safe
119	West Godavari	AKIVEEDU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
120	West Godavari	UNDI	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
121	West Godavari	GANAPAVARAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
122	West Godavari	PALACOLE	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
123	West Godavari	VEERAVASARAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
124	West Godavari	ACHANTA	1473.99	2923.69	0	3117.78	7515.46	375.8	7139.66	1800.70	0	11.13	1811.79	82.37	5256.6	25.37641848	safe
125	West Godavari	ATTILI	1446.1	3528.69	43.83	3561.21	8579.83	429.01	8150.82	192.51	0	15.33	207.83	88.66	7869.68	2.549804805	safe
126	West Godavari	BHEEMAVARAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
127	West Godavari	IRAGAVARAM	1433.96	3597.85	14.67	3721.2	8767.68	438.37	8329.31	468.45	15.12	52.65	536.21	91.5	7754.22	6.437628087	safe
128	West Godavari	KALLA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
129	West Godavari	MOGALTHUR	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity

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130	West Godavari	NARASAPURAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
131	West Godavari	PALAKODERU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
132	West Godavari	PENTAPADU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
133	West Godavari	PENUGONDA	1184.74	3051.72	0	3059.77	7296.23	364.81	6931.42	42.48	0	15.56	57.99	89.37	6799.62	0.836625 107	safe
134	West Godavari	PENUMANTRA	1375.41	3411.1	19.95	3625.38	8431.84	421.56	8010.28	10.08	0	17.77	27.78	80.83	7919.48	0.346804 356	safe
135	West Godavari	PODURU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
136	West Godavari	TADEPALLIGUDEM	3678.71	1851.19	124.47	1646.57	7300.94	365.04	6935.9	1067.08	15.84	247.05	1329.93	471.54	5381.46	19.17458 441	safe
137	West Godavari	TANUKU	1419	3185.17	32.44	3400.28	8036.89	401.83	7635.06	279.72	70.56	114.41	464.69	226.14	7058.64	6.086265 203	safe
138	West Godavari	YELAMANCHILI	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
139	Krishna	GANNAVARAM	855.2	4858.94	388.06	1568.68	7670.88	383.56	7287.32	1632.52	16.8	175.19	1824.53	186.04	5451.93	25.03705 066	safe
140	Krishna	UNGUTURU	1498.83	6629.56	738.15	2156.16	11022.7	551.15	10471.55	2937.32	104.4	243.12	3284.83	243.19	7186.66	31.36909 054	safe
141	Krishna	NAGAYALANKA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
142	Krishna	MOVVA	1407.13	6873.49	1016.45	3003.99	12301.0 6	615.05	11686.01	3111.59	103.5	52.19	3267.32	152.48	8318.39	27.95924 357	safe
143	Krishna	MOPIDEVI	848.44	5008.22	882	2233.75	8972.41	448.62	8523.79	2204.63	36	56.01	2296.61	74.5	6208.69	26.94353 099	safe
144	Krishna	MACHILIPATNAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
145	Krishna	KANKIPADU	1190.6	10434.1 7	686.74	6225.76	18537.2 7	926.88	17610.39	5591.53	84.64	55.82	5731.97	125.27	11808.99	32.54879 648	safe
146	Krishna	GUDUR (KRISHNA)	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
147	Krishna	GHANTASALA	1030.99	5530.97	741.84	2370.19	9673.99	483.68	9190.31	3036.03	46.8	16.07	3098.9	91.49	6016.02	33.71921 078	safe
148	Krishna	BANTUMILLI	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
149	Krishna	VUYYURU	1048.63	6500.9	633.27	3357.63	11540.4 3	577.03	10963.4	3654.96	271.8	19.98	3946.72	119.65	6917	35.99905 139	safe
150	Krishna	KRUTHIVENNU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
151	Krishna	BAPULAPAD	1950.56	4159.11	646	1736.92	8492.59	424.62	8067.97	2867.10	189.1	243.00	3299.2	265.67	4746.08	40.89256 653	safe
152	Krishna	NANDIVADA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
153	Krishna	PEDAPARUPUDI	1014.25	7126.64	602.73	4528.13	13271.7 5	663.57	12608.18	3285.87	22.1	57.87	3365.81	60.96	9239.28	26.69544 692	safe
154	Krishna	THOTLAVALLURU	1573.78	1895.47	790.43	1509.02	5768.7	288.43	5480.27	2009.27	25.2	163.75	2198.16	166.35	3279.52	40.11043 252	safe
155	Krishna	PENAMALURU	896.05	2790.99	462.11	2170.06	6319.21	315.96	6003.25	2203.16	20.7	510.13	2733.97	510.13	3269.28	45.54149 836	safe

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156	Krishna	CHALLAPALLE	910.09	4860.48	739.05	2965.96	9475.58	473.78	9001.8	2087.13	2.7	42.22	2132.07	107.85	6804.1	23.68492968	safe
157	Krishna	GUDLAVALLERU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
158	Krishna	PAMIDIMUKKALA	1186.91	6848.31	736.02	3717.79	12489.03	624.45	11864.58	5656.60	37.8	119.26	5813.66	128.57	6041.62	49.00013317	safe
159	Krishna	PEDANA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
160	Krishna	AVANIGADA	340.37	3170.99	340.09	1950.53	5801.98	290.12	5511.86	979.34	8.1	72.86	1060.29	83.48	4440.95	19.23651907	safe
161	Krishna	KODURU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
162	Krishna	GUDIVADA	1239.72	5171.25	675.51	2222.07	9308.55	465.44	8843.11	845.13	35.1	98.23	978.45	229.69	7733.2	11.0645463	safe
163	Krishna	PAMARRU	1171.38	8484.36	768.31	5674.37	16098.42	804.96	15293.46	1808.64	7.2	111.74	1927.59	147.47	13330.14	12.60401505	safe
164	Bapatla	VETAPALEM	889.18	239.17	89.89	174.54	1392.78	69.64	1323.14	315.67	0	164.08	479.75	169.9	837.57	36.25844582	safe
165	Bapatla	KORISAPADU	506.88	547.32	12.56	533.29	1600.05	80	1520.05	294.08	0	109.82	403.9	113.72	1112.27	26.57149436	safe
166	Bapatla	YEDDANAPUDI	310.51	64.37	7.78	59.25	441.91	22.11	419.8	52.88	0	19.64	72.5	20.34	346.61	17.27012863	safe
167	Bapatla	CHINAGANJAM	1734.29	163.59	81.86	407.67	2387.41	119.37	2268.04	159.20	0	106.67	265.87	110.45	1998.38	11.72245639	safe
168	Bapatla	VEMURU	1504.48	3357.72	47.06	1056.64	5965.9	298.29	5667.61	786.51	6.2464	99.62	892.37	103.17	4771.71	15.74508479	safe
169	Bapatla	AMRUTHALUR	590.29	1075.96	25.28	327.68	2019.21	100.96	1918.25	403.08	0	35.65	438.74	36.91	1478.24	22.87188844	safe
170	Bapatla	BALLIKURUVA	893.76	998.34	30.94	1007.31	2930.35	146.55	2783.8	638.59	0	121.98	760.56	126.31	2018.91	27.32092823	safe
171	Bapatla	JANAKAVARAMPANGULU	591.28	366.14	17.93	341.69	1317.04	65.86	1251.18	198.02	0	93.22	291.23	96.51	956.68	23.27642705	safe
172	Bapatla	BAPATLA	1884.02	1736.98	165.56	648.52	4435.08	221.75	4213.33	1747.77	6.2464	109.44	1863.47	113.32	2464.47	44.22796221	safe
173	Bapatla	BHATTIPROLU	1472.43	4375.16	57.55	822.05	6727.19	336.34	6390.85	1904.71	6.2464	115.07	2026.04	119.16	4360.72	31.70219924	safe
174	Bapatla	CHERUKUPALLE	1137.54	3424.25	71.23	439.75	5072.77	253.65	4819.12	1577.50	0	138.27	1715.8	143.18	3098.41	35.60401069	safe
175	Bapatla	KARLAPALEM	789.14	1403.29	82.98	567.19	2842.6	142.13	2700.47	719.76	12.4928	67.62	799.86	70.02	1898.21	29.61928849	safe
176	Bapatla	KOLLUR	1649.1	4076	26.55	1086.19	6837.84	341.91	6495.93	3075.66	3.1232	130.30	3209.07	134.92	3376.85	49.40124047	safe
177	Bapatla	NAGARAM	1207.34	2849.8	99.73	133.5	4290.37	214.51	4075.86	96.36	0	57.53	153.89	59.57	3919.94	3.775644895	safe
178	Bapatla	NIZAMPATNAM	525.54	385.58	48.59	35.43	995.14	49.75	945.39	114.84	0	41.90	156.74	43.39	787.16	16.57940109	safe
179	Bapatla	PITTALAVANIPALEM	824.35	1857.61	92.2	340.98	3115.14	155.75	2959.39	444.86	0	82.00	526.86	84.9	2429.63	17.80299318	safe
180	Bapatla	REPALLE	688.62	1139.86	35.97	331.35	2195.8	109.78	2086.02	130.50	6.2464	136.17	272.91	141.01	1813.64	13.08280841	safe
181	Bapatla	TSUNDUR	1301.51	3657.6	89.72	1351.55	6400.38	320.02	6080.36	907.09	0	113.66	1020.78	117.69	5055.57	16.7881507	safe

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182	Bapatla	ADDANKI	1568.89	666.4	55.45	700.92	2991.66	149.58	2842.08	605.73	2.1424	205.56	813.45	212.84	2021.36	28.6216433	safe
183	Bapatla	CHIRALA	1415.13	101.41	106.03	510.22	2132.79	106.64	2026.15	413.81	12.4928	403.92	830.2	418.25	1393.87	40.97426153	safe
184	Bapatla	INKOLLU	190.31	207.48	9.16	291.73	698.68	34.94	663.74	23.63	0	35.71	59.35	36.97	603.13	8.941754301	safe
185	Bapatla	KARAMCHEDU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
186	Bapatla	MARTUR	697.73	680.21	30.01	690.83	2098.78	104.94	1993.84	576.10	0	165.88	741.94	171.76	1246.03	37.21161176	safe
187	Bapatla	PARCHUR	146.39	16.06	5.69	39.26	207.4	10.37	197.03	0.00	0	19.33	19.34	20.02	177.01	9.815764097	safe
188	Bapatla	SANTHAMAGULURU	799.49	181.87	33.08	167.91	1182.35	59.12	1123.23	305.14	0	136.31	441.47	141.14	676.93	39.30361547	safe
189	Guntur	PEDAKAKANI	638.91	731.37	58.5	217.29	1646.07	82.3	1563.77	68.13	4.088	152.85	225.07	188.79	1302.75	14.39278155	safe
190	Guntur	TADIKONDA	125.59	606.25	30.41	479.61	1241.86	62.1	1179.76	87.61	0	59.73	147.34	61.3	1030.85	12.48898081	safe
191	Guntur	THADEPALLE	597.21	1787	159.56	588.73	3132.5	156.64	2975.86	889.61	18.25	118.19	1026.05	124.29	1943.71	34.47910856	safe
192	Guntur	KAKUMANU	12.79	122.15	2.84	10.43	148.21	7.41	140.8	0.00	0	5.56	5.56	5.71	135.09	3.948863636	safe
193	Guntur	PRATHIPADU	292.26	180.83	61.34	10.9	545.33	27.26	518.07	32.51	5.3655	43.14	81.02	48.83	431.35	15.63881329	safe
194	Guntur	MANGALAGIRI	910.5	3893.73	154.95	2773.9	7733.08	386.67	7346.41	980.82	22.63	289.30	1292.76	324.57	6018.37	17.59716651	safe
195	Guntur	PONNUR	1895.8	1265.65	322.46	1013.28	4497.19	224.86	4272.33	1184.99	0	227.87	1412.86	242.09	2845.26	33.07001098	safe
196	Guntur	THULLUR	950.76	41.29	254.45	21.15	1267.65	63.38	1204.27	32.88	1.825	142.99	177.7	154.96	1014.6	14.75582718	safe
197	Guntur	TENALI	1279.58	6963.18	318.43	2248.79	10809.98	540.48	10269.5	964.64	27.74	352.83	1345.21	387.19	8889.93	13.0990798	safe
198	Guntur	CHEBROLE	1374	1692.62	121.72	851.88	4040.22	202	3838.22	391.20	61.32	231.99	684.5	239.8	3145.9	17.83378754	safe
199	Guntur	MEDIKONDURU	218.05	597.75	33.74	10.76	860.3	43.02	817.28	18.63	6.132	56.61	81.38	90.62	701.89	9.957419734	safe
200	Guntur	VATTICHERUKURU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
201	Guntur	DUGGIRALA	1825.82	3697.09	192.1	2959.45	8674.46	433.75	8240.71	1131.93	11.49	250.55	1393.98	257.14	6840.13	16.91577546	safe
202	Guntur	KOLLIPARA	1398.44	3163.37	213.23	2275.73	7050.77	352.56	6698.21	1776.14	0	227.90	2004.03	235	4687.07	29.91888878	safe
203	Guntur	PEDANANDIPADU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
204	Guntur	PHIRANGIPURAM	310.65	626.39	66.75	41.03	1044.82	52.22	992.6	218.33	0	92.72	311.05	110.37	663.91	31.33689301	safe
205	Guntur	GUNTUR EAST	219.34	363.39	27.51	238.63	848.87	42.43	806.44	9.69	24.2725	54.73	88.68	56.25	716.23	10.99647835	safe
206	Guntur	GUNTUR WEST	365.17	527.8	46.54	20.86	960.37	48.03	912.34	40.09	35.2225	254.64	329.94	261.33	575.7	36.16414933	safe
207	Nandyal	DHONE	1423.72	671.11	37.16	732.41	2864.4	143.22	2721.18	780.72	16.5	6.77	803.97	6.75	1917.21	29.54490331	safe

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208	Nandyal	NANDYAL	665.94	2495.53	3.38	265.92	3430.77	171.58	3259.19	148.38	13.5	44.46	206.29	44.5	3052.9	6.329486774	safe
209	Nandyal	ALLAGADDA	1230.22	1832.36	0	1195.86	4258.44	212.92	4045.52	596.61	6.6	32.80	635.92	32.77	3409.6	15.71911645	safe
210	Nandyal	ATMAKUR (KURNOOL)	547.53	420.6	0	305.93	1274.06	63.7	1210.36	276.48	0.6	13.81	290.94	13.79	919.42	24.03747645	safe
211	Nandyal	BANAGANAPALLE	1282.28	1302.8	5.85	1439	4029.93	201.48	3828.45	681.17	9	35.50	725.42	35.56	3103.03	18.94813828	safe
212	Nandyal	BANDI ATMAKUR	605.91	4989.74	0	773.59	6369.24	318.47	6050.77	417.74	0	11.77	429.42	11.8	5621.35	7.096947992	safe
213	Nandyal	BETHAMCHERLA	799.49	407.85	0	476.96	1684.3	84.22	1600.08	818.44	28.2	2.92	849.57	2.9	750.51	53.09547023	safe
214	Nandyal	CHAGALAMARRI	891.83	2429.62	0	1241.68	4563.13	228.17	4334.96	717.16	0.6	14.40	732.23	14.47	3602.73	16.89127466	safe
215	Nandyal	DORNIPADU	250.93	512.14	3.89	542.74	1309.7	65.48	1244.22	30.28	0	7.92	38.2	7.92	1206.02	3.070196589	safe
216	Nandyal	GADIVEMULA	707.29	837.74	0	632.17	2177.2	108.88	2068.32	554.29	4.8	11.66	570.78	11.71	1497.54	27.59631005	safe
217	Nandyal	GOSPADU	388.43	2305.82	0	1279.1	3973.35	198.66	3774.69	91.18	0	26.72	117.83	26.75	3656.86	3.121580845	safe
218	Nandyal	JUPADU BUNGALOW	525.76	839.24	0.68	511.37	1877.05	93.83	1783.22	454.01	0	10.21	464.23	10.21	1318.99	26.03324323	safe
219	Nandyal	KOILKUNTALA	787.36	1153.07	1.46	204.17	2146.06	107.29	2038.77	126.59	2.4	23.18	152.08	23.18	1886.69	7.45939954	safe
220	Nandyal	KOLIMIGUNDLA	1097.55	200.33	0	215.59	1513.47	75.67	1437.8	313.23	6.6	10.35	330.09	10.39	1107.71	22.95799138	safe
221	Nandyal	KOTHAPALLE	1204.88	420.76	0	234.33	1859.97	93.01	1766.96	423.35	0	16.85	440.11	16.87	1326.85	24.90775117	safe
222	Nandyal	MAHANANDI	377.43	1849.94	0	266.42	2493.79	124.7	2369.09	942.92	1.5	13.22	957.64	13.22	1411.45	40.42227184	safe
223	Nandyal	MIDTHUR	549.02	316.65	0	274.09	1139.76	57.02	1082.74	251.89	1.2	19.91	273.04	19.95	809.7	25.21750374	safe
224	Nandyal	NANDI KOTKUR	566.59	196.75	0	175	938.34	46.91	891.43	333.88	1.8	15.52	351.19	15.54	540.24	39.39625097	safe
225	Nandyal	OWK	1152.44	1510.95	0.48	798	3461.87	173.09	3288.78	398.32	26.4	29.38	454.13	29.38	2834.65	13.80846393	safe
226	Nandyal	PAGIDYALA	384.88	367.11	0	205.9	957.89	47.89	910	152.53	0	12.38	164.92	12.38	745.08	18.12307692	safe
227	Nandyal	PAMULAPADU	539.45	547.55	0	282.53	1369.53	68.48	1301.05	366.16	0.6	11.48	378.23	11.5	922.82	29.07113485	safe
228	Nandyal	PANYAM	753.88	1876.89	3.53	1340.42	3974.72	198.74	3775.98	526.88	1.5	18.07	546.39	18.03	3229.59	14.4701508	safe
229	Nandyal	PEAPALLY	1299.27	750.33	9.59	770.57	2829.76	141.5	2688.26	1006.73	4.8	6.26	1017.86	6.31	1670.4	37.86315312	safe
230	Nandyal	RUDRAVARAM	1427.83	5265.26	0	2158.78	8851.87	442.61	8409.26	769.23	0	31.06	800.29	31.1	7608.97	9.516770798	safe
231	Nandyal	SANJAMALA	1084.94	1058.11	0	428.99	2572.04	128.62	2443.42	399.90	0	28.71	428.64	28.76	2014.78	17.54262468	safe
232	Nandyal	SIRVEL	824.84	856.06	0	769.97	2450.87	122.56	2328.31	490.09	1.8	18.13	510.02	18.13	1818.29	21.90515868	safe
233	Nandyal	SRISAILAM	111.49	0	0	0	111.49	5.57	105.92	0.00	0	0.00	0	0	105.92	0	safe

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234	Nandyal	UYVALAWADA	660.54	2754.65	0	1124.31	4539.5	226.98	4312.52	67.39	0	21.06	88.47	21.06	4224.05	2.051468747	safe
235	Nandyal	VELGODE	509.99	2219.69	0	468.27	3197.95	159.89	3038.06	85.34	0	10.93	96.25	10.93	2941.81	3.168140195	safe
236	Palnadu	MACHAVARAM	1002.55	3310.84	10.52	399.03	4722.94	236.14	4486.8	1461.27	0	153.28	1614.57	156.44	2869.08	35.98488901	safe
237	Palnadu	KAREMPUDI	780.15	3513.32	38.5	1302.18	5634.15	281.72	5352.43	1759.26	9.198	161.70	1930.18	165.04	3418.9	36.06175139	safe
238	Palnadu	ROMPICHERLA (GUNTUR)	957.37	227.32	19.5	106	1310.19	65.53	1244.66	147.25	5.2144	190.87	343.36	194.81	897.37	27.58665017	safe
239	Palnadu	RENTACHINTALA	582.59	5403.87	32.43	550.09	6568.98	328.47	6240.51	1361.56	4.088	153.25	1518.9	156.4	4718.47	24.33935688	safe
240	Palnadu	AMARAVATHI	779.56	1640.23	10.52	65.33	2495.64	124.79	2370.85	396.28	0	119.78	516.04	122.24	1852.34	21.76603328	safe
241	Palnadu	ACHAMPETA	1520.93	526.49	69.24	181.82	2298.48	114.92	2183.56	875.96	3.8288	158.91	1038.72	162.17	1141.58	47.57002326	safe
242	Palnadu	BELLAMKONDA	1084.12	1022.91	52.67	306.25	2465.95	123.29	2342.66	622.30	0	128.34	750.64	130.99	1589.38	32.04220843	safe
243	Palnadu	BOLLAPALLE	2176.71	635.83	224.86	254.82	3292.22	164.6	3127.62	2552.70	1.985	189.26	2743.95	193.19	1054.69	87.73284478	semi critical
244	Palnadu	CHILAKALURIPET	263.89	41.71	7.51	44.44	357.55	17.88	339.67	123.84	13	38.16	175.03	38.94	167.14	51.52942562	safe
245	Palnadu	DACHEPALLE	852.65	4326.73	14.98	248.15	5442.51	272.12	5170.39	989.76	11.91	220.35	1222.01	224.91	3943.81	23.63477417	safe
246	Palnadu	DURGI	1198.31	567.93	38.96	489.35	2294.55	114.74	2179.81	1151.67	0	181.29	1332.94	185.05	1030.45	61.14936623	safe
247	Palnadu	EDLAPADU	322.23	347.6	34.44	307.26	1011.53	50.56	960.97	99.68	0	162.60	262.28	165.99	695.3	27.29325577	safe
248	Palnadu	GURAZALA	1305.43	2012.96	51.86	891.89	4262.14	213.12	4049.02	1323.06	0	188.79	1511.85	192.69	2533.29	37.33866466	safe
249	Palnadu	IPURU	781.58	3697.24	46.61	1021.37	5546.8	277.35	5269.45	1317.35	0	140.94	1458.28	143.85	3819.09	27.67423545	safe
250	Palnadu	KROSURU	1383.72	540.56	21.59	226.14	2172.01	108.62	2063.39	828.33	0	172.26	1000.57	175.83	1059.27	48.49156001	safe
251	Palnadu	MACHERLA	1053.71	3146.01	45.04	4094.67	8339.43	416.97	7922.46	1782.72	0	167.57	1950.27	171.03	6066.5	24.61697503	safe
252	Palnadu	MUPPALLA	733.53	321.47	33.15	186.24	1274.39	63.71	1210.68	132.41	0	146.69	279.07	149.72	928.58	23.05068226	safe
253	Palnadu	NADENDLA	295.76	563.96	8.89	146.41	1015.02	50.75	964.27	124.06	0	165.08	289.18	168.5	671.65	29.98952576	safe
254	Palnadu	NAKARIKALLU	626.33	999.2	53.51	1265.74	2944.78	147.23	2797.55	630.72	0.9572	111.42	743.13	113.72	2120.43	26.56360029	safe
255	Palnadu	NARASARAOPETA	985.23	2436.82	64.26	713.46	4199.77	209.99	3989.78	294.28	11.7688	136.07	442.11	138.88	3552.66	11.08106211	safe
256	Palnadu	NUZENDLA	1295.57	5005.06	101.36	400.89	6802.88	340.15	6462.73	785.63	1.985	162.56	950.27	165.94	5509.07	14.70384806	safe
257	Palnadu	PEDAKURAPADU	615.52	4025.58	4.74	60.83	4706.67	235.36	4471.31	249.31	0	153.87	403.2	157.03	4064.96	9.017491518	safe
258	Palnadu	PIDUGURALLA	1293.13	505.76	26.42	519.13	2344.44	117.22	2227.22	1134.25	0	151.25	1285.5	154.38	938.59	57.71769291	safe
259	Palnadu	RAJUPALEM	524.55	2703.45	18.49	354.47	3600.96	180.04	3420.92	993.92	0	107.50	1101.45	109.73	2379.27	32.19747904	safe

Sl.No	District	Assessment Unit Name	Recharge from Rainfall I-MON	Recharge from Other Sources-MON	Recharge from Rainfall I-NM	Recharge from Other Sources-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
260	Palnadu	SATTENAPALLE	743.9	1989.36	32.26	162.22	2927.74	146.38	2781.36	379.20	0	189.41	568.6	193.33	2208.84	20.4432364	safe
261	Palnadu	SAVALYAPURAM	473	262.69	39.07	271.57	1046.33	52.33	994	277.70	1.985	104.20	383.91	106.36	607.94	38.62273642	safe
262	Palnadu	VELDURTHI	1401.27	212.15	57	252.39	1922.81	96.15	1826.66	2246.34	0	195.77	2442.11	199.81	96.05	133.6926412	over exploited
263	Palnadu	VINUKONDA	626.77	5392.54	56.81	4399.41	10475.53	523.8	9951.73	824.55	0	153.67	978.3	156.85	8970.25	9.83045159	safe
264	Konaseema	I POLAVARAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
265	Konaseema	KAPILESWARAPURAM	1195.54	5274.45	154.05	4906.82	11530.86	576.57	10954.29	454.21	0	69.27	523.48	71.72	10428.37	4.77876704	safe
266	Konaseema	AINAVILLI	1225.88	2879.86	179.89	2661.12	6946.75	347.34	6599.41	749.92	2.88	70.15	822.94	72.65	5773.97	12.46990261	safe
267	Konaseema	ALAMURU	987.23	3077.81	101.18	2831.37	6997.59	349.89	6647.7	1071.69	2.88	74.86	1149.44	77.51	5495.61	17.2907923	safe
268	Konaseema	ALLAVARAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
269	Konaseema	AMALAPURAM	950.87	2070.42	156.61	1927.97	5105.87	255.32	4850.55	271.74	0	142.99	414.69	148.04	4430.79	8.549339766	safe
270	Konaseema	AMBAJIPETA	812.56	978.87	96.87	907.41	2795.71	139.78	2655.93	582.77	0	77.17	659.97	79.92	1993.21	24.84892298	safe
271	Konaseema	ATREYAPURAM	1373.99	4410.34	106.21	3602.44	9492.98	474.68	9018.3	1483.60	0	58.11	1541.69	60.18	7474.55	17.09512879	safe
272	Konaseema	KATRENKONA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
273	Konaseema	KOTHAPETA	844.32	2794.03	88.07	2412.22	6138.64	306.93	5831.71	891.98	0	80.46	972.47	83.3	4856.4	16.67555485	safe
274	Konaseema	MALIKIPURAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
275	Konaseema	MAMIDIKUDURU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
276	Konaseema	MANDAPETA	1163.16	5748.51	119.18	5219.37	12250.22	612.52	11637.7	975.43	5.44	144.02	1124.9	149.15	10507.66	9.665999295	safe
277	Konaseema	MUMMIDIVARAM	1221.04	2378.59	192.77	2293.93	6086.33	304.3	5782.03	134.40	0	69.80	204.21	72.26	5575.35	3.531804574	safe
278	Konaseema	P GANNAVARAM	1020.29	2308.59	160.48	2112.05	5601.41	280.09	5321.32	265.83	0.96	77.54	344.31	80.31	4974.23	6.470387047	safe
279	Konaseema	PAMARRU (EAST GODAVARI)	1318.66	3835.16	178.84	3823.57	9156.23	457.81	8698.42	255.43	1.92	68.79	326.12	71.24	8369.85	3.749186634	safe
280	Konaseema	RAMACHANDRAPURAM	1223.69	4747.42	173.34	4653.87	10798.32	539.95	10258.37	538.16	6.62	121.03	665.84	125.34	9588.23	6.49069979	safe
281	Konaseema	RAVULAPALEM	815.25	2748.71	75.68	2318.72	5958.36	297.95	5660.41	977.20	7.7	80.55	1065.44	83.4	4592.11	18.82266479	safe
282	Konaseema	RAYAVARAM	866.71	4371.21	120	3965.67	9323.59	466.2	8857.39	578.75	7.92	69.35	656.01	71.81	8198.92	7.406357855	safe
283	Konaseema	RAZOLE	913.52	1096.16	174.51	1882.47	4066.66	203.32	3863.34	26.70	0	88.41	115.12	91.55	3745.07	2.979805039	safe
284	Konaseema	SAKHINETIPALLE	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
285	Konaseema	UPPALAGUPTAM	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity

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286	Y.S.R Kadapa	KHAJIPET	1726.12	430.09	15.21	500.33	2671.75	133.57	2538.18	990.80	28.8	483.68	1503.25	483.7	1034.93	59.22550804	safe
287	Y.S.R Kadapa	PRODDUTUR	1161.97	425.48	97.74	709.03	2394.22	119.71	2274.51	866.17	21.6	259.52	1147.29	259.53	1290.05	50.44119393	safe
288	Y.S.R Kadapa	LINGALA	1510.72	412.73	0	541.02	2464.47	123.2	2341.27	1352.51	0	183.22	1535.72	183.22	816.26	65.59345996	safe
289	Y.S.R Kadapa	CHINTHA KOMMADINNE	2363.97	370.21	0	382.04	3116.22	155.83	2960.39	584.19	19.2	257.06	860.42	257.07	2145.17	29.06441381	safe
290	Y.S.R Kadapa	MUDDANUR	1495.4	502.26	22.36	508.56	2528.58	126.43	2402.15	689.12	0.648	175.21	864.96	175.21	1537.19	36.00774306	safe
291	Y.S.R Kadapa	CHENNUR	731.88	313.75	0	154.07	1199.7	59.99	1139.71	489.86	8.64	146.27	644.76	146.28	595.52	56.57228593	safe
292	Y.S.R Kadapa	CUDDAPAH MANDAL	449.09	177.69	0	89.51	716.29	35.81	680.48	20.86	122.76	180.14	323.76	180.15	427.18	47.57818011	safe
293	Y.S.R Kadapa	RAJU PALEM	991.35	688.89	0	1685.76	3366	168.3	3197.7	311.66	3.6	121.18	436.44	121.17	2761.7	13.6485599	safe
294	Y.S.R Kadapa	KALASAPADU	4804.31	393.04	0	427.74	5625.09	281.28	5343.81	1366.10	1.8	282.66	1650.55	282.66	3717.51	30.88713858	safe
295	Y.S.R Kadapa	S MYDUKUR	2501.23	346.62	64.12	445.35	3357.32	167.88	3189.44	1572.75	3.42	573.94	2150.11	573.95	1430.36	67.41340173	safe
296	Y.S.R Kadapa	VONTIMITTA	2100.97	289.97	0	321.28	2712.22	135.62	2576.6	678.71	0	241.54	920.27	241.53	1661.62	35.71644803	safe
297	Y.S.R Kadapa	MYLAVARAM	1503.82	846.43	87.24	872.11	3309.6	165.46	3144.14	742.24	22.32	209.79	974.37	209.79	2169.77	30.99003225	safe
298	Y.S.R Kadapa	VALLUR	946.92	485.31	0	548.45	1980.68	99.03	1881.65	786.85	5.4	121.67	913.91	121.68	984.79	48.56960646	safe
299	Y.S.R Kadapa	CHAKNAYAPET	1565.25	404.4	23.44	458.25	2451.34	122.6	2328.74	1127.57	3.6	391.77	1522.96	391.75	838.3	65.39845582	safe
300	Y.S.R Kadapa	VEMULA	852.81	402.79	0	481.1	1736.7	86.83	1649.87	1237.77	0	121.77	1359.55	121.77	453.47	82.40346209	semi critical
301	Y.S.R Kadapa	VEMPALLE	909.85	352.91	0	496.58	1759.34	87.95	1671.39	789.67	13.32	174.52	977.5	174.51	693.9	58.48425562	safe
302	Y.S.R Kadapa	BRAHMAMGARIMATTAM	1842.46	1971.57	0	1105.61	4919.64	245.99	4673.65	1072.74	9.396	575.28	1657.44	575.28	3059.76	35.46350283	safe
303	Y.S.R Kadapa	ATLUR	1138.4	294.17	0	336.75	1769.32	88.45	1680.87	473.76	0	190.79	664.58	190.78	1025.52	39.53785837	safe
304	Y.S.R Kadapa	B KODUR	760.63	791.42	0	909.91	2461.96	123.1	2338.86	653.54	3.6	479.62	1136.78	479.62	1212.82	48.60402076	safe
305	Y.S.R Kadapa	BADVEL	1194.48	856.68	0	898.91	2950.07	147.49	2802.58	591.31	7.2	397.32	995.86	397.31	1823.71	35.53368682	safe
306	Y.S.R Kadapa	CHAPAD	942.63	141.91	0	234.25	1318.79	65.95	1252.84	661.59	0	337.50	999.12	337.5	329.62	79.74841161	semi critical
307	Y.S.R Kadapa	DUVVUR	1409.5	315.12	0	357.11	2081.73	104.1	1977.63	1040.92	3.6	420.25	1464.79	420.23	614.24	74.06795002	semi critical
308	Y.S.R Kadapa	GOPAVARAM	1013.82	229.97	0	353.93	1597.72	79.9	1517.82	327.98	21.6	23.26	372.8	23.26	1145.02	24.56154221	safe
309	Y.S.R Kadapa	JAMMALAMADUGU	1162.57	415.53	144.64	918.94	2641.68	132.08	2509.6	593.77	7.2	92.34	693.29	92.33	1816.31	27.62551801	safe
310	Y.S.R Kadapa	KAMALAPURAM	1463.39	641.59	0	921.17	3026.15	151.32	2874.83	1267.42	10.8	224.42	1502.62	224.42	1528.77	52.26813412	safe
311	Y.S.R Kadapa	KONDAPURAM	1546.01	799.58	73.79	725.53	3144.91	157.26	2987.65	1303.11	0	342.97	1646.07	342.99	1417.34	55.09581109	safe

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312	Y.S.R Kadapa	PEDDAMUDIUM	1234.15	738.39	7.46	2017.76	3997.76	199.89	3797.87	383.20	2.88	120.24	506.3	120.24	3297.46	13.33115667	safe
313	Y.S.R Kadapa	PENDIMARRI	4545.48	461.92	13.87	585.56	5606.83	280.35	5326.48	1121.43	0	320.70	1442.15	320.69	3891.69	27.07510401	safe
314	Y.S.R Kadapa	PORUMAMILLA	2121.21	763.39	0	917.71	3802.31	190.12	3612.19	1188.98	11.988	638.50	1839.48	638.5	1792.18	50.92423156	safe
315	Y.S.R Kadapa	PULIVENDLA	1054.95	165.73	0	309.31	1529.99	76.52	1453.47	982.41	24.192	182.39	1189.03	182.38	542.97	81.80629803	semi_critical
316	Y.S.R Kadapa	SIDHOUT	2157.25	258.71	0	376.86	2792.82	139.64	2653.18	1160.36	1.944	214.76	1377.1	214.76	1343.94	51.90375323	safe
317	Y.S.R Kadapa	SIMHADRIPURAM	1892.91	1233.54	0	3998.12	7124.57	356.26	6768.31	1795.95	2.52	172.26	1970.78	172.27	4801.16	29.11775613	safe
318	Y.S.R Kadapa	SRI AVADHUTHA KASINAYANA	1895.17	657.94	0	738.48	3291.59	164.62	3126.97	1165.45	0	344.23	1509.7	344.23	1630.87	48.27996431	safe
319	Y.S.R Kadapa	THANDUR	1063.57	1483.47	0	3747.1	6294.14	314.71	5979.43	1068.18	3.672	95.42	1167.27	95.43	4826.51	19.52142596	safe
320	Y.S.R Kadapa	VEERAPUNAYUNIPALLE	1777.65	473.11	0	543.08	2793.84	139.71	2654.13	781.20	0.432	36.62	818.27	36.62	1882	30.83006484	safe
321	Y.S.R Kadapa	YERRAGUNTALA	1272.65	580.91	0	856.53	2710.09	135.53	2574.56	476.38	64.296	162.34	703.01	162.35	1954.6	27.30602511	safe
322	Sri Potti Sriramulu Nellore	ALLUR	1795.53	2825.65	507.64	6103.64	11232.46	561.64	10670.82	392.93	5.8	71.18	469.9	107.6	10164.49	4.403597849	safe
323	Sri Potti Sriramulu Nellore	ANANTHASAGARAM	1246.87	1425.57	349.53	2892.97	5914.94	295.75	5619.19	1412.32	22.15	76.42	1510.87	108.63	4076.1	26.8876831	safe
324	Sri Potti Sriramulu Nellore	ANUMASAMUDRAMPETA	1249.59	710.85	399.94	1081.32	3441.7	172.09	3269.61	1081.53	11.1	78.37	1170.97	87.53	2089.5	35.81375149	safe
325	Sri Potti Sriramulu Nellore	ATMAKUR (NELLORE)	1224.03	1267.52	422.56	2689.78	5603.89	280.21	5323.67	779.55	20.6	67.72	867.88	84.44	4439.06	16.30228771	safe
326	Sri Potti Sriramulu Nellore	BOGOLE	1617.65	528.24	460.28	2518.72	5124.89	256.23	4868.66	573.02	0	73.21	646.23	102.97	4192.66	13.27326205	safe
327	Sri Potti Sriramulu Nellore	BUCCHIREDDIPALEM	966.68	2306.43	266.37	2654.67	6194.15	309.71	5884.44	1830.36	0	56.55	1886.93	172.34	3982.18	32.06643283	safe
328	Sri Potti Sriramulu Nellore	CHEJERLA	1228.44	805.3	401.96	1832.34	4268.04	213.43	4054.61	520.81	5.55	67.50	593.83	79.04	3463.21	14.64579824	safe
329	Sri Potti Sriramulu Nellore	DAGADARTHI	1388.37	1780.78	398.92	3457.06	7025.13	351.25	6673.88	760.38	33.3	91.15	884.85	95.57	5784.62	13.25840441	safe
330	Sri Potti Sriramulu Nellore	DUTTALUR	1652.94	352.13	588.99	346.56	2940.62	147.06	2793.56	754.58	12.95	56.30	823.82	71.99	1954.04	29.48996979	safe
331	Sri Potti Sriramulu Nellore	INDUKURPET	1943.02	1098.9	490.73	1933.51	5466.16	273.31	5192.85	3596.30	26.1	55.17	3677.57	127.85	1446.35	70.81987733	semi_critical
332	Sri Potti Sriramulu Nellore	JALADANKI	2078.71	813.92	655.61	2866.1	6414.34	320.72	6093.63	798.11	12.95	58.04	869.1	107.73	5174.84	14.26243471	safe
333	Sri Potti Sriramulu Nellore	KALIGIRI	2110.56	647.04	701.41	942.8	4401.81	220.09	4181.72	1190.77	33.3	80.36	1304.48	116.41	2851.9	31.19481936	safe
334	Sri Potti Sriramulu Nellore	KALUVOYA	2580.48	741.32	810.36	1573.94	5706.1	285.3	5420.8	668.14	1.85	99.46	769.46	108.99	4641.84	14.19458383	safe
335	Sri Potti Sriramulu Nellore	KAVALI	2872.16	444.36	841.05	1584.43	5742	287.09	5454.91	714.91	51.2	441.31	1207.44	514.08	4389.42	22.13492065	safe
336	Sri Potti Sriramulu Nellore	KODAVALUR	1065.76	2433.94	193.24	3753.58	7446.52	372.3	7074.22	1480.85	31.9	63.57	1576.34	115.66	5445.81	22.28288066	safe
337	Sri Potti Sriramulu Nellore	KONDAPURAM (NELLORE)	2442.65	477.22	577.66	738.75	4236.28	211.81	4024.47	919.68	9.25	101.35	1030.26	121.26	2974.31	25.59989266	safe

Sl.No	District	Assessment Unit Name	Recharge from Rainfall I-MON	Recharge from Other Sources-MON	Recharge from Rainfall I-NM	Recharge from Other Sources-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
338	Sri Potti Sriramulu Nellore	KOVUR	1010.42	1472.94	255.29	2171.57	4910.22	245.5	4664.72	2213.63	29	47.12	2289.76	165.32	2256.77	49.08676191	safe
339	Sri Potti Sriramulu Nellore	MANUBOLU	1460.29	942.88	338.45	3540.96	6282.58	314.13	5968.45	359.09	18.5	79.48	457.04	86.76	5504.13	7.657599544	safe
340	Sri Potti Sriramulu Nellore	MARRIPADU	2571.83	505.73	720.69	697.29	4495.54	224.79	4270.75	1249.47	24.05	88.43	1361.97	111.47	2885.75	31.89065152	safe
341	Sri Potti Sriramulu Nellore	MUTHUKUR	1725.1	777.83	670.89	2666	5839.82	292.01	5547.81	1771.87	179.8	71.52	2023.23	129.42	3472.59	36.46898506	safe
342	Sri Potti Sriramulu Nellore	NELLORE URBAN	3109.16	1442.97	762.06	3350.6	8664.79	433.27	8231.52	281.47	187.9	2361.25	2830.69	2486.47	7154.48	34.38842401	safe
343	Sri Potti Sriramulu Nellore	PODLAKUR	2739.42	1605.07	924.63	2735.39	8004.51	400.25	7604.25	1278.67	42.55	138.48	1459.75	171.64	6111.37	19.19650196	safe
344	Sri Potti Sriramulu Nellore	RAPUR	2971.68	1019.21	898.04	970.09	5859.02	293	5566.02	523.49	20.35	151.39	695.25	180.74	4841.45	12.49097201	safe
345	Sri Potti Sriramulu Nellore	SANGAM	874.14	1544.2	297.69	2212.75	4928.78	246.48	4682.3	846.30	22.15	98.34	966.76	102.32	3714.57	20.64711787	safe
346	Sri Potti Sriramulu Nellore	SEETHARAMAPURAM	771.21	230.97	413.48	175.29	1590.95	79.55	1511.4	401.40	1.85	82.33	485.59	88.29	1019.87	32.12849014	safe
347	Sri Potti Sriramulu Nellore	SYDAPURAM	2015.58	607.33	605.97	734.4	3963.28	198.19	3765.09	903.08	37	133.77	1073.93	155.71	2672.29	28.52335535	safe
348	Sri Potti Sriramulu Nellore	THOTAPALLIGUDUR	1746.68	1191.4	520.87	2642.18	6101.13	305.05	5796.08	1744.78	5.8	79.91	1830.49	115.62	3934.86	31.58151716	safe
349	Sri Potti Sriramulu Nellore	UDAYAGIRI	2829.48	461.41	1047.58	488.68	4827.15	241.36	4585.79	1162.25	16.65	67.75	1246.67	108.44	3298.46	27.1855013	safe
350	Sri Potti Sriramulu Nellore	VARIKUNTAPADU	1532.95	297.89	596.5	577.07	3004.41	150.25	2854.16	851.14	7.4	109.23	967.8	119.33	1877.28	33.90840037	safe
351	Sri Potti Sriramulu Nellore	VENKATACHALAM	2342.67	1489.26	593.12	4033.25	8458.3	422.96	8035.34	489.00	16.35	127.51	632.82	132.94	7397.1	7.875460155	safe
352	Sri Potti Sriramulu Nellore	VIDAVALUR	2171.33	2481.64	451.81	4024.86	9129.64	456.49	8673.15	1954.98	0	57.37	2012.35	101.96	6616.2	23.20206615	safe
353	Sri Potti Sriramulu Nellore	VINJAMUR	1484.06	444.21	489.51	549.29	2967.07	148.34	2818.73	1058.91	33.3	59.84	1152.04	97.98	1628.54	40.87088866	safe
354	Sri Potti Sriramulu Nellore	GUDLURU	2436.19	314.4	686.39	582.83	4019.81	200.98	3818.83	553.63	0	60.37	613.95	104.31	3160.94	16.07691361	safe
355	Sri Potti Sriramulu Nellore	KANDUKUR	1714.61	216.66	600.24	501.38	3032.89	151.68	2881.21	501.78	0	325.22	826.98	343.26	2036.18	28.70252429	safe
356	Sri Potti Sriramulu Nellore	LINGASAMUDRAM	1384.86	207.05	412.18	380.55	2384.64	119.27	2265.37	260.63	0	34.95	295.56	47.51	1957.27	13.04687534	safe
357	Sri Potti Sriramulu Nellore	ULAVAPADU	2458.23	726.53	660.79	503.38	4348.93	217.44	4131.49	843.48	66.6	34.08	944.16	113.7	3107.72	22.85277224	safe
358	Sri Potti Sriramulu Nellore	VOLETIVARIPALEM	1532.06	352.53	498.26	619.25	3002.1	150.11	2851.99	387.65	0	44.43	432.05	88.09	2376.3	15.14907135	safe
359	Vizianagaram	GAJAPATHINAGARAM	747.65	3160.17	78.52	1365.76	5352.1	267.59	5084.51	1269.53	0	78.40	1347.95	78.4	3767.5	26.51091256	safe
360	Vizianagaram	PUSAPATIREGA	1004.49	3476.8	44.38	2729.68	7255.35	362.76	6892.59	3697.77	29.26	44.50	3771.53	44.5	3501.38	54.71861811	safe
361	Vizianagaram	GARIVIDI	1217.27	1917.79	55.75	1592.78	4783.59	239.14	4544.44	1856.53	0	47.80	1904.32	47.8	2676.74	41.90439306	safe
362	Vizianagaram	VIZIANAGARAM MANDAL	384.79	1995.16	27.43	1838.07	4245.45	212.29	4033.17	429.27	1.4	62.00	492.67	62	3540.5	12.21545335	safe
363	Vizianagaram	MENTADA	1186.74	3988.49	103.4	3423.36	8701.99	435.17	8266.81	788.26	5.76	60.60	854.61	60.6	7412.2	10.33784495	safe

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364	Vizianagaram	BOBBILI	1271.95	5600.41	213.48	2052.35	9138.19	456.98	8681.21	1263.57	11.88	86.40	1361.83	86.4	7322.48	15.68709892	safe
365	Vizianagaram	RAJAM	423.65	2293.85	45.27	981.49	3744.26	187.24	3557.01	939.38	0	65.80	1005.19	65.8	2606.7	28.25940889	safe
366	Vizianagaram	REGIDIAMADALAVALASA	1052.63	2717.73	101.91	969.85	4842.12	242.12	4600	923.64	0	77.80	1001.48	77.8	3623.07	21.77130435	safe
367	Vizianagaram	SANTHAKAVITI	928.53	1520.7	101.64	481.37	3032.24	151.63	2880.61	720.60	0	66.00	786.59	66	2141.21	27.30636914	safe
368	Vizianagaram	VANGARA	738.91	770.65	74.85	480.93	2065.34	103.29	1962.05	504.56	0	56.80	561.35	56.8	1400.7	28.610382	safe
369	Vizianagaram	BADANGI	517.48	2996.16	80.57	2176.52	5770.73	288.56	5482.17	1227.99	9.3	36.50	1273.78	36.5	4210.49	23.23495988	safe
370	Vizianagaram	BHOGHAPURAM	821.59	710.24	44.74	464.89	2041.46	102.07	1939.39	703.80	0.8	58.80	763.39	58.8	1244.43	39.36237683	safe
371	Vizianagaram	BONDAPALLE	858.24	3014.6	101.6	1846.91	5821.35	291.07	5530.27	701.94	0	79.40	781.34	79.4	4748.93	14.12842411	safe
372	Vizianagaram	CHEEPURUPALLE	636.56	2089.21	36.6	1827.4	4589.77	229.49	4360.28	2841.69	0.8	41.70	2884.18	41.7	1821.16	66.14666948	safe
373	Vizianagaram	DATTIRAJERU	1249.42	2411.5	123.71	1583.49	5368.12	268.38	5099.74	1796.63	9	41.50	1847.12	41.5	3278	36.21988572	safe
374	Vizianagaram	DENKADA	476.72	2352.23	37.61	1280.86	4147.42	207.38	3940.04	1316.23	9.36	67.80	1393.4	67.8	2595.46	35.3651232	safe
375	Vizianagaram	GANTYADA	1852.98	1998.58	139.93	1733.35	5724.84	286.23	5438.61	1102.58	0	111.60	1214.18	111.6	4231.7	22.32518971	safe
376	Vizianagaram	GURLA	1322.21	1335.61	80.79	1389.86	4128.47	206.47	3922.01	1677.79	0	44.60	1722.39	44.6	2366.15	43.91600225	safe
377	Vizianagaram	JAMI	900.64	1186.37	129.68	924.8	3141.49	157.03	2984.46	1187.91	5.6	72.80	1266.31	72.8	2076.72	42.43012136	safe
378	Vizianagaram	KOTHAVALASA	988.69	1964.36	126.08	2064.93	5144.06	257.21	4886.85	892.80	5.5	81.80	980.1	81.8	3973.97	20.05586421	safe
379	Vizianagaram	LAKKAVARAPUKOTA	928.93	2313.95	87.38	2361.85	5692.11	284.67	5407.44	667.95	0	103.60	771.55	103.6	4635.89	14.26830441	safe
380	Vizianagaram	MERAKAMUDIDAM	897.61	3591.38	89.06	3275.26	7853.31	392.71	7460.59	1361.83	15.84	69.60	1447.25	69.6	6059.08	19.39859984	safe
381	Vizianagaram	NELLIMARLA	801.07	2687.64	59.29	2197.72	5745.72	287.33	5458.38	1866.25	0	34.40	1900.65	34.4	3813.02	34.820771	safe
382	Vizianagaram	RAMABHADRAPURAM	1065.55	1920.15	131.81	1799.53	4917.04	245.83	4671.22	1240.10	0	23.60	1263.7	23.6	3573.19	27.05288982	safe
383	Vizianagaram	SRUNGAVARAPUKOTA	821.64	2927.55	111.5	2152.16	6012.85	300.6	5712.25	1158.67	2.88	113.40	1274.95	113.4	4461.45	22.31957635	safe
384	Vizianagaram	THERLAM	923.34	2982.21	149.52	2631.52	6686.59	334.36	6352.22	1113.32	0	80.60	1193.92	80.6	5187.66	18.7953188	safe
385	Vizianagaram	VEPADA	977.8	2621.9	161.83	3036.05	6797.58	339.83	6457.75	638.70	2.88	101.00	742.58	101	5722.96	11.49905153	safe
386	Kurnool	YEMMIGANUR	2163.33	738.32	26.98	489.99	3418.62	170.89	3247.73	939.20	2.1	470.80	1412.1	482.65	2044.38	43.4795996	safe
387	Kurnool	VELDURTHI (KURNOOL)	998.52	460.3	0	432.22	1891.04	94.51	1796.53	422.92	2.1	199.22	624.27	204.24	1167.22	34.74865435	safe
388	Kurnool	ORVAKAL	1035.83	220.23	30.85	232.12	1519.03	75.93	1443.1	795.20	9.13	208.03	1012.33	213.28	426.63	70.14967778	semi_critical
389	Kurnool	PEDDA KADUBUR	1125.35	1818.28	39.03	1504.02	4486.68	224.33	4262.35	816.21	0	201.66	1017.9	206.74	3248.24	23.8811923	safe

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390	Kurnool	KURNOOL RURAL	1212.36	510.51	0	218.77	1941.64	97.1	1844.54	427.10	10.5	205.12	642.71	210.28	1196.68	34.84391772	safe
391	Kurnool	MANTRALAYAM	1261.24	544.82	0	175.11	1981.17	99.07	1882.1	377.17	0	174.52	551.71	178.92	1325.96	29.31353276	safe
392	Kurnool	ADONI	2835.28	1580.36	91.48	822.06	5329.18	266.46	5062.72	752.20	6.3	578.25	1336.76	592.86	3795.95	26.40398837	safe
393	Kurnool	DEVANAKONDA	2331.9	316.25	43.83	327.13	3019.11	150.96	2868.15	750.06	0	253.60	1003.66	259.99	1858.09	34.99328836	safe
394	Kurnool	ALUR	2200.82	117.57	85.41	110.13	2513.93	125.73	2388.2	292.90	2.1	197.46	492.45	202.45	1890.75	20.62013232	safe
395	Kurnool	HALAHARVI	902.3	1210.71	16.75	204.31	2334.07	116.7	2217.37	242.78	0	146.39	389.19	150.11	1824.44	17.55187452	safe
396	Kurnool	HOLAGUNDA	1351.26	1863.63	39.94	930.45	4185.28	209.28	3976	404.73	0	153.81	558.56	157.69	3413.58	14.04828974	safe
397	Kurnool	C BELAGAL	1264.89	1061.72	7.59	738.11	3072.31	153.62	2918.69	531.02	2.1	163.28	696.39	167.4	2218.18	23.85967677	safe
398	Kurnool	KOSIGI	1158.77	986.77	0	554.47	2700.01	135.05	2564.96	989.35	0	240.28	1229.62	246.35	1331.26	47.93914915	safe
399	Kurnool	MADDIKERA EAST	1305.48	114.38	2.51	98.35	1520.72	76.04	1444.68	218.08	0	131.13	349.25	134.42	1092.14	24.17490378	safe
400	Kurnool	GUDUR (KURNOOL)	688.9	140.3	0	110.22	939.42	46.98	892.44	281.19	2.1	132.99	416.28	136.34	472.83	46.64515262	safe
401	Kurnool	TUGGALI	1767.65	186.21	0	182.61	2136.47	106.82	2029.65	338.51	6.45	211.68	556.63	217.02	1467.67	27.42492548	safe
402	Kurnool	ASPARI	1949.52	180.56	48.99	156.94	2336.01	116.81	2219.2	497.26	0.365	231.71	729.33	237.56	1484	32.86454578	safe
403	Kurnool	PATTIKANDA	1464.35	109.93	24.31	111.35	1709.94	85.5	1624.44	366.76	2.1	190.79	559.64	195.59	1060.02	34.45125705	safe
404	Kurnool	KRISHNAGIRI	1920.2	521.99	0	297.32	2739.51	136.99	2602.52	784.74	0	165.72	950.47	169.91	1647.86	36.52114105	safe
405	Kurnool	CHIPPAGIRI	1009.02	756.13	60.91	162.49	1988.55	99.43	1889.12	187.91	0	116.69	304.62	119.63	1581.54	16.12496824	safe
406	Kurnool	KOWTHALAM	1660.7	2271.51	0	1301.72	5233.93	261.7	4972.23	727.95	0	263.40	991.35	270.03	3974.24	19.93773418	safe
407	Kurnool	KURNOOL URBAN	57.63	16.66	0	0.25	74.54	3.73	70.81	14.31	1.05	0.00	15.36	0	55.45	21.69185143	safe
408	Kurnool	KALLUR	1926.85	154.37	1.14	182.94	2265.3	113.28	2152.02	617.26	21	168.25	806.54	172.48	1341.23	37.47827622	safe
409	Kurnool	KODUMUR	1631.52	219.14	33.37	287.47	2171.5	108.59	2062.91	589.37	2.83	265.55	857.74	272.24	1198.47	41.57912851	safe
410	Kurnool	NANDAVARAM	1733.27	953.7	13.08	370.57	3070.62	153.54	2917.08	457.83	1.05	216.28	675.17	221.72	2236.46	23.14540568	safe
411	Kurnool	GONEGANDLA	1386	584.35	0	905.95	2876.3	143.83	2732.47	1171.36	0	245.88	1417.25	252.07	1309.04	51.86699214	safe
412	Kakinada	YELESWARAM	886.24	1469.52	175.88	1154.13	3685.77	184.3	3501.47	1308.34	23.9976	93.07	1425.41	96.37	2072.76	40.70890226	safe
413	Kakinada	GOLLAPROLU	721.27	2886.43	140.97	1873.77	5622.44	281.13	5341.31	425.63	120.0688	79.93	625.6	82.77	4712.87	11.71248252	safe
414	Kakinada	ROUTHULAPUDI	1307.81	1937.29	259.86	1596.45	5101.41	255.1	4846.31	907.66	35.56	92.44	1035.69	95.72	3807.44	21.37069234	safe
415	Kakinada	KAKINADA (RURAL)	557.75	1950.27	141.07	1666.75	4315.84	215.78	4100.04	85.30	94.2544	128.38	307.95	132.93	3787.53	7.510902333	safe

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416	Kakinada	KIRLAMPUDI	525.18	3616.14	128.93	2992.02	7262.27	363.15	6899.12	936.45	92.7404	97.31	1126.49	100.77	5769.17	16.32802444	safe
417	Kakinada	KOTANANDURU	420.57	3430.67	100.06	512.83	4464.13	223.21	4240.92	450.80	104.4932	40.51	595.79	41.95	3643.66	14.04860266	safe
418	Kakinada	GANDEPALLE	1474.84	1176.61	189.73	930.69	3771.87	188.6	3583.27	1827.06	60.6652	57.97	1945.69	60.02	1635.53	54.29928529	safe
419	Kakinada	JAGGAMPETA	1185.66	3159.31	220.93	1253.89	5819.79	290.99	5528.8	808.12	71.7454	89.89	969.79	93.07	4555.83	17.54069599	safe
420	Kakinada	KAJULURU	20.53	286.73	4.87	4.24	316.37	15.82	300.55	0.00	0	0.00	0	0	300.55	0	safe
421	Kakinada	KAKINADA (URBAN)	115.23	51.48	29.92	45.93	242.56	12.13	431.52	0.00	12.886	251.63	264.51	259.2	181.52	61.29727475	safe
422	Kakinada	KARAPA	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
423	Kakinada	KOTHAPALLEE	1062.67	2797.67	223.76	2748.71	6832.81	341.65	6491.16	229.85	165.8404	88.75	484.41	91.88	6003.63	7.462610689	safe
424	Kakinada	PEDAPUDI	894.19	6452.07	201.25	5814.71	13362.22	668.1	12694.12	566.88	379.9436	80.07	1026.84	82.91	11664.43	8.08909952	safe
425	Kakinada	PEDDAPURAM	1616.76	2569.74	344.71	2173.25	6704.46	335.22	6369.22	1116.55	73.446	133.52	1323.5	137.96	5041.29	20.77962451	safe
426	Kakinada	PITHAPURAM	808.54	5941.59	179.17	768.22	7697.52	384.88	7312.64	713.46	49.741	143.84	907.09	148.96	6400.43	12.40441209	safe
427	Kakinada	PRATHIPADU (EAST GODAVARI)	785.77	3742.55	180.1	1130.43	5838.85	291.97	5546.88	869.40	54.3108	65.34	989.01	67.67	4555.55	17.83002336	safe
428	Kakinada	SAMALKOTA	1619.05	5495.88	324.28	4968.45	12407.66	620.4	11787.26	461.04	249.307	151.74	862.09	157.14	10919.78	7.313743822	safe
429	Kakinada	SANKHAVARAM	643.57	1704.43	142.28	710.79	3201.07	160.07	3041	603.01	47.5632	63.67	714.22	65.95	2324.48	23.48635317	safe
430	Kakinada	THALLAREVU	0	0	0	0	0	0	0	0.00	0	0.00	0	0	0		salinity
431	Kakinada	THONDANGI	1054.79	3483.66	325.5	1903.72	6767.67	338.41	6429.26	1029.32	157.032	98.20	1284.54	101.68	5141.23	19.9795933	safe
432	Kakinada	TUNI	964.19	7501.75	173.61	539.48	9179.03	458.92	8720.11	1001.56	85.174	120.72	1207.4	124.99	7508.42	13.84615561	safe
433	Chittoor	YADAMARI	1372.88	211.96	35.57	286.36	1906.77	95.36	1811.41	854.18	11.52	86.79	952.5	163.8	815.7	52.58334667	safe
434	Chittoor	NINDRA	1126.21	596.02	0	870.29	2592.52	129.64	2462.88	2072.82	4.44	97.40	2174.64	103.02	282.63	88.29662834	semi_critical
435	Chittoor	PENUMURU	1069.5	439.17	22.71	445.56	1976.94	98.84	1878.1	1132.98	52.56	167.39	1352.95	170.68	521.88	72.03823013	semi_critical
436	Chittoor	SODAM	2031.28	416.4	0	573.94	3021.62	151.08	2870.54	1319.45	2.88	77.68	1400.03	99.23	1448.97	48.77235642	safe
437	Chittoor	KUPPAM	1973.77	864.05	118.84	890.47	3847.13	192.34	3654.79	1883.30	49.83	300.23	2233.43	397.07	1442.8	61.10966704	safe
438	Chittoor	SRIRANGARAJAPURAM	1305.07	579.53	0	773.28	2657.88	132.93	2524.95	1540.21	18.52	108.73	1667.44	145.21	826.8	66.03853542	safe
439	Chittoor	PALAMANER	1400.07	217.14	81.04	280.32	1978.57	98.92	1879.65	608.19	79.2	78.60	766.02	111.95	1108.85	40.75333174	safe
440	Chittoor	BAIREDDI PALLE	1820.95	459.34	108.72	568.82	2957.83	147.92	2809.91	1608.94	8.64	124.59	1742.15	178.09	1121.95	62.00020641	safe
441	Chittoor	PEDDA PANJANI	1918.35	809.79	59.89	1019.78	3807.81	190.38	3617.43	2546.49	3.6	166.12	2716.2	201.22	869.3	75.08645641	semi_critical

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442	Chittoor	VENKATAGIRI KOTA	1950.08	762.69	95.57	888.44	3696.78	184.86	3511.92	2401.41	20.88	297.64	2719.96	365.03	815.53	77.44937242	semi_critical
443	Chittoor	GUDI PALLE	1245.9	915.75	52.69	958.58	3172.92	158.65	3014.27	2222.22	15.54	103.90	2341.64	150.73	655.13	77.68514433	semi_critical
444	Chittoor	GANGAVARAM	2062.21	399.02	58.68	509.62	3029.53	151.47	2878.06	1459.17	18.045	100.78	1577.98	158.72	1242.46	54.82790491	safe
445	Chittoor	BANGARUPALEM	4300.95	304.4	74.32	431.48	5111.15	255.57	4855.58	1503.54	73.38	136.36	1713.28	231.62	3047.04	35.28476516	safe
446	Chittoor	VEDURU KUPPAM	2502.77	509.05	0	701.33	3713.15	185.68	3527.47	1576.12	16.56	101.46	1694.14	154.16	1782.52	48.02705622	safe
447	Chittoor	CHITTOOR MANDAL	1754.53	302.49	72.46	419.65	2549.13	127.44	2421.69	1167.42	268.99	319.76	1756.2	413.4	806.76	72.51960408	semi_critical
448	Chittoor	PALASAMUDRAM	829.91	285.53	0	361.54	1476.98	73.85	1403.13	701.71	2.88	60.25	764.82	68.2	630.36	54.50813538	safe
449	Chittoor	VIJAYA PURAM	2239.18	295.54	0	472.25	3006.97	150.35	2856.62	983.28	5.76	58.01	1047.03	85.93	1781.66	36.65275745	safe
450	Chittoor	CHOWDEPALLE	1497.57	889.82	59.84	823.81	3271.04	163.54	3107.5	2312.27	24.6	161.06	2497.94	169.06	822.21	80.3842317	semi_critical
451	Chittoor	PUTHALAPATTU	1590.71	489.94	0	669.33	2749.98	137.52	2612.46	2051.74	33.82	148.26	2233.82	165.96	371.17	85.50638096	semi_critical
452	Chittoor	NAGARI	1560.9	415.54	0	545.11	2521.55	126.12	2395.43	1006.19	444.984	118.35	1569.5	171.57	789.12	65.52059547	safe
453	Chittoor	GANGADHARA NELLORE	1845.59	428.18	32.84	594.43	2901.04	145.11	2755.93	1327.10	30.24	189.11	1546.44	232.23	1166.36	56.11318139	safe
454	Chittoor	RAMA KUPPAM	1398.07	580.02	69.5	686.48	2734.07	136.73	2597.34	1724.58	7.2	127.31	1859.05	181.19	713.12	71.57514996	semi_critical
455	Chittoor	SOMALA	3450.4	363.51	7.51	504.2	4325.62	216.32	4109.3	1168.63	16.56	84.15	1269.35	125.56	2798.53	30.88968924	safe
456	Chittoor	SANTHI PURAM	924.41	722.54	52.03	828.22	2527.2	126.31	2400.89	1461.29	18	142.25	1621.56	204.03	760.96	67.53995393	safe
457	Chittoor	IRALA	1817.74	248.89	4.26	341.63	2412.52	120.65	2291.87	637.52	14.4	152.48	804.38	177.47	1462.47	35.09710411	safe
458	Chittoor	PUNGANUR	1734	701.38	137.54	833.93	3406.85	170.36	3236.49	1639.16	57.56	212.06	1908.83	289.38	1293.46	58.97839944	safe
459	Chittoor	THAVANAMPALLE	1604.2	306.08	28.32	474.57	2413.17	120.7	2292.47	1578.36	20.52	107.73	1706.58	157.15	536.48	74.44284985	semi_critical
460	Chittoor	GUDIPALA	1466.42	275.88	46.96	336.53	2125.79	106.3	2019.49	833.94	20.9304	119.41	974.25	167.98	1014.99	48.24237803	safe
461	Chittoor	KARVETINAGAR	2576.09	530.71	0	739.12	3845.92	192.29	3653.63	1534.83	10.08	159.09	1704	180.31	1928.4	46.63854851	safe
462	Chittoor	PULICHERLA	1456.15	474.21	27.7	605.6	2563.66	128.18	2435.48	1548.92	10.08	121.72	1680.72	135.09	759.47	69.00980505	safe
463	Chittoor	ROMPICHERLA	1372.26	336.14	0	414.98	2123.38	106.18	2017.2	848.01	4.32	54.69	907.02	83.43	1081.45	44.96430696	safe
464	Annamayya	OBULAVARIPALLE	712.65	551.21	47.72	552.41	1863.99	93.2	1770.79	869.66	21.5	134.35	1025.53	162.14	735.98	57.91369954	safe
465	Annamayya	RAJAMPET	1042.96	479.41	33.3	477.18	2032.85	101.64	1931.2	1055.98	16.314	188.87	1261.15	305.68	668.05	65.30395609	safe
466	Annamayya	GALIVEEDU	2489.29	389.06	128.87	476.51	3483.73	174.19	3309.54	511.24	0	233.47	744.69	237.48	2560.83	22.50131438	safe
467	Annamayya	PEDDAMANDYAM	1384.36	549.29	142.59	541.37	2617.61	130.87	2486.74	793.23	0	34.18	827.4	126.11	1577.38	33.27247722	safe

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468	Annamayya	RAMASAMUDRAM	1023.94	1095.92	92.59	1126.77	3339.22	166.96	3172.26	1749.03	0	68.11	1817.12	134.06	1289.19	57.28155952	safe
469	Annamayya	GURRAMKONDA	1725.92	545.98	192.25	719.96	3184.11	159.2	3024.91	934.07	2.188	82.63	1018.87	141.88	1946.79	33.68265502	safe
470	Annamayya	THAMBALLAPALLE	1856.61	717.69	100.95	709.11	3384.36	169.23	3215.13	1161.27	0	226.30	1387.55	257.38	1796.5	43.15688635	safe
471	Annamayya	PULLAMPETA	560.17	362.01	35.92	374.58	1332.68	66.63	1266.05	606.33	0	216.40	822.72	223.76	521.01	64.98321551	safe
472	Annamayya	PEDDATHIPPASAMUDRA	2084.25	909.75	164.06	935.95	4094.01	204.69	3889.32	2091.64	0	20.90	2112.55	154.82	1657.35	54.31669289	safe
473	Annamayya	CHITVEL	1043.99	236.45	52.79	274.74	1607.97	80.38	1527.59	977.37	0	160.48	1137.86	174.04	381.22	74.48726425	semi_critical
474	Annamayya	LAKKIREDDIPALLE	1539.25	472.21	30.53	520.67	2562.66	128.12	2434.54	987.32	0	131.81	1119.14	131.81	1315.4	45.96925908	safe
475	Annamayya	KALIKIRI	1480.85	449.11	102.81	508.1	2540.87	127.06	2413.81	1234.03	0	58.50	1292.53	140.12	1039.65	53.5472966	safe
476	Annamayya	KURABALAKOTA	1350.82	509.24	112.79	520.94	2493.79	124.68	2369.11	1179.14	0	58.54	1237.67	114.79	1075.19	52.24198117	safe
477	Annamayya	MULAKALACHERUVU	1196.6	593.71	88.8	602.3	2481.41	124.06	2357.35	1106.56	0	94.24	1200.8	140.9	1118.19	50.93855388	safe
478	Annamayya	B KOTHAKOTA	1691.47	376.9	138.55	330.28	2537.2	126.89	2410.31	1072.00	8.482	23.33	1103.81	167.07	1162.76	45.79535412	safe
479	Annamayya	KALAKADA	1736.24	380.11	123.97	398.84	2639.16	131.97	2507.19	1253.98	0	43.81	1297.81	105.66	1147.54	51.76352809	safe
480	Annamayya	KAMBHAMVARIPALLE	2124.25	547.53	149.83	575.11	3396.72	169.84	3226.88	1722.74	0	75.85	1798.61	129.03	1394.36	55.73836027	safe
481	Annamayya	MANDAPALLE	2441.36	1131.02	230.56	1096.39	4899.33	244.97	4654.36	1526.68	14.048	198.14	1738.88	319.41	2811.07	37.36023857	safe
482	Annamayya	NIMMANAPALLE	1460.56	753.28	152.47	830.82	3197.13	159.86	3037.27	1384.86	0	40.15	1425.01	96.28	1556.14	46.91746206	safe
483	Annamayya	PILERU	1363.45	1232.14	85.52	1318.75	3999.86	199.99	3799.87	268.27	0	35.73	304.01	189.14	3342.45	8.00053686	safe
484	Annamayya	VALMIKIPURAM	2597.08	999.83	196.38	1054.65	4847.94	242.4	4605.54	1450.73	75.108	32.89	1558.71	125.46	2954.26	33.84423976	safe
485	Annamayya	CHINNAMANDEM	833.44	256.05	44.87	265.27	1399.63	69.99	1329.64	989.26	0	93.29	1082.57	100.36	263.38	81.41827863	semi critical
486	Annamayya	KODUR	812.43	201.76	65.58	261.92	1341.69	67.11	1274.58	454.81	0	113.22	568.01	114.15	705.64	44.56448399	safe
487	Annamayya	NANDALUR	600.95	323.6	19.69	346.85	1291.09	64.57	1226.52	410.14	0	93.74	503.86	116.66	699.73	41.08045527	safe
488	Annamayya	PENAGALURU	1007.55	567.09	15.57	569.82	2160.03	107.98	2052.05	733.35	0	264.47	997.83	281.41	1054.56	48.62600814	safe
489	Annamayya	RAMAPURAM	1209.08	702.77	30.03	702.69	2644.57	132.25	2512.32	778.58	0.108	137.03	915.72	138.27	1595.37	36.44917845	safe
490	Annamayya	ROYACHOTI	1441	904.92	42.34	919.64	3307.9	165.38	3142.52	712.10	0	163.87	875.95	261.98	2174.65	27.8741265	safe
491	Annamayya	SAMBEPALLE	1548.39	1032.38	69.9	1006.84	3657.51	182.86	3474.65	1238.21	0	132.47	1370.68	177.27	2059.17	39.44800196	safe
492	Annamayya	T SUNDUPALLE	2258.26	1167.24	73.05	1240.34	4738.89	236.93	4501.96	1772.12	0	211.30	1983.41	224.08	2511.38	44.05658869	safe
493	Annamayya	VEERABALLE	1982.02	728.37	71.87	710.14	3492.4	174.64	3317.76	795.56	0	137.43	932.96	145.87	2376.35	28.12017747	safe

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494	Eluru	KALIDINDI	0	1845.37	0	252.27	2097.64	104.88	1992.75	0.00	0	0.00	0	0	1992.75	0	salinity
495	Eluru	KUKUNOOR	5174.38	4534.71	0	2368.78	12077.87	603.89	11473.99	865.04	0	247.91	1112.93	289.58	10350.56	9.699590116	safe
496	Eluru	NUZVID	1316.45	1628.53	393.04	472.06	3810.08	190.53	3619.55	897.43	176.4	54.38	1128.37	314.62	2331.7	31.17431725	safe
497	Eluru	BHIMADOLE	2923.97	2573.54	0	1820.97	7318.48	365.93	6952.54	1634.23	0	214.82	1849.09	274.68	5083.14	26.59589157	safe
498	Eluru	AGIRIPALLE	1099.5	2413.36	329.95	713.55	4556.36	227.83	4328.53	1703.22	14.4	76.12	1793.71	129.47	2497.5	41.4392415	safe
499	Eluru	KAIKALUR	0	2581.7	0	390.38	2972.08	148.62	2823.46	0.00	0	0.00	0	0	2823.46	0	salinity
500	Eluru	POLAVARAM	2230.8	3856.5	0	1861.97	7949.27	397.42	7551.85	845.47	0	355.73	1201.25	422.09	6288.62	15.90669836	safe
501	Eluru	DENDULURU	1577.96	2728.38	0	1954.51	6260.85	313.02	5947.83	1150.26	35.64	722.70	1908.62	745.09	4052.55	32.08935023	safe
502	Eluru	NIDAMARRU	0	1809.77	0	6.32	1816.09	90.81	1725.29	0.00	0	0.00	0	0	1725.29	0	salinity
503	Eluru	CHATRAI	1051.86	1100.33	258.12	653.2	3063.51	153.19	2910.32	796.56	0	48.32	844.86	143.31	1991.25	29.02979741	safe
504	Eluru	MANDAVALLI	0	2732.22	0	1258.16	3990.38	199.49	3790.88	0.00	0	0.00	0	0	3790.88	0	salinity
505	Eluru	MUDINAPALLE	0	1006.81	0	1805.85	2812.66	140.6	2672.05	0.00	0	0.00	0	0	2672.05	0	salinity
506	Eluru	MUSUNURU	1722.24	1256.57	528.07	418.29	3925.17	196.35	3728.88	1096.64	77.4	302.08	1476.14	302.99	2264.11	39.58668555	safe
507	Eluru	BUTTAYAGUDEM	4334.74	2641.79	0	1866.77	8843.3	442.12	8401.18	963.09	0	926.23	1889.23	976.72	6495.7	22.48767435	safe
508	Eluru	CHINTALAPUDI	2420.88	2096.79	111.56	2189.75	6818.98	340.98	6478	2209.40	4.68	725.77	2939.71	841.9	3524.4	45.3799012	safe
509	Eluru	DWARAKA TIRUMALA	2097.01	5703.65	0	3729.33	11529.99	576.58	10953.41	2592.24	56.16	861.45	3509.71	863.66	7558.5	32.04216769	safe
510	Eluru	ELURU	1772.14	3756.11	0	382.57	5910.82	295.55	5615.27	116.62	0	254.41	370.98	338.21	5160.47	6.606627998	safe
511	Eluru	JANGAREDDIGUDEM	2461.66	5954.63	0	1016.22	9432.51	471.63	8960.87	2527.40	39.6	674.38	3241.36	803.59	5590.31	36.17238058	safe
512	Eluru	JEELUGUMILLI	2854.85	4903.73	0	2437.64	10196.22	509.88	9686.38	1731.70	0	910.76	2642.44	910.77	7073.94	27.27995391	safe
513	Eluru	KAMAVARAPUKOTA	1886.99	2200.63	0	1238.55	5326.17	266.28	5059.88	1767.24	0	597.92	2365.16	659.75	2769.15	46.74340103	safe
514	Eluru	KOYYALAGUDEM	1982.98	2341.2	0	821.86	5146.04	257.31	4888.73	1822.07	4.68	478.98	2305.69	548.52	2544.47	47.16337372	safe
515	Eluru	LINGAPALEM	1456.51	3414.71	0	2530.45	7401.67	370.22	7031.52	2876.19	0	274.04	3150.19	410.01	3866.37	44.80098186	safe
516	Eluru	PADAVEGI	2049.37	8631.47	0	3245.48	13926.32	696.35	13229.97	2805.24	2.16	855.27	3662.64	887.23	9641.59	27.68441652	safe
517	Eluru	PEDAPADU	1932.19	4759.75	0	1273.58	7965.52	398.29	7567.22	1049.02	22.68	357.35	1429.03	364.51	6255.75	18.88447805	safe
518	Eluru	T NARASAPURAM	3528.2	7382.87	0	3815.42	14726.49	736.33	13990.16	3964.37	0	687.53	4651.88	691.88	9375.31	33.25108505	safe
519	Eluru	UNGUTURU (WEST GODAVARI)	2192.45	4091.19	0	2676.03	8959.67	448.01	8511.66	3319.67	68.04	378.86	3766.54	435.99	5154.3	44.2515326	safe

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520	Eluru	VELAIRPAD	2613.69	944.82	0	419.95	3978.46	198.9	3779.56	270.59	0	193.60	464.16	217.8	3323.69	12.28079459	safe
521	Tirupati	CHANDRAGIRI	3649.37	660.11	54.55	812.2	5176.23	258.83	4917.4	1426.81	43.55	60.98	1531.34	155.07	3404.52	31.14125351	safe
522	Tirupati	PICHATUR	1396.85	798.75	0	858.85	3054.45	152.72	2901.73	1323.68	11.05	37.76	1372.49	85.95	1509.16	47.29902506	safe
523	Tirupati	KOTA	2516.01	1098.08	0	1790.47	5404.56	270.22	5134.34	1998.04	6.56	44.59	2049.21	146.3	2993.15	39.91184846	safe
524	Tirupati	BUCHINAIDU KHANDRI	2832.47	870.89	0	1926.65	5630.01	281.49	5348.52	925.44	28.65	52.98	1007.07	95.83	4298.6	18.82894707	safe
525	Tirupati	OJILI	2261.56	631.32	0	1614.86	4507.74	225.4	4282.34	1207.00	52.65	126.12	1385.76	136.26	2886.41	32.35987801	safe
526	Tirupati	SULLURPETA	7858.98	840.01	0	2674.58	11373.57	568.64	10804.93	1168.96	22.87	323.90	1515.7	358.04	9371.52	14.0278558	safe
527	Tirupati	VARADAIHPALEM	2281.86	828.78	6.5	2134.94	5252.08	262.61	4989.47	939.15	73.95	59.21	1072.29	124.19	3852.18	21.49106017	safe
528	Tirupati	SRIKALAHASTI	3406.74	1599.92	0	2500.37	7507.03	375.38	7131.65	2721.73	88.9	163.49	2974.09	655.78	4205.19	41.70269152	safe
529	Tirupati	PUTTUR	1780.42	423.3	0	538.55	2742.27	137.1	2605.17	1237.83	64.55	190.24	1492.59	417.03	1270.83	57.29338201	safe
530	Tirupati	CHINNAGOTTIGALLU	1178.49	839.42	0	987.8	3005.71	150.29	2855.42	1397.97	10.2	32.14	1440.31	69.51	1377.74	50.44126608	safe
531	Tirupati	TIRUPATI	927.66	325.78	0	410.35	1663.79	83.21	1580.58	871.23	54.4	92.80	1018.47	399	542.84	64.43647269	safe
532	Tirupati	RAMACHANDRAPURAM-17	1039.34	535.89	0	639.82	2215.05	110.73	2104.32	1070.36	23.8	32.16	1126.28	83.22	993.43	53.52227798	safe
533	Tirupati	PELLAKUR	1557.71	1031.07	0	2123.45	4712.23	235.65	4476.58	1654.93	17.5	95.09	1767.52	105.54	2700.38	39.48371301	safe
534	Tirupati	TIRUPATI URBAN	763.76	128.68	0	162.64	1055.08	52.74	1002.34	115.55	23.5	289.59	428.64	2425.87	560.97	42.7639324	safe
535	Tirupati	RENIGUNTA	1078.9	573.55	0	728.36	2380.81	119.04	2261.77	871.61	116.45	223.91	1212	289.8	1192.44	53.58635051	safe
536	Tirupati	VENKATAGIRI	2418.88	780.3	0	1331.36	4530.54	226.56	4303.98	1382.53	11.3	106.09	1499.99	424.35	2796.38	34.85123072	safe
537	Tirupati	GUDUR	2109.27	676.93	0	1568.97	4355.17	217.78	4137.39	1574.33	58.91	513.04	2146.27	651.66	2243.98	51.87497432	safe
538	Tirupati	K V B PURAM	3508.59	848.03	0	1105.71	5462.33	273.13	5189.2	2234.31	16.8	84.10	2335.25	174.68	3085.38	45.00211979	safe
539	Tirupati	SATYAVEDU	1745.28	688.47	0	1335.96	3769.71	188.49	3581.22	883.05	99	56.46	1038.48	139.71	2461.71	28.99793925	safe
540	Tirupati	PAKALA	1360.55	683.22	26.89	859.77	2930.43	146.55	2783.88	1791.55	17	50.39	1858.91	152.19	830.02	66.77407072	safe
541	Tirupati	TADA	4672.14	358.28	0	1755.27	6785.69	339.29	6446.4	582.39	93.6	58.92	734.9	145.32	5625.11	11.40016133	safe
542	Tirupati	NAGALAPURAM	1205.74	783.63	0	819.95	2809.32	140.46	2668.86	1371.90	20.4	42.74	1435.06	92.34	1185.4	53.77052374	safe
543	Tirupati	DORAVARISATRAM	3047.06	840.85	0	2779.87	6667.78	333.42	6334.36	1227.22	17.82	102.45	1347.51	115.14	4974.16	21.27302521	safe
544	Tirupati	NARAYANAVANAM	1025.11	483.95	0	554.38	2063.44	103.2	1960.24	1008.56	58.85	34.85	1102.27	135.12	973.97	56.23137983	safe
545	Tirupati	VAKADU	4483.13	372.7	0	1444.18	6300.01	315.01	5985	1461.64	8.9	54.99	1525.53	112.27	4426.31	25.48922306	safe

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546	Tirupati	YARRAVARIPALEM	1015.99	1650.76	0	1803.85	4470.6	223.53	4247.07	1348.18	5.95	31.20	1385.32	73.83	2819.12	32.61825211	safe
547	Tirupati	YERPEDU	2808.02	1065.24	0	1484.65	5357.91	267.91	5090	2824.34	54.86	80.70	2959.91	153.69	2228.27	58.15147348	safe
548	Tirupati	THOTTAMBEDU	1626.73	1415.13	0	1975.6	5017.46	250.89	4766.57	1931.77	59.5	70.09	2061.33	122.2	2656.39	43.24556232	safe
549	Tirupati	VADAMALAPETA	1611.87	609.7	0	723.36	2944.93	147.29	2797.64	1358.50	11.05	47.10	1416.68	91.63	1349.59	50.63839522	safe
550	Tirupati	BALAYAPALLE	2056.15	1591.34	0	2923.87	6571.36	328.57	6242.79	1431.56	3.24	161.32	1596.06	177.42	4630.61	25.56645346	safe
551	Tirupati	CHILLAKUR	3848.63	770.45	0	2285.01	6904.09	345.22	6558.87	1306.74	205.74	57.19	1569.7	145.73	4900.63	23.93247617	safe
552	Tirupati	CHITTAMUR	3649.89	936.4	0	4475.78	9062.07	453.1	8608.97	1847.78	12.96	88.68	1949.44	123.35	6626.17	22.64428846	safe
553	Tirupati	DAKKILI	2549.49	1126.03	0	1953.91	5629.43	281.5	5347.93	2148.21	0.72	114.77	2263.7	153.61	3058.7	42.32852711	safe
554	Tirupati	NAIDUPETA	2093.99	896.14	0	1233.04	4223.17	211.16	4012.01	1654.76	101.61	226.13	1982.46	353.08	2109.64	49.41313706	safe
555	Srikakulam	HIRAMANDALAM	1251.1	1409.29	134.97	480.2	3275.56	163.78	3111.78	691.35	12.5925	160.70	864.63	575.4	2261.27	27.78570464	safe
556	Srikakulam	LAXMINARASAMPET	1114.39	1330.37	102.07	378.54	2925.37	146.28	2779.09	310.40	0	115.98	426.37	116.03	2352.64	15.3420724	safe
557	Srikakulam	KAVITI	1089.3	1396.67	28.39	347.35	2861.71	143.07	2718.64	583.12	0	169.63	752.72	559.27	1925.31	27.6873731	safe
558	Srikakulam	AMADALAVALASA	1261.03	1644.88	70.17	573.77	3549.85	177.53	3372.32	561.38	0	370.04	931.4	383.88	2562.89	27.61896854	safe
559	Srikakulam	BURJA	917.63	2353.91	63.31	991.31	4326.16	216.39	4109.77	545.30	0	158.34	703.62	184.73	3476.98	17.12066612	safe
560	Srikakulam	ETCHERLA	2652.66	1189.03	165.94	673.1	4680.73	234.04	4446.69	1835.73	23.579	346.39	2205.69	376.01	2472.45	49.6029631	safe
561	Srikakulam	GANGUVARI SINGADAM	1078.82	2131.96	91.11	1633.82	4935.71	246.79	4688.92	2327.97	0	270.93	2598.91	288.18	2351.62	55.42662276	safe
562	Srikakulam	GARA	2152.82	2928.3	188.58	1539.84	6809.54	340.48	6469.06	3373.88	0	223.64	3597.49	241.99	2980.94	55.61070697	safe
563	Srikakulam	ICHCHAPURAM	277.43	1257.07	12.03	484.65	2031.18	101.55	1929.63	434.79	0	399.88	834.7	420.15	1233.04	43.25699746	safe
564	Srikakulam	JALUMURU	1084.58	2759.17	137.26	1213.28	5194.29	259.75	4934.54	852.60	0	179.97	1032.55	206.08	3883.45	20.92494944	safe
565	Srikakulam	KANCHILI	866.73	1507.23	60.2	285.85	2720.01	136.04	2583.97	321.67	0	276.14	597.82	311.27	1982.79	23.13571752	safe
566	Srikakulam	KOTABOMMAL	1251.96	2573.62	90.05	977.72	4893.35	244.66	4648.69	1558.06	0.292	291.80	1850.16	321.63	2809.85	39.79959946	safe
567	Srikakulam	KOTHURU	1746.77	2349.92	214.11	350.2	4661	233.09	4427.91	740.30	0	226.87	967.14	244.84	3447.72	21.84190736	safe
568	Srikakulam	LAVERU	924.14	1637.2	72.25	2030.34	4663.93	233.24	4430.69	3867.65	0.073	228.56	4096.3	263.68	1218.17	92.45286851	critical
569	Srikakulam	MANDASA	1258.85	2743.5	85.74	274.3	4362.39	217.99	4144.39	318.31	0	157.65	475.93	261.45	3575.45	11.48371654	safe
570	Srikakulam	MELIAPUTTI	620.35	2550.32	53.33	538.3	3762.3	188.12	3574.18	595.16	1.7155	232.61	829.47	267.84	2709.51	23.20728111	safe
571	Srikakulam	NANDIGAM	469.76	2867.59	29.5	821.59	4188.44	209.42	3979.03	740.14	0	384.88	1124.89	399.75	2990.77	28.27045788	safe

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572	Srikakulam	NARASANNAPETA	1213.07	2195.14	102.68	714.18	4225.07	211.23	4013.84	791.49	0	273.01	1064.48	928.4	2944.07	26.52023997	safe
573	Srikakulam	PALASA	1005.1	1559.05	40.26	263	2867.41	143.36	2724.04	567.32	0.73	237.09	805.13	1157.79	1943.91	29.5564676	safe
574	Srikakulam	PATHAPATNAM	437.94	1830.05	36.86	649.32	2954.17	147.68	2806.49	801.44	0	177.21	978.57	467.58	1879.48	34.86810928	safe
575	Srikakulam	POLAKI	1317.2	2500.46	130.25	1319.7	5267.61	263.38	5004.23	1281.00	0	219.02	1500.04	239.8	3483.43	29.97544078	safe
576	Srikakulam	PONDURU	1216.73	1444.3	80.35	1233.94	3975.32	198.76	3776.55	2074.88	0	257.89	2332.76	301.82	1915.55	61.76960453	safe
577	Srikakulam	RANASTALAM	2112.1	1578.36	143.55	1897.74	5731.75	286.61	5445.15	7468.23	330.928345	396.56	8195.69	654.2	249.3	150.5135763	over exploited
578	Srikakulam	SANTHABOMMALI	354.61	4534.19	21.27	1581.88	6491.95	324.62	6167.33	296.66	0	102.90	399.56	139.93	5732.88	6.478654458	safe
579	Srikakulam	SARAVAKOTA	1997.94	2692.27	126.12	917.86	5734.19	286.7	5447.48	435.04	0	162.72	597.74	189.87	4822.58	10.97278007	safe
580	Srikakulam	SARUBUJJILI	676.99	1551.46	56.15	548.13	2832.73	141.61	2691.12	295.60	0	138.90	434.49	140.32	2255.2	16.14532239	safe
581	Srikakulam	SOMPETA	1389.92	1416.45	101	492.52	3399.89	170.05	3229.84	1207.42	0	206.73	1414.15	305.3	1854.37	43.78390261	safe
582	Srikakulam	SRIKAKULAM MANDAL	1423.93	1715.96	77.23	389.19	3606.31	180.29	3426.02	880.34	0	400.22	1280.57	405.26	2208.5	37.37777363	safe
583	Srikakulam	TEKKALI	338.5	7516.86	18.14	4349.1	12222.6	611.18	11611.41	421.84	0	218.13	639.98	749.73	10962.8	5.511647595	safe
584	Srikakulam	VAJRAPUKOTHURU	1085.03	1483.9	90.21	383.05	3042.19	152.13	2890.06	523.59	0	302.87	826.49	457.72	2211.27	28.59767617	safe
585	NTR	REDDIGUDEM	1028.92	2006.61	139.71	749.54	3924.78	196.25	3728.53	1942.72	21.45	40.63	2004.82	41.87	1724.07	53.76971622	safe
586	NTR	NANDIGAMA	1178.79	1311.32	49.27	675.68	3215.06	160.75	3054.31	1184.41	52.47	92.75	1329.61	95.6	1721.85	43.53225442	safe
587	NTR	MYLAVARAM (KRISHNA)	1311.91	1607.91	142.15	673.86	3735.83	186.79	3549.04	1667.70	125.4	60.67	1853.79	62.53	1693.39	52.23356175	safe
588	NTR	VATSAVAI	1065.95	1061.36	27.21	443.5	2598.02	129.9	2468.12	894.20	46.2	61.74	1002.13	63.63	1464.08	40.60296906	safe
589	NTR	VISSANNAPET	1578.95	1651.21	90.69	891.21	4212.06	210.59	4001.47	1755.89	84.15	56.68	1896.7	58.42	2103.03	47.40008047	safe
590	NTR	KANCHIKA CHERLA	882.36	570.32	47.45	239.17	1739.3	86.97	1652.33	444.72	47.85	64.08	556.65	66.03	1093.74	33.68879098	safe
591	NTR	VIJAYAWADA NORTH	58.81	1017.84	9.39	525.35	1611.39	80.57	1530.82	0.00	0	277.17	277.16	278.43	1252.4	18.10532917	safe
592	NTR	CHANDARLAPADU	1160.11	1169.69	81.02	456.78	2867.6	143.39	2724.21	697.50	52.8	79.91	830.24	82.35	1894.58	30.47635828	safe
593	NTR	IBRAHIMPATNAM	1562.94	1207.77	139.22	825.53	3735.46	186.8	3548.66	689.17	83.82	87.63	860.64	90.31	2746.56	24.25253476	safe
594	NTR	VIJAYAWADA WEST	174.74	11.95	18.6	2.71	208	10.4	197.6	0.00	0	130.38	130.37	130.97	66.64	65.97672065	safe
595	NTR	VIJAYAWADA RURAL	1346.86	4016.44	152.57	2065.59	7581.46	379.09	7202.37	680.93	163.68	91.71	936.33	94.49	6263.24	13.00030407	safe
596	NTR	VIJAYAWADA EAST	146.83	518.9	23.44	259.45	948.62	47.43	901.19	0.00	0	353.19	353.18	354.8	546.4	39.1904038	safe
597	NTR	TIRUVURU	1202.64	2659.23	21.42	1165.62	5048.91	252.46	4796.45	2358.60	107.25	60.32	2526.19	62.17	2295.91	52.66791064	safe

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598	NTR	G KONDURU	1345.6	1739.27	88.94	748.49	3922.3	196.11	3726.19	1137.52	118.47	51.87	1307.86	53.45	2439.59	35.0991227	safe
599	NTR	VEERULLAPADU	1403.8	709.66	57.79	467.45	2638.7	131.96	2506.74	939.02	13.2	47.93	1000.19	49.41	1508.29	39.90002952	safe
600	NTR	PENUGANCHIPROLU	907.28	1432.95	43.67	815.85	3199.75	159.98	3039.77	1529.12	58.74	39.78	1627.64	41	1443.09	53.54484056	safe
601	NTR	A KONDURU	1207.91	1122.02	14.82	918.12	3262.87	163.15	3099.72	1898.51	3.3	42.17	1943.99	43.45	1162.14	62.71501942	safe
602	NTR	GAMPALAGUDEM	1319.68	1506.53	46.76	829.44	3702.41	185.1	3517.31	1471.44	79.2	63.84	1614.44	65.79	1905.31	45.8998496	safe
603	NTR	JAGGAYYAPETA	1783.45	902.07	15.61	479.31	3180.44	159.03	3021.41	1038.01	75.9	114.78	1228.66	118.32	1816.71	40.66511993	safe
604	NTR	VIJAYAWADA CENTRAL	113.25	1867.6	18.08	927.27	2926.2	146.31	2779.89	0.00	0	615.25	615.25	618.06	2161.83	22.1321707	safe
605	Anakapalli	NATHAVARAM	1197.35	4035.66	828.41	230.9	6292.32	314.63	5977.69	346.28	0	131.18	477.45	150.46	5480.97	7.987199069	safe
606	Anakapalli	DEVARAPALLE (ANAKAPALLI)	370.07	2651.52	269.87	658.17	3949.63	197.49	3752.14	527.67	0	163.29	690.98	187.29	3037.16	18.41562415	safe
607	Anakapalli	CHODAVARAM	1082.72	1836.04	666.35	304	3889.11	194.44	3694.67	1337.22	8.802	304.83	1650.91	349.63	2000.02	44.68355767	safe
608	Anakapalli	YELAMANCHILI (ANAKAPALLI)	747.99	1147.54	499.94	212.96	2608.43	130.44	2477.99	494.53	3.402	248.10	746.03	284.57	1695.49	30.10625547	safe
609	Anakapalli	PARAVADA	788.43	243.92	479.9	188.17	1700.42	85.03	1615.39	246.19	66.6	153.07	465.85	175.57	1127.05	28.83823721	safe
610	Anakapalli	GOLUGONDA	663.51	984.42	373.82	256.93	2278.68	113.93	2164.75	311.50	0	156.57	468	179.59	1673.74	21.61912461	safe
611	Anakapalli	NARSIPATNAM	555.01	1702.95	491.39	85.96	2835.31	141.76	2693.55	170.89	7.2	256.51	434.59	294.21	2221.27	16.13446938	safe
612	Anakapalli	MAKAVARAPALEM	730.83	1340.32	530.99	292.81	2894.95	144.76	2750.19	579.14	1.8	175.87	756.8	201.73	1967.53	27.51809875	safe
613	Anakapalli	ROLUGUNTA	666.21	1394.67	494.26	197.57	2752.71	137.62	2615.09	390.93	4.536	146.30	541.8	167.79	2051.79	20.7182162	safe
614	Anakapalli	RAMBILLI	731.63	1106.53	479.49	427.49	2745.14	137.27	2607.87	262.31	6.84	164.55	433.76	188.74	2158.35	16.63273093	safe
615	Anakapalli	RAVIKAMATHAM	1097.78	1998.15	696.54	317.32	4109.79	205.5	3904.29	446.50	2.268	167.46	616.24	192.07	3274.07	15.78366361	safe
616	Anakapalli	K KOTAPADU	1229.46	1743.77	646.34	255.55	3875.12	193.77	3681.35	520.22	0	177.98	698.21	204.11	2966.52	18.96614014	safe
617	Anakapalli	ANAKAPALLE	1336.9	1557.77	765.5	298.99	3959.16	197.99	3761.17	762.02	28.8	315.34	1106.14	361.69	2610.42	29.40946567	safe
618	Anakapalli	ATCHUTAPURAM	1006.99	365.18	627.12	130.13	2129.42	106.49	2022.93	646.94	5.4	202.65	854.95	232.44	1138.89	42.26295522	safe
619	Anakapalli	BUTCHAYYAPETA	987.73	1592.36	580.55	423.05	3583.69	179.17	3404.52	789.08	0	201.52	990.63	231.12	2384.31	29.09749392	safe
620	Anakapalli	CHEEDIKADA	516.44	2241.42	439.56	257.41	3454.83	172.74	3282.1	592.48	241.095	140.97	974.49	161.68	2286.94	29.69105146	safe
621	Anakapalli	KASIMKOTA	1175.23	859.57	683.87	530.49	3249.16	162.46	3086.7	1283.76	25.1532	190.83	1499.77	218.89	1684.17	48.5881362	safe
622	Anakapalli	KOTAURATLA	866.08	1438.37	568.21	249.52	3122.18	156.09	2966.09	856.28	0	163.07	1019.33	187.06	1923.94	34.36611836	safe
623	Anakapalli	MADUGULA	996.15	3232.73	763.13	459.91	5451.92	272.69	5179.23	1509.37	0	219.10	1728.51	251.28	3438.51	33.3738799	safe

Sl.No	District	Assessment Unit Name	Recharge from Rainfall I-MON	Recharge from Other Source s-MON	Recharge from Rainfall I-NM	Recharge from Other Source s-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
624	Anakapalli	MUNAGAPAKA	667.5	445.14	378.49	245.1	1736.23	86.82	1649.41	811.98	3.6	148.41	964	170.22	875.65	58.44514099	safe
625	Anakapalli	NAKKAPALLE	1649.43	1528.15	1010.82	219.69	4408.09	220.38	4187.71	485.28	8.7012	260.31	754.32	298.57	3395.13	18.01270862	safe
626	Anakapalli	PAYAKARAOPETA	1067.14	1420.53	691.73	67.26	3246.66	162.35	3084.31	761.89	30.096	253.86	1045.81	291.17	2001.2	33.90742176	safe
627	Anakapalli	S RAYAVARAM	1123.71	2184	709.71	351.73	4369.15	218.45	4150.7	953.56	29.664	213.45	1196.67	244.81	2992.14	28.8305587	safe
628	Anakapalli	SABBAVARAM	1189.4	1145.05	747.45	228.63	3310.53	165.56	3144.97	633.33	4.59	213.88	851.82	245.32	2261.72	27.08515503	safe
629	East Godavari	CHAGALLU	1090.53	908.8	168.06	1169.01	3336.4	166.83	3169.57	2092.81	21.6	163.56	2277.98	170.4	884.75	71.87031679	semi_critical
630	East Godavari	RANGAMPETA	1582.29	849.78	172.14	571.33	3175.54	158.78	3016.76	2168.66	10.5696	57.61	2236.85	60.03	983.52	74.14742969	semi_critical
631	East Godavari	THALLAPUDI	1097.35	912.64	112.75	883.65	3006.39	150.38	2856.01	1015.09	0	144.67	1159.7	150.7	1703.17	40.60560012	safe
632	East Godavari	RAJANAGARAM	2104.44	704.8	275.24	539.22	3623.7	181.22	3442.48	2053.76	64.8828	120.00	2238.66	125.01	1229.84	65.03044317	safe
633	East Godavari	RAJAHMUNDURY (URBAN)	150.61	1.97	14.78	1.3	168.66	8.43	160.23	0.00	13.212	0.00	13.21	0	147.02	8.244398677	safe
634	East Godavari	GOKAVARAM	970.18	3871.17	190.06	1317.06	6348.47	317.46	6031.01	566.37	26.3736	188.46	781.19	196.33	5241.93	12.95288849	safe
635	East Godavari	SEETHANAGARAM	1271.61	3281.11	153.98	2534.23	7240.93	362.03	6878.9	1075.39	133.4412	82.12	1290.93	85.54	5625.08	18.7665179	safe
636	East Godavari	ANAPARTHI	719.41	3951.9	85.66	3577.24	8334.21	416.71	7917.5	468.91	161.1864	14.24	644.34	14.84	7272.56	8.138174929	safe
637	East Godavari	BICCAVOLU	1059.87	4938.07	128.7	4489.67	10616.31	530.8	10085.51	681.15	220.6404	38.75	940.53	40.36	9143.36	9.325557161	safe
638	East Godavari	KADIAM	829.69	4132.29	101.03	3891.66	8954.67	447.74	8506.93	2697.72	158.544	103.66	2959.95	107.99	5653.19	34.79457337	safe
639	East Godavari	KORUKONDA	1057.86	6824.22	158.1	5612.77	13652.95	682.64	12970.31	1426.30	113.6232	93.28	1633.18	97.17	11333.23	12.59168054	safe
640	East Godavari	RAJAHMUNDURY RURAL	922.92	215.02	79.23	117.49	1334.66	66.73	1267.93	572.59	71.3448	38.74	682.68	40.36	602.55	53.84208907	safe
641	East Godavari	DEVARAPALLE	2022.13	814.97	124.51	887.01	3848.62	192.44	3656.18	2378.73	18.72	190.91	2588.36	198.86	1160.55	70.79410751	semi_critical
642	East Godavari	GOPALAPURAM	1735.07	1622.02	149.83	1291.21	4798.13	239.94	4558.19	2839.49	28.08	146.27	3013.85	152.18	1734.13	66.11944653	safe
643	East Godavari	KOVVUR	1411.27	1407.68	156	2046.65	5021.6	251.07	4770.53	2465.55	27.36	262.28	2755.2	273.23	2079.24	57.75458911	safe
644	East Godavari	NALLAJERLA	3347.98	716.74	145.13	899.6	5109.45	255.48	4853.97	3250.59	10.08	206.74	3467.42	214.9	1524.48	71.43472251	semi_critical
645	East Godavari	NIDADAVOLE	1543.55	1428.38	130.55	2078.23	5180.71	259	4921.71	2315.74	13.05936	113.97	2442.78	118.72	2633.46	49.63274959	safe
646	East Godavari	PERAVALI	1113.24	1974.43	124.49	2166.73	5378.89	268.95	5109.94	1486.76	15.12	92.67	1594.54	96.53	3511.55	31.20467168	safe
647	East Godavari	UNDRAJAVARAM	1339.55	3157.89	134.6	3527.67	8159.71	407.99	7751.72	1333.17	12.96	99.57	1445.7	103.72	6334.82	18.65005444	safe
648	Sri Sathya Sai	CHILAMATHUR	839.22	894.76	2.74	1063.14	2799.86	139.98	2659.88	1294.23	16.2	217.73	1528.18	311.02	1176.78	57.4529678	safe
649	Sri Sathya Sai	HINDUPUR	1221.63	1219.19	42.77	1876.97	4360.56	218.04	4142.52	5569.57	30.096	243.66	5843.32	745.24	700.1	141.0571343	over_exploited

Sl.No	District	Assessment Unit Name	Recharge from Rainfall I-MON	Recharge from Other Source s-MON	Recharge from Rainfall I-NM	Recharge from Other Source s-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/ Semi critical/Safe)
650	Sri Sathya Sai	OBULADEVARACHERUVU	1102.14	521.69	0	571.79	2195.62	109.83	2085.79	751.86	0	308.45	1060.3	340.65	1041.88	50.83445601	safe
651	Sri Sathya Sai	AGALI	726.27	697.77	43.75	828.79	2296.58	114.8	2181.78	1079.98	0.072	1099.01	2179.06	1099.01	198.02	99.87533115	critical
652	Sri Sathya Sai	SOMANDEPALLE	1221.36	987.31	0	1210.16	3418.83	170.95	3247.88	1973.36	7.2	235.87	2216.46	260.18	1341.54	68.24328485	safe
653	Sri Sathya Sai	TANAKAL	1589.04	1450.04	12.36	2176.45	5227.89	261.34	4966.55	6259.82	0	403.06	6662.87	419.09	1086.1	134.1548963	over_exploited
654	Sri Sathya Sai	AMADAGUR	1117.12	547.14	22.51	629.83	2316.6	115.81	2200.79	1394.66	0	213.84	1608.5	230.01	761.8	73.08739135	semi_critical
655	Sri Sathya Sai	AMARAPURAM	1273.42	673.02	26.45	807.34	2780.23	139.01	2641.22	1567.92	0	181.44	1749.35	222.85	874.85	66.23265006	safe
656	Sri Sathya Sai	BATHALAPALLE	1394.68	1083.66	0	1173.05	3651.39	182.58	3468.81	1275.82	0	155.52	1431.32	168.26	2024.76	41.26256555	safe
657	Sri Sathya Sai	BUKKAPATNAM	1166.75	642.22	0.28	2925.75	4735	236.78	4498.23	857.06	0	102.38	959.45	147.33	3493.83	21.32950071	safe
658	Sri Sathya Sai	CHENNE KOTHAPALLE	948.52	877.16	12.16	2184.88	4022.72	201.14	3821.57	1383.05	0	119.23	1502.28	176.99	2295.03	39.31054514	safe
659	Sri Sathya Sai	DHARMAVARAM	2356.59	1547.61	0	1642.28	5546.48	277.34	5269.14	1799.20	3.6	117.94	1920.71	593.36	2873.01	36.45205859	safe
660	Sri Sathya Sai	GANDLAPENTA	771.73	458.58	0	497.41	1727.72	86.39	1641.33	1448.22	0	1289.52	2737.75	1302.07	274.85	166.8007043	over_exploited
661	Sri Sathya Sai	GORANTLA	1091.16	1733.21	0	1906.74	4731.11	236.56	4494.55	1436.49	1.944	200.88	1639.3	349.28	2724.66	36.47306182	safe
662	Sri Sathya Sai	GUDIBANDA	705.3	1026.45	8.88	1276.91	3017.54	150.87	2866.67	1994.39	0	308.45	2302.87	308.47	872.47	80.33258101	semi_critical
663	Sri Sathya Sai	KADIRI	1258.83	720.41	13.11	973	2965.35	148.26	2817.09	609.94	45.936	129.60	785.48	190.28	1970.92	27.88267326	safe
664	Sri Sathya Sai	KANAGANAPALLE	2382.62	1395.39	10.87	2375.96	6164.84	308.22	5856.62	1819.83	0	132.19	1952.01	185.9	3850.9	33.32997531	safe
665	Sri Sathya Sai	KOTHACHERUVU	1114.52	1260.47	0	2218.42	4593.41	229.66	4363.75	2441.89	0	115.34	2557.24	181.89	2156.04	58.60189058	safe
666	Sri Sathya Sai	LEPAKSHI	762.95	750.63	6.38	1016.98	2536.94	126.86	2410.08	1676.15	3.6	171.07	1850.83	176.37	568.17	76.79537609	semi_critical
667	Sri Sathya Sai	MADAKASIRA	1500.7	1027.44	171.57	1336.37	4036.08	201.81	3834.28	1793.98	1.8	300.67	2096.45	439.87	1911.25	54.67649728	safe
668	Sri Sathya Sai	MUDIGUBBA	3178.08	1270.87	57.87	1812.8	6319.62	316.01	6003.61	2111.63	0	276.05	2387.69	346.5	3550.06	39.77090451	safe
669	Sri Sathya Sai	NALLACHERUVU	796.1	547.69	0	771.65	2115.44	105.76	2009.68	1344.70	0	162.00	1506.71	185.61	538.46	74.97263246	semi_critical
670	Sri Sathya Sai	NALLAMADA	1118.62	737.92	0	826.62	2683.16	134.15	2549.01	892.00	0	260.50	1152.49	262.25	1394.76	45.21323965	safe
671	Sri Sathya Sai	NAMBULIPULIKUNTA	1654.39	779.91	0	956.5	3390.8	169.59	3221.21	2255.93	0	242.35	2498.27	266.81	960.55	77.5568808	semi_critical
672	Sri Sathya Sai	PARIGI	915.63	1671.93	26.89	1193.5	3807.95	190.38	3617.57	770.45	10.917	217.73	999.1	277	2570.61	27.61798666	safe
673	Sri Sathya Sai	PENU KONDA	2383.84	804.35	49.81	1131.87	4369.87	218.52	4151.35	1321.60	10.8	387.50	1719.92	397.59	2431.02	41.43037807	safe
674	Sri Sathya Sai	PUTTAPARTHI	1251.56	1103.69	0.61	1880.08	4235.94	211.82	4024.12	1604.96	0	130.90	1735.85	227.53	2197.17	43.13613908	safe
675	Sri Sathya Sai	RAMAGIRI	1922.8	1376.8	0	1880.13	5179.73	258.99	4920.74	1652.19	0	147.74	1799.92	179.68	3088.89	36.57823823	safe

Sl.No	District	Assessment Unit Name	Recharge from Rainfall-MON	Recharge from Other Sources-MON	Recharge from Rainfall-NM	Recharge from Other Sources-NM	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Irrigation Use (Ham)	Industrial Use (Ham)	Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (OE/Critical/Semi critical/Safe)
676	Sri Sathya Sai	RODDAM	1972.02	970.21	74.69	1186.65	4203.57	210.2	3993.37	2252.59	0	388.80	2641.42	452.57	1450.21	66.14513556	safe
677	Sri Sathya Sai	ROLLA	723.75	1255.88	11.56	1596.49	3587.68	179.38	3408.3	3758.31	0	318.82	4077.15	318.81	403.72	119.6241528	over_exploited
678	Sri Sathya Sai	TADIMARRI	1152.68	1129.6	0	1266.52	3548.8	177.45	3371.35	1864.60	0	134.78	1999.41	156.04	1394.47	59.30591603	safe
679	Sri Sathya Sai	TALUPULA	1463.76	811.68	0	1051.65	3327.09	166.39	3160.69	1798.36	0	191.81	1990.16	224.75	1245.35	62.96599793	safe

Annexure VII: Minutes of the Meeting of the SLC Committee.

Minutes of Meeting of 2nd State Level Committee (SLC) on Ground Water Resource Assessment 2024, Andhra Pradesh

The 2nd State Level Committee (SLC) meeting has been convened at Conference Hall, Office of the ENC (IW), Water Resources Department, Jalasoudha Irrigation Compound, Governerpeta, Vijayawada on 22.10.2024 at 11:00 hrs for approval of Ground Water Resource Assessment of Andhra Pradesh State for the year 2024. The meeting is chaired by Sri. G.Sai Prasad, IAS, Special Chief Secretary to Government, Water Resources Department (WRD), Govt. of Andhra Pradesh & Chairman, State Level Committee (SLC).

At the outset, Sri G. Krishnamurthy, Regional Director, Central Ground Water Board, Southern Region and Member Secretary, SLC welcomed Sri. G.Sai Prasad, IAS, Special Chief Secretary, Water Resources Department (WRD) & Chairman, State Level Committee (SLC) and the Members of SLC and representatives from different departments. He briefed about completion of Ground Water resources assessment of Andhra Pradesh as per time lines stipulated by Ministry of Jal Shakti, Govt. of India. He has highlighted the computations in IN-GRESS software for the fourth consecutive year. He briefed about the agenda of the meeting and requested the Chairman, SLC to preside over the proceedings.

Sri. M. John Satya Raju, Director, GW & WA Department, Government of Andhra Pradesh welcomed all the participants and appraised the committee on Ground water Resources Assessment. He thanked all the Members of SLC for providing data, support of CGWB, timely assessment by all 26 districts and compilation at HO, Vijayawada. He further expressed his thanks to the Special Chief Secretary, WRD for his continuous support and guidance.

Sri N. Srinivasu, Joint Director of the Ground Water & Water Audit Department, Government of Andhra Pradesh, delivered a detailed Power Point presentation on procedure adopted in computations and explained the assessment unit wise results to all members. During the session, Sri G. Sai Prasad, IAS, Special Chief Secretary and Chairman of the State Level Committee (SLC), asked several questions to clarify key aspects, such as the components of basic equation involved in recharge and extraction calculations. He asked certain details on the storage capacity of aquifers in TMC and the total runoff available in the state. He also suggested naming principal aquifers / sub aquifers based on their geographical locations.

Furthermore, he instructed to do the calculation of recharge from the HNSS Canal and the surface water irrigation within the HNSS canal command area. He proposed measures to enhance recharge feasibility in irrigated areas through Aquifer Recharge Systems (ARS). He stressed the importance of controlling groundwater extraction in Over-Exploited, Critical, and Semi-Critical (OCS) villages, basins, and mandals.

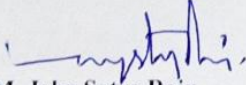
During the presentation, it was explained that the groundwater resources in the state have been assessed on a watershed basis and apportioned to mandals. The total annual groundwater recharge for the state is estimated at 27,799 MCM, with an annual extractable resource of 26,411 MCM. The current annual groundwater extraction for all uses stands at 7,879 MCM, resulting in a

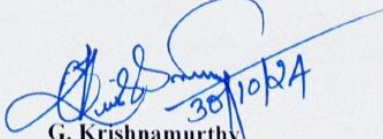
stage of groundwater extraction of 29.83%. (In Command area: 22.58 % and in Non-Command area: 37.40%) Out of 679 assessment units (mandals), 09 units have been categorized as 'Over-exploited', 02 units as 'Critical', 38 units as 'Semi-Critical', 591 units as 'Safe', and 39 units as 'Saline'. 300 villages are identified as Over Exploited Villages for the purpose of notification under APWALTA. The groundwater extraction for all uses increased slightly to 7,880 MCM in 2024, compared to 7,478 MCM in 2023. A detailed comparison of the stage of groundwater extraction was discussed, highlighting that Sri Satya Sai district has the highest extraction rate at 61%, while Alluri Sitharama Raju district has the lowest at 1%, with the state's average stage of extraction being 29.83%. Compared to the 2023 assessment, the total annual groundwater recharge for the state showed a negligible decrease from 27,830 MCM to 27,800 MCM, a difference of 30 MCM. This slight reduction is attributed to decreased recharge from sources such as canals, surface water irrigation, groundwater irrigation, tanks, ponds, and water conservation structures. Following the presentation, the State Level Committee (SLC) members engaged in discussions and sought clarifications on various aspects of the computations.

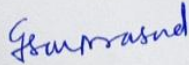
Sri G. Sai Prasad, IAS, Special Chief Secretary, Water Resources Department (WRD), Government of Andhra Pradesh and Chairman of the State Level Committee (SLC), appreciated the efforts of the officers from the AP Ground Water Department (GW&WAD) and the Central Ground Water Board (CGWB) for completing the assessment within the stipulated time frame. He advised the GW&WAD and CGWB scientists to prepare Groundwater recharge plans at aquifer level to improve the Groundwater recharge. Sri Venkateswara Rao, Engineer-in-Chief (Irrigation), interacted and shared his insights on the Groundwater recharge in different aquifers and aquifer disposition etc.

With this, the SLC approved the Ground Water Resource Assessment 2024 for Andhra Pradesh State.

The meeting ended with vote of thanks by Sri. T Raja Babu, Sc-D (GP) & Technical Secretary, CGWB, Southern Region, Hyderabad.


M. John Satya Raju
Director, Ground Water &
Water Audit Department, Govt.
of Andhra Pradesh


G. Krishnamurthy
Regional Director, Central Ground
Water Board, Southern Region,
Hyderabad
क्षेत्रीय निदेशक / Regional Director
केन्द्रीय भूमि जल बोर्ड / Central Ground Water Board
दक्षिणी क्षेत्र / Southern Region
हैदराबाद / Hyderabad-500 068


G. Sai Prasad, IAS
Special Chief Secretary, Water Resources Department, Govt. of AP
& Chairman, SLC

Chairman, SLC

Sri. G. Sai Prasad, IAS, Special Chief Secretary, WRD, Govt. of Andhra Pradesh

Members/Representatives of the Committee:

S/Sri/Smt./Kum.

S. No.	Name of the Member	Designation	Department
1.	Venkateswara Rao	Engineer-in-Chief, Irrigation	WRD
2.	M. John Satya Raju	Director	GW & WAD
3.	R. Sasidhar Reddy	JC	PR & RD
4.	Dr. D. Koteswara Rao	Professor & HOD	Civil Engg, JNTU Kakinada
5.	Dr. K Chandra Sekhar	Professor & HOD	Agronomy, ANGRAU, Guntur
6.	N. Samuel Raju	GM/ Co-ordination	AP Transco, Vidhut Saudha, Vijayawada
7.	P. Ravi Kumar	Dy EE/ Planning	AP Transco, Vidhut Saudha, Vijayawada
8.	Dr. M. J. Nandan	Chief Scientist	CSIR-NGRI, Hyderabad
9.	T. Vamsi Mohan	AD	Dept of Industries, Mangalagiri
10.	R. Kannan	Scientist	APSAC, Vijayawada
11.	K. Sowjanya	DD	DE & S
12.	M. Rambabu	AD	DE & S
13.	M. Lakshminarayana	MD	APSIDC
14.	G. Kajeswaramma	AD	Dept of Agriculture
15.	YVS Prasad	JC OSD/ APMIP	APMIP & Horticulture
16.	J. Gunakar Rao	SE	Hydrology
17.	Y. Srinivas	CE & SE	Minor Irrigation & CADA
18.	G Krishnamurthy	RD, SR & Member Secretary	CGWB, SR, Hyderabad

Other Participants:

S. No.	Name of the Member	Designation	Department
19.	N. Srinivasu	Joint Director	GW & WA
20.	T. Raja Babu	Scientist D	CGWB, SR Hyderabad
21.	Ravi Kumar Gumma	Scientist D	CGWB, CHQ Faridabad
22.	B. Gopalu	DD	GW & WA
23.	K. Lakshmi	DD	GW & WA
24.	P. Kodanda Rao	DD	GW & WA
25.	M. V Raja Kumar	DEE	Hydrology
26.	E. Sunil Kumar	AD	GW & WA
27.	M.V.Subba Rao	AD	GW & WA
28.	J. Bala Balaji	AEE	WRD
29.	L. Anusha	AHG	GW & WA
30.	K.Siva Vara Prasad	AE	GW & WA
31.	T. Madhav	Scientist-C	CGWB SR Hyderabad
32.	LN Damodara	Scientist-C & OIC	CGWB APSUO, Visakhapatnam
33.	D. Anantha Rao	AHG	CGWB APSUO Visakhapatnam
34.	Nilima Patra	AHG	CGWB APSUO Visakhapatnam
35.	S S Shravanthi	Young Professional	CGWB APSUO Visakhapatnam